Differential Geometry

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1.1 Topological manifolds

1.1.1 Definition

A **topological manifold of dimension** n (or an n**-manifold**) is a topological space M such that

(1) M is Hausdorff (every two points have disjoint open neighborhoods),

(2) M is second-countable (there is a countable basis for M),

(3) M is **locally Euclidean** of dimension n: every point is contained in a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

The empty set is a topological manifold of every dimension, but for the most part we will ignore this special case. It can be shown that the dimension of a topological manifold is well-defined: its dimension is unique (other than the empty space). This can be shown via standard tools in algebraic geometry.

The most basic example of an n-manifold is $\mathbb{R}^n$ itself: it is clearly locally Euclidean and Hausdorff, and the set of all balls with rational center and radius forms a countable basis.

1.1.2 Definition

Let M be an n-manifold. A **coordinate chart** on M is a pair (U, φ) with $U \subseteq M$ and $\varphi: U \to \mathbb{R}^n$ an embedding (equivalently $\varphi: U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism). U is the **coordinate domain** or **coordinate neighborhood** of the chart, φ is the **(local) coordinate map** the coordinates of φ (i.e. the composition with the projections $\mathbb{R}^n \to \mathbb{R}$) are called the **local coordinates** on U.

By the definition of a manifold, every point in M is contained in the domain of some chart. A chart (U, φ) is **centered** at $p \in U$ if $\varphi(u) = 0$. Clearly every point in M is the center of some chart.

1.1.3 Example

Let $U \subseteq \mathbb{R}^n$ be open and $f: U \to \mathbb{R}^k$ continuous. Define its **graph**:

$$\Gamma(f) = \left\{ (x, y) \in \mathbb{R}^n \times \mathbb{R}^k \;\middle|\; x \in U,\ f(x) = y \right\},$$

and endow it with the subspace topology of $\mathbb{R}^n \times \mathbb{R}^k$.

Now consider the projection $\pi_1: \mathbb{R}^{n+k} \to \mathbb{R}^n$ and its restriction to $\Gamma(f)$: $\varphi: \Gamma(f) \to U$. This is a continuous map, and it has a continuous inverse $\varphi^{-1}(x) = (x, f(x))$. Thus φ forms a homeomorphism between $\Gamma(f)$ and U, meaning $\Gamma(f)$ is an n-manifold.

1.1.4 Example

For an integer $n \geq 0$, consider the unit n-sphere S^n of all vectors $x \in \mathbb{R}^{n+1}$ with unit norm, endowed with the subspace topology. As a subspace of $\mathbb{R}^{n+1}$, it is Hausdorff and second-countable. It is also locally Euclidean of dimension n, and thus an n-manifold.

Indeed, define for each $i = 1, \ldots, n+1$:

$$U_i^+ = \left\{ (x^1, \ldots, x^{n+1}) \in \mathbb{R}^{n+1} \;\middle|\; x^i > 0 \right\}, \qquad U_i^- = \left\{ (x^1, \ldots, x^{n+1}) \in \mathbb{R}^{n+1} \;\middle|\; x^i < 0 \right\}$$

And consider $f: B^n \to \mathbb{R}$ by $f(u) = \sqrt{1 - |u|^2}$, where B^n is the unit ball on $\mathbb{R}^n$ (all vectors with norm < 1). Then $U_i^+ \cap S^n$ is the graph of the function

$$x^i = f(x^1, \ldots, \widehat{x}^i, \ldots, x^{n+1}),$$

where the hat means the indicated index is removed, viewed as a function $\mathbb{R}^n \to \mathbb{R}$. Similarly $U_i^- \cap S^n$ is the graph of $x^i = -f(x^1, \dots, \widehat{x}^i, \dots, x^{n+1})$.

Thus each $U_i^\pm \cap S^n$ is locally Euclidean of dimension n, as per our previous example. The maps $\varphi_i^\pm : U_i^\pm \cap S^n \to B^n$ from the example are given by

$$\varphi_i^\pm(x^1, \dots, x^{n+1}) = (x^1, \dots, \widehat{x}^i, \dots, x^{n+1}).$$

These are all local homeomorphisms to an open subset of $\mathbb{R}^n$; and furthermore every point in S^n is in some $U_i^\pm \cap S^n$, thus S^n is indeed locally Euclidean of dimension n.

1.1.5 Example

The n**-dimensional real projective space**, denoted $\mathbb{RP}^n$ is the quotient of $\mathbb{R}^{n+1} - 0$ by equating scalar multiples of vectors. Equivalently, it is the set of 1-dimensional subspaces of $\mathbb{R}^{n+1}$. Its quotient topology is given by the natural map $\pi : \mathbb{R}^{n+1} - 0 \to \mathbb{RP}^n$. Recall that this means the open sets are those $U \subseteq \mathbb{RP}^n$ for which $\pi^{-1}(U) \subseteq \mathbb{R}^{n+1} - 0$ is open (equivalently if $\pi^{-1}(U) \cup 0 \subseteq \mathbb{R}^{n+1}$ is open).

$\mathbb{RP}^n$ is an n-manifold. It can be shown to be second-countable (using any countable basis of $\mathbb{R}^{n+1}$) and Hausdorff. We show now local Euclidean-ness.

For $i = 1, \dots, n+1$ define $\widetilde{U}_i \subseteq \mathbb{R}^{n+1}$ to be the set of all vectors whose ith component is nonzero, and define $U_i = \pi(\widetilde{U}_i)$. Notice that $\pi^{-1}(U_i) = \widetilde{U}_i$, which is open, thus $U_i \subseteq \mathbb{RP}^n$ is open and $\pi|_{\widetilde{U}_i} : \widetilde{U}_i \to U_i$ is a quotient map. Now define $\varphi_i : U_i \to \mathbb{R}^n$ by

$$\varphi_i[\overline{x}] = \frac{1}{x^i}(x^1, \dots, \widehat{x}^i, \dots, x^n),$$

where $[\overline{x}]$ is the equivalence class of $\overline{x} \in \mathbb{R}^{n+1}$. φ_i is continuous since $\varphi_i \circ \pi$ is, and it is a quotient map.

φ_i has an inverse:

$$\varphi_i^{-1}(\overline{u}) = [u^1, \dots, u^{i-1}, 1, u^i, \dots, u^n],$$

and this map is continuous so $\varphi_i : U_i \to \mathbb{R}^n$ is a homeomorphism. And as before, $U_1, \dots, U_{n+1}$ cover $\mathbb{RP}^n$ so it is locally Euclidean of dimension n.

1.1.6 Proposition

The product of manifolds is a manifold. More specifically if $M_1, \dots, M_k$ are manifolds of dimension $n_1, \dots, n_k$ respectively, then their product space $M_1 \times \cdots \times M_k$ is a manifold of dimension $n_1 + \cdots + n_k$.

Proof

It is clearly Hausdorff and second countable. Let $\overline{p} \in \prod_1^k M_i$ and let (U_i, φ_i) be a coordinate chart for $p_i \in M_i$. These assimilate into a coordinate chart $(\prod_1^k U_i, \prod_1^k \varphi_i : \prod_1^k U_i \to \mathbb{R}^{n_1+\cdots+n_k})$. ∎

Recall that the n**-torus** is the n-fold product $T^n = S^1 \times \cdots \times S^1$, and by this proposition and the above example is an n-manifold.

We now turn to some topological properties of topological manifolds. Recall the following definitions:

1.1.7 Definition

A topological space X is

(1) **connected** if it cannot be written as the disjoint union of two non-empty open subsets (equivalently it has no non-trivial clopen sets);

(2) **path connected** if every two points in X can be connected by a path (a continuous $\gamma : [0, 1] \to X$ whose endpoints are the given points);

(3) **locally path connected** if X has a basis of path-connected open subsets.

Furthermore, X's **connected components** is the set of its maximally connected subsets. Its **path components** is the set of its maximally path connected subsets.

1.1.8 Proposition

Let M be a topological manifold, then

(1) M is locally path connected,

(2) M is connected iff it is path connected,

(3) The connected and path components of M coincide,

(4) M has countably many components, all of which is an open subset of M and a connected topological manifold.

Proof

Every point in M is contained in a coordinate ball (a coordinate chart whose image is a ball). Since balls are path-connected, the coordinate ball is as well. Thus we can form a basis of coordinate balls, which are all path connected.

(2) and (3) are true in any locally path connected space. Furthermore in any locally path connected space, its components are open. Thus the components form an open cover of M, and must have a countable subcover by second-countability. But components are disjoint, and thus form a minimal cover; the cover must be countable already. In general an open subset of a manifold is itself a manifold clearly. ■

1.1.9 Definition

A topological space X is **locally compact** if every point has a **compact neighborhood**. That is to say, for every $x \in X$ there is an open U and compact K such that $x \in U \subseteq K$.

A **precompact** set in X is a subset $Y \subseteq X$ whose closure is compact.

1.1.10 Lemma

Every topological manifold M has a countable basis of precompact coordinate balls.

Proof

First consider the case where M can be covered by a single chart, $\varphi\colon M \to \widehat{U} \subseteq \mathbb{R}^n$. Consider the set of all balls with rational center and radius $B_q(x)$ such that $B_{q'}(x) \subseteq \widehat{U}$ for some $q' > q$. These form a countable precompact basis of $\widehat{U}$ (they are precompact because their closure in $\mathbb{R}^n$ is contained in the closed ball of the same radius, which in turn is contained in $B_{q'}(x)$). φ lifts this precompact basis to a precompact basis of M, as desired.

M can be covered by countably many charts: take all the charts and then take a countable subcover, which exists by second-countability. Each of these charts U_i has a countable precompact basis, and if $V \subseteq U_i$ is precompact then it is precompact in M. Thus the countable union of these countable precompact bases is a countable precompact basis for M, as desired. ■

1.1.11 Corollary

A topological manifold is locally compact.

Proof

The basis of precompact coordinate balls gives the desired result: let $p \in U$ be a precompact coordinate ball, then $p \in U \subseteq \overline{U}$ is compact. ■

1.1.12 Definition

Let X be a topological space. Then a collection $\mathcal{X}$ of subsets of X is **locally finite** if every $x \in X$ has a neighborhood that intersects at most finitely many subsets in $\mathcal{X}$. Given a cover $\mathcal{U}$ of X, a **refinement** of $\mathcal{U}$ is another cover $\mathcal{V}$ such that every set in $\mathcal{V}$ is contained in some set in $\mathcal{U}$. X is **paracompact** if every open cover of X admits an open locally finite refinement.

If $\mathcal{X}$ is locally finite, then so is $\overline{\mathcal{X}} = \left\{ \overline{X} \mid X \in \mathcal{X} \right\}$ and we have

$$\overline{\bigcup \mathcal{X}} = \bigcup \overline{X}$$

1.1.13 Theorem

Topological manifolds are paracompact. Moreso, if $\mathcal{U}$ is an open cover of M and $\mathcal{B}$ is a basis, there is a locally finite open refinement of $\mathcal{U}$ consisting of elements from $\mathcal{B}$.

Proof

The above lemma tells us that M can be covered by countably many compact subsets. Recall that the union of finitely many compact subsets is compact, and so we can construct an ascending chain of compact subsets $(K_i)_i$ whose union is M. We can also require that $K_i \subseteq K_{i+1}^\circ$.

Define $V_i = K_{i+1} - K_i^\circ$ and $W_i = K_{i+2}^\circ - K_{i-1}$ for all i (let $K_i = \emptyset$ for negative i). Then V_i is compact, and contained in the open set W_i. For $x \in V_i$ there is some $X_x \in \mathcal{X}$ containing x, and since $\mathcal{B}$ is a basis there is a $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq X_x \cap W_i$. Consider then the set of all such B_x as x ranges over V_i, this forms an open cover of V_i and thus has a finite subcover.

The union of all these finite subcovers as i ranges over all integers forms a countable open cover of M that refines $\mathcal{X}$. The finite subcover of V_i consists of sets contained in W_i, and $W_i \cap W_{i'} = \emptyset$ except when $|j - j'| \leq 2$, thus this is a locally finite open cover of M refining $\mathcal{X}$. ■

1.2 Smooth structures

Thus far we have restricted our attention to *topological* manifolds. But in this course we want to do analysis on manifolds, so we must now consider smooth structures on manifolds.

Recall that a function $f\colon U \to V$ for $U \subseteq \mathbb{R}^n, V \subseteq \mathbb{R}^m$ is C^k (for a potentially infinite nonnegative integer k) if it is continuous, and each of its component functions can be continuously differentiated at least k times. That is to say, $f = (f_1, \ldots, f_m)$ and each of its partial derivatives

$$\frac{\partial^\alpha f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}$$

exist and are continious, for $\alpha = \alpha_1 + \cdots + \alpha_n \leq k$.

We are mostly concerned with C^∞ functions; those which can be continuously differentiated an arbitrary number of times. Such functions are also called **smooth**. A smooth function with a smooth inverse is called a **diffeomorphism**.

1.2.1 Definition

Let M be an n-manifold. Let $(U, \varphi), (V, \psi)$ be two non-disjoint charts. Consider then the map $\psi \circ \varphi^{-1} \colon \varphi(U \cap V) \to \psi(U \cap V)$, this is called the **transition map** from φ to ψ. As the composition of homeomorphisms, it itself is a homeomorphism. Two charts $(U, \varphi), (V, \psi)$ are **smoothly compatible** if they are either disjoint or their transition map is a diffeomorphism.

An **atlas** for M is a collection of charts whose domains cover M. A **smooth atlas** for M is an atlas whose charts are all smoothly compatible.

To show that an atlas is smooth, we need only show that each transition map is smooth (and not a diffeomorphism). This is because its inverse is a transition map itself: $(\psi \circ \varphi^{-1})^{-1} = \varphi \circ \psi^{-1}$.

1.2.2 Definition

Let M be a manifold, then a **smooth structure** on M is a maximal smooth atlas (i.e. one not contained in any other atlas). A **smooth manifold** is a pair $(M, \mathcal{A})$ where $\mathcal{A}$ is a smooth structure on a topological manifold M.

1.2.3 Proposition

Let M be a topological manifold.

(1) An atlas for M is contained in a unique maximal smooth atlas (smooth structure),

(2) Two smooth atlases for M are contained in the same smooth structure iff their union is a smooth atlas.

Proof

Let $\mathcal{A}$ be a smooth atlas for M, and define $\overline{\mathcal{A}}$ to be the set of all charts smoothly compatible with every chart in $\mathcal{A}$. We claim that $\overline{\mathcal{A}}$ is itself a smooth atlas; i.e. that every two charts in it are smoothly compatible.

Let $(U, \varphi), (V, \psi)$ be charts in $\overline{\mathcal{A}}$, and let $x = \varphi(p) \in \varphi(U \cap V)$. Then take some chart (W, θ) from $\mathcal{A}$ such that $p \in W$. Now (U, φ) and (V, ψ) are both smoothly compatible with (W, θ), so $\theta \circ \varphi^{-1}, \psi \circ \theta^{-1}$ are smooth when defined. In particular since $p \in U \cap V \cap W$, $\psi \circ \varphi^{-1}$ is smooth, as the composition of these two smooth functions, in a neighborhood of p. This is true for every $p \in U \cap V$, so $\psi \circ \varphi^{-1}$ is smooth as desired. Thus $\overline{\mathcal{A}}$ is indeed a smooth atlas. It is also clearly maximal: any chart not in it by definition is not smoothly compatible with some chart in $\mathcal{A} \subseteq \overline{\mathcal{A}}$.

If $\mathcal{B}$ is any other smooth structure containing $\mathcal{A}$, then its charts are smoothly compatible with $\mathcal{A}$ and thus by definition $\mathcal{B} \subseteq \overline{\mathcal{A}}$. By maximality, this is equality. So we have shown the first point.

The second point is clear: if $\mathcal{A}, \mathcal{B}$ are contained in the same smooth structure, then their union is as well and thus must be a smooth atlas. Conversely if $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas, it is contained in a unique smooth structure, and this smooth structure contains $\mathcal{A}$ and $\mathcal{B}$. ■

1.2.4 Definition

Let $\mathcal{A}$ be a smooth atlas for M, and $\mathcal{B}$ a smooth atlas for N. A map $f \colon M \to N$ is **smooth** if for every chart $p \in M$ there are charts $(U, \varphi) \in \mathcal{A}, (V, \psi) \in \mathcal{B}$ with $p \in U$ and $f(U) \subseteq V$ such that the composite $\psi \circ f \circ \varphi^{-1} \colon \varphi(U) \to \psi(V)$ is smooth. A bijection f is a **diffeomorphism** if it is smooth and has a smooth inverse.

Note that this definition should seemingly talk about smooth structures, and not just smooth atlases. But this is unnecessary:

1.2.5 Proposition

$f\colon (M,\mathcal{A}) \to (N,\mathcal{B})$ is smooth iff $f\colon (M,\overline{\mathcal{A}}) \to (N,\overline{\mathcal{B}})$ is.

Proof

Clearly the right implies the left. Now let (U,φ) be a chart compatible with $\mathcal{A}$ and (V,ψ) compatible with $\mathcal{B}$. For $p \in \varphi(U)$ let (W,θ) be a chart in $\mathcal{A}$ with p in its domain, and (W',θ') a chart in $\mathcal{B}$ with $f(p)$ in its domain. Then we can write

$$\psi \circ f \circ \varphi^{-1} = (\psi \circ \theta'^{-1}) \circ (\theta' \circ f \circ \theta^{-1}) \circ (\theta \circ \varphi^{-1}),$$

a composition of smooth maps, and is thus smooth. So $\psi \circ f \circ \varphi^{-1}$ is smooth on a neighborhood of any element in its domain, thus smooth as desired. ∎

Thus we can characterize smoothness by any compatible smooth atlas. An immediate result is the following (expanding definitions):

1.2.6 Lemma

Two smooth atlases $\mathcal{A}, \mathcal{B}$ for M are smoothly compatible iff the identity $(M,\mathcal{A}) \to (M,\mathcal{B})$ is a diffeomorphism.

Note that two atlases may give diffeomorphic manifolds without them being smoothly compatible. Necessarily the diffeomorphism must be different from the identity.

So instead of talking about maximal smooth atlases, we can concentrate on ordinary smooth atlases, while keeping in mind that distinct smooth atlases can give the same smooth structure.

1.2.7 Example

Clearly Euclidean space $\mathbb{R}^n$ can be given a natural smooth structure: the one induced by the atlas $\mathrm{id}_{\mathbb{R}^n}$. This is called the **standard smooth structure**. These are not the only smooth structures possible: $\varphi(x) = x^3$ is a homeomorphism $\mathbb{R} \to \mathbb{R}$ and thus defines a smooth atlas $(\mathbb{R},\varphi)$. Since φ is not a diffeomorphism, it is not smoothly compatible with the standard structure. Despite this $(\mathbb{R},\varphi)$ is diffeomorphic to the standard structure, just via a non-identity map.

1.2.8 Example

Let V be a finite-dimensional real vector space, so that it is isomorphic to $\mathbb{R}^n$. Any two norms on V are equivalent, i.e. they induce the same topology. Thus V can be given a smooth structure by lifting $\mathbb{R}^n$'s, and this agrees with any norm-induced topology.

For example, the space of matrices $M_{n\times m}(\mathbb{R})$ has a natural smooth structure. Similarly for $M_{n\times m}(k)$ for any finite-dimensional $\mathbb{R}$-algebra k (e.g. the complex numbers or quaternions).

1.2.9 Example

Let M be a manifold and $U \subseteq M$ be open. Then for an atlas $\mathcal{A}$ of M, we can restrict it to U by

$$\mathcal{A}_U = \{(U \cap V, \varphi|_{U\cap V}) \mid (V,\varphi) \in \mathcal{A}\}.$$

If $\mathcal{A}$ is a smooth structure, this is just all of the charts $(V,\varphi) \in \mathcal{A}$ such that $V \subseteq U$. This is called an **open submanifold** of M.

1.2.10 Example

Consider the general linear group $\mathrm{GL}_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$. This is the preimage of $\mathbb{R} - 0$ under det, and is thus an open subset of $M_n(\mathbb{R})$. As discussed, this means that $\mathrm{GL}_n(\mathbb{R})$ is an open submanifold of $M_n(\mathbb{R})$.

More generally, consider the subspace of $M_{n\times m}(\mathbb{R})$ of matrices of full rank. This is also an open subspace: if A has full rank, then consider its submatrix given by its linearly independent columns (or rows). We can then define a continuous map $\det\colon M_{n\times m}(\mathbb{R}) \to \mathbb{R}$ by mapping a matrix to the determinant of this square matrix. Then the inverse image of $\mathbb{R} - 0$ is an open neighborhood of A, and all matrices in it are of full rank. So the set of matrices of full rank is indeed an open submanifold.

1.2.11 Example

We showed previously that S^n is a topological n-manifold with charts $(U_i^{\pm}, \varphi_i^{\pm})$ for $i = 1, \dots, n+1$ where $\varphi_i^{\pm}$ removes the ith coordinate, and its inverse maps $\overline{x}$ to the coordinate with $\sqrt{1 - |x|^2}$. Thus for $i \neq j$, the transition $\varphi_i^{\pm} \circ (\varphi_j^{\pm})^{-1}$ maps x to the coordinate with x_i removed and $\pm\sqrt{1-|x|^2}$ in the jth coordinate. This is smooth. For $i = j$ we just get the identity, which is smooth.

Thus S^n is a smooth n-manifold.

1.2.12 Example

Similar to before the n-dimensional topological manifold $\mathbb{R}\mathrm{P}^n$ is smooth. Its atlas is given by maps (U_i, φ_i) where $U_i \subseteq \mathbb{R}\mathrm{P}^n$ is the set of all $[x]$ where $x_i \neq 0$ and $\varphi_i[x]$ is $1/x^i$ times the vector x with its ith coordinate removed. Then $\varphi_j \circ \varphi_i^{-1}$ maps x to $1/x^j$ times the vector obtained by removing the jth coordinate of x and adding 1 to its ith coordinate. This is smooth.

Recall that if $M_1, \dots, M_n$ were topological manifolds with charts (U_i, φ_i), then $M_1 \times \cdots \times M_n$ was a topological manifold with charts $(U_1 \times \cdots \times U_n, \varphi_1 \times \cdots \varphi_n)$. If these are smooth atlases, then these charts are smoothly compatible, since

$$(\psi_1 \times \cdots \times \psi_n) \circ (\varphi_1 \times \cdots \times \varphi_n)^{-1} = (\psi_1 \circ \varphi_1^{-1}) \times \cdots \times (\psi_n \circ \varphi_n^{-1})$$

which is smooth. So we have proven the following.

1.2.13 Proposition

The finite product of smooth manifolds is a smooth manifold, whose dimension is the sum of the dimensions of the manifolds.

For example, the n-torus is a smooth manifold of dimension n.

1.2.14 Lemma

Let M be a set and suppose we have a collection of subsets $\{U_i\}_{i\in I}$ with maps $\varphi_i\colon U_i \to \mathbb{R}^n$ for some n. Further suppose the following:

(1) for each i, $\varphi_i(U_i)$ is open and φ_i is injective;

(2) for each i, j, $\varphi_i(U_i \cap U_j)$ and $\varphi_j(U_i \cap U_j)$ are open in $\mathbb{R}^n$;

(3) for $\varphi_i \cap \varphi_j^{-1}\colon \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is smooth;

(4) $\{U_i\}_i$ has a countable subcover;

(5) whenever $p \neq q \in M$ are distinct, there is either a U_i containing them both, or disjoint U_i, U_j such that $p \in U_i$ and $q \in U_j$,

then M has a unique smooth manifold structure such that (U_i, φ_i) are all smooth charts.

Proof

We first need to give M a topology: take $\varphi_i^{-1}(V)$ to be a basis for $V \subseteq \mathbb{R}^n$ open. In particular $U_i = \varphi_i^{-1}(\mathbb{R}^n)$ is open. To show that this is a basis, we need to show that the intersection $\varphi_i^{-1}(V_i) \cap \varphi_j^{-1}(V_j)$ can be written as a union of base sets:

$$\varphi_i^{-1}(V_i) \cap \varphi_j^{-1}(V_j) = \varphi_i^{-1}(V_i \cap (\varphi_j \cap \varphi_i)^{-1}(V_j)),$$

which is a base set, since $(\varphi_j \cap \varphi_i)^{-1}(V_j)$ is an open subset of the open subset $\varphi_i(U_i \cap U_j)$ and is thus open. Note that this is the unique topology making each φ_i a homeomorphism; thus far we have made no choices.

Then M is Hausdorff: for $p \neq q$ either they are in the same U_i which is Hausdorff (as it is Euclidean), or in distinct U_i, U_j which are open. And M is second countable: as U_i are Euclidean, they are second countable, and M is thus second countable as the union of countably many U_is.

So we have shown that M is a topological n-manifold. And then (U_i, φ_i) are all smoothly compatible charts because we assumed the transition maps were smooth. ■

Note:

We could state a similar result for topological manifolds, whose proof is the same as above. Instead of assuming the transition maps are smooth, we relax this condition to just require they are continuous. Then M has a unique topological manifold structure making (U_i, φ_i) all charts.

1.2.15 Example

Let K be a finite-dimensional real vector space of dimension n. For $0 \leq k \leq n$, define $G_k(V)$ to be the set of all k-dimensional subspaces of V. Using the abovbe lemma, we can turn this into a smooth manifold called the **Grassmanian manifold**. For $U \leq V$ of dimension k, let W be a complementary subspace such that $V = U \oplus W$. Then define $\Gamma_{U,W}\colon \hom(U, W) \to G_k(V)$ by $\Gamma_{U,W}(L) = \{u + Lu \mid u \in U\}$.

Then $(\Gamma_{U,W} \hom(U, W), \Gamma_{U,W}^{-1})$ satisfy the conditions above, turning $G_k(V)$ into a $k(n-k)$-dimensional smooth manifold (since $\hom(U, W)$ has dimension $k(n-k)$). The proof of this is not trivial, and can be found in my homework solutions.

For $k = 1$, notice that $G_1(\mathbb{R}^{n+1}) \cong \mathbb{RP}^n$.

1.3 Manifolds with boundary

Define the **closed n-dimensional upper half-space** $\mathbb{H}^n \subseteq \mathbb{R}^n$ as the set of all points whose nth coordinate is nonnegative:

$$\mathbb{H}^n = \{\overline{x} \in \mathbb{R}^n \mid x_n \geq 0\}.$$

Its interior and boundary are

$$\mathrm{Int}\mathbb{H}^n = \{\overline{x} \in \mathbb{R}^n \mid x_n > 0\}, \qquad \partial\mathbb{H}^n = \{\overline{x} \in \mathbb{R}^n \mid x_n = 0\}.$$

1.3.1 Definition

An n-dimensional **(topological) manifold with boundary** is a second-countable Hausdorff space M such that every point in M lies in a neighborhood homeomorphic to an open subset of $\mathbb{H}^n$.

A **chart** for M is a pair (U, φ) for $U \subseteq M$ open and $\varphi\colon U \to \varphi(U)$ a homeomorphism to an open subset $\varphi(U)$ of $\mathbb{H}^n$. Call a chart a **boundary chart** if $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$, and an **interior chart** otherwise (if its image lies in the interior of $\mathbb{H}^n$). An **atlas** for M is a collection of charts whose domains cover M.

Call $p \in M$ a **boundary point** if it lies in the domain of some boundary chart (U, φ) such that $\varphi(p) \in \partial\mathbb{H}^n$, and an **interior point** if it lies in the domain of some interior chart. Define the set of boundary and interior points to be

$$\partial M \text{ and } \mathrm{Int}M$$

respectively.

Open subsets of $\mathbb{H}^n$ are of the form $\mathbb{H}^n \cap U$ for an open subset $U \subseteq \mathbb{R}^n$. In particular we can construct a basis of $\mathbb{H}^n$ by $\mathbb{H}^n \cap B_r(x)$. We can refine this basis by considering only balls of the form $B_r(x)$ for $B_r(x) \subseteq \mathbb{H}^n$ and $B_r(x) \cap \mathbb{H}^n$ for $x \in \partial\mathbb{H}^n$, called **half-balls**. If (U, φ) is a chart with $\varphi(U)$ a half-ball, call U a **coordinate half-ball**.

Note that every point is either a boundary point or an interior point: suppose $p \in M$ is neither on the boundary or interior. So it must be in the domain of a boundary chart (U, φ), but $\varphi(p) \notin \partial\mathbb{H}^n$. But then we can take a ball around $\varphi(p)$ so that it doesn't intersect the boundary of $\mathbb{H}^n$, and lift it to M; this gives an interior chart with p in its domain, so p is an interior point.

Note that an ordinary manifold is also a manifold with boundary, but without boundary points. This is because open subsets in $\mathbb{R}^n$ are all homeomorphic to open subsets of $\mathbb{H}^n$: for example a ball $B_r(x) \subseteq \mathbb{R}^n$ is homeomorphic to the ball $B_r(x') \subseteq \mathrm{Int}\mathbb{H}^n$ where x' is just x but the last coordinate is raised so that $B_r(x')$ lies within the interior of the half-space. Thus we can view manifolds with boundaries as a generalization of ordinary manifolds. We will still say "manifolds with or without boundary" when discussing manifolds with boundary to emphasize the point that a result holds for ordinary manifolds as well.

As it turns out, the boundary and interior of a manifold are topological invariants.

1.3.2 Theorem

If M is a topological manifold with boundary, then every point is an interior or boundary point, but not poth. That is,

$$M = \partial M \sqcup \mathrm{Int}M.$$

This can be proven using standard tools from algebraic topology.

Note that the concept of a manifold boundary and interior is distinct (yet related) to the notion of a topological boundary and interior. The notion of a topological boundary and interior is *relative*: it is defined relative to a larger topological space. While manifold boundaries and interior are not. To see how this can manifest, consider the manifold $\mathbb{R}$. It has an empty boundary as a manifold, and also as a topological space (as is true for all topological spaces, relative to themselves). But viewed as a topological subspace of $\mathbb{R}^2$ say, its boundary is itself.

Thus it is important to distinguish manifold and topological boundaries and interiors.

1.3.3 Proposition

Let M be a topological n-manifold with boundary.

(1) $\mathrm{Int}M$ is an open subset of M and a topological n-manifold, without boundary.

(2) ∂M is a closed subset of M, and a topological $(n-1)$-manifold, without boundary.

(3) M is a topological manifold iff ∂M is empty.

Proof

For $p \in \mathrm{Int}M$, the domain of its interior chart also lies in $\mathrm{Int}M$ by definition, so $\mathrm{Int}M$ must be open in M. It is clearly a topological n-manifold, as its charts are to $\mathbb{R}^n$.

Since ∂M is the complement of the open $\mathrm{Int}M$, it is closed. For $p \in \partial M$ consider its boundary chart mapping p to $\partial\mathbb{H}^n$, (U, φ). Then the restriction of φ to $U \cap \partial M$ is a homeomorphism to an open subset of $\partial\mathbb{H}^n \cong \mathbb{R}^{n-1}$. Indeed, φ must map boundary points to boundary points, otherwise we could lift an open ball to show that a boundary point is also an interior point. So ∂M is locally Euclidean of dimension $n-1$, as desired.

Point (3) follows directly from the invariance of the boundary and point (1). ■

$\mathrm{Int}M$ being a submanifold of M is true for all open subsets.

1.3.4 Proposition

Let M be a topological n-manifold with boundary, and $U \subseteq M$ open. Then U is a topological n-manifold with boundary $\partial U = U \cap \partial M$.

Proof

For $p \in U$ consider its chart in M and restrict it to U. $p \in U$ is then a boundary point in U iff it is a boundary point in M. ■

1.3.5 Proposition

Let M be a topological n-manifold with boundary. Then

(1) M has a countable basis of precompact coordinate balls and half-balls.

(2) M is locally compact.

(3) M is paracompact.

(4) M is locally path-connected.

(5) M has countably many components, each of which is an open subset of M

Recall that a function $f\colon A \to \mathbb{R}^m$ for $A \subseteq \mathbb{R}^n$ arbitrary is **smooth** if A has an open neighborhood U such that f can be extended to $U \to \mathbb{R}^m$ and this is smooth. In particular if $U \subseteq \mathbb{H}^n$ is open, then $f\colon U \to \mathbb{R}^m$ is smooth if there is an open subset $\widetilde{U} \subseteq \mathbb{R}^n$ such that $U = \widetilde{U} \cap \mathbb{H}^n$ and a lift $\widetilde{f}\colon W \to \mathbb{R}^m$ of f, which is smooth. Notice that if f is smooth, then f's restriction to $U \cap \operatorname{Int} M$ is smooth in the usual sense.

So say we have a topological manifold with boundary M, and let $(U, \varphi), (V, \psi)$ be charts. Then we have the transition map $\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \to \psi(U \cap V)$, which is a homeomorphism. To say that this is smooth is to say $\varphi(U \cap V)$ has an open neighborhood $W \subseteq \mathbb{R}^n$ to which we can extend $\psi \circ \varphi^{-1}$, and this lift is smooth.

1.3.6 Definition

Let M be a manifold with boundary. Then an **atlas** for M is a set of charts (U_i, φ_i) whose domain covers M. A **smooth atlas** for M is an atlas for which all the transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth in the sense explained above. A **smooth structure** for M is a maximally smooth atlas. A **smooth manifold with boundary** is a manifold M along with a choice of a smooth structure $\mathcal{A}$.

As before, a smooth atlas uniquely determines a unique smooth structure.

1.3.7 Definition

Let M, N be manifolds with boundary. A map $F\colon M \to N$ is **smooth** if for every point $p \in M$, there are smooth charts $(U, \varphi) \in M$ and $(V, \psi) \in N$ such that $p \in U$ and $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1} \circ \varphi(U) \to \psi(V)$ is smooth.

A **diffeomorphism** is a smooth map with a smooth inverse.

As before, a function is smooth relative to a smooth atlas iff it is smooth relative to the smooth structure induced by that atlas. And two smooth atlases define the same smooth structure iff the identity is a diffeomorphism with respect to them.

1.3.8 Proposition

If M is a smooth manifold with boundary, and $U \subseteq M$ is open, then it is also a smooth manifold with boundary.

Proof

Consider the construction above showing U is a topological manifold with boundary, the smoothness of the chart given is evident. ■

Note:
Consider our equivalent definition of a smooth manifold via a covering of a set by injections into $\mathbb{R}^n$. If we replace $\mathbb{R}^n$ with $\mathbb{H}^n$, then this gives an equivalent definition of smooth manifolds with boundary.

Interestingly, the product of manifolds with boundary is not necessarily a manifold with boundary. This is because the product of half-spaces is not a half-space. But we can take the product of a manifold with boundary with a manifold (without boundary).

1.3.9 Proposition

Let $M_1, \dots, M_n$ be manifolds (without boundary), and N a manifold with boundary. Then their direct product $M_1 \times \cdots \times M_n \times N$ is a manifold with boundary

$$\partial(M_1 \times \cdots \times M_n \times N) = M_1 \times \cdots \times M_n \times \partial N.$$

2 Smooth Maps

2.1 Smooth maps

2.1.1 Definition

For two smooth manifolds M, N, we define $C^\infty(M, N)$ to be the collection of smooth maps $M \to N$. In the case of $N = \mathbb{R}$, we just denote $C^\infty(M, \mathbb{R})$ by $C^\infty(M)$.

2.1.2 Proposition

A smooth map between manifolds with or without boundary is continuous.

Proof

Let $F\colon M \to N$ be smooth, then for $p \in M$ take smooth charts (U, φ) and (V, ψ) for M, N respectively such that $p \in U$ and $F(U) \subseteq V$. Then $\psi \circ F \circ \varphi^{-1}\colon \varphi(U) \to \psi(V)$ is smooth, and thus $F|_U$ is continuous as the composition

$$\psi^{-1} \circ (\psi \circ F \circ \varphi^{-1}) \circ \varphi.$$

Thus F is continuous at p for all points $p \in M$, as desired. ■

It is easy to see that a map $F\colon M \to N$ is smooth iff it is continuous and $\psi \circ F \circ \varphi^{-1}$ is smooth for all coordinate charts.

2.1.3 Proposition

Let $F\colon M \to N$ be a map, then

(1) if every point has a neighborhood in which F is smooth, then F is smooth.

(2) if F is smooth, its restriction to any open subset is smooth.

This proposition has the useful corollary:

2.1.4 Corollary

Let $\{U_i\}_i$ be an open cover of M, and let $F_i\colon U_i \to N$ be a collection of smooth maps. If the maps agree on overlaps: $F_i|_{U_i \cap U_j} = F_j|_{U_i \cap U_j}$, then there exists a unique extension of all these maps to a smooth map $F\colon M \to N$.

2.1.5 Proposition

The composition of smooth maps is smooth.

Proof

If $F\colon M \to N$ and $G\colon N \to P$ are smooth, then

$$\theta \circ (G \circ F) \circ \varphi^{-1} = (\theta \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}),$$

the composition of the relevant functions, is smooth. ■

Thus the category of smooth manifolds with smooth maps between them forms a category (the identity is clearly smooth), and the hom-sets are $C^\infty(M, N)$. The isomorphisms in this category are the diffeomorphisms. Other maps are the constant maps: for $c \in N$ the map $M \to N$ mapping every point to c, and the inclusion of an open submanifold in a larger manifold.

The sum of vector-valued smooth functions is also smooth, i.e. $C^\infty(M, \mathbb{R}^n)$ is a real vector space. When $n = 1$ we can also multiply (pointwise) these smooth maps, so that $C^\infty(M)$ becomes a real algebra.

2.1.6 Proposition

The product of manifolds (of which at most one has boundary) defines a product in the category of smooth manifolds, with the usual projection maps.

2.1.7 Example

Consider the map $\varepsilon\colon \mathbb{R} \to S^1$ given by $\varepsilon(t) = e^{2\pi i t}$. This is a smooth map.

Thus $\varepsilon^n\colon \mathbb{R} \to T^n$ given by $\varepsilon^n(t) = (\varepsilon(t), \ldots, \varepsilon(t))$ is smooth as well. This is because T^n is the product of S^1s, and each of the projections of ε^n is smooth.

2.1.8 Example

The inclusion S^n in $\mathbb{R}^{n+1}$ is smooth, as is the quotient map $\mathbb{R}^{n+1} - 0 \to \mathbb{R}\mathrm{P}^n$. Thus their composition $S^n \to \mathbb{R}\mathrm{P}^n$ is a smooth bijection (but not a diffeomorphism, the spaces are not even homeomorphic).

Diffeomorphisms inherit the usual properties of isomorphisms (compositions of diffeomorphisms are diffeomorphisms, the product of diffeomorphisms is a diffeomorphism, etc). If $F\colon M \to N$ is a diffeomorphism, and $U \subseteq M$ is an open submanifold, then the restriction of F to $U \to F(U)$ is a diffeomorphism.

2.2 Partitions of unity

2.2.1 Lemma

Define $f\colon \mathbb{R} \to \mathbb{R}$ by

$$f(t) = \exp(-1/t), \text{ for } t > 0 \text{ and } 0 \text{ otherwise},$$

then f is smooth.

Proof

Clearly f is smooth on $\mathbb{R} - 0$, so we need only show that it has continuous derivatives at the origin. It is sufficient to show it derivatives of all order. We show by induction that the kth derivative of f at $t > 0$ is

$$f^{(k)}(t) = p_k(t)\frac{\exp(-1/t)}{t^{2k}},$$

for some polynomial p_k. This is clearly true for $k = 0$ with $p_0 = 1$. Then

$$f^{(k+1)}(t) = p_k'(t)\frac{\exp(-1/t)}{t^{2k}} + p_k(t)\frac{t^{-2}\exp(-1/t)}{t^{2k}} - 2kp_k(t)\frac{\exp(-1/t)}{t^{2k+1}}.$$

Simplifying gives us the desired result with $p_{k+1}(t) = t^2p_k'(t) + p_k(t) - 2ktp_k(t)$.
Now we show that $f^{(k)}(0) = 0$ for all $k \geq 0$. We show this by induction; for $k = 0$ this is clearly true. To show that it is true for $k + 1$ we show that $f^{(k)}$ has both one-sided derivatives at 0 and they coincide. Clearly the left derivative is zero. The right derivative is

$$\lim_{t \to 0^+} \frac{p_k(t)\frac{\exp(-1/t)}{t^{2k}}}{t} = p_k(0) \cdot \lim_{t \to 0^+} \frac{\exp(-1/t)}{t^{2k+1}} = 0,$$

as desired. Thus $f^{(k)}$ is differentiable, as desired. ■

2.2.2 Lemma

Let $r_1 < r_2$ be two real numbers. Then there is a smooth function $h\colon \mathbb{R} \to [0, 1]$ such that $h(t) = 1$ for $t \leq r_1$ and $h(t) = 0$ for $t \geq r_2$.

Proof

Take f as in the previous lemma and define

$$h(t) = \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}$$ ■

Such a function is often called a **cutoff function**.

2.2.3 Lemma

Let $r_1 < r_2$ be real numbers. Then there is a smooth function $H\colon \mathbb{R}^n \to [0, 1]$ such that $H(x) = 1$ on

$\overline{B}_{r_1}(0)$, and $H(x) = 0$ on $\mathbb{R} - B_{r_2}(0)$.

Proof

Define $H(x) = h(|x|)$. ■

Such a function is called a **smooth bump function**.

2.2.4 Definition

Let X be a topological space and $f\colon X \to \mathbb{R}^n$ be a function. Its **support** is defined to be

$$\operatorname{supp} f = \overline{\{x \in X \mid f(x) \neq 0\}},$$

the closure of the preimage of $\mathbb{R}^n - 0$. If $\operatorname{supp} f \subseteq A$, then we say f is **supported** in A. f is **compactly supported** if $\operatorname{supp} f$ is compact.

Note every (vector-valued) function on a compact space is compactly supported.

2.2.5 Definition

Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover for a topological space X. A **partition of unity** subordinate to $\mathcal{U}$ is a collection $\{\varphi_i\}_{i \in I}$ of continuous functions $\varphi_i\colon X \to [0,1]$, such that

(1) $\operatorname{supp} \varphi_i \subseteq U_i$ for all $i \in I$;

(2) the set $\{\operatorname{supp} \varphi_i\}_i$ is **locally finite**: every $x \in X$ has a neighborhood intersecting only finitely many of the sets $\operatorname{supp} \varphi_i$;

(3) $\sum_i \varphi_i(x) = 1$ for all $x \in X$.

Note that the last condition is well-defined, since only finitely many $\varphi_i(x)$ are non-zero by the second condition.

2.2.6 Theorem

Suppose M is a smooth manifold with or without boundary, and $\mathcal{U} = \{U_i\}_i$ an open cover of M. Then there is a partition of unity subordinate to $\mathcal{U}$.

Proof

We show for the case that M has no boundary. Consider for each U_i a basis of regular coordinate balls: these are coordinate balls B such that $B \subseteq B'$ another coordinate ball with coordinate map φ such that $\varphi(B) = B_r$, $\varphi(\overline{B}) = \overline{B}_r$, and $\varphi(B') = B_{r'}$ for $r < r'$. For a regular coordinate ball B_i, let (B_i', φ_i) be this extension. Define $f_i\colon M \to R$ by

$$f_i = \begin{cases} H_i \circ \varphi_i & \text{on } B_i' \\ 0 & \text{on } M - \overline{B}_i \end{cases},$$

where H_i is the smooth bump function positive in B_{r_i} and zero elsewhere, as above. $H_i \circ \varphi_i$ is zero on $B_i' - \overline{B}_i$, so this is well-defined and smooth. Note that $\operatorname{supp} f_i = \overline{B}_i$.

Define $f\colon M \to R$ by $f(x) = \sum_i f_i(x)$ which is well-defined since $\overline{B}_i$ is locally finite, and thus also smooth. $f(x)$ is nonnegative and positive on B_i, which cover M, so $f(x)$ is positive. Thus define $g_i(x) = f_i(x)/f(x)$; this is smooth. Immediately we have that $g_i(x) \in [0,1]$ and $\sum_i g_i(x) = 1$.

These g_is have support $\overline{B}_i$, and these form a locally finite cover of the corresponding U_α. For each i, choose an α_i such that $\overline{B}_i \subseteq U_{\alpha_i}$, and then let φ_α be the sum of all the g_is where $\alpha_i = \alpha$. By local finiteness, we have that φ_α's support is

$$\mathrm{supp}\varphi_\alpha = \overline{\bigcup_{\alpha_i=\alpha} B_i} = \bigcup_{\alpha_i=\alpha} \overline{B}_i \subseteq U_\alpha,$$

as desired. The sum over φ_α is just the sum over g_is, so 1, as desired. ■

2.2.7 Definition

Let X be a topological space, $A \subseteq X$ a closed subset, and $A \subseteq U$ open. Then a continuous $\varphi\colon X \to [0,1]$ is a **bump function for A supported in U** if $\varphi = 1$ on A and $\mathrm{supp}\varphi \subseteq U$.

2.2.8 Corollary

Let M be a smooth manifold with or without boundary. Then for every closed $A \subseteq M$ and open $A \subseteq U$, there is a bump function for A supported in U.

Proof

Let $\{\varphi_0, \varphi_1\}$ be a partition of unity subordinate to $\{U, M - A\}$. Then φ_0 has the desired property: $\mathrm{supp}\varphi_0 \subseteq U$ and since $\varphi_0 + \varphi_1 = 1$ and $\varphi_1 = 0$ on A, this means $\varphi_0 = 1$ on A. ■

Let M, N be manifolds and $A \subseteq M$ be arbitrary. A map $F\colon A \to N$ is called **smooth** similar to before: if for every $p \in A$ there is a neighborhood $p \in U$ and a smooth function $\widetilde{F}\colon W \to N$ agreeing with F on $W \cap A$.

2.2.9 Lemma

Let M be a smooth manifold with or without boundary, $A \subseteq M$ closed, and $f\colon A \to \mathbb{R}^k$ smooth. For any open subset U containing A, there is a smooth function $\widetilde{f}\colon M \to \mathbb{R}^k$ restricting to f on A such that $\mathrm{supp}\widetilde{f} \subseteq U$.

Proof

For $p \in A$ let W_p be a neighborhood of p and $\widetilde{f}_p\colon W_p \to \mathbb{R}^k$ restricting to f. Replacing W_p with $W_p \cap U$, we may assume that $W_p \subseteq U$. Then $\{W_p \mid p \in A\} \cup \{M - A\}$ is an open cover of M and thus has a smooth partition of unity $\{\psi_p\}_{p \in A} \cup \{\psi_0\}$ which is subordinate to it: $\mathrm{supp}\psi_p \subseteq W_p$ and $\mathrm{supp}\psi_0 \subseteq M - A$.

We can extend $\psi_p \widetilde{f}_p$ to all of M by defining it to be 0 on $M - \mathrm{supp}\psi_p$. This is well-defined since $\widetilde{f}_p$ is zero in the overlap of $M - \mathrm{supp}\psi_p$ and W_p, thus it is also smooth. Now define

$$\widetilde{f}(x) = \sum_{p \in A} \psi_p(x) \widetilde{f}_p(x),$$

this is well-defined by local finiteness, and is thus smooth.
If $x \in A$ then $\psi_0(x) = 0$ and $\widetilde{f}_p(x) = f(x)$ for every p such that $\psi_p(x) \neq 0$ (since then $x \in W_p \cap A$, in which $\widetilde{f}_p$ agrees with f). Thus,

$$\widetilde{f}(x) = \sum_{p \in A} \psi_p(x) \widetilde{f}_p(x) = \left(\psi_0(x) + \sum_{p \in A} \psi_p(x) \right) f(x) = f(x)$$

as desired.

And furthermore
$$\operatorname{supp}\widetilde{f} = \overline{\bigcup_{p\in A} \operatorname{supp}\psi_p} = \bigcup_{p\in A} \overline{\operatorname{supp}\psi_p} \subseteq U,$$
as desired. ■

Another application of partitions of unity is the following.

2.2.10 Definition

Let X be a topological space. Then an **exhaustion function** is a continuous map $f\colon X \to \mathbb{R}$ such that $f^{-1}(-\infty, c]$ is compact in X for every $c \in \mathbb{R}$.

We can show (via a partition of unity, shown in homework) that every smooth manifold with or without boundary admits a smooth nonnegative exhaustion function.

2.3 Einstein summation notation

Sums of the form $\sum_i a^i x_i$ are abundant in differential geometry. Thus common notation, introduced by Einstein, is to write such sums as $a^i x_i$, omitting the summation symbol and leaving the bounds implicit (or explained explicitly). The general rule is this: if in an expression the same index name (e.g. i) occurs twice in any monomial term, once as an upper index and once as a lower index, then that expression is to be taken as implicitly summed over all relevant values of i.

3 Tangent Vectors

3.1 Geometric tangent vectors

3.1.1 Definition

For $a \in \mathbb{R}^n$ define the **geometric tangent space** at a to be $\mathbb{R}^n_a = \{a\} \times \mathbb{R}^n$, endowed with the vector-space structure induced by the obvious bijection $\mathbb{R}^n_a \cong \mathbb{R}^n$. We denote a vector $(a, v) \in \mathbb{R}^n_a$ by v_a or $v|_a$.

For a geometric tangent vector v_a, we can define the **directional derivative** at a in the direction v:

$$D_v|_a\colon C^\infty(\mathbb{R}^n) \to \mathbb{R}, \qquad D_v|_a f = D_v f(a) = \left.\frac{d}{dt}\right|_{t=0} f(a + tv).$$

$D_v|_a$ is a linear map clearly, and it is also a **derivation**: by the product rule,

$$D_v|_a(fg) = f(a)D_v|_a(g) + g(a)D_v|_a(f).$$

Let $e_1, \ldots, e_n$ form the standard basis for $\mathbb{R}^n$, the chain rule gives us that for $v_a = v^i e_i|_a$ (Einstein summation), then
$$D_v|_a f = v^i \frac{\partial f}{\partial x^i}(a).$$
(Note that to use the summation convention on the right-hand side, we must view that the upper index in the denominator as a lower index.) More succinctly,

$$D_v|_a f = v \cdot \nabla f(a).$$

This leads to the following definition.

3.1.2 Definition

A map $w\colon C^\infty(\mathbb{R}^n) \to \mathbb{R}$ is called a **derivation** at $a \in \mathbb{R}^n$ if it is $\mathbb{R}$-linear, and for all f, g:

$$w(fg) = f(a) \cdot w(g) + g(a) \cdot w(f).$$

Define $T_a\mathbb{R}^n$ to be the set of all derivations at a.

Clearly $T_a\mathbb{R}^n$ has the natural structure of a linear space, via pointwise addition and multiplication, and all directional derivatives are derivations. More surprisingly, $T_a\mathbb{R}^n$ is naturally isomorphic to $\mathbb{R}^n_a$.

3.1.3 Lemma

Let $w \in T_a\mathbb{R}^n$ be a derivation at a.

(1) w is zero on constant functions.

(2) If $f(a) = g(a) = 0$ then $w(fg) = 0$.

Proof

By linearity, it is sufficient to prove (1) on the constant function $f = 1$. Notice that $f^2 = f$, so

$$w(f) = w(f^2) = f(a)w(f) + f(a)w(f) = 2w(f),$$

so $w(f) = 0$, as desired. The second point is clear. ■

3.1.4 Proposition

Let $a \in \mathbb{R}^n$, then the map $\mathbb{R}^n_a \to T_a\mathbb{R}^n$ given by $v_a \mapsto D_v|_a$ is a linear isomorphism.

Proof

Firstly, clearly this is linear (for a quick proof recall that $D_v|_a = v \cdot \nabla|_a$, and inner products are bilinear). To show that it is injective, suppose $D_v|_a = 0$. In particular $D_v|_a(x^i) = 0$ for all $i = 1, \ldots, n$ (x^i denoting the ith projection). But then

$$D_v|_a(x^i) = v \cdot \nabla x^i(a) = v \cdot e_i = v^i,$$

where e_i is the ith basis vector. So $v^i = 0$ for all i, meaning $v = 0$. Thus the map has trivial kernel and is thus injective.

To show surjectivity, notice that we would expect $w = D_v|_a$ where v has components $w(x^i)$, as per the above computation. So let $v = w(x^i)e_i$, and we show that $D_v|_a = w$. Let $f \in C^\infty(\mathbb{R}^n)$, and recall by Taylor's theorem we have

$$f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a)(x^i - a^i) + \sum_{i,j=1}^n (x^i - a^i)(x^j - a^j) \int_0^1 (1-t) \frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a))\, dt.$$

Notice that $x^i - a^i$ are all zero on a, so by the above lemma w maps the right sum to zero ($f = x^i - a^i$ and g is $x^j - a^j$ times the integral). w maps constants to zero, so

$$w(f) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(a) \cdot w(x^i) = v \cdot \nabla f(a) = D_v|_a(f),$$

as desired. ■

3.1.5 Corollary

For any $a \in \mathbb{R}^n$, the n derivations

$$\left.\frac{\partial}{\partial x^1}\right|_a, \ldots, \left.\frac{\partial}{\partial x^n}\right|_a, \qquad \left.\frac{\partial}{\partial x^i}\right|_a (f) = \frac{\partial f}{\partial x^i}(a)$$

form a basis for $T_a\mathbb{R}^n$.

This is simply since these are the derivations corresponding to the standard basis under the natural isomorphism from the above result.

3.2 Tangent vectors on manifolds

In Euclidean space we defined tangent vectors via a geometric construction, and showed that it was natural isomorphic to a more algebraic definition. For general smooth manifolds we do not have the benefit of a geometric construction, and so we turn to the algebraic definition as above.

3.2.1 Definition

Let M be a smooth manifold (with or without boundary) and $p \in M$. A **derivation** at p is a linear map $v: C^\infty(M) \to \mathbb{R}$ satisfying

$$v(fg) = f(p)v(g) + g(p)v(f)$$

for all f, g. Denote the space of derivations at p by T_pM: this is a linear space as before.

As before, derivations map constant functions to zero and if p is a zero of both f and g then $v(fg) = 0$.

3.2.2 Definition

The tangent space construction is functorial: if M, N are smooth manifolds, $p \in M$, and $F: M \to N$ is a smooth map, we define F's **differential** at p to be the map

$$dF_p: T_pM \to T_{F(p)}N$$

defined by mapping a derivation $v \in T_pM$ to the derivation $dF_p(v) \in T_{Fp}N$ given by $dF_p(v)(f) = v(f \circ F)$.

dF_p is well-defined, since $f \circ F$ is a composition of smooth maps $M \to N \to \mathbb{R}$ and thus is smooth. $dF_p(v)$ is a derivation at $F(p)$ because clearly it is linear, and for $f, g \in C^\infty(N)$:

$$\begin{aligned} dF_p(v)(fg) = v((fg) \circ F) = v((f \circ F)(g \circ F)) = \\ (f \circ F)(p)v(g \circ F) + (g \circ F)(p)v(f \circ F) = f(F(p))dF_p(v)(f) + g(F(p))dF_p(v)(g) \end{aligned}$$

as desired. dF_p is also clearly linear.

To show functorality we need to show that it preserves identities and composition. Clearly it does map id_M to id_{T_pM}, and if $M \xrightarrow{F} N \xrightarrow{G} P$ are smooth then

$$d(G \circ F)_p(v): f \mapsto v(f \circ G \circ F), d(G)_{Fp} \circ d(F)_p(v): f \mapsto d(F_p)(v)(f \circ G) = v(f \circ G \circ F),$$

as desired. In particular, for a diffeomorphism F, $dF_p: T_pM \to T_{Fp}M$ is an isomorphism with inverse $d(F^{-1})_{Fp}$.

3.2.3 Proposition

Let M be smooth and $p \in M$. If $f, g \in C^\infty(M)$ agree on some neighborhood of p, then $v(f) = v(g)$ for all derivations $v \in T_pM$.

Proof

Setting $h = f - g$ and using linearity, it is sufficient to show that v is zero on maps which are zero in a neighborhood of p. Since h is zero in a neighborhood U of p, $\operatorname{supp} h \subseteq M - U \subseteq M - p$, and thus we can take $\psi \in C^\infty(M)$ to be a bump function which is 1 on $\operatorname{supp} h$ and supported in $M - p$. So $\psi h = h$ always, since $\psi = 1$ when h is nonzero. Thus $v(\psi h) = v(h)$, and since $\psi(p) = h(p) = 0$, $v(\psi h) = 0$ as desired. ■

Thus T_pM is determined by the behavior of maps in neighborhoods of p (this will lead to an equivalent definition of T_pM as we will see soon). In particular, we have the following:

3.2.4 Proposition

Let $U \subseteq M$ be an open submanifold and let $\iota: U \to M$ be the inclusion map. Then for $p \in U$, the differential $d\iota_p: T_pU \to T_pM$ is an isomorphism.

Proof

To show injectivity, suppose $d\iota_p(v) = 0$ for $v \in T_pU$. Now let $f \in C^\infty(U)$ and let $p \in B \subseteq U$ be a neighborhood such that $\overline{B} \subseteq U$. Then we can extend f to $\widetilde{f}: M \to U$ such that $\widetilde{f}$ agrees with f on $\overline{B}$. By the above proposition, since $\widetilde{f}$ and f agree on the neighborhood B of p, $v(f) = v(\widetilde{f}|_U)$. Restriction is just composition with ι, i.e. we have

$$v(f) = v(\widetilde{f}|_U) = v(\widetilde{f} \circ \iota) = d\iota_p(v)(\widetilde{f}) = 0$$

since $d\iota_p(v) = 0$. Thus $v(f) = 0$ for all $f \in C^\infty(U)$, so $v = 0$ as desired. Thus $d\iota_p$ is injective.

To show surjectivity, for $w \in T_pM$ define $v \in T_pU$ by $v(f) = w(\widetilde{f})$ for the extension above. This is well-defined since any extension agree on a neighborhood of p. This is a valuation, as $\widetilde{f} + \widetilde{g}$, $c\widetilde{f}$, and $\widetilde{f} \cdot \widetilde{g}$ are extensions of $f + g$, cf, fg respectively. Thus linearity follows and

$$v(fg) = w(\widetilde{f}\widetilde{g}) = f(p)w(\widetilde{f}) + g(p)w(\widetilde{g}) = f(p)v(f) + g(p)v(g)$$

as desired. Finally since $f \in C^\infty(M)$ is an extension of $f \circ \iota$, $d\iota_p(v)(f) = v(f \circ \iota) = w(f)$, showing surjectivity. ■

3.2.5 Proposition

Let M be a smooth manifold without boundary of dimension n, and $p \in M$. Then T_pM is an n-dimensional real vector space.

Proof

Let (U, φ) be a smooth chart containing p to an open ball $B \subseteq \mathbb{R}^n$. In particular φ forms a diffeomorphism $U \cong B$, which is in turn diffeomorphic to $\mathbb{R}^n$, and the above proposition shows us that

$$T_pM \cong T_pU \cong T_{\varphi p}\widehat{U} \cong T_{\varphi p}\mathbb{R}^n,$$

which is n-dimensional, as we showed. ■

What about manifolds with boundary? For an interior point $p \in \operatorname{Int} M$ we have $T_pM \cong T_p\operatorname{Int} M$ which is n-dimensional (as $\operatorname{Int} M$ has no boundary). So we need to consider only boundary points. In particular we need to consider $T_a\mathbb{H}^n$ for $a \in \partial\mathbb{H}^n$.

3.2.6 Lemma

Let $\iota\colon \mathbb{H}^n \to \mathbb{R}^n$ be the inclusion map. Then for any $a \in \partial\mathbb{H}^n$, $d\iota_a\colon T_a\mathbb{H}^n \to T_a\mathbb{R}^n$ is an isomorphism.

Proof

To show injectivity suppose $d\iota_a(v) = 0$ for $v \in T_a\mathbb{H}^n$. Then for $f \in C^\infty(\mathbb{H}^n)$ we can extend f to $\widetilde{f} \in C^\infty(\mathbb{R}^n)$ (since $\mathbb{H}^n \subseteq \mathbb{R}^n$ is closed). This means that $\widetilde{f} \circ \iota = f$ and so

$$0 = d\iota_a(v)(\widetilde{f}) = v(\widetilde{f} \circ \iota) = v(f),$$

meaning $v = 0$, so $d\iota_a$ is injective.

Suppose $w \in T_a\mathbb{R}^n$, then define $v \in T_a\mathbb{H}^n$ by $v(f) = w(\widetilde{f})$ for any extension $\widetilde{f}$ of f. Now it is not immediately clear that this is well-defined, since two extensions may not agree on a neighborhood of $p \in \mathbb{R}^n$. But we know that $w = w^i \partial/\partial x^i$ with respect to the standard basis of $T_a\mathbb{R}^n$, and so

$$v(f) = w^i \frac{\partial \widetilde{f}}{\partial x^i}(a).$$

By continuity, the partial derivatives of $\widetilde{f}$ are determined by those of f in a neighborhood of $a \in \mathbb{H}^n$. So for any extension, the partial derivatives at a are the same, so this is indeed well-defined. The proof that v is a derivation and $d\iota_a(v) = w$ is the same as before. ■

3.2.7 Proposition

Let M be a manifold with or without boundary of dimension n. Then for each $p \in M$, T_pM is n-dimensional.

Proof

This is true as explained for interior points. For a boundary point $p \in \partial M$, let (U, φ) be a coordinate half-ball, φ being a diffeomorphism to $B \subseteq \mathbb{H}^n$. Then B is in turn diffeomorphic to $\mathbb{H}^n$ and we have

$$T_pM \cong T_pU \cong T_pB \cong T_p\mathbb{H}^n \cong T_p\mathbb{R}^n,$$

which is n-dimensional. ■

For a Euclidean space $V \cong \mathbb{R}^n$, we can naturally identify T_pV with V itself, as we did with $\mathbb{R}^n$. The construction is the same: map $v \in V$ to the derivation

$$D_v|_p(f) = \left.\frac{d}{dt}\right|_{t=0} f(a + tv).$$

Then the map $V \to T_pV$ given by $v \mapsto D_v|_p$ is a natural isomorphism. The argument for it being an isomorphism is the same as before. Naturality just means that it commutes with linear maps in the sense that if $L\colon V \to W$ is linear, the diagram below commutes

$$\begin{tikzcd} V \arrow[r, "\cong"'] \arrow[d, "L"'] & T_pV \arrow[d, "d_pL"] \\ W \arrow[r, "\cong"] & T_{Lp}W \end{tikzcd}$$

In particular, for an open submanifold of a vector space, its tangent space is naturally isomorphic to that vector space. For example, $\mathrm{GL}_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$ is an open submanifold and as such $T_X\mathrm{GL}_n(\mathbb{R}) \cong M_n(\mathbb{R})$ for every $X \in \mathrm{GL}_n(\mathbb{R})$.

3.2.8 Proposition

Let $M_1, \dots, M_n$ be smooth manifolds (of which at most one has boundary) and let $M = M_1 \times \cdots \times M_n$ be their direct product. Let $\pi_i \colon M \to M_i$ be the projection maps. Then for every $p = (p_1, \dots, p_n) \in M$ the differentials assemble into an isomorphism,

$$(d\pi_1, \dots, d\pi_n) \colon T_pM \to \bigoplus_{i=1}^{n} T_{p_i}M_i.$$

Proof

Let $\alpha = (d\pi_1, \dots, d\pi_n)$, and define the inclusion maps $\iota_i \colon M_i \to M$ which maps $\iota_i(q)$ to be the point in M whose ith coordinate is q and for $i \neq j$ the jth coordinate is p_j. ι_i is smooth which gives us maps $(d\iota_i)_{p_i} \colon T_{p_i}M_i \to T_pM$. Furthermore $\pi_i \circ \iota_i = \mathrm{id}_{M_i}$ and $\pi_j \circ \iota_i$ is constant, so $d(\pi_j \circ \iota_i) = \delta_{ij}$.
Further define $\beta \colon \bigoplus_{i=1}^n T_{p_i}M_i \to T_pM$ by $\beta(v_1, \dots, v_n) = \sum_{i=1}^n (d\iota_i)_{p_i}(v_i)$. Then

$$\alpha \circ \beta(v_1, \dots, v_n) = \sum_{i=1}^{n} \alpha(d\iota_i)_{p_i}(v_i),$$

and by functorality

$$= \sum_{i=1}^{n} (d(\pi_1\iota_i)(v_i), \dots, d(\pi_n\iota_i)(v_i)) = \sum_{i=1}^{n} (0, \dots, v_i, \dots, 0) = (v_1, \dots, v_n).$$

So $\alpha \circ \beta$ is the identity, and the dimensions of T_pM and $\bigoplus T_{p_i}M_i$ are equal, so they are isomorphisms.

■

3.3 Alternative definitions

There exist many equivalent definitions of the tangent space of a manifold at a point.

3.3.1 Definition

Let M be a smooth manifold and $p \in M$. Consider then the set of all smooth functions $f \colon U \to \mathbb{R}$ where $p \in U \subseteq M$ is an open neighborhood of p; denote this data by (f, U). We define an equivalence of these functions, where (f, U) and (g, V) are identified if they are equal on a neighborhood of p. Let $C_p^\infty(M)$ denote the set of all these equivalence classes, called **germs**, we denote the germ of a function f at p by $[f]_p$.
Note that the space of germs can naturally be endowed with a real algebra structure. A **derivation** of $C_p^\infty(M)$ is a linear map $v \colon C_p^\infty(M) \to \mathbb{R}$ satisfying:

$$v[fg]_p = f(p)v[g]_p + g(p)v[f]_p.$$

We can then define the tangent space of M at p to be the space of all such derivations, which can be given a natural real vector space structure.

If we denote by D_pM the space of all derivations of $C_p^\infty(M)$, it is not hard to see that D_pM and T_pM are naturally isomorphic. A derivation of $C^\infty(M)$ defines a derivation of $C_p^\infty(M)$ and vice versa in a unique way. The benefit of this definition via germs is that it is local, and highlights the locality of the tangent space. In analytic manifolds (i.e. C^k ones), this is definition gives the desired properties of a tangent space, as the global definition does not reduce to a local one, since bump functions may not exist.

A more geometric definition can also be given.

3.3.2 Definition

Let M be a smooth manifold and $p \in M$. Consider the collection of all smooth curves centered at p: all smooth maps $\gamma\colon J \to M$ where $J \subseteq \mathbb{R}$ is an open interval containing 0 and $\gamma(0) = p$. Again we define an equivalence on such curves: consider γ_1, γ_2 equivalent if for every open neighborhood $p \in U \subseteq M$ and every smooth $f\colon U \to \mathbb{R}$, $(f \circ \gamma_1)'(0) = (f \circ \gamma_2)'(0)$. Let $\mathcal{V}_p(M)$ be the collection of all these curves.

$\mathcal{V}_p$ does not have an obvious linear structure, but we can give it one, and it is naturally isomorphic to T_pM. We can also naturally define the differential of a smooth map $F\colon M \to N$ to be

$$dF_p\colon \mathcal{V}_p(M) \to \mathcal{V}_{F(p)}(N), \qquad [\gamma] \mapsto [F \circ \gamma].$$

This is well-defined: if $\gamma_1 \sim \gamma_2$ then for $f\colon V \to \mathbb{R}$ smooth in a neighborhood of $F(p)$, $f \circ F$ is smooth in a neighborhood of p, and thus $(f \circ F \circ \gamma_1)'(0) = (f \circ F \circ \gamma_2)'(0)$ as desired.

3.3.3 Proposition

Let M be a smooth manifold, with or without boundary, and $p \in M$. Then $\Psi\colon \mathcal{V}_p(M) \to T_p(M)$ defined by $\Psi[\gamma](f) = (f \circ \gamma)'(0)$ is a bijection.

Proof

Clearly Ψ is well-defined, and it maps to derivations: clearly $\Psi[\gamma]$ is linear, and it is a derivation:

$$\Psi[\gamma](fg) = ((f \circ \gamma) \cdot (g \circ \gamma))'(0) = f(\gamma(0)) \cdot (g \circ \gamma)'(0) + g(\gamma(0)) \cdot (f \circ \gamma)'(0) = f(p)\Psi[\gamma](g) + g(p)\Psi[\gamma](f).$$

Ψ is also injective; we can view this via the equivalence of derivations on $C_p^\infty(M)$ and $C^\infty(M)$. If $\Psi[\gamma_1] = \Psi[\gamma_2]$, then they define the same derivation on $C_p^\infty(M)$ and this clearly shows that $\gamma_1 \sim \gamma_2$.
Now let (U, φ) be a coordinate ball centered at p, so that $T_pM \cong T_pU \cong T_0B \cong \mathbb{R}^n$. For $v \in \mathbb{R}^n$, consider its tangent vector in T_0B given by $D_v|_0$, which we can pull back via $d\varphi_0^{-1}$ to give the tangent vector

$$d\varphi_0^{-1}(D_v|_0) \in T_pU, \qquad d\varphi_0^{-1}(D_v|_0)(f) = D_v|_0(f \circ \varphi^{-1}).$$

This corresponds naturally to a tangent vector in T_pM (which maps $f \in C^\infty(M)$ to its restriction to U and then via the above formula). Define the curve

$$\gamma_v(t) = \varphi^{-1}(tv)$$

on an open interval $(-\varepsilon, \varepsilon)$ so that $tv \in U$ for all $t \in (-\varepsilon, \varepsilon)$. Then

$$\Psi[\gamma_v](f) = (f \circ \gamma_v)'(0) = \frac{d}{dt}\bigg|_0 (f \circ \varphi^{-1}(t \cdot v)) = D_v|_0(f \circ \varphi^{-1}),$$

meaning $\Psi[\gamma_v] = d\varphi_0^{-1}(D_v|_0)$, so that Ψ is surjective as desired. ∎

Ψ can then be pulled back to give $\mathcal{V}_p(M)$ a vector space structure naturally isomorphic to T_pM.

This definition of the tangent space allows us to simply formalize the concept of a **velocity vector**. Let $\gamma\colon J \to M$ be a smooth curve, then its velocity at 0 is its equivalence class in $\mathcal{V}_{\gamma(0)}(M)$. Its velocity at any $t_0 \in J$ is the equivalence class of $\gamma_{t_0}(t) = \gamma(t_0 + t)$ in $\mathcal{V}_{\gamma(t_0)}(M)$.

3.4 Computations in coordinates

Let M be a smooth manifold without boundary and (U, φ) be a coordinate chart, so that

$$d\varphi_p\colon T_pM \to T_{\varphi(p)}\mathbb{R}^n$$

is an isomorphism for $p \in U$. We know that the partial derivatives evaluated at $\varphi(p)$, $\partial/\partial x^i\big|_{\varphi(p)}$ for $1 \le i \le n$, forms a basis for $T_{\varphi(p)}\mathbb{R}^n$. Then their preimages under $d\varphi_p$ forms a basis for T_pM. Let us denote by $\partial/\partial x^i\big|_p$ these preimages, i.e.

$$\frac{\partial}{\partial x^i}\bigg|_p = (d\varphi_p)^{-1}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right) = d\varphi^{-1}_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right).$$

Unwinding definitions, this valuation acts by

$$\frac{\partial}{\partial x^i}\bigg|_p f = \frac{\partial}{\partial x^i}\bigg|_{\varphi(p)} (f \circ \varphi^{-1}) = \frac{\partial \widehat{f}}{\partial \partial x^i}(\widehat{p}),$$

where $\widehat{f} = f \circ \varphi^{-1}$ is the coordinate representation of f, and $\widehat{p} = (p^1, \dots, p^n) = \varphi(p)$ is the coordinate representation of p.

When M is a manifold with boundary, we need only consider the case of boundary points: for interior points we take interior charts. For a boundary point $p \in \partial M$, the differential becomes an isomorphism

$$d\varphi_p \colon T_pM \to T_{\varphi(p)}\mathbb{H}^n,$$

and we can compose this with the differential $d\iota_{\varphi(p)} \colon T_{\varphi(p)}\mathbb{H}^n \to T_{\varphi(p)}\mathbb{R}^n$. The preimage of $\partial/\partial x^i_a$ under $d\iota_a$ is the one-sided partial derivative. Then the preimages of $\partial/\partial x^i\big|_{\varphi(p)}$ of $d(\iota\varphi)_p$ gives us $\partial/\partial x^i\big|_p$ defined by

$$\frac{\partial}{\partial x^i}\bigg|_p f = \frac{\partial \widehat{f}}{\partial \partial x^i}(\widehat{p}),$$

where the partial derivative is one-sided.

So $\frac{\partial}{\partial x^i}\big|_p$ for $1 \le i \le n$ forms a basis for T_pM, so any valuation $v \in T_pM$ can be uniquely written as

$$v = v^i \, \frac{\partial}{\partial x^i}\bigg|_p .$$

The basis $\frac{\partial}{\partial x^i}\big|_p$ is called a **coordinate basis** for T_pM, and $(v^1, \dots, v^n)$ the **components** of v. Notice that if we let $\pi_i \colon \mathbb{R}^n \to \mathbb{R}$ be the ith projection,

$$v(\pi_i \circ \varphi) = \left(v^j \, \frac{\partial}{\partial x^j}\bigg|_p\right)(\pi_i \circ \varphi) = v^j \frac{\partial x^i}{\partial x^j}(\widehat{p}) = v^i.$$

Oftentimes $\pi_i \circ \varphi$ is identified with x^i, so we can write this concisely as

$$v^i = v(x^i),$$

giving an easy method of computing the components of v relative to any coordinate basis.

3.4.1 Proposition

Let $p \in M$ be a point in a smooth manifold. Then $\partial/\partial x^1\big|_p, \dots, \partial/\partial x^n\big|_p$ form a basis for T_pM, and for $v \in T_pM$ we have

$$v = v^i \, \frac{\partial}{\partial x^i}\bigg|_p, \qquad \text{where } v^i = v(x^i).$$

Now how can we represent the differential of a smooth map $F \colon M \to N$ in coordinates? To do so, we must compute the components of $(dF_p)(\partial/\partial x^i\big|_p)$ in N.

First suppose that $F \colon U \to V$ is a map from open Euclidean subsets $U \subseteq \mathbb{R}^n, V \subseteq \mathbb{R}^m$, and let $p \in U$. Let $(x^1, \dots, x^n)$ denote the coordinates in the domain and $(y^1, \dots, y^m)$ in the codomain. Then let us see how dF_p maps the coordinate basis:

$$dF_p\left(\frac{\partial}{\partial x^i}\bigg|_p\right) f = \frac{\partial}{\partial x^i}\bigg|_p (f \circ F) = \frac{\partial f}{\partial y^j}(F(p)) \frac{\partial F^j}{\partial x^i}(p) = \left(\frac{\partial F^j}{\partial x^i}(p) \, \frac{\partial}{\partial y^j}\bigg|_{F(p)}\right)(f),$$

where the second equality is due to the chain rule. Thus

$$dF_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \frac{\partial F^j}{\partial x^i}(p)\cdot \left.\frac{\partial}{\partial y^j}\right|_{F(p)},$$

meaning that the representation of F relative to the coordinate bases is the matrix

$$\begin{pmatrix} \frac{\partial F^1}{\partial x^1}(p) & \cdots & \frac{\partial F^1}{\partial x^n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial F^n}{\partial x^1}(p) & \cdots & \frac{\partial F^n}{\partial x^n}(p) \end{pmatrix},$$

the **Jacobian** of F at p.

Now in general suppose $F\colon M \to N$ is a smooth map between smooth manifolds, with $p \in M$. Let (U,φ) be a chart in M containing p, and (V,ψ) a chart in N containing $F(p)$. Then let $\widehat{F} = \psi\circ F\circ\varphi^{-1}\colon \varphi(U\cap F^{-1}(V)) \to \psi(V)$ be its coordinate representation, $\widehat{p} = \varphi(p)$ be the coordinate representation of p. Then since $F\circ\varphi^{-1} = \psi^{-1}\circ\widehat{F}$,

$$dF_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = dF_p\left(d(\varphi^{-1})_{\widehat{p}}\left.\frac{\partial}{\partial x^i}\right|_{\widehat{p}}\right) = d(\psi^{-1})_{\widehat{F}(\widehat{p})}\left(d\widehat{F}_p\left(\left.\frac{\partial}{\partial x^i}\right|_{\widehat{p}}\right)\right).$$

We just computed the representation of $\widehat{F}$:

$$= d(\psi^{-1}_{\widehat{F}(\widehat{p})})\left(\frac{\partial \widehat{F}^j}{\partial x^i}(\widehat{p})\left.\frac{\partial}{\partial y^j}\right|_{\widehat{F}(\widehat{p})}\right) = \frac{\partial \widehat{F}^j}{\partial x^i}\left.\frac{\partial}{\partial y^j}\right|_{F(p)},$$

where the last equality is due to the definition of $\partial/\partial y^j\big|_{F(p)}$. Thus the representation of dF_p is the Jacobian of $\widehat{F}$ at $\widehat{p}$.

3.4.2 Proposition

Let $F\colon M \to N$ be a map of smooth manifolds, with or without boundary, and $p \in M$. Then the matrix representation of dF_p with respect to the coordinate bases of M, N is the Jacobian of its coordinate representation $\widehat{F}$ at the coordinate $\widehat{p}$ of p.

Now how are two coordinate bases related? Suppose (U,φ) and (V,ψ) be two charts with $p \in U\cap V$. Let x^i be the coordinate functions of φ and $\widetilde{x}^i$ those of ψ, so we can represent $v \in T_pM$ with respect to either $\partial/\partial x^i\big|_p$ or $\partial/\partial\widetilde{x}^i\big|_p$.

Let us denote the transition map $\psi\circ\varphi^{-1}\colon \varphi(U\cap V) \to \psi(U\cap V)$ by

$$\psi\circ\varphi^{-1}(x) = (\widetilde{x}^1(x),\ldots,\widetilde{x}^n(x)).$$

Then from our computation above,

$$d(\psi\circ\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}_{\varphi(p)}\right) = \frac{\partial\widetilde{x}^j}{\partial x^i}(\varphi(p))\left.\frac{\partial}{\partial\widetilde{x}^j}\right|_{\psi(p)}.$$

Now, by definition

$$\left.\frac{\partial}{\partial x^i}\right|_p = d(\varphi^{-1})_{\varphi(p)}\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right) = d(\psi^{-1})_{\psi(p)}\circ d(\psi\circ\varphi^{-1})_{\varphi(p)}\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right) = d(\psi^{-1})_{\psi(p)}\left(\frac{\partial\widetilde{x}^j}{\partial x^i}(\varphi(p))\left.\frac{\partial}{\partial\widetilde{x}^j}\right|_{\psi(p)}\right),$$

and then again by the same definition,

$$= \frac{\partial\widetilde{x}^j}{\partial x^i}(\varphi(p))\left.\frac{\partial}{\partial\widetilde{x}^j}\right|_p.$$

So if $v = v^i\,\partial/\partial x^i\big|_p$ then

$$v = v^i\frac{\partial}{\partial x^i}_p = v^i\frac{\partial\widetilde{x}^j}{\partial x^i}(\varphi(p))\cdot\left.\frac{\partial}{\partial\widetilde{x}^j}\right|_p,$$

meaning v's components with respect to ψ are

$$\widetilde{v}^j = \frac{\partial\widetilde{x}^j}{\partial x^i}(\varphi(p))\cdot v^i.$$

3.4.3 Example

Let φ be the standard chart of $\mathbb{R}^2$, and ψ be the **polar coordinates**: it is the map on $\mathbb{R}^2 - 0$ mapping $(r\cos\theta, r\sin\theta)$ to (r,θ). So here $\widetilde{x}_1, \widetilde{x}_2$ in the above discussion are r, θ respectively.
We see that

$$\left.\frac{\partial}{\partial r}\right|_p = \frac{\partial x}{\partial r}(\widehat{p})\left.\frac{\partial}{\partial x}\right|_p + \frac{\partial y}{\partial r}(\widehat{p})\left.\frac{\partial}{\partial y}\right|_p, \qquad \left.\frac{\partial}{\partial \theta}\right|_p = \frac{\partial x}{\partial \theta}(\widehat{p})\left.\frac{\partial}{\partial x}\right|_p + \frac{\partial y}{\partial \theta}(\widehat{p})\left.\frac{\partial}{\partial y}\right|_p.$$

Now, $x = r\cos\theta$ so that $\frac{\partial x}{\partial r} = \cos\theta$, and so on. So we have

$$\left.\frac{\partial}{\partial r}\right|_p = \cos\theta\left.\frac{\partial}{\partial x}\right|_p + \sin\theta\left.\frac{\partial}{\partial y}\right|_p, \qquad \left.\frac{\partial}{\partial \theta}\right|_p = -r\sin\theta\left.\frac{\partial}{\partial x}\right|_p + r\cos\theta\left.\frac{\partial}{\partial y}\right|_p.$$

So for example, if p's representation in polar coordinates is $(r,\theta) = (2, \pi/2)$ we see that

$$\left.\frac{\partial}{\partial r}\right|_p = \left.\frac{\partial}{\partial y}\right|_p, \qquad \left.\frac{\partial}{\partial \theta}\right|_p = -2\left.\frac{\partial}{\partial x}\right|_p.$$

3.5 The tangent bundle

3.5.1 Definition

Let M be a smooth manifold with or without boundary. Then we define M's **tangent bundle** to be the disjoint union

$$TM = \coprod_{p\in M} T_pM.$$

We denote an element in TM by (p, v) or v_p for $p \in M$ and $v \in T_pM$. There is an obvious **projection map** $\pi\colon TM \to M$ mapping $(p, v) \mapsto p$.

When $M = \mathbb{R}^n$ we see that quite naturally

$$T\mathbb{R}^n = \coprod_{a\in\mathbb{R}^n} T_a\mathbb{R}^n \cong \coprod_{a\in\mathbb{R}^n} \mathbb{R}^n_a = \coprod_{a\in\mathbb{R}^n} \{a\}\times\mathbb{R}^n = \mathbb{R}^n\times\mathbb{R}^n.$$

Obviously by similar considerations in general TM is in bijection with $M\times\mathbb{R}^n$, but in general this does not lift to an isomorphism with richer structure; as we will see now TM has a natural topology making it into a smooth manifold.

3.5.2 Proposition

For a smooth n-manifold M, TM has a natural topology making it into a smooth manifold of dimension $2n$ and for which the projection $\pi\colon TM \to M$ is smooth. The boundary of TM is

$$\partial TM = \coprod_{p\in\partial M} T_pM.$$

Proof

For a chart (U, φ) of M, notice that $\pi^{-1}(U) \subseteq TM$ is the collection of all tangent vectors tangent to points in U. Let $x^1, \dots, x^n$ be the coordinate functions of φ, and define $\widetilde{\varphi}\colon \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\widetilde{\varphi}\left(v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = (x^1(p), \dots, x^n(p), v^1, \dots, v^n),$$

the vector whose first n components are $\varphi(p)$, and then $v^1, \dots, v^n$. The image of $\widetilde{\varphi}$ is $\varphi(U) \times \mathbb{R}^n$, which is open in $\mathbb{R}^{2n}$, and $\widetilde{\varphi}$ is a bijection onto this: we can explicitly write its inverse as

$$\widetilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) = v^i \left.\frac{\partial}{\partial x^i}\right|_{\varphi^{-1}(\overline{x})}.$$

Now suppose we have two smooth charts $(U, \varphi), (V, \psi)$ with corresponding charts $(\pi^{-1}(U), \widetilde{\varphi}), (\pi^{-1}(V), \widetilde{\psi})$ in TM. Then we have

$$\widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n,$$

which is open in $\mathbb{R}^{2n}$, and similar for $\widetilde{\psi}$. And the transition map $\widetilde{\psi} \circ \widetilde{\varphi}^{-1}\colon \varphi(U \cap V) \times \mathbb{R}^n \to \psi(U \cap V) \times \mathbb{R}^n$ can be computed explicitly:

$$\widetilde{\psi} \circ \widetilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) = \widetilde{\psi}\left(v^i \left.\frac{\partial}{\partial x^i}\right|_{\varphi^{-1}(x^1, \dots, x^n)}\right) = (\widetilde{x}^1, \dots, \widetilde{x}^n, \widetilde{v}^1, \dots, \widetilde{v}^n).$$

Where $\widetilde{x} = \psi \circ \varphi^{-1}(x)$ is smooth, and $\widetilde{v}^i$ are given as above,

$$\widetilde{v}^i = \frac{\partial \widetilde{x}^i}{\partial x^j}(x) v^j = \frac{\partial \pi_i \circ \psi \circ \varphi^{-1}}{\partial x^j} v^j,$$

which is smooth as well.

A countable cover $\{U_i\}$ of M by coordinate domains covers TM by $\{\pi^{-1}U_i\}$. For the Hausdorff condition, for v_p, u_q, either p and q are in the same coordinate domain U, in which case $v_p, u_q \in \pi^{-1}U$, or they have disjoint coordinate domains $p \in U, q \in V$ so that $v_p \in \pi^{-1}U, u_q \in \pi^{-1}V$ are disjoint as well. Thus this construction endows TM with a smooth structure, of dimension $2n$.

π is then smooth, as

$$\psi \circ \pi \circ \widetilde{\varphi}^{-1}(\overline{x}, \overline{v}) = \psi \circ \varphi^{-1}(\overline{x})$$

is smooth.

If $p \in \partial M$ then its boundary chart (U, φ) maps vectors in T_pM to vectors in $\varphi(p) \times \mathbb{R}^n$, on the boundary. All other points are mapped to the interior. ■

The tangent bundle is functorial: for $F\colon M \to N$ smooth, we define its **global differential** $dF\colon TM \to TN$ to be the map whose restriction to $T_pM \subseteq TM$ is dF_p; it maps $v_p \in TM$ to $dF_p(v)$.

3.5.3 Proposition

The global differential is smooth, and functorial.

Proof

Let (U, φ) be a chart for M and (V, ψ) a chart for N. We need to show that $\widetilde{\psi} \circ dF \circ \widetilde{\varphi}^{-1}$ is smooth: let $\varphi(p) = \overline{x}$, then

$$\begin{aligned}
\widetilde{\psi} \circ dF \circ \widetilde{\varphi}^{-1}(x^1, \dots, x^n, v^1, \dots, v^n) &= \widetilde{\psi} \circ dF \left(v^i \left. \frac{\partial}{\partial x^i} \right|_p \right) \\
&= \widetilde{\psi} \left(v^i \frac{\partial \widehat{F}^j}{\partial x^i} (\psi(p)) \left. \frac{\partial}{\partial \widetilde{x}^j} \right|_{F(p)} \right) \\
&= \left(\widehat{F}^1(x), \dots, \widehat{F}^n(x), v^i \frac{\partial \widehat{F}^1}{\partial x^i}(\widetilde{x}), \dots, v^i \frac{\partial \widehat{F}^n}{\partial x^i}(\widetilde{x}) \right),
\end{aligned}$$

where $\widehat{F} = \psi \circ F \circ \varphi^{-1}$ is the coordinate representation, and $\widetilde{x} = \psi \circ \varphi^{-1}(x)$ is the transition function. This is smooth, as desired.

Functorality is clear from the functorality of local differentials. ■

Now notice that if M has a global chart φ, then the construction of TM gives us a global chart $\widetilde{\varphi}: TM \to \varphi(M) \times \mathbb{R}^n$. Thus TM is diffeomorphic to $\varphi(M) \times \mathbb{R}^n$, which is diffeomorphic to $M \times \mathbb{R}^n$. So we do have that $TM \cong M \times \mathbb{R}^n$ when M can be covered with a global chart (i.e. is diffeomorphic to $\mathbb{R}^n$ or $\mathbb{H}^n$). In general we will see that this is not true.

3.6 Velocity vectors of curves

3.6.1 Definition

A **curve** on a smooth manifold M is a smooth map $\gamma: J \to M$, from an interval $J \subseteq \mathbb{R}$. For a point $t_0 \in J$, we define γ's **velocity** at t_0 to be

$$\gamma'(t_0) = d\gamma \left(\left. \frac{d}{dt} \right|_{t_0} \right) \in T_{\gamma(t_0)}(M),$$

where $d/dt \in T_{t_0}(\mathbb{R})$ is the standard basis vector.

Spelled out, $\gamma'(t_0)$ is the derivation given by

$$\gamma'(t_0)(f) = \frac{d(f \circ \gamma)}{dt}(t_0) = (f \circ \gamma)'(t_0).$$

If $t_0 \in J$ is an endpoint, this derivative is taken as one-sided. Now let (U, φ) be a chart containing $\gamma(t_0)$, so that $\varphi\gamma(t_0) = t_0$. Then we see

$$\gamma'(t_0) = d\gamma \left(\left. \frac{d}{dt} \right|_{t_0} \right) = \frac{\partial \widehat{\gamma}^i}{\partial t}(t_0) \left. \frac{\partial}{\partial x^i} \right|_{\gamma(t_0)},$$

where $\widehat{\gamma} = \varphi \circ \gamma$ is its coordinate representation, with projections $\widehat{\gamma}^i = \pi_i \circ \widehat{\gamma}$. Thus $\gamma'(t_0)$ is given by essentially the same formula as in Euclidean space: by taking the partial derivatives of each component.

3.6.2 Proposition

Let M be a smooth manifold, with or without boundary, and $p \in M$. Then every $v \in T_pM$ is the velocity of some smooth curve in M.

Proof

For an interior point p, let (U, φ) be an interior chart centered at p. For $v \in T_pM$, let $v = v^i \, \partial/\partial x^i\big|_p$ be its coordinate representation with respect to φ, let $\overline{v} = (v^1, \dots, v^n)$. Then define

$$\gamma\colon (-\varepsilon, \varepsilon) \to M, \qquad \gamma(t) = \varphi^{-1}(t\overline{v})$$

for $\varepsilon > 0$ small enough so that $t\overline{v} \in \varphi(U)$ for all t in the domain. This is a smooth curve with $\gamma(0) = p$, and its velocity at 0 is

$$\gamma'(0) = \frac{\partial \varphi\gamma^i}{\partial t}(0) \left.\frac{\partial}{\partial x^i}\right|_p = \frac{\partial t v^i}{\partial t}(0) \left.\frac{\partial}{\partial x^i}\right|_p = v^i \left.\frac{\partial}{\partial x^i}\right|_p = v,$$

as desired.

If $p \in \partial M$ then take a smooth chart as before and consider its coordinate representation again. The above formula is only defined when $tv^n \geq 0$, so we take the appropriate subinterval of $(-\varepsilon, \varepsilon)$ so that γ is defined, and we get the same result. ■

3.6.3 Proposition

Let $F\colon M \to N$ be smooth, $\gamma\colon J \to M$ a smooth curve. Then for $t_0 \in J$, the velocity of the composite curve $F \circ \gamma$ is given by

$$(F \circ \gamma)'(t_0) = dF(\gamma'(t_0)).$$

Proof

By definition and functorality,

$$(F \circ \gamma)'(t_0) = d(F \circ \gamma) \left.\frac{\partial}{\partial t}\right|_{t_0} = dF \circ d\gamma \left.\frac{\partial}{\partial t}\right|_{t_0} = dF(\gamma'(t_0)).$$ ■

While this may seem like a good way to compute the velocity vectors of composite curves, it turns out that in general it is more helpful to view this proposition as a method of computing differentials:

3.6.4 Corollary

Let $F\colon M \to N$ be smooth, $p \in M$, and $v \in T_pM$. Then for any curve $\gamma\colon J \to M$ where $0 \in J$ and whose velocity at 0 is v, we have that

$$dF_p(v) = (F \circ \gamma)'(0).$$

4 Submersions, Immersions, and Embeddings

4.1 Maps of constant rank

4.1.1 Definition

Let $F\colon M \to N$ be smooth, and $p \in M$. Then F's **rank** at p is the rank of the linear transformation $dF_p\colon T_pM \to T_{Fp}N$.

Taking the coordinate bases of the tangent spaces, F's rank is just the rank of the Jacobian of F at p. Equivalently it is the dimension of dF_p's image in $T_{Fp}N$. A matrices rank is bound by the dimensions of its domain and codomain, so $\operatorname{rank}_p F \leq \min\{\dim M, \dim N\}$. If F's rank at p is equal to this minimum, we say that F has **full rank** at p.

4.1.2 Definition

$F\colon M \to N$ has **constant rank** if its rank is equal at every point in M. F is a **smooth submersion** if its differential is surjective at every point, equivalently if F has constant rank $\dim N$. F is a **smooth immersion** if its differential is injective at every point, equivalently if F has constant rank $\dim M$.

4.1.3 Proposition

Let $F\colon M \to N$ be smooth and at $p \in M$, F has full rank. Then there is a neighborhood U of p such that $F|_U$ has full constant rank. That is to say, if F has full rank at a point, there is a neighborhood where F is a smooth immersion or submersion.

Proof

Let (U, φ) and (V, ψ) be charts of M and N respectively, and let $\widehat{F}$ be F's coordinate representation. Then the map $p \mapsto J\widehat{F}_{\varphi(p)}$ is continuous. The space of full-rank matrices is open, so its preimage under this map is open. This gives a neighborhood of p where F has full rank. ■

4.1.4 Example

Let $M = M_1 \times \cdots \times M_n$ be the direct product of smooth manifolds, and consider the projection $\pi = \pi_i\colon M \to M_i$: this is a smooth submersion. Indeed, for $p \in M$ let (U_j, φ_j) be charts containing p^i, so that π's representation is

$$\widehat{\pi} = \varphi_i \circ \pi \circ (\varphi_1^{-1}, \ldots, \varphi_n^{-1}),$$

which maps $\overline{x}$ to x^i: i.e. $\widehat{\pi}$ is the projection of $\mathbb{R}^{n_1 + \cdots + n_k} \to \mathbb{R}^{n_i}$. The Jacobian of this has full rank, as desired.

4.1.5 Example

Let M be smooth, and $\pi\colon TM \to M$ is a smooth submersion. Indeed, if (U, φ) is a chart of M and $(\pi^{-1}(U), \widetilde{\varphi})$ is its corresponding chart in TM, then

$$\widehat{\pi} = \varphi \circ \pi \circ \widetilde{\varphi}^{-1}\colon (\overline{x}, \overline{v}) \mapsto \overline{x},$$

which clearly has full rank.

4.2 Local diffeomorphisms

4.2.1 Definition

A smooth map $F\colon M \to N$ is a **local diffeomorphism** if for every $p \in M$, there is a neighborhood $p \in U$ such that $F(U) \subseteq N$ is open and $F|_U\colon U \to F(U)$ is a diffeomorphism.

4.2.2 Theorem

Suppose $F\colon M \to N$ is smooth between manifolds without boundary. If dF_p is invertible at some $p \in M$, there are connected neighborhoods $p \in U_0$ and $F(p) \in V_0$ such that $F|_{U_0}\colon U_0 \to V_0$ is a diffeomorphism.

Proof

dF_p being an isomorphism implies M and N have the same dimension. Take a chart centered at p, (U, φ), and a chart containing $F(U)$, (V, ψ), centered at $F(p)$. Then $\widehat{F} = \psi \circ F \circ \varphi^{-1} \colon \varphi(U) \to \psi(V)$ has $\widehat{F}(\widehat{p}) = 0$ and $\widehat{F}$'s Jacobian at $\widehat{p}$ is invertible. The inverse function theorem (the ordinary Euclidean one) implies that there are connected neighborhoods $\widehat{U}_0 \subseteq \varphi(U)$ and $\widehat{V}_0 \subseteq \psi(V)$ where $\widehat{F}|_{\widehat{U}_0} \colon \widehat{U}_0 \to \widehat{V}_0$ is a diffeomorphism. Lifting these to M and N shows that $F|_{U_0} \colon U_0 \to V_0$ is a diffeomorphism for $U_0 = \varphi^{-1}\widehat{U}_0$ and $V_0 = \psi^{-1}\widehat{V}_0$. ■

4.2.3 Corollary

A smooth map between manifolds without boundary is a local diffeomorphism iff its differential is invertible at every point.

The following are obvious properties of local diffeomorphisms.

4.2.4 Proposition

(1) The composition of local diffeomorphisms is a local diffeomorphism;

(2) Every finite product of local diffeomorphisms is a local diffeomorphism;

(3) Every local diffeomorphism is a local homeomorphism and an open map;

(4) The restriction of a local diffeomorphism to an open submanifold with or without boundary is a local diffeomorphism;

(5) Every diffeomorphism is a local diffeomorphism;

(6) A bijective local diffeomorphism is a diffeomorphism;

Proof

The only non-obvious claim is the last. Suppose $F \colon M \to N$ is a bijective local diffeomorphism, then $F^{-1} \colon N \to M$ is smooth. This is because for every $F(p) \in N$, there is a neighborhood $F(p) \in V$ where $F^{-1}|_V$ is smooth. Thus by the gluing lemma, F^{-1} is smooth. ■

4.2.5 Proposition

Let $F \colon M \to N$ be a smooth map between manifolds without boundary.

(1) F is a local diffeomorphism iff it is a smooth immersion and a smooth submersion.

(2) If $\dim M = \dim N$ then if F is either a smooth immersion or submersion, then F is a local diffeomorphism.

Proof

The first point is clear, and if F is a smooth immersion or submersion, then it must be the other (as injectivity or surjectivity of a linear map between spaces of the same dimension must be invertible). ■

4.2.6 Example

Recall the map $\varepsilon\colon \mathbb{R} \to S^1$ by $\varepsilon(t) = \exp(2\pi i t)$. Then ε's representations are $t \mapsto \cos(2\pi t), \sin(2\pi t)$ which are local diffeomorphisms. So ε is a local diffeomorphism, and its n-fold product $\varepsilon^n\colon \mathbb{R}^n \to T^n$ is as well.

4.2.7 Theorem (Rank theorem)

Suppose M, N are smooth manifolds without boundary with dimensions m, n, and $F\colon M \to N$ is a map of constant rank r. Then for every $p \in M$ there exist smooth charts (U, φ) centered at p and (V, ψ) centered at $F(p)$ containing $F(U)$, relative to which F has coordinate representation

$$\widehat{F}(x^1, \ldots, x^r, x^{r+1}, \ldots, x^n) = (x^1, \ldots, x^j, 0, \ldots, 0).$$

In particular, if F is a smooth submersion then $\widehat{F}\colon \mathbb{R}^n \to \mathbb{R}^r$ is projection, and if F is a smooth immersion then $\widehat{F}\colon \mathbb{R}^n \to \mathbb{R}^r$ is inclusion (appending zeroes).

Proof

Without loss of generality, we can assume $F\colon U \to V$ is a smooth map between open subsets of Euclidean space, $U \subseteq \mathbb{R}^n, V \subseteq \mathbb{R}^m$. The Jacobian $JF(0)$ having rank r means that it has some $r \times r$ minor with nonzero determinant. By reordering coordinates we can assume this is the upper-left minor. Let us label the standard coordinates as $(x, y) \in \mathbb{R}^r \times \mathbb{R}^{m-r}$ in the domain and $(v, w) \in \mathbb{R}^r \times \mathbb{R}^{n-r}$ in the codomain.
Then $F(x, y) = (Q(x, y), R(x, y))$ for smooth maps $Q\colon U \to \mathbb{R}^r$ and $R\colon U \to \mathbb{R}^{n-r}$, and our assumption means that $JQ(0)$ is non-singular. Define $\varphi\colon U \to \mathbb{R}^m$ by $\varphi(x, y) = (Q(x, y), y)$ so that its Jacobian is

$$J\varphi(0,0) = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j}(0) & \frac{\partial Q^i}{\partial y^j}(0) \\ 0 & \delta_{ij} \end{pmatrix}.$$

This is non-singular and thus φ restricts to a diffeomorphism in a neighborhood of 0, say U_0 and $\varphi(U_0) = \widetilde{U}_0$. We can assume that $\widetilde{U}_0$ is an open cube, potentially by shrinking these two sets.
Write $\varphi^{-1}(x, y) = (A(x, y), B(x, y))$ for $A\colon \widetilde{U}_0 \to \mathbb{R}^r$ and $B\colon \widetilde{U}_0 \to \mathbb{R}^{m-r}$. Then we have

$$(x, y) = \varphi(A(x, y), B(x, y)) = (Q(A(x, y), B(x, y)), B(x, y)),$$

so $B(x, y) = y$ and φ^{-1} has the form $(x, y) \mapsto (A(x, y), y)$. On the other hand, we have

$$(x, y) = \varphi^{-1}\varphi(x, y) = \varphi^{-1}(Q(x, y), y) = (A(Q(x, y), y), y),$$

and so $A(Q(x, y), y) = x$ and $Q(A(x, y), y) = x$.
Thus

$$F \circ \varphi^{-1}(x, y) = (Q(A(x, y), y), R(A(x, y), y)) = (x, \widetilde{R}(x, y))$$

for $\widetilde{R}(x, y) = R(A(x, y), y)$. The Jacobian of this map is

$$J(F \circ \varphi^{-1}) = \begin{pmatrix} \delta_{ij} & 0 \\ \frac{\partial \widetilde{R}^i}{\partial x^j} & \frac{\partial \widetilde{R}^i}{\partial y^j} \end{pmatrix},$$

which has rank r as the product of F and φ^{-1}'s Jacobians, the latter of which is invertible. The first r columns here are obviously linearly independent, and thus the rank of this map is r iff $\partial \widetilde{R}^i / \partial y^j$ are zero on $\widetilde{U}_0$. Since $\widetilde{U}_0$ is a cube, this implies $\widetilde{R}$ is independent of $(y^1, \ldots, y^{m-r})$ so that if we define $S(x) = \widetilde{R}(x, 0)$ then $R(x, y) = S(x)$ for all y and we have

$$F \circ \varphi^{-1}(x, y) = (x, S(x)).$$

Now we want to define a smooth chart in some neighborhood of 0 in V such that

$$\psi \circ F \circ \varphi^{-1} = \psi(x, S(x)) = (x, 0).$$

Let $V_0 = \left\{(v, w) \in V \;\middle|\; (v, 0) \in \widetilde{U}_0\right\}$, which is open. Then $F \circ \varphi^{-1}(\widetilde{U}_0) \subseteq V_0$ since $\widetilde{U}_0$ is an open cube, and thus $F(U_0) \subseteq V_0$. Define $\psi\colon V_0 \to \mathbb{R}^n$ by $\psi(v, w) = (v, w - S(v))$, which has inverse $\psi^{-1}(x, y) = (s, t + S(s))$. And then

$$\psi \circ F \circ \varphi^{-1}(x, y) = \psi(x, S(x)) = (x, 0)$$

as desired. ∎

4.2.8 Corollary

Let $F\colon M \to N$ be a smooth map between manifolds without boundary, and M is connected. Then F has constant rank iff for every $p \in M$, F has a coordinate representation which is linear.

Proof

If F has a linear representation in a neighborhood of each point, then F has constant rank in a neighborhood of each point. By connectedness, F has constant rank.

Conversely, if F has constant rank, then apply the rank theorem as above. ■

4.2.9 Theorem (Global rank theorem)

If $F\colon M \to N$ is as above of constant rank, then

(1) if F is surjective, it is a smooth submersion;

(2) if F is injective, it is a smooth immersion;

(3) if F is bijective, it is a diffeomorphism.

Proof

Let m, n be the dimensions of M, N respectively, and r the rank of F. If F is not a smooth submersion then $r < n$. Consider then a coordinate representation of F with respect to $(U, \varphi), (V, \psi)$ which is a projection. Without loss of generality, $F(\overline{U}) \subseteq V$.

Then $F(\overline{U})$ is a compact subset of $\left\{ y \in V \;\middle|\; y^{r+1} = \cdots = r^n = 0 \right\}$, and is thus closed in N. But it can't contain any open subset of N, and thus is nowhere dense. If we choose countably many charts covering M, (U_i, φ_i), then $F(M)$ is the union of the nowhere dense sets $F(\overline{U}_i)$, and thus has empty interior. So F cannot be surjective.

If F is not a smooth immersion, then for $p \in M$ we have a coordinate representation as in the rank theorem. But such a coordinate representation is not injective, so F cannot be.

The last point is because the previous two points imply F is a bijective local diffeomorphism, thus it is a diffeomorphism. ■

4.3 Embeddings

4.3.1 Definition

A **smooth embedding** is a smooth map $F\colon M \to N$ which is a topological embedding (a homeomorphism onto its image), and a smooth immersion.

A smooth embedding is not just a smooth topological embedding!

4.3.2 Example

The inclusion of an open subset of a manifold in the manifold is a smooth embedding.

4.3.3 Example

For $M = M_1 \times \cdots \times M_n$ and $p_i \in M_i$, the maps

$$\iota_j \colon M_j \to M, \qquad \iota_j(q) = (p_1, \dots, q, \dots, p_n)$$

are smooth embeddigs. They can be given coordinate representations $\overline{x} \mapsto (\overline{x}, 0)$ which is clearly a smooth embedding.

4.3.4 Proposition

Suppose M, N are smooth manifolds and $F \colon M \to N$ is an injective smooth immersion. If any of the following hold, it is a smooth embedding.

(1) F is open or closed;

(2) F is proper (preimages of compact subsets are compact);

(3) M is compact;

(4) M has empty boundary and $\dim M = \dim N$.

Proof

If F is open or closed and injective, it is a topological embedding, proving (1). If F is proper, it is closed. If M is compact, then F is proper (since preimages of compact subspaces are closed in M and thus compact).

Now suppose M has empty boundary and $\dim M = \dim N$. Since dF_p is an isomorphism, we must have $F(M) \subseteq \mathrm{Int} N$. Thus $F \colon M \to \mathrm{Int} N$ is a local diffeomorphism, and thus open. So F is a composition of open maps, and by (1) is a smooth embedding. ■

4.3.5 Theorem (Local embedding theorem)

If M, N are manifolfds with or without boundary and $F \colon M \to N$ is smooth, then F is a smooth immersion iff every point in M has a neighborhood U such that $F|_U \colon U \to N$ is a smooth embedding.

Proof

If every point has a neighborhood where F is a smooth embedding, then F has full rank and is thus a smooth immersion.

Conversely if F is a smooth immersion, then the rank theorem shows that F has a coordinate representation of the form $F(\overline{x}) = (\overline{x}, 0)$, which is injective, so $F|_U$ is injective. So we have that for some neighborhood $F|_U$ is injective. Now if $V \subseteq U$ is precompact, then $F|_{\overline{V}}$ is a topological embedding as it is injective and proper. Thus $F|_V$ is a topological embedding and a smooth immersion, as desired. ■

4.4 Embedded and immersed submanifolds

4.4.1 Definition

Let M be a manifold, then an **embedded submanifold** is a subset $S \subseteq M$ which is a manifold without boundary such that the inclusion $S \hookrightarrow M$ is a smooth embedding. For an embedded submanifold $S \subseteq M$, its **codimension** is $\dim M - \dim S$.

Note:
Embedded submanifolds are also called **regular submanifolds**.

Note that $S \hookrightarrow M$ being a smooth embedding is the same as saying that it is a smooth immersion, and S has the subspace topology.

4.4.2 Proposition

The embedded submanifolds of M with codimension 0 are precisely the open subsets of M.

Proof

For $\iota : U \hookrightarrow M$ the inclusion map, we need only show it is a smooth immersion. And ι's coordinate representation relative to any point in U is just the identity (take the same chart), which is a smooth immersion.

Conversely, if $\iota : U \hookrightarrow M$ is a smooth embedding of codimension 0, then it is a local diffeomorphism. Thus it is open, and so U must be an open subset of M. ■

4.4.3 Proposition

Let $F : M \to N$ be a smooth embedding, then $S = F(M)$ has a unique smooth structure making it an embedded submanifold of N diffeomorphic to M.

Proof

We need to give S the subspace topology relative to N, and then by assumption it is homeomorphic to M. We need to lift the smooth structure of M to S, that is take charts $(F(U), \varphi \circ F^{-1})$ for a chart (U, φ) for M. This is a smooth structure, and it turns F into a diffeomorphism onto S, and $\iota : S \hookrightarrow N$ into a smooth embedding. ■

4.4.4 Example

Consider the product manifold $M \times N$ where M doesn't have boundary. Then for $p \in N$, $M \times \{p\}$ is an embedded submanifold diffeomorphic to M. Indeed, $F : M \to M \times N$ given by $x \mapsto (x, p)$ is a smooth embedding with image $M \times \{p\}$.

This embedded submanifold is called a **slice** of the product manifold.

4.4.5 Definition

Let M be a smooth manifold, then an **immersed submanifold** is a subset $S \subseteq M$ endowed with a smooth structure making the inclusion $\iota : S \hookrightarrow M$ a smooth immersion. The **codimension** of an immersed submanifold is defined still to be $\dim M - \dim S$.

Since the inclusion is not necessarily a topological embedding, S's topology need not be the subspace topology.

4.4.6 Proposition

Suppose $F : M \to N$ is an injective smooth immersion, and let $S = F(M)$. Then S has a unique topology and smooth structure making it an immersed submanifold of N, and $F : M \to S$ a diffeomorphism onto its image.

Proof

The proof is similar to the one above. Give S the topology where $U \subseteq S$ is open iff $F^{-1}U \subseteq M$ is. Then charts $(F(U), \varphi \circ F^{-1})$ for a chart (U, φ) for M. ■

4.4.7 Proposition

Suppose $S \subseteq M$ is an immersed submanifold. Then if any of the following hold, S is embedded:

(1) S has codimension 0,

(2) The inclusion map $S \hookrightarrow M$ is proper (preimages of compact sets are compact),

(3) S is compact.

Proof

If S is compact, then the inclusion map is proper (closed subsets of compact sets are compact), so (2) implies (3). We need only prove the first two points.

If S has codimension 0, then a smooth immersion is a smooth submersion is a local diffeomorphism. Thus the inclusion map is an injective local diffeomorphism, thus a topological embedding. ■

4.4.8 Proposition

If $S \subseteq M$ is an immersed submanifold, then for every $p \in S$ there is a neighborhood $p \in U \subseteq S$ which is an embedded submanifold of M.

Proof

We showed above that smooth immersions are locally smooth embeddings. ■

Note that if $S \subseteq M$ is an immersed submanifold and $F: M \to N$ is smooth, then

$$F \circ \iota = F|_S : S \to N$$

is smooth. Thus we can restrict smooth maps to immersed submanifolds. But we can't always restrict the codomains of smooth maps, as the topology of an immersed submanifold differs, and thus the restriction may not even be continuous.

4.4.9 Theorem

Suppose $F: M \to N$ is smooth and $S \subseteq N$ is an immersed submanifold containing F's image. If the restriction $F: M \to S$ is continuous, it is smooth.

Proof

Let $p \in M$ arbitrary, and let $q = f(p) \in S$. Then $\iota: S \to N$ locally has the representation $\overline{x} \mapsto (\overline{x}, 0)$, let $(V, \psi) \subseteq S$ and $(W, \widetilde{\psi}) \subseteq N$ be charts centered at q giving this representation. This means that $\widetilde{\psi} \circ \iota \circ \psi^{-1}: \overline{x} \mapsto (\overline{x}, 0)$. If we let π be the projection $(\overline{x}, \overline{y}) \mapsto \overline{x}$, then we have

$$\pi \circ \widetilde{\psi} \circ \iota \circ \psi^{-1} = \mathrm{id} \Longrightarrow \psi = \pi \circ \widetilde{\psi} \circ \iota.$$

Since F is continuous, $U = F^{-1}(V_0)$ is an open neighborhood of p in M. Choose a smooth chart (U_0, φ) centered at p with $p \in U_0 \subseteq U$. Then if we denote by $\widetilde{F}: M \to S$ the restriction onto the codomain, we have $\iota \circ \widetilde{F} = F$, and

$$\psi \circ \widetilde{F} \circ \varphi^{-1} = \pi \circ \widetilde{\psi} \circ \iota \circ \widetilde{F} \circ \varphi^{-1} = \pi \circ \widetilde{\psi} \circ F \circ \varphi^{-1},$$

which is smooth, as F and π are. ■

4.4.10 Corollary

Suppose $F: M \to N$ is smooth and $S \subseteq N$ is an embedded submanifold containing F's image. Then the restriction of F to $M \to S$ is smooth.

Proof

The restriction of a continuous map to its image is continuous with respect to the subspace topology, which S has. ■

Note that if $S \subseteq M$ is an immersed submanifold, then by definition $d\iota_p: T_pS \to T_pM$ is an injection for every $p \in S$. Thus we can and will identify T_pS with its image in T_pM. Recall that for $f \in C^\infty(M)$ and $v \in T_pS$,

$$d\iota_p v(f) = v(f \circ \iota) = v(f|_S).$$

4.4.11 Proposition

Let $S \subseteq M$ be an immersed submanifold, with a point $p \in S$. Then the tangent space $T_pS \subseteq T_pM$ is characterized by

$$T_pS = \{v \in T_pM \mid vf = 0 \text{ for all smooth real-valued functions } f \text{ which are zero on } S\}.$$

Proof

If $v \in T_pS$ then clearly $d\iota_p v(f) = v(f|_S) = 0$ if $f|_S = 0$. So clearly the lefthand side is contained in the right. Conversely if $v(f) = 0$ for all $f|_S = 0$, then we need to find a origin for v in T_pS.

Let $(U, \varphi) \subseteq S$ and $(V, \psi) \subseteq M$ be charts centered at p for which ι's coordinate representation is $\overline{x} \mapsto (\overline{x}, 0)$ for $\overline{x} = (x^1, \ldots, x^k)$. Then we see that $T_pS \subseteq T_pM$ is spanned by $\partial/\partial x^1\big|_p, \ldots, \partial/\partial x^k\big|_p$. Now we write v in terms of the entire basis set,

$$v = \sum_{i=1}^{n} v^i \left.\frac{\partial}{\partial x^i}\right|_p.$$

Consider a bump function h supported in U which is equal to 1 in a neighborhood of p, and define $f(x) = h(x)x^j$ for $j > k$. Then f vanishes on S, but is identically 1 in a neighborhood of p so that

$$0 = v(f) = \sum_{i=1}^{n} v^i \frac{\partial h(x)x^j}{\partial x^i}(p) = v^j,$$

as desired. ■

4.4.12 Definition

Let $U \subseteq \mathbb{R}^n$ be open, then a **k-slice** of U is a subset of the form

$$S = \{(x^1, \dots, x^n) \in U \mid x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

for constants $c^{k+1}, \dots, c^n$. Note that a k-slice is homeomorphic to an open subset of $\mathbb{R}^k$ (by projecting onto the first k coordinates).

In more generality, if M is a smooth manifold with chart (U, φ), a subset $S \subseteq U$ is a **k-slice** of U if $\varphi(S)$ is a k-slice of $\varphi(U)$.

A subset $S \subseteq M$ satisfies the **local k-slice condition** if each point of S is contained in a chart U of M for which $S \cap U$ is a k-slice of U. Such a chart is called a **slice chart** for S in M and the corresponding coordinates are **slice coordinates**.

4.4.13 Theorem (Local slice criterion)

Let M be a smooth n-manifold and $S \subseteq M$ a subset endowed with the subspace topology. Then S is an embedded k-submanifold of M iff it satisfies the local k-slice condition.

Proof

If S is an embedded k-submanifold, then the inclusion $S \hookrightarrow M$ is an immersion and thus for any $p \in S$ there are smooth charts (U, φ) for S and (V, ψ) for M such that the inclusion $\iota|_U \colon U \to V$ has coordinate representation $(x^1, \dots, x^k) \mapsto (x^1, \dots, x^k, 0, \dots, 0)$.

Let $\varepsilon > 0$ small enough so that both U and V contain a coordinate ball around p of radius ε, denoted $U_0 \subseteq U$ and $V_0 \subseteq V$. It follows that $U_0 = \iota(U_0)$ is a slice of V_0, since $\psi\iota(U_0) = \psi\iota\varphi^{-1}(\varphi(U_0))$ is the projection of the k-dimensional coordinate ball $B^k_\varepsilon(p)$ into n-dimensional Euclidean space. This is a k-slice of $B^n_\varepsilon(p) = \psi(V_0)$.

Since S has the subspace topology, $U_0 = S \cap W$ for some open $W \subseteq M$. Let $V_1 = V_0 \cap W$, so $(V_1, \psi|_{V_1})$ is a smooth chart for M containing p such that $V_1 \cap S = V_0 \cap W \cap S = V_0 \cap U_0 = U_0$, which is a k-slice of V_0 and thus V_1. So S satisfies the local slice condition.

Conversely, if a subspace S satisfies the local k-slice condition, we construct a smooth atlas. Let $\pi \colon \mathbb{R}^n \to \mathbb{R}^k$ denote the projection onto the first k coordinates. Let (U, φ) be a slice chart for S in M, and define

$$V = U \cap S, \qquad \widehat{V} = \pi \circ \varphi(V), \qquad \psi = \varphi \circ \varphi|_V \colon V \to \widehat{V}.$$

Since U is a slice chart, $\varphi(V)$ is a k-slice of $\varphi(U)$, and thus π is a homeomorphism from $\varphi(V)$ onto its image in $\mathbb{R}^k$. Thus (V, ψ) is a chart for S; it is a topological k-manifold.

We show that this atlas is smooth: let $(U, \varphi), (U', \varphi')$ be two slice charts for S, and let $(V, \psi), (V', \psi')$ be the corresponding charts for S. Then

$$\psi' \circ \psi^{-1} = \pi \circ \varphi' \circ \varphi^{-1} \circ j,$$

where j is the map $\mathbb{R}^k \to \mathbb{R}^n$ obtained by appending $c^{k+1}, \dots, c^n$, which is smooth. Thus $\psi' \circ \psi^{-1}$ is smooth, as desired.

The inclusion is an immersion, since its local representation with respect to (U, φ) and (V, ψ) is just appending $c^{k+1}, \dots, c^n$, which is clearly an immersion. ∎

4.4.14 Definition

Let $\Phi \colon M \to N$ be any map and $c \in N$. Then Φ's preimage on c, $\Phi^{-1}c$, is called a **level set** of Φ.

4.4.15 Proposition

Let $\Phi\colon M \to N$ be smooth with constant rank r. Then each level set of Φ is an embedded submanifold of M of codimension r.

Proof

Let $m = \dim M, n = \dim N, k = m - r$, and choose $c \in N$. Let $S = \Phi^{-1}c \subseteq M$ be the level set at c. Then for every $p \in S$, there are smooth charts centered at $p \in (U, \varphi)$ and $c \in (V, \psi)$ such that Φ has the representatiion

$$\psi \circ \Phi \circ \varphi^{-1}\colon (x^1, \dots, x^m) \mapsto (x^1, \dots, x^r, 0, \dots, 0).$$

Then for $q \in U \cap S$, we have that $\psi \circ \Phi(q) = \psi(c) = 0$ meaning $\varphi(q) = (x^1, \dots, x^r, x^{r+1}, \dots, x^m)$ where $x^1, \dots, x^r = 0$. Thus we can define $\widetilde{\varphi}\colon U \cap S \to \mathbb{R}^k$ as the projection of φ onto its last k coordinates. These are homeomorphisms as they just map $\varphi(S \cap U) \cong 0 \times \mathbb{R}^k \cong \mathbb{R}^k$, and they are smoothly compatible. Thus they form a smooth atlas for S.

The inclusion is a smooth embedding, as $\varphi \circ \iota \circ \widetilde{\varphi}^{-1}\colon \mathbb{R}^k \to \mathbb{R}^n$ is just the inclusion and thus has full rank. ■

4.4.16 Corollary

If $\Phi\colon M \to N$ is a smooth submersion, then each level set of Φ is an embedded submanifold of M with codimension $\dim N$.

4.4.17 Definition

If $\Phi\colon M \to N$ is smooth, then a point $p \in M$ is a **regular point** of Φ if Φ's rank at p is $\dim N$, and otherwise is a **critical point**.

A point $c \in N$ is a **regular value** if each $p \in \Phi^{-1}c$ is a regular value, and a **critical value** otherwise.

If $c \in N$ is a regular value, then $\Phi^{-1}c$ is called a **regular level set**. That is, it is a level set all of whose points are regular.

Note that if $\Phi^{-1}c = \emptyset$, then c is a regular value. If Φ is a smooth submersion, then every point is regular (these are of course equivalent), and thus all of its level sets are regular. We can then improve the above corollary.

4.4.18 Corollary

Every regular level set of a smooth map is an embedded submanifold of the domain whose codimension is equal to the dimension of the codomain.

Proof

Let $\Phi\colon M \to N$ be smooth, and $c \in N$ a regular value. Let $U \subseteq M$ be the set of points $p \in M$ where $d\Phi_p = \dim N$. For $p \in U$ we can take a coordinate representation of Φ in a neighborhood of U, call it $\widehat{\Phi}$. Then $J\widehat{\Phi}(p)$ has full rank, and then must have full rank in a neighborhood of p, since $q \mapsto J\widehat{\Phi}(q)$ is continuous and the set of matrices of full rank is open. Thus U is an open set, and thus an embedded submanifold of M, and contains $\Phi^{-1}c$ by assumption.

Thus $\Phi|_U\colon U \to N$ is a smooth submersion, and by the above corollary $\Phi^{-1}c \subseteq U$ is an embedded submanifold. An embedded submanifold of an embedded submanifold is an embedded submanifold (the composition of smooth embeddings is a smooth embedding), so $\Phi^{-1}c$ is an embedded submanifold of M. ■

4.4.19 Proposition

Let $S \subseteq M$ be a subset of the m-dimensional manifold M. Then S is an embedded k-submanifold iff every point of S lies in a neighborhood U of M such that $U \cap S$ is a level set of a smooth submersion $\Phi\colon U \to \mathbb{R}^{m-k}$.

Proof

If S is an embedded k-submanifold and $x \in S$, let $(x^1, \ldots, x^k)$ be slice coordinates for S in an open neighborhood $x \in U \subseteq M$. Define $\Phi\colon U \to \mathbb{R}^{m-k}$ by $\Phi(x^1, \ldots, x^m) = (x^{k+1}, \ldots, x^m)$, which is clearly a smooth submersion. Then $\Phi^{-1}0 = \left\{(x^1, \ldots, x^m) \;\middle|\; x^1 = \cdots = x^k = 0\right\} = S \cap U$ as desired.

Conversely if $\Phi\colon U \to \mathbb{R}^{m-k}$ is a submersion, then its level sets are embedded submanifolds of M. So $S \cap U$ is an embedded submanifold of M, and thus satisfies the local slice condition, so it is an embedded submanifold of M. ■

4.4.20 Definition

Suppose $S \subseteq M$ is an embedded submanifold. If $U \subseteq M$ is an open subset and $\Phi\colon U \to N$ is a smooth map such that $S \cap U$ is a regular level set of Φ, then Φ is called a **local defining map for** S.

4.4.21 Corollary

Every embedded submanifold admits a local defining map in a neighborhood of each of its points.

4.5 Submersions

A **section** of a map $\pi\colon M \to N$ is a map $\sigma\colon N \to M$ which is a right inverse: $\pi \circ \sigma = \mathrm{id}_N$. If our maps are continuous or smooth, we speak of continuous or smooth sections. A **local section** is a continuous map $\sigma\colon U \to M$ defined in an open subset of N where $\pi \circ \sigma_U = \mathrm{id}_U$.

4.5.1 Theorem

If $\pi\colon M \to N$ is smooth, then it is a smooth submersion iff every point of M is in the image of a smooth local section.

Proof

If π is a smooth submersion, then it has the local representation $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^n)$ around any point $p \in M$. For $\varepsilon > 0$ small enough, the coordinate cube $C_\varepsilon = \left\{x \;\middle|\; \left|x^i\right| < \varepsilon \text{ for } i = 1, \ldots, m\right\}$ is an open neighborhood centered at p whose image under π is the coordinate cube $C'_\varepsilon = \left\{y \;\middle|\; \left|y^i\right| < \varepsilon \text{ for } i = 1, \ldots, n\right\}$. Then defining $\sigma\colon C'_\varepsilon \to C_\varepsilon$ by $\sigma(y) = (y, 0)$ gives a local section of π and $p = \sigma(0)$ as desired.

Conversely let $\sigma\colon U \to M$ be a local section with p in its image, say $p = \sigma(q)$. Then since $\pi \circ \sigma = \mathrm{id}_U$, we have that $d\pi_p \circ d\sigma_q = \mathrm{id}_{T_qN}$, which means $d\pi_p$ must be surjective. ■

This allows us to extend the definition of a smooth submersion to a topological one: a **topological submersion** is a continuous map $X \to Y$ such that every point of X is in the image of a local section of the map. This theorem shows that a smooth submersion is just a smooth topological submersion.

4.5.2 Definition

Recall that a continuous map $\pi: X \to Y$ is a **quotient map** if it is surjective and Y's topology is the quotient topology induced by π. That is, $U \subseteq Y$ is open iff $\pi^{-1}U \subseteq X$ is.

Note that a quotient map is just a surjective, continuous, and open (equivalently closed) map.

4.5.3 Proposition

Let $\pi: M \to N$ be a smooth submersion. Then π is an open map, and if it is surjective it is a quotient map.

Proof

Let $W \subseteq M$ open, and $q \in \pi(W)$. Then for any $p \in W$ such that $\pi(p) = q$, there is a neighborhood U of q and a local section $\sigma: U \to N$ such that $\sigma(q) = p$. (Note that p being in the image of σ requires $\sigma(q) = p$, since if $\sigma(r) = p$ then $r = \pi\sigma(r) = \pi(p) = q$.) Then for $y \in \sigma^{-1}(W)$, since $\sigma(y) \in W$ we have that $y \in \pi(W)$. And then $\sigma^{-1}(W)$ is an open subset of $\pi(W)$ containing q, and so $\pi(W)$ is open.

Thus π is an open map. Surjective open continuous maps are quotient maps. ∎

4.5.4 Theorem

Let $\pi: M \to N$ be a surjective smooth submersion. Then for any manifold P, a map $F: P \to N$ is smooth iff $F \circ \pi: M \to P$ is smooth.

Proof

If F is smooth then so is $F \circ \pi$ is. Conversely if $F \circ \pi$ is smooth, let $q \in N$ and since π is surjective, there is a $p \in M$ with $\pi(p) = q$. And then there is a local section $\sigma: U \to M$ with U a neighborhood of q, and $\sigma(q) = p$. Then $\pi \circ \sigma = \mathrm{id}_U$ implies

$$F|_U = F|_U \circ \mathrm{id}_U = F|_U \circ (\pi \circ \sigma) = (F \circ \pi) \circ \sigma,$$

which is smooth. So for every point in N, there is a neighborhood around it in which F is smooth. Thus F is smooth. ∎

4.5.5 Theorem

Let $\pi: M \to N$ be a surjective smooth submersion. If $F: M \to P$ is smooth which is constant on the fibers of π, there is a unique map $\widetilde{F}: N \to P$ such that $F = \widetilde{F} \circ \pi$.

Proof

As a quotient space, there is a continuous $\widetilde{F}$ satisfying this. Then since $\widetilde{F} \circ \pi$ is smooth, so is $\widetilde{F}$. ∎

5 Sard's Theorem

5.1 Sets of measure zero

5.1.1 Definition

Recall that a subset $S \subseteq \mathbb{R}^n$ is said to be of **measure zero** if for every $\varepsilon > 0$, S can be covered by a countable collection of open rectangles where the sum of their volume is less than ε. Equivalently, S literally has zero measure with respect to the Lebesgue measure on $\mathbb{R}^n$.

5.1.2 Lemma

Suppose that $A \subseteq \mathbb{R}^n$ is measurable (including Borel subsets), such that for every $c \in \mathbb{R}$, A's intersection with $\{c\} \times \mathbb{R}^{n-1}$ has $(n-1)$-dimension measure zero. Then A has measure zero.

Proof

A is measurable and so its indicator function χ_A is integrable. By Fubini's theorem,

$$\mu(A) = \int_{\mathbb{R}^n} \chi_A \, d(x^1, \ldots, x^n) = \int_{\mathbb{R}} \int_{\mathbb{R}^{n-1}} \chi_A(c, x^2, \ldots, x^n) \, d(x^2, \ldots, x^n) \, dx^1,$$

where μ is the Lebesgue measure. Now the integral of $\chi_A(c, \overline{x})$ is the measure of the intersection of A with $\{c\} \times \mathbb{R}^{n-1}$, which we assumed to be zero. Thus the integral is zero and A has measure zero. ■

5.1.3 Proposition

Let A be open or closed in $\mathbb{R}^{n-1}$ or $\mathbb{H}^{n-1}$, and $f \colon A \to \mathbb{R}$ continuous. Then f's graph has measure zero in $\mathbb{R}^n$.

Proof

Note that f's graph is closed: it is the preimage of 0 under the map $g \colon \mathbb{R}^n \to \mathbb{R}$ mapping $(\overline{x}, a)$ to $f(\overline{x}) - a$. For $c \in \mathbb{R}$, f's graph's intersection with $\{c\} \times \mathbb{R}^{n-1}$ is the graph of the function mapping $x \in \mathbb{R}^{n-2}$ to $f(c, x)$, i.e. the graph of the restriction of f to $\mathbb{R}^{n-2}$. We can then induct, showing that for every $c \in \mathbb{R}$ these have measure zero, and then appeal to the above lemma. ■

5.1.4 Corollary

Every proper affine subspace of $\mathbb{R}^n$ has measure zero.

Proof

If S is an affine subspace of $\mathbb{R}^n$ of dimension $n-1$, it is the graph of an affine function $\mathbb{R}^{n-1} \to \mathbb{R}$. By the above proposition, it has measure zero. ■

5.1.5 Proposition

Suppose $A \subseteq \mathbb{R}^n$ has measure zero and $F \colon A \to \mathbb{R}^n$ is smooth. Then $F(A)$ has measure zero.

Proof

For $p \in A$, F extends to an open ball around p. By potentially shrinking the ball, we can extend F to its closure, a compact set $\overline{U}$. A can be covered by countably many $A \cap \overline{U}$s, so it is sufficient to show that $F(A \cap \overline{U})$ has measure zero.

Since $\overline{U}$ is compact, $|JF(x)|$ is bound by some constant C on it (this is true for all continuous real-valued functions on compact sets). Then we see that $|F(x) - F(y)| \leq C\,|x - y|$ for all $x, y \in \overline{U}$. Since A has measure zero, for any $\varepsilon > 0$ we can cover $A \cap \overline{U}$ with balls $\{B_j\}$ of total measure $< \varepsilon$. By Lipschitz continuity, $F(B_j)$'s measure is bound by C times B_j's, so that

$$\mu F(A \cap \overline{U}) \leq \sum_j \mu F(B_j) \leq \sum_j C\mu(B_j) < C\varepsilon.$$

Since ε is arbitrary, we have the desired result. ■

Note:

In the above proof we used the fact that smooth functions from a compact set are Lipschitz continuous. Indeed, if $F\colon \overline{U} \to \mathbb{R}^n$ is smooth (even C^1), then let $|JF(x)| \leq C$ for all $x \in C$, then

$$F(y) - F(x) = \int_0^1 \frac{d}{dt} F(x + t(y - x))\,dt$$

by the fundamental theorem of calculus. Now,

$$\frac{d}{dt} F(x + t(y - x)) = (y - x) JF(x + t(y - x))$$

so we have that

$$|F(y) - F(x)| = \left| \int_0^1 (y - x) JF(x + t(y - x))\,dt \right| \leq \int_0^1 |y - x|\,|JF(x + t(y - x))|\,dt.$$

We can bound the Jacobian by C, giving us

$$\leq |y - x| \int_0^1 C = C\,|y - x|$$

as desired.

5.1.6 Definition

If M is a smooth n-manifold, with or without boundary, a subset $A \subseteq M$ has **measure zero** if for every smooth chart (U, φ) for M, $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has measure zero.

5.1.7 Lemma

Let $(U_\alpha, \varphi_\alpha)$ be a collection of smooth charts covering M. If $A \subseteq M$ is a subset where $\varphi(A \cap U_\alpha)$ has measure zero for all α, then A has measure zero.

Proof

Let (V, ψ) be an arbitrary smooth chart, we need to show $\psi(A \cap V)$ has measure zero. For each α, consider

$$\psi(A \cap V \cap U_\alpha) = (\psi \circ \varphi_\alpha)^{-1} \circ \varphi_\alpha(A \cap V \cap U_\alpha),$$

which has measure zero by the above proposition. Since $\psi(A \cap V)$ is covered by countably many of these, it has measure zero. ■

5.1.8 Proposition

Let $A \subseteq M$ be a subset with measure zero. Then $M - A$ is dense in M.

Proof

If not, then there is an open subset U contained in A Let (V, φ) be a chart contained in $U \subseteq V$, then $\varphi(A \cap V) = \varphi(V)$, which is nonempty open in $\mathbb{R}^n$. But open subsets of $\mathbb{R}^n$ cannot have measure zero (they contain an open ball/rectangle which by definition has positive measure), a contradiction. ■

5.1.9 Theorem

Let $F\colon M \to N$ be smooth, and $A \subseteq M$ of measure zero. Then $F(A) \subseteq N$ has measure zero.

Proof

Let $(U_\alpha, \varphi_\alpha)$ cover M, and let (V, ψ) be a chart for N. Then $\psi(F(A) \cap V)$ can be covered by $\psi \circ F \circ \varphi_\alpha^{-1}(\varphi_\alpha(A \cap U_\alpha \cap F^{-1}(V)))$. Now $\psi \circ F \circ \varphi_\alpha^{-1}$ is a smooth Euclidean map, and $\varphi_\alpha(A \cap U_\alpha \cap F^{-1}(V))$ has measure zero by assumption, so these coverings have measure zero. Thus so does $\psi(F(A) \cap V)$ as desired. ■

5.1.10 Definition

Let $F\colon M \to N$ be smooth. A point $p \in M$ is a **regular point** of F if F has rank $\dim N$ at p, otherwise it is a **critical point**. The image of a regular point is called a **regular value**, and the image of a critical point is a **critical value**.

In other words, a point is a regular point iff $dF_p\colon T_pM \to T_{Fp}N$ is a surjection. Notice that if $\dim M < \dim N$, every point is critical.

5.1.11 Theorem (Sard's theorem)

Let $F\colon M \to N$ be smooth map. Then the set of critical values has measure zero in N.

Proof

Let $m = \dim M, n = \dim N$, we induct on m. For $m = 0$ it is clear: either $n = 0$ for which F has no critical points, or $n > 0$ and all points are critical but M is countable and thus $F(M) \subseteq N$ has measure zero.

By covering F with countably many smooth charts, we can reduce this to the case that F is a smooth map from an open subset U of $\mathbb{R}^m$ or $\mathbb{H}^m$ to $\mathbb{R}^n$. Let $C \subseteq U$ be the set of critical points of F. Define

$$C_k = \{x \in C \mid \text{For } 1 \leq i \leq k, \text{ all of the } i\text{th partial derivatives of } F \text{ vanish at } x\}.$$

Then C_k is decreasing, $C \supseteq C_1 \supseteq C_2 \supseteq \cdots$, and by continuity they are all closed in U.

Now we prove in three parts that $F(C)$ has measure zero.

(1) Firstly, $F(C - C_1)$ has measure zero. Because C_1 is closed in U, we can take $U = U - C_1$ and assume $C_1 = \emptyset$. Let $a \in C$, so we assumed that F's first partial derivatives are not all 0 at a. Without loss of generality, $\partial F / \partial^1 x^1(a) \neq 0$. So we can define new smooth coordinates $(u, v) = (u, v^2, \ldots, v^m)$ in a neighborhood of a, where $u = F^1$ and $v^i = x^i$. This is because the Jacobian of these coordinates is invertible at a and thus locally a diffeomorphism. Let V_a be the neighborhood of a using these coordinates, and potentially shrinking it, we can assume $\overline{V}_a$ is compact in U.

In these coordinates, F has representation

$$F(u, v^2, \dots, v^m) = (u, F^2(u, v), \dots, F^n(u, v)),$$

so its Jacobian is

$$JF(u, v) = \begin{pmatrix} 1 & 0 \\ * & \frac{\partial F^i}{\partial v^j} \end{pmatrix}.$$

Thus $C \cap \overline{V}_a$ consists of points the $(n-1) \times (m-1)$ matrix $\frac{\partial F^i}{\partial v^j}$ has rank $< n-1$.

We wish to show that $F(C \cap \overline{V}_a)$ has measure zero in N. This set is compact, and thus measurable, so it is sufficient to show that its intersection with the hyperplane $y^1 = c$ has $n-1$-dimensional measure zero. For $c \in \mathbb{R}$, define $B_c = \{v \mid (c, v) \in \overline{V}_a\} \subseteq \mathbb{R}^{m-1}$, and define $F_c \colon B_c \to \mathbb{R}^{n-1}$ by

$$F_c(v) = (F^2(c, v), \dots, F^n(c, v)).$$

Because $F(c, v) = (c, F_c(v))$, the critical values of $F|_{\overline{V}_a}$ which lie in the hyperplane $y^1 = c$ are precisely the points of the form (c, w) where w is a critical value of F_c. By induction, these have $n-1$-dimensional measure zero, so $F(C \cap \overline{V}_a)$ has measure zero.

We can cover U with countably many sets of the form $\overline{V}_a$, and thus C by countably many sets of the form $C \cap \overline{V}_a$. Thus $F(C)$ is the countable union of sets of measure zero, and thus has measure zero.

(2) For the second step, we claim $F(C_k - C_{k+1})$ has measure zero for each k. As before, we can take $U = U - C_{k+1}$ and assume that every point in C_k has a $k+1$th partial derivative which does not vanish. Let $a \in C_k$, and denote by $y \colon U \to \mathbb{R}$ some kth partial derivative of F which has a nonvanishing partial derivative at a. Since $a \in C_k$ we have that $y(a) = 0$.

Then a is a regular point of y, so it has a neighborhood V_a consisting of regular points of y. Let Y be the zero set of y in V_a, i.e. $Y = y^{-1}0$. Since V_a are regular points of y, by definition Y is a regular level set, and thus an embedded submanifold of V_a of codimension 1. Since V_a is open, Y is an embedded submanifold of M of dimension $m-1$.

For any $p \in C_k \cap V_a$, dF_p is not surjective (by assumption, p is critical), so then neither is $d(F|_Y)_p = (dF_p)|_{T_pY}$. So $F(C_k \cap V_a)$ is contained in the critical values of $F|_Y \colon Y \to \mathbb{R}^n$, which by induction has measure zero. Since U can be covered by countably many V_a, the result follows.

(3) Now, notice that the intersection of all $C - C_1$ and $C_k - C_{k+1}$ need not be all of C: we may have points in C all of whose partial derivatives vanish. So for our final step, we claim that for $k > m/n + 1$, $F(C_k)$ has measure zero.

For $a \in U$, let $E \subseteq U$ be a closed cube containing a. Since U can be covered by countably many such cubes, we will show $F(C_k \cap E)$ has measure zero for all such closed cubes.

Since E is compact, we can find a value $A > 0$ which bounds the absolute values of all of F's $k+1$th partial derivatives in E. Let R be the side length of E, and let K be some integer to be chosen. Then we can subdivide E into K^m subcubes of side length R/K, denoted by $(E_1, \dots, E_{K^m})$. If E_i is a cube and there is an $a_i \in E_i \cap C_k$, then for all $x \in E_i$ we have

$$|F(x) - F(a_i)| \leq A' \, |x - a_i|^{k+1}$$

for some value A' dependent only on A, k, m. This means that $F(E_i)$ is contained in a ball of radius $A'(R/K)^{k+1}$, so of measure $A''(R/K)^{nk+n}$. Then $F(C_k \cap E)$ is contained in the union of K^m balls of this radius. So its measure is bound by

$$K^m \cdot A''(R/K)^{nk+n} = A'' R^{nk+n} K^{nk+n-m}.$$

By assumption $nk + n - m < 1$, so this can be made arbitrarily small by taking large enough K, so $F(C_k \cap E)$ has measure zero.

Now, we just note that

$$C = (C - C_1) \cup \bigcup_k (C_k - C_{k+1}) \cup \bigcup_{k > m/n+1} C_k,$$

a countable union of sets of measure zero, and thus C has measure zero. ■

Note:
In the above proof we had a point where we had a bound like

$$|F(x) - F(a)| \leq A\,|x-a|^{k+1}.$$

We obtain this bound by considering each of F's coordinate's Taylor expansions.
If $f\colon W \subseteq \mathbb{R}^n \to \mathbb{R}$ is smooth and defined in some convex subset W, then take f's kth order Taylor expansion $P_k(x)$ around $a \in W$. Its remainder can be bound by

$$|f(x) - P_k(x)| \leq \frac{n^{k+1}M}{(k+1)!}\,|x-a|^{k+1},$$

as well as

$$|P_k(x) - f(a)| \leq \frac{kn^k M}{k!}\,|x-a|^k$$

(Just take a good look at the remainder and polynomial expressions, here M is a bound on the partial derivatives.) Then we have

$$|f(x) - f(a)| \leq |f(x) - P_k(x)| + |P_k(x) - f(a)| \leq A\,|x-a|^{k+1}$$

for some absolute constant A dependent only on n, k, and M (a bound on the partial derivatives). We then get that

$$|F(x) - F(a)| = \sqrt{\sum_i |F^i(x) - F^i(a)|^2} \leq nA\,|x-a|^{k+1}.$$

5.2 Transversality

5.2.1 Definition

Let $F\colon M \to N$ be a smooth map and $S \subseteq N$ an embedded submanifold. We say F is **transverse** to S if for every $x \in F^{-1}(S)$, $T_{Fx}S$ and $dF_x(T_xM)$ span $T_{Fx}N$.

Two embedded submanifolds $S, S' \subseteq M$ **intersect transversally** if for each $p \in S \cap S'$, T_pS and T_pS' span T_pM.

Note that S, S' intersect transversally iff the inclusion of one in M is transverse to the other.

Furthermore if F is a submersion, then dF_x is surjective and thus F is transverse to every embedded submanifold.

5.2.2 Theorem

Let M, N be smooth manifolds, $S \subseteq N$ an embedded submanifold.

(1) If $F\colon M \to N$ is transverse to S then $F^{-1}S$ is an embedded submanifold of M whose codimension in M is equal to S's codimension in N.

(2) If $S' \subseteq N$ is an embedded submanifold that intersects S transversally, $S \cap S'$ is an embedded submanifold of N whose codimension is equal to the sum of the codimensions of S and S'.

Proof

The second point follows from the first, by taking F to be the inclusion $S' \hookrightarrow N$. Then $F^{-1}S = S \cap S'$ is an embedded submanifold of S' of codimension $\operatorname{codim}_N S$ by the first point. The composition of embeddings is an embedding, so $F^{-1}S$ is embedded in N, and its codimension is

$$\begin{aligned}\operatorname{codim}_N(S\cap S') = \dim N - \dim S' + \dim S' - \dim(S\cap S') &= \operatorname{codim}_N S' + \operatorname{codim}_{S'}(S\cap S')\\ &= \operatorname{codim}_N S' + \operatorname{codim}_N S'.\end{aligned}$$

Let $n = \dim N$ and $k = \operatorname{codim}_N S$. For $x \in F^{-1}(S)$, we find a neighborhood U of $F(x)$ in M and a local defining function $\varphi\colon U \to \mathbb{R}^k$ such that $S \cap U = \varphi^{-1}0$. If we can show that 0 is a regular value of $\varphi \circ F$, because $F^{-1}(S) \cap F^{-1}(U)$ is the zero set of $\varphi \circ F|_{F^{-1}U}$.

For $z \in T_0\mathbb{R}^k$ and $p \in (\varphi\circ F)^{-1}0$, 0 being a regular value of φ means that there is a vector $y \in T_{Fp}N$ such that $d\varphi_{Fp}y = z$. F being transverse to S means that $y = y_0 + dF_p v$ for some $y_0 \in T_{Fp}S$ and $v \in T_pM$. Because φ is constant on $S \cap U$, it follows that $d\varphi_{Fp}y_0 = 0$ so

$$z = d\varphi_{Fp}y = d\varphi_{Fp}y_0 + d\varphi_{Fp}dF_pv = d(\varphi\circ F)_pv.$$

So $\varphi \circ F$ is indeed a submersion. ■

Note that this proof showed that $x \in F^{-1}S$ is a regular point of F.

5.2.3 Proposition

If $F\colon M \to N$ is transverse to $S \subseteq N$, then the points of $F^{-1}S$ are regular points of F.

Since submersions are transverse to every embedded submanifold, the following corollary is immediate.

5.2.4 Corollary

If $S \subseteq N$ is an embedded submanifold, and $F\colon M \to N$ is a submersion, then $F^{-1}S$ is an embedded submanifold of M with

$$\operatorname{codim}_M F^{-1}S = \operatorname{codim}_N S.$$

5.2.5 Definition

Suppose N, M, S are smooth manifolds, and for each $s \in S$ we are given a map $F_s\colon N \to M$. The collection $\{F_s\}_{s\in S}$ is called a **smooth family of maps** if $F\colon M \times S \to N$ defined by $F(m,s) = F_s(m)$ is smooth.

The notion of a smooth family of maps is a sort of higher-dimensional analogue of homotopy.

5.2.6 Proposition

If $\{F_s\}_{s\in S}$ is a smooth family of maps where S is connected, then for every $s_1, s_2 \in S$, F_{s_1}, F_{s_2} are homotopic.

Proof

Since S is connected, let $\gamma\colon I \to S$ be a path from s_1 to s_2. Then $H\colon M \times I \to N$ given by $H(m,t) = F_{\gamma(t)}(m)$ is a homotopy from F_{s_1} to F_{s_2}. ■

5.2.7 Lemma

If $F\colon N \to M$ is transverse to $X \subseteq M$ embedded, let $W = F^{-1}X$. Then for every $p \in W$, $dF_p(T_pW) = T_{Fp}X$ and $T_pW = (dF_p)^{-1}(T_{Fp}X)$. Thus if $X, X' \subseteq M$ intersect transversally then $T_p(X \cap X') = T_pX \cap T_pX'$ for all $p \in X \cap X'$.

Proof

The second point is due to $X \cap X'$ being the preimage of X' under the inclusion $X \hookrightarrow M$. Then $(d\iota_p)^{-1}(T_pX') = T_pX \cap T_pX'$.
The first point is because F maps W into X so $dF_p(T_pW) \subseteq T_{Fp}X$ for all $p \in W$. Now by rank-nullity,

$$\dim dF_p(T_pW) = \dim T_pW - \dim \ker dF_p = \dim W - \dim \ker dF_p.$$

Now recall that p is a regular point of F, so dF_p is surjective and thus

$$\dim \ker dF_p = \dim T_pN - \dim T_{Fp}M = \dim N - \dim M,$$

so

$$\dim dF_p(T_pW) = \dim W - \dim N + \dim M = \dim M - \operatorname{codim}_N W = \dim X,$$

since $\operatorname{codim}_M X = \operatorname{codim}_N W$, and thus considering dimensions $dF_p(T_pW) = T_{Fp}X$.
And so $T_pW \subseteq (dF_p)^{-1}(T_{Fp}X)$, as well, and considering dimensions they are equal. Indeed, $\dim T_pW = \dim W$ and in general if $T\colon V \to U$ is a surjective linear map, and $U' \subseteq U$ a subspace, then

$$\dim T^{-1}U' = \dim V - \dim U' + \dim T(U').$$

This is because $T|_{T^{-1}U'}\colon T^{-1}U' \to U'$ has kernel $\ker T$, and thus by rank-nullity

$$\dim T^{-1}U' = \dim \ker T + \dim U' = \dim V - \dim U + \dim U'.$$

dF_p is surjective onto its image $dF_p(T_pM)$, so

$$\dim (dF_p)^{-1}T_{Fp}X = \dim T_pN - \dim dF_p(T_pN) + \dim T_{Fp}X = \dim N - \dim dF_p(T_pN) + \dim X.$$

Since $p \in F^{-1}X$, it is a regular point of F, and so $dF_p(T_pN) = T_pM$. Thus this is equal to

$$= \dim N - \dim M + \dim X = \dim W.$$

■

5.2.8 Definition

A subset of a manifold is said to contain **almost every element** if its complement has measure zero.

5.2.9 Theorem (Parametric transversality)

Suppose N, M are smooth manifolds, $X \subseteq M$ an embedded submanifold, and $\{F_s\}_{s\in S}$ a smooth family of maps from N to M. If $F\colon N \times S \to M$ is transverse to X, then for almost every $s \in S$, F_s is transverse to X.

Proof

By assumption $W = F^{-1}(X)$ is an embedded submanifold of $N \times S$. Let $\pi\colon N \times S \to S$ be projection onto the second component. We will show that if $s \in S$ is a regular value of $\pi|_W$, then F_s is transverse to X. By Sard's theorem almost every $s \in S$ is a regular value, so the theorem will be proven.

If $s \in S$ is a regular value of $\pi|_W$, then let $p \in F_s^{-1}(X)$ be arbitrary, and set $q = F_s(p)$. We want to show that $T_qM = T_qX + d(F_s)(T_pN)$. By our hypothesis on F,

$$T_qM = T_qX + dF(T_{(p,s)}(N \times S)),$$

and since s is a regular value of $\pi|_W$ and $(p,s) \in W$, we have

$$T_sS = d\pi(T_{(p,s)}W).$$

The above lemma showed that

$$dF_{(p,s)}(T_{(p,s)}W) = T_qX.$$

We want to show that F_s is transverse to X, i.e.

$$T_qM = T_qX + dF_s(T_pN).$$

So let $w \in T_qM$ arbitrary, we want $v \in T_qX$ and $y \in T_pN$ such that $w = v + dF_s(y)$. Since F is transverse to X, there is $v_1 \in T_qX$ and $(y_1, z_1) \in T_pN \times T_sS \cong T_{(p,s)}(N \times S)$ such that

$$w = v_1 + dF(y_1, z_1).$$

And then since $T_sS = d\pi(T_{(p,s)}W)$ there is a $(y_2, z_2) \in T_{(p,s)}W$ such that $d\pi(y_2, z_2) = z_1$. Since π is a projection, this means $z_2 = z_1$. By linearity,

$$dF(y_1, z_1) = dF(y_2, z_1) + dF(y_1 - y_2, 0).$$

Now,

$$dF(y_2, z_1) = dF(y_2, z_2) \in dF(T_{(p,s)}W) = T_qX.$$

On the other hand let $\iota_s \colon N \to N \times S$ be the slice map $\iota_s(n) = (n, s)$ so $F_s = F \circ \iota_s$ and $d\iota_s(y_1 - y_2) = (y_1 - y_2, 0)$ so

$$dF(y_1 - y_2, 0) = dF \circ d\iota_s(y_1 - y_2) = dF_s(y_1 - y_2).$$

And so all in all we have

$$w = (v_1 + dF(y_2, z_1)) + dF_s(y_1 - y_2),$$

where $v_1, dF(y_2, z_1) \in T_qX$ and $y_1 - y_2 \in T_pN$, as desired. ■

5.2.10 Theorem (Transversality homotopy)

If $F \colon N \to M$ is smooth and $X \subseteq M$ is embedded, then F is homotopic to a smooth map $G \colon N \to M$ which is transverse to X.

Proof

The idea of the proof is to define a smooth map $H \colon N \times S \to M$ which is transverse to X, where S is connected, and $H_s = F$ for some $s \in S$. Then by parametric transversality there is a $t \in S$ such that H_t is transverse to X. And $F = H_s$ and $G = H_t$ are homotopic, as desired.

We omit the details. ■

6 Lie Groups

6.1 Basic definitions

6.1.1 Definition

A **Lie group** is a smooth manifold G without boundary, along with smooth maps

$$m: G \times G \to G, \qquad i: G \to G$$

turning G into a group with multiplicatio given by m and inversion given by i.

Note:

A Lie group is a special type of **topological group**, which is a group which is also a topological space making multiplication and inversion into continuous functions.

Note that G is a Lie group iff the map $(g, h) \mapsto gh^{-1}$ is smooth. This map is a composition of multiplication and inversion, so if G is a Lie group it is smooth, and conversely we note that it being smooth implies i is smooth, and then that m is smooth.

For a Lie group G and $g \in G$, define **left translation** and **right translation** by

$$L_g, R_g: G \to G, \qquad L_g(h) = gh, \quad R_g(h) = hg.$$

These are smooth maps, as they are simply compositions of m with a suitable inclusion $G \to G \times G$. Moreso, they are diffeomorphisms: the inverse of L_g is $L_{g^{-1}}$ and similar for right translations.

6.1.2 Example

Euclidean space $\mathbb{R}^n$ is a Lie group under addition, clearly.

6.1.3 Example

The general linear group $\mathrm{GL}_n(\mathbb{R})$ is an open submanifold of the Euclidean space $\mathrm{Mat}_n(\mathbb{R})$. Furthermore it is a Lie group, with multiplication and inversion given by usual multiplication and inversion of matrices. These are smooth, as the coordinates of multiplication are polynomial in their inputs and inversion can be written as

$$A^{-1} = 1/\det(A) \cdot \mathrm{adj}(A),$$

the product of smooth functions (the coordinates of the adjugate are determinants of minors).

Denote by $\mathrm{GL}_n^+(\mathbb{R})$ the subgroup of $\mathrm{GL}_n(\mathbb{R})$ with positive determinant. This is an open subspace of $\mathrm{GL}_n(\mathbb{R})$ (the preimage of $(0, \infty)$) and the restriction of multiplication and inversion to it is still smooth (this is true for embedded submanifolds in general).

Continuing the above example, if G is a Lie group and $H \subseteq G$ is a subgroup which is also an open set, then it is also a Lie group.

6.1.4 Example

The complex general linear group $\mathrm{GL}_n(\mathbb{C})$ is the group of invertible matrices in $\mathrm{Mat}_n(\mathbb{C})$. This is an open subset of $\mathrm{Mat}_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$ and thus a $2n^2$-dimensional Lie group.

6.1.5 Definition

If G and H are Lie groups, a **Lie group morphism** $G \to H$ is a smooth map which is also a group morphism. A **Lie group isomorphism** is a Lie group morphism which is also a diffeomorphism (and in particular a group isomorphism).

6.1.6 Example

Consider $\mathbb{R}$ as a Lie group under addition, and $\mathbb{R}^\times$ a Lie group under multiplication. (We need not show that these are Lie groups again, $\mathbb{R}$ is Euclidean and $\mathbb{R}^\times = \mathrm{GL}_1(\mathbb{R})$.) Then the exponential map, $\exp\colon \mathbb{R} \to \mathbb{R}^\times$, is a Lie group morphism. Its image is $\mathbb{R}^+$, and it has an inverse $\log\colon \mathbb{R}^+ \to \mathbb{R}$, so that $\mathbb{R}^+$ (with multiplication) and $\mathbb{R}$ (with addition) are isomorphic Lie groups.

6.1.7 Example

The determinant $\det\colon \mathrm{GL}_n(\mathbb{C}) \to \mathbb{R}^\times$ is a Lie group morphism.

Most importantly, if G is a Lie group and $g \in G$, then **conjugation** by g,

$$C_g\colon G \to G, \qquad C_g(h) = ghg^{-1}$$

is a Lie group morphism. It is clearly smooth, as a composition of multiplication and inversion, and it is an isomorphism with inverse $C_{g^{-1}}$. Recall that $H \subseteq G$ is **normal** if $C_g(H) = H$ for all $g \in G$.

6.1.8 Theorem

Every Lie group morphism has constant rank.

Proof

Let $F\colon G \to H$ be a Lie group morphism, with $e, \widetilde{e}$ the respective units of the groups. Now let $g_0 \in G$ and notice that

$$F(L_{g_0}(g)) = F(g_0 g) = F(g_0)F(g) = L_{F(g_0)}F(g)$$

so that $F \circ L_{g_0} = L_{F(g_0)} \circ F$. Thus we get that

$$dF_{g_0} \cdot d(L_{g_0})_e = d(L_{F(g_0)})_{\widetilde{e}} \cdot dF_e.$$

Since left translations are diffeomorphism, they do not change rank, and thus dF_{g_0} and dF_e have the same rank, meaning F has constant rank. ■

6.1.9 Corollary

A Lie group morphism is an isomorphism iff it is bijective.

Proof

A bijective Lie group morphism is a bijective immersion (by constant rank), thus a diffeomorphism.

■

6.2 Lie subgroups

6.2.1 Definition

Let G be a Lie group. A **Lie subgroup** of G is a subgroup $H \leq G$ endowed with a smooth structure making it a Lie group and an immersed submanifold.

6.2.2 Proposition

Let G be a Lie group, and $H \leq G$ a subgroup which is an embedded submanifold. Then H is also a Lie group, and thus a Lie subgroup of G.

Proof

Since H is embedded, multiplication $G \times G \to G$ restricts both on its domain (for all submanifolds) and its codomain (as H is embedded), to give H's multiplication $H \times H \to H$. Same for inversion. ■

6.2.3 Lemma

Let G be a Lie group and $H \subseteq G$ an open subgroup. Then H is an embedded Lie subgroup. Moreso, H is also closed, and thus a union of connected components of G.

Proof

By the above proposition, H is an embedded Lie subgroup. Furthermore, each coset gH is open in G as the image of H under the diffeomorphism L_g. Then $G - H = \bigcup_{g \in G-H} gH$ is open, and so H is closed. Clopen sets are unions of connected components. ■

6.2.4 Proposition

Let G be a Lie group, and $W \subseteq G$ an open, then

(1) W generates an open subgroup of G,

(2) if W is a connected neighborhood of the identity, it generates a connected open subgroup of G,

(3) if G is connected, then W generates G.

Proof

Denote for arbitrary sets A, B,

$$AB = \{ab \mid a \in A, b \in B\}, \qquad A^{-1} = \left\{a^{-1} \mid a \in A\right\}.$$

For each $k > 0$ define by W_k the set of all elements in G that can be written as a product of k or fewer elements of $W \cup W^{-1}$. Then W's generated subgroup is $\bigcup_{k>0} W_k$.
Note that $W_1 = W \cup W^{-1}$ (we do not allow empty products, defined to be the identity, so if $e \notin W$ then it is not in W_1, but is in W_2). This is open, as $W^{-1} = i(W)$ is the diffeomorphic image of an open set. And inductively we have

$$W_k = W_1 W_{k-1} = \bigcup_{g \in W_1} L_g(W_{k-1})$$

is open. Thus the subgroup generated by W is a union of open sets and thus open.
If W is a connected neighborhood of e, then so is W^{-1} (as it is the diffeomorphic image of a connected set, and contains e). Then $W_1 = W \cup W^{-1}$ is also a connected neighborhood of e, and inductively so are $W_k = m(W_1 \times W_{k-1})$, thus so is their union. This is because $W_1 \times W_{k-1}$ is connected, and the continuous image of a connected space is connected, and the union of connected spaces with nontrivial intersection is connected (each W_k contains e).
If G is connected, then the subgroup W generates is open by point (1), and thus closed. Only trivial subspaces of a connected space are clopen, so W generates G. ■

For a Lie group G, the connected component containing the identity is called the **identity component**.

6.2.5 Proposition

Let G_0 be the identity component of G. Then G_0 is a normal subgroup of G, and it is the only connected open subgroup. Every connected component of G is diffeomorphic to G_0.

Proof

Any connected open neighborhood of the identity generates G_0 by the above proposition. It is normal since $C_g(G_0)$ is a connected neighborhood of the identity, and thus contained in G_0 for all g. And L_g maps G_0 diffeomorphically to the connected component of $g \in G$. ■

6.2.6 Proposition

Let $F\colon G \to H$ be a Lie group morphism, then $\ker F$ is a properly embedded Lie subgroup of G, whose codimension is equal to F's rank.

Proof

$\ker F$ is the level set $F^{-1}e$, and since F has constant rank, it is a properly embedded submanifold of G. We already showed that subgroups which are embedded submanifolds of Lie groups are Lie subgroups. ■

6.2.7 Proposition

If $F\colon G \to H$ is an injective Lie group morphism, its image has a unique smooth manifold structure turning it into a Lie subgroup of H with F an isomorphism onto its image.

Proof

F is injective of constant rank, and thus an immersion. Thus $F(G)$ has a unique smooth manifold structure turning it into an immersed submanifold of H diffeomorphic to G. Since it is diffeomorphic to G, it is also a Lie group and a subgroup (for obvious algebraic reasons). Thus it is a Lie subgroup. ■

6.2.8 Example

The set $\mathrm{SL}_n(\mathbb{R})$ of $n \times n$ real matrices of determinant 1 is a properly embedded Lie subgroup of $\mathrm{GL}_n(\mathbb{R})$, as the kernel of the determinant. The rank of the determinant map is 1, so $\mathrm{SL}_n(\mathbb{R})$ has dimension $n^2 - 1$.

6.2.9 Example

The set $\mathrm{SL}_n(\mathbb{C})$ of $n \times n$ complex matrices of determinant 1 is the kernel of the determinant map to $\mathbb{C}^\times$. This is a surjective map, and thus has rank 2 (since $\mathbb{C}^\times$ does). Thus $\mathrm{SL}_n(\mathbb{C})$ is a properly embedded submanifold of $\mathrm{GL}_n(\mathbb{C})$ of dimension $2n^2 - 2$.

6.2.10 Theorem

Let $H \subseteq G$ be a Lie subgroup. Then H is closed iff it is embedded.

Proof

If H is embedded, let g be a limit point of H (recall that in Hausdorff spaces, a set is closed iff it contains its limit points). Then there is a sequence of elements $h_i \in H$ converging to g. Let (U, φ) be a chart for G containing the identity such that ι's coordinate representation is concatenation with 0. Then we can take W to be a smaller neighborhood so that $\overline{W} \subseteq U$. We can take a neighborhood V of e such that $g_1 g_2^{-1} \in W$ whenever $g_1, g_2 \in V$. Consider the smooth map $G \times G \to G$ by $g_1, g_2 \mapsto g_1 g_2^{-1}$ and then take W's preimage, which is open and thus contains a neighborhood $V_1 \times V_2$ of the identity, and we can shrink them to give $V_1 = V_2 = V$.

Since $h_i g^{-1} \to e$, we can assume that $h_i g^{-1} \in V$ for all i (since eventually this is true). Thus

$$h_i h_j^{-1} = (h_i g^{-1})(h_j g^{-1})^{-1} \in W$$

for all i, j. Fixing j and letting $i \to \infty$, we see that $h_i h_j^{-1} \to g h_j^{-1} \in \overline{W} \subseteq U$. By our assumption, $H \cap U$ is closed and thus $g h_j^{-1} \in H$, so $g \in H$.

Conversely let H be a closed Lie subgroup with $m = \dim H, n = \dim G$. If $m = n$ then it must be embedded as full-rank immersed submanifolds are embedded. So assume $m < n$.

It is sufficient to show that for some $h_1 \in H$ there is a neighborhood $h_1 \in U_1 \subseteq G$ such that $H \cap U_1$ is an embedded submanifold of U_1. For any other $h \in H$, consider the right translation $L_{hh_1^{-1}}\colon G \to G$ which maps H to H and U_1 to a neighborhood U_1' of h. Then $U_1' \cap H$ is embedded in U_1'. So for every $h \in H$ there is a neighborhood $U \subseteq G$ such that $U \cap H$ is embeddable in U. Then by the local slice criterion, H is embeddable in G.

Every immersion is locally an embedding, so let $h_1 \in V \subseteq G$ such that $\iota|_V\colon V \cap H \to V$ is an embedding.

TODO: Above

6.3 Lie group actions

Recall that a (left) group action of a group G on a set X is a map $\theta\colon G \times X \to X$ written by juxtaposition satisfying

$$g_1(g_2 x) = (g_1 g_2)x, \qquad ex = x.$$

A right group action is defined similarly.

6.3.1 Definition

Let G be a topological group and X a topological space. A **continuous (left) group action** of G on X is a group action $\theta\colon G \times X \to X$ which is continuous. A **smooth group action** is a group action of a Lie group on a smooth manifold which is smooth.

There is a duality between left and right group actions: a left group action defines a right group action by $xg = g^{-1}x$ and vice versa.

Note that regular group actions can be equivalently characterized as group morphisms $G \to \mathrm{Sym}(X)$ where $\mathrm{Sym}(X)$ is the set of permutations of X. For continuous group actions, since the category of topological spaces doesn't in general have a right adjoint to products, there isn't in general a topological space like $\mathrm{Sym}(X)$.

But $\cdot \times X$ does have a right adjoint if X is locally compact and Hausdorff. Its right adjoint is Z^X, the set of continuous maps $X \to Z$, given the compact-open topology: for compact $K \subseteq X$ and open $U \subseteq Z$, we add base sets $V(K, U)$, which are continuous maps $f\colon X \to Z$ where $f(K) \subseteq U$. Let $\mathrm{Aut}(X)$ be the topological group of automorphisms of X, viewed as a subspace of X^X. In this case, a continuous group action is the same as a continuous group morphism

$$G \to \mathrm{Aut}(X).$$

But when $X = M$ is a smooth manifold, we then need to give $\mathrm{Aut}(M)$ a Lie group structure.

For a group action $\theta\colon G \times M \to M$, for $g \in G$ we write the map $\theta_g\colon M \to M$. These are smooth maps, as θ is, and satisfy

$$\theta_g \circ \theta_h = \theta_{gh}, \qquad \theta_e = \mathrm{id}_M.$$

Thus θ_g is a diffeomorphism with inverse $\theta_{g^{-1}}$.

6.3.2 Definition

For a group action of G on M, define

(1) the **orbit** of $p \in M$ to be $G \cdot p = \{gp \mid g \in G\}$,

(2) the **stabilizer** of $p \in M$ to be $G_p = \{g \in G \mid gp = p\}$ (also called the **isotropy group**).

The isotropy group is indeed a subgroup of G. The action is

(1) **transitive** if for every $p, q \in M$ there is a $g \in G$ such that $gp = q$, equivalently M has a single orbit.

(2) **free** if every isotropy group is trivial: if $gp = p$ then $g = e$.

6.3.3 Example

If G is a Lie group and M a smooth manifold, the **trivial action** of G on M is $gp = p$ for all $g \in G, p \in M$.

6.3.4 Example

$\mathrm{GL}_n(\mathbb{R})$ has a natural action on $\mathbb{R}^n$ by matrix multiplication: $(A, v) \mapsto Av$. This is clearly a group action, and smooth since its components are all polynomially dependent on the inputs. $\mathrm{GL}_n(\mathbb{R})$ almost acts transitively: for every $v, u \in \mathbb{R}^n$ which are nonzero there is an invertible matrix A such that $Av = u$. So this action has two orbits: $\mathbb{R}^n - 0$ and 0.

If $H \subseteq G$ is a Lie subgroup, and G acts on M, then we can restrict the action to H, and it remains smooth. This is because $H \times M$ is an immersed submanifold of $G \times M$.

6.3.5 Example

Left translation defines a smooth action of a Lie group on itself (right translation defines a right smooth action). This is transitive, as $g_2 g_1^{-1}$ maps g_1 to g_2, and it is free. The restriction of this to a Lie subgroup remains free and smooth, but not necessarily transitive.

6.3.6 Example

Conjugation forms a smooth group action of a Lie group on itself.

6.3.7 Definition

If G acts smoothly on two manifold M and N, a G**-equivariant** map $F\colon M \to N$ is a smooth map which is equivariant in the usual way:

$$F(g \cdot p) = g \cdot F(p)$$

for all $g \in G, p \in M$.

If the actions of G on M and N are given by θ, φ respectively, this is the same as saying that the following diagram commutes for all $g \in G$:

$$\begin{tikzcd} M \arrow[r, "F"'] \arrow[d, "\theta_g"'] & N \arrow[d, "\varphi_g"] \\ M \arrow[r, "F"] & N \end{tikzcd}$$

6.3.8 Theorem

Let G act smoothly on M and N such that it acts transitively on M. If $F\colon M \to N$ is G-equivariant, it has constant rank.

Proof

For $p, q \in M$ let $g \in G$ be such that $gp = q$ (by transitivity). Then we have $\varphi_g \circ F = F \circ \theta_g$ and thus

$$d(\varphi_g)_{F(p)} \cdot dF_p = dF_q \cdot d(\theta_g)_p,$$

and since the actions are diffeomorphisms, we have that dF_p and dF_q have the same rank, as desired. ∎

For a smooth action $\theta\colon G \times M \to M$, consider the smooth maps for $p \in M$ defined by

$$\theta^{(p)}\colon G \to M, \qquad \theta^{(p)}(g) = gp.$$

This is called the **orbit map**, as its image is the orbit $G \cdot p$. The preimage $(\theta^{(p)})^{-1}(p)$ is the isotropy group G_p.

6.3.9 Corollary

For any smooth action θ and $p \in M$, $\theta^{(p)}\colon G \to M$ is smooth and has constant rank. Thus the isotropy group $G_p = (\theta^{(p)})^{-1}(p)$ is a properly embedded Lie subgroup of G. If $G_p = \{e\}$, then $\theta^{(p)}$ is injective and thus a smooth immersion, so $G \cdot p$ is an immersed submanifold of M.

Proof

$\theta^{(p)}$ is equivariant relative to the group action of G on itself and G on M:

$$\theta^{(p)}(g_1 g_2) = g_1 g_2 p = g_1 \theta^{(p)}(g_2),$$

and since G acts transitively on above, this has constant rank as above. Thus G_p is a level set of a constant rank map, and thus a properly embedded smooth manifold, and thus a Lie subgroup. If G_p is trivial, then $\theta^{(p)}$ is clearly injective, and thus a smooth immersion, and the result follows. ∎

6.3.10 Example

A matrix $A \in \mathrm{Mat}_n(\mathbb{R})$ is **orthogonal** if it preserves the Euclidean dot product:

$$(Ax) \cdot (Ay) = x \cdot y,$$

or equivalently its columns are orthonormal, i.e. $A^\top A = I_n$. Let $\mathrm{O}(n)$ be the set of all orthogonal real $n \times n$ matrices. So define $\Phi\colon \mathrm{GL}_n(\mathbb{R}) \to \mathrm{Mat}_n(\mathbb{R})$ by $\Phi(A) = A^\top A$, so that $\mathrm{O}(n)$ is the level set $\Phi^{-1} I_n$.

To show that $\mathrm{O}(n)$ is an embedded submanifold of the general linear group, we will show that it is equivariant for some group action. There is the clear transitive smooth action of $\mathrm{GL}_n(\mathbb{R})$ on itself, and define its action on $\mathrm{Mat}_n(\mathbb{R})$ by a sort of conjugation:

$$A \cdot X = AXA^\top.$$

Then Φ is equivariant:

$$\Phi(AB) = (AB)(AB)^\top = ABB^\top A^\top = A \cdot \varphi(B),$$

as desired.

$\mathrm{O}(n)$ is compact, as it is closed and bounded in $\mathrm{Mat}_n(\mathbb{R})$: closed as the preimage of a closed set, and bounded since norms of orthogonal matrices are

$$|A|^2 = \mathrm{tr}(A^\top A) = \mathrm{tr}(I_n) = n,$$

so bound by $\sqrt{n}$.

To compute $\mathrm{O}(n)$'s dimension we need to compute Φ's rank, which we can do at any matrix of our choosing, so take I_n. To do this, we compute $d\Phi$'s image on tangent vectors, viewed as velocity curves. So for $B \in \mathrm{Mat}_n(\mathbb{R})$ we consider the curve $\gamma\colon (-\varepsilon, \varepsilon) \to \mathrm{GL}_n(\mathbb{R})$ by $\gamma(t) = I_n + tB$. Since $T_{I_n}\mathrm{GL}_n(\mathbb{R}) \cong \mathrm{Mat}_n(\mathbb{R})$, all curves are uniquely of this form. Then

$$d\Phi_{I_n}(B) = \frac{d}{dt}\bigg|_{t=0} \Phi \circ \gamma(t) = \frac{d}{dt}\bigg|_{t=0} (I_n + tB)^\top (I_n + tB) = B^\top + B.$$

Thus $d\Phi$'s image is precisely the symmetric matrices, which has dimension $n + n(n-1)/2 = n(n+1)/2$. And so $\mathrm{O}(n)$'s dimension is $n^2 - n(n+1)/2 = n(n-1)/2$.

6.3.11 Example

The **special orthogonal group** of degree n is the set of real orthogonal matrices of determinant 1. That is, $\mathrm{SO}(n) = \mathrm{O}(n) \cap_n (\mathbb{R})$. Notice that for $A \in \mathrm{O}(n)$ we have

$$1 = \det I_n = \det(A^\top A) = \det(A)^2,$$

so that $\mathrm{SO}(n)$ is the set of orthogonal matrices of positive determinant, thus an open submanifold of $\mathrm{O}(n)$.

6.4 Quotient manifolds

6.4.1 Definition

Let G be a Lie group acting on a smooth manifold M. Define the **orbit space** as usual to be the set of all orbits:

$$M/G = \{G \cdot p \mid p \in M\}.$$

It can be equivalently viewed as the quotient of M by the equivalence that two points are identified if they have the same orbit. This is made into a topological space via the quotient topology: let $\pi\colon M \to M/G$ be the projection $p \mapsto G \cdot p$, then $U \subseteq M/G$ is open iff $\pi^{-1}U \subseteq M$ is.

6.4.2 Lemma

The projection map $\pi\colon M \to M/G$ is an open map.

Proof

We need to show that for $U \subseteq M$ open, $\pi^{-1} \circ \pi(U) \subseteq M$ is open. We have

$$\pi^{-1} \circ \pi(U) = \{p \in M \mid G \cdot p = G \cdot q \text{ for some } q \in U\} = \bigcup_{g \in G} gU.$$

Where $gU = \theta_g(U)$, which is open as the diffeomorphic image of an open set. Thus $\pi^{-1} \circ \pi(U)$ is open, as desired. ∎

6.4.3 Definition

A map between topological spaces is **proper** if its inverse image of compact subsets are compact. A smooth action of G on M is **proper** if the map $\Theta\colon G \times M \to M \times M$ given by $\Theta(g, p) = (gp, p)$ is proper.

Note that proper continuous maps $\Theta\colon X \to Y$ where Y is locally compact and Hausdorff (e.g. a manifold), then Θ is closed.

6.4.4 Lemma

If a topological group acts continuously and properly on a topological manifold, then its orbit space is Hausdorff.

Proof

Let G be a topological group acting continuously and properly on M, let $\Theta\colon G \times M \to M \times M$ be the proper map, and $\pi\colon M \to M/G$ the quotient map. Define the relation $\mathcal{O} \subseteq M \times M$ by

$$\mathcal{O} = \Theta(G \times M) = \{(gp, p) \mid g \in G, p \in M\},$$

so that $(p, q) \in \mathcal{O}$ iff they are in the same orbit. Since Θ is proper and thus closed, $\mathcal{O} \subseteq M \times M$ is closed. Now if $Gp \neq Gq$ so that $(p, q) \notin \mathcal{O}$ there is an open neighborhood $V \times U$ containing (p, q) disjoint from $\mathcal{O}$. Then $\pi(U), \pi(V)$ are neighborhoods of $\pi(p), \pi(q)$, since π is open, and they are disjoint. ■

6.4.5 Lemma

Let G be a topological group acting continuously on M. Then the following are equivalent:

(1) the action is proper;

(2) if (p_i) is a sequence in M and (g_i) a sequence in G such that both (p_i) and (g_ip_i) converge, then a subsequence of (g_i) converges;

(3) for every compact $K \subseteq M$, the set $G_K = \{g \in G \mid (gK) \cap K \neq \emptyset\}$ is compact.

Proof

If Θ is proper, and we have sequences as in the first point, then let U, V be precompact neighborhoods of $p = \lim p_i$ and $q = \lim g_ip_i$ respectively. Then eventually $\Theta(g_i, p_i)$ lie in the compact set $\overline{V} \times \overline{U}$, we can assume they always do. Then $(g_i, p_i) \in \Theta^{-1}(\overline{V} \times \overline{U})$ is compact, and thus there is a subsequence of (g_i, p_i) which converges, and so there is a subsequence of g_i which converges.

Now suppose we have the second point, and let $K \subseteq M$ be compact. Let $g_i \in G_K$ be any sequence of points, so that there are $p_i \in (g_iK) \cap K$, meaning $p_i \in K$ and $g_i^{-1}p_i \in K$. There are then subsequences of p_i and $g_i^{-1}p_i$ which converge, and thus by assumption g_i^{-1} has a subsequence which converges. Thus g_i has a subsequence which converges in G, let its limit be g. Then $g \in G_K$ since if p is the limit of p_i, we have $p \in (gK) \cap K$.

Finally assume that G_K is always compact for compact K. Suppose $L \subseteq M \times M$ is compact, and let $K = \pi_1(L) \cup \pi_2(L) \subseteq M$ where $\pi_i\colon M \times M \to M$ are the projections. These are compact as the continuous image of compact sets, then

$$\Theta^{-1}(L) \subseteq \Theta^{-1}(K \times K) = \{(g, p) \mid p \in K, gp \in K\} \subseteq G_K \times K.$$

L is closed and thus compact, so $\Theta^{-1}(L)$ is a closed subset of a compact space, and thus compact. ■

6.4.6 Corollary

Every smooth action by a compact Lie group is proper.

Proof

If $(p_i), (g_i)$ are sequences as in the above lemma, then a subsequence of (g_i) converges because G is compact and every sequence has a convergent subsequence. ■

6.4.7 Proposition

If θ is a smooth action of G on M, then for every $p \in M$ the map $\theta^{(p)}$ is proper. Thus the orbit $G\cdot p = \theta^{(p)}(G)$ is closed in M. If G_p is trivial, then $\theta^{(p)}$ is a smooth embedding and thus $G \cdot p$ is a properly embedded submanifold.

Furthermore, the isotropy groups G_p are compact.

Proof

For $K \subseteq M$ compact, $(\theta^{(p)})^{-1}(K)$ is a closed subset contained in the compact set $G_{K\cup\{p\}}$, and is thus compact. The rest follows, as $\theta^{(p)}$ is then closed, and if it is also an injection it is a homeomorphism onto its image (as it is closed and continuous).

The isotropy groups G_p are just $G_{\{p\}}$, compact by assumption. ■

6.4.8 Theorem

If G acts freely and properly on M, then M/G is a topological manifold of dimension $\dim M - \dim G$ which has a unique smooth structure making $\pi\colon M \to M/G$ into a smooth submersion.

Proof

The proof is long, but can be found in Lee's book for example. ■

6.4.9 Theorem

Quotienting is functorial. If G acts freely and properly on both M and N, and $F\colon M \to N$ is G-equivariant, there is a unique map $\widetilde{F}\colon M/G \to N/G$ making the following diagram commute

$$\begin{tikzcd} M \arrow[r, "F"'] \arrow[d, "\pi_M"'] & N \arrow[d, "\pi_N"] \\ M/G \arrow[r, "\widetilde{F}"] & N/G \end{tikzcd}$$

Proof

Note that $\pi_N \circ F$ is a smooth map which is constant on the fibers of π_M. This is because the fibers of π_M are of the form $\pi_M^{-1}(G\cdot p) = G \cdot p$, and $\pi_N \circ F(G\cdot p) = \pi_N(G \cdot F(p)) = G \cdot F(p)$. Thus there exists a unqiue smooth $\widetilde{F}\colon M/G \to N/G$ such that $\widetilde{F} \circ \pi_M = \pi_N \circ F$, as desired. ■

Note that $\widetilde{F}$ is given by $\widetilde{F}(G \cdot p) = G \cdot F(p)$.

6.4.10 Corollary

If $F\colon G \to H$ is a Lie group morphism, then $\mathrm{im}F \subseteq H$ is a Lie subgroup.

Proof

We already proved this for injective F, so we reduce this to that case by quotienting F by its kernel. So we show that $\widetilde{F}\colon G/\ker F \to H$ is a Lie group morphism; it is injective and has image $\mathrm{im}F$, so this will be sufficient. Notice that $\ker F$ is closed, and thus its action on G is proper: if $k_n \in \ker F$ and $g_n \in G$ are sequences such that $g_n, k_n g_n$ converge in G, then k_n converges in G, and since $\ker F$ is closed its limit is in $\ker F$. It is also free, as it inherits this from G's action on itself, and it is smooth (again as the restriction of G's multiplication).

Thus $G/\ker F$ is a quotient manifold. It is a Lie group, since the obvious map $G\times G \to (G/\ker F)\times(G/\ker F)$ is smooth, surjective, and $\ker F$-equivariant, and the diagram

$$\begin{tikzcd}
G \times G \arrow[r, "\text{mult}"'] \arrow[d] & G \arrow[d] \\
(G/\ker F) \times (G/\ker F) \arrow[r, "\text{mult}"] & G/\ker F
\end{tikzcd}$$

commutes. Similar for inversion. And $\widetilde{F}$ obviously preserves group structure, and is smooth (compose with the projection), so it is an injective Lie group morphism as desired. ■

7 Vector Fields

7.1 Vector fields on manifolds

7.1.1 Definition

Let M be a smooth manifold and TM its tangent bunldle. A **vector field** is a section of the map $\pi\colon TM \to M$, meaning a continuous map $X\colon M \to TM$ where $\pi \circ X = \mathrm{id}_M$. A vector field is **smooth** if X is. A **rough vector field** is any set-theoretic function $X\colon M \to TM$ satisfying $\pi \circ X = \mathrm{id}_M$.

For a vector field X and $p \in M$, we write its value at p as X_p instead of $X(p)$. The requirement that $\pi \circ X = \mathrm{id}_M$ just means that $X(p) \in T_pM$ for all $p \in M$.

If X is a rough vector field on M, then for any point $p \in M$ and a chart (U, φ) containing p, we can write X_p with respect to this chart:

$$X_p = X^i(p) \left.\frac{\partial}{\partial x^i}\right|_p .$$

So we have defined functions $X^i\colon U \to \mathbb{R}$, called the **component functions** of X.

7.1.2 Proposition

Let X be a rough vector field on a smooth manifold M, and let (U, φ) be a chart for M. The restriction of X to U is smooth iff its component functions with respect to it are.

Proof

Consider the chart for TM corresponding to (U, φ), i.e. $(\pi^{-1}U, \widetilde{\varphi})$. Then X's representation with respect to these coordinates are

$$\widetilde{\varphi} \circ X \circ \varphi^{-1}(\widehat{p}) = \widetilde{\varphi}\left(X^i(p) \left.\frac{\partial}{\partial x^i}\right|_p \right) = (\widehat{p}, X^1(p), \dots, X^n(p)),$$

which is clearly smooth iff X^i are. ■

7.1.3 Example

If $(U, (x^i))$ is any coordinate chart for M, then the map

$$p \mapsto \left.\frac{\partial}{\partial x^i}\right|_p$$

is a smooth vector field on U, called the ith **coordinate vector field**. It is also denoted $\partial/\partial x^i$.

7.1.4 Example

The vector field V on $\mathbb{R}^n$ whose value at x is

$$V_x = x^i \left.\frac{\partial}{\partial x^i}\right|_x$$

is smooth, as its coordinates are linear. It is called the **Euler vector field**.

If $U \subseteq M$ is open, then identifying T_pU with T_pM, we can identify TU with the open subset $\pi^{-1}U \subseteq TM$. So a vector field on U can be viewed either as a map $U \to TU$ or $U \to TM$. The restriction of a vector field on M to U is smooth when the vector field is.

Let $A \subseteq M$ arbitrary, then a **vector field along** A is a continuous map $X\colon A \to TM$ satisfying $\pi \circ X = \mathrm{id}_A$, i.e. $X_p \in T_pM$ for each $p \in A$. It is a smooth vector field if it can be extended to a smooth vector field at every point $p \in A$.

7.1.5 Lemma

If $A \subseteq M$ is closed and $X\colon A \to TM$ is a smooth vector field along A, then for any open $A \subseteq U$, there is a smooth vector field $\widetilde{X}$ on M such that $\widetilde{X}|_A = X$ and $\mathrm{supp}\widetilde{X} \subseteq U$.

Proof

This proof is similar to the usual extension lemma for Euclidean-valued smooth maps. For $p \in A$ let W_p be a neighborhood of p and $\widetilde{X}^p\colon W_p \to TM$ a smooth vector field agreeing with X on $W_p \cap A$. Then $\{W_p\} \cup \{M - A\}$ is an open cover of M and thus has a subordinate partition of unity, $\{\varphi_p\} \cup \{\varphi_0\}$.

Define $\widetilde{X}\colon M \to TM$ for $x \in M$ by

$$\widetilde{X}_x = \sum_{p \in A} \varphi_p(x) \widetilde{X}^p_x.$$

This is a smooth map, as for a neighborhood of x we can consider the coordinates of $\widetilde{X}^p_x$, and this is just a (finite) smooth sum of them. If $x \in A$ then $\widetilde{X}^p_x = X_x$ and $\varphi_0(x) = 0$ so that $\widetilde{X}_x = X_x$. Furthermore,

$$\mathrm{supp}\widetilde{X}_x = \overline{\bigcup_{p \in A} \mathrm{supp}\varphi_p} \subseteq \bigcup \overline{\mathrm{supp}\varphi_p} \subseteq U$$

as desired. ■

7.1.6 Corollary

For any $p \in M$ and $v \in T_pM$, there is a smooth vector field on M which maps p to v.

Proof

The vector field along $\{p\}$ mapping $p \mapsto v$ is smooth: it can be extended to any chart containing p (as it then has constant components). By the above lemma it can be extended. ∎

7.1.7 Definition

Let M be a smooth manifold, then the set of its vector fields is denoted $\mathfrak{X}(M)$.

Notice that $\mathfrak{X}(M)$ is a real linear space: for $X, Y \in \mathfrak{X}(M)$ we can define their sum by

$$(X+Y)_p = X_p + Y_p,$$

and for $a \in \mathbb{R}$ we can define

$$(aX)_p = aX_p.$$

These are smooth vector fields by considering their components in each open chart. Clearly these form a real linear space with the zero vector field being the zero vector.
But moreso, they actually form a $C^\infty(M)$-module. For $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$ define

$$fX : M \to TM, \qquad (fX)_p = f(p)X_p.$$

Again this is smooth by considering components in each chart.
Thus when we wrote above

$$X_p = X^i(p) \left.\frac{\partial}{\partial x^i}\right|_p$$

we could write this relation as a linear combination of fields above $C^\infty(M)$:

$$X = X^i \frac{\partial}{\partial x^i}.$$

More than just being a $C^\infty(M)$-module, smooth vector fields are also derivations in a sense to be explained. For an open set $U \subseteq M$, $f \in C^\infty(U)$, and $X \in \mathfrak{X}(M)$, we define the function $Xf : U \to \mathbb{R}$ by

$$(Xf)(p) = X_p f,$$

recalling that $X_p \in T_pM$ is a derivation at p. This is a smooth map, again considering components.

Note:
While the notations fX and Xf may seem similar, they are distinct. We could write them more distinctly as $f \cdot X$ and $X(f)$: the former is multiplication in the $C^\infty(M)$ module, while the second is a sort of evaluation.

Because the actions of tangent vectors are locally determined, for $V \subseteq U$ we have that

$$(Xf)|_V = X(f|_V).$$

7.1.8 Proposition

Let $X\colon M \to TM$ be a rough vector field, then the following are equivalent:

(1) X is smooth,

(2) for every $f \in C^\infty(M)$, the function Xf is smooth on M,

(3) for every open $U \subseteq M$ and $f \in C^\infty(U)$, Xf is smooth on U.

Proof

If X and f are smooth, then so is Xf. Indeed, for any chart $(U,(x^i))$ and $x \in U$, we have that

$$Xf(x) = \left(X^i(x) \left. \frac{\partial}{\partial x^i} \right|_x \right) f = X^i(x) \frac{\partial f}{\partial x^i}(x),$$

which is a smooth function of x, since X^i and f are smooth.
To prove the second implication, let $f \in C^\infty(U)$ and $p \in U$, let ψ be a bump function identically equal to one in a neighborhood of p and supported in U. Then $\widetilde{f} = \psi f$ is identically equal to f in a neighborhood of p and supported in U. By assumption, $X\widetilde{f}$ is smooth, and equal to Xf in a neighborhood of p. Thus Xf is smooth.
For the final implication, if Xf is smooth for any $f \in C^\infty(U)$, let $(U,(x^i))$ be a chart for M and notice that

$$Xx^i = X^j \frac{\partial}{\partial x^j}(x^i) = X^i$$

is then smooth for all i, meaning X is smooth. ∎

Note then that a vector field $X \in \mathfrak{X}(M)$ defines a map $C^\infty(M) \to C^\infty(M)$ by $f \mapsto Xf$. This map is clearly $\mathbb{R}$-linear, and since $X_p \in T_pM$ we see that

$$X(fg) = fXg + gXf$$

for all $f, g \in C^\infty(M)$, by simply plugging in p. Such a map is called a **derivation**:

7.1.9 Definition

A **derivation** of a commutative R-algebra A is a map $D\colon A \to A$ which is R-linear (a module morphism) and satisfies

$$D(fg) = fD(g) + gD(f).$$

7.1.10 Proposition

The derivations of $C^\infty(M)$ are precisely the smooth vector fields over M.

Proof

As we showed, a smooth vector field is a derivation. Conversely let D be a derivation of $C^\infty(M)$; we want to find an $X \in \mathfrak{X}(M)$ such that $Xf = Df$ for all $f \in C^\infty(M)$. For $p \in M$, we must have that

$$X_pf = (Df)(p),$$

so simply define X_p to be this. Since D is a derivation, this is a rough vector field. Furthermore, $Xf = Df$

is smooth for all f (since $Df \in C^\infty(M)$), and by the above proposition means that X is a smooth vector field. ■

Thus we can naturally identify $\mathfrak{X}(M)$ with derivations of $C^\infty(M)$.

7.2 Vector fields and smooth maps

7.2.1 Definition

Let $F\colon M \to N$ be a smooth map, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Then X and Y are **F-related** if for all $p \in M$,

$$dF_p(X_p) = Y_{F(p)}.$$

Not every vector field on M is F-related to a vector field on N. If F is not surjective, then there isn't a clear way to define Y on points outside of F's image, and if F is not injective we may have conflicting definitions.

7.2.2 Proposition

$X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F-related iff for every smooth real-valued f defined on an open subset of N,

$$X(f \circ F) = (Yf) \circ F.$$

Proof

For a $p \in M$ and $f \in C^\infty(U)$ for $F(p) \in U \subseteq N$ open

$$X(f \circ F)(p) = X_p(f \circ F) = dF_p(X_p)(f)$$

and

$$((Yf) \circ F)(p) = (Yf)(F(p)) = Y_{F(p)}(f).$$

Thus equality holds iff they are F-related. ■

7.2.3 Proposition

If F is a diffeomorphism, then for every $X \in \mathfrak{X}(M)$ there is a unique vector field on N which is F-related to X. This is called the **pushforward** of X by F, denoted F_*X.

Proof

If Y is F-related to X, we must have

$$Y_q = dF_{F^{-1}q}(X_{F^{-q}}).$$

This definition is a rough vector field, and it is smooth as the composition

$$Y\colon N \xrightarrow{F^{-1}} M \xrightarrow{M} TM \xrightarrow{dF} .$$ ■

If $S \subseteq M$ is an immersed submanifold, then a vector field on M doesn't necessarily restrict to one on S. We say that a vector field $X \in \mathfrak{X}(M)$ is **tangent to S at** $p \in S$ if $X_p \in T_pS \subseteq T_pM$. It is **tangent to S** if it is tangent to S at every point of S.

7.2.4 Proposition

$X \in \mathfrak{X}(M)$ is tangent to S iff $X(f)|_S = 0$ for all $f \in C^\infty(M)$ such that $f|_S = 0$.

Proof

Recall that

$$T_pS = \{v \in T_pM \mid vf = 0 \text{ for } f \in C^\infty(M) \text{ when } f|_S = 0\},$$

and the result follows. ■

Let $S \subseteq M$ be immersed and $\iota\colon S \to M$ the immersion. If $Y \in \mathfrak{X}(M)$ is *iota*-related to a vector field $X \in \mathfrak{X}(S)$, then $Y_p = d\iota_p(X_p)$ (the inclusion $T_pS \subseteq T_pM$) is in T_pS and thus tangent to S.

7.2.5 Proposition

If $S \subseteq M$ is immersed and $Y \in \mathfrak{X}(M)$ is tangent to S, then there is a unique smooth vector field on S which is ι-related to Y. We denote this vector field by $Y|_S$.

Proof

Since ι is an immersion, $d\iota_p$ is an injection for all $p \in S$ and thus $Y|_S$ is unique if it exists. To show it exists, let $p \in S$ and take a neighborhood $p \in V \subseteq S$ such that $\iota|_V$ is an embedding. Let $(U, (x^i))$ be a slice chart for V centered at p so that $V \cap S$ is the subset where $x^{k+1} = \cdots = x^n = 0$, and $(x^1, \ldots, x^k)$ form local coordinates for S in $V \cap U$. Now if $Y = Y^1 \frac{\partial}{\partial x^1} + \cdots + Y^n \frac{\partial}{\partial x^n}$ in U, then X must have the coordinate representation $Y^1 \frac{\partial}{\partial x^1} + \cdots + Y^k \frac{\partial}{\partial x^k}$, which is clearly smooth. ■

7.3 Lie brackets

Vector fields $X \in \mathfrak{X}(U)$ are maps $C^\infty(M) \to C^\infty(M)$, and thus we can compose vector fields: for $X, Y \in \mathfrak{X}(M)$ we have a linear map $XY\colon C^\infty(M) \to C^\infty(M)$. But the composition of two vector fields need not be a vector field itself. We can get something closed though.

7.3.1 Definition

For two vector fields $X, Y \in \mathfrak{X}(M)$, define their **Lie bracket** to be the map $[X, Y]\colon C^\infty(M) \to C^\infty(M)$ given by

$$[X, Y] = XY - YX.$$

7.3.2 Lemma

The Lie bracket of two vector fields is a vector field.

Proof

Let $f, g \in C^\infty(M)$ be real-valued, then

$$\begin{aligned}
[X, Y](fg) &= XY(fg) - YX(fg) = X(fYg + gYf) - Y(fXg + gXf) \\
&= fXYg + YgXf + gXYf + YfXg - fYXg - XgYf - gYXf - XfYg \\
&= fXYg + gXYf - fYXg - gYXf \\
&= f[X, Y](g) + g[X, Y](f).
\end{aligned}$$

As desired. ■

By definition,

$$[X,Y]_p f = X_p(Yf) - Y_p(Xf).$$

7.3.3 Proposition

Let $X, Y \in \mathfrak{X}(M)$ with coordinate representations $X = X^i \frac{\partial}{\partial x^i}$ and $Y = Y^j \frac{\partial}{\partial x^j}$ in some local coordinate chart (x^i) for M. Then their Lie bracket has the representation

$$[X,Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j},$$

or more concisely

$$[X,Y] = (XY^j - YX^j)\frac{\partial}{\partial x^j}.$$

Proof

$[X,Y]$ is smooth, and thus determined by its actions locally. So it is sufficient to compute its components in a single smooth chart, here we have

$$[X,Y]f = X^i \frac{\partial}{\partial x^i}\left(Y^j \frac{\partial f}{\partial x^j} \right) - Y^j \frac{\partial}{\partial x^j}\left(X^i \frac{\partial f}{\partial x^i} \right) = X^i \frac{\partial Y^j}{\partial x^i}\frac{\partial f}{\partial x^j} + X^i Y^j \frac{\partial^2 f}{\partial x^i x^j} - Y^j \frac{\partial X^i}{\partial x^j}\frac{\partial f}{\partial x^i} - Y^j X^i \frac{\partial^2 f}{\partial x^j x^i}.$$

By symmetry of the second partial derivative, this is equal to

$$[X,Y](f) = X^i \frac{\partial Y^j}{\partial x^i}\frac{\partial f}{\partial x^j} - Y^j \frac{\partial X^i}{\partial x^j}\frac{\partial f}{\partial x^i}.$$

This gives the desired result. ■

Notice then for the coordinate vector fields $\partial/\partial x^i$, we have

$$\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right] = \left(\frac{\partial}{\partial x^i}\delta_{tj} - \frac{\partial}{\partial x^j}\delta_{it} \right)\frac{\partial}{\partial x^t} = 0,$$

since derivations of constants are zero.

7.3.4 Proposition

The Lie bracket satisfies the following properties.

(1) The Lie bracket is bilinear:

$$[aX + bY, Z] = a[X,Z] + b[Y,Z], \qquad [Z, aX + bY] = a[Z,X] + b[Z,Y].$$

(2) The Lie bracket is antisymetric:

$$[X,Y] = -[Y,X].$$

(3) The Lie bracket satisfies the **Jacobi identity**:

$$[X,[Y,Z]] + [Z,[X,Y]] + [Y,[Z,X]] = 0.$$

(4) For $f, g \in C^\infty(M)$:

$$[fX, gY] = fg[X,Y] + (fXg)Y - (gYf)X.$$

Proof

By standard computations. ■

7.3.5 Proposition

Let $F\colon M \to N$, $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \mathfrak{X}(N)$ such that X_i is F-related to Y_i for $i = 1, 2$. Then $[X_1, X_2]$ is F-related to $[Y_1, Y_2]$.

Proof

We have that

$$X_1X_2(f \circ F) = X_1((Y_2f) \circ F) = (Y_1Y_2f) \circ F,$$

and similarly

$$X_2X_1(f \circ F) = (Y_2Y_1) \circ F.$$

Thus

$$[X_1, X_2](f \circ F) = X_1X_2(f \circ F) - X_2X_1(f \circ F) = (Y_1Y_2f - Y_2Y_1f) \circ F = ([Y_1, Y_2]f) \circ F,$$

as desired. ■

7.3.6 Corollary

If $F\colon M \to N$ is a diffeomorphism, and $X_1, X_2 \in \mathfrak{X}(M)$ then $F_*[X_1, X_2] = [F_*X_1, F_*X_2]$.

Proof

By above, and the uniqueness of F-relationships for diffeomorphisms. ■

7.3.7 Corollary

If $S \subseteq M$ is immersed, and Y_1, Y_2 are smooth vector fields on M tangent to S, then so is $[Y_1, Y_2]$.

Proof

Being tangent to S is the same as being ι-related to some vector field on S. ■

7.4 Lie algebras of Lie groups

7.4.1 Definition

Let G be a Lie group. A vector field $X \in \mathfrak{X}(G)$ is **left invariant** if it is L_g-related to itself for all $g \in G$. Explicitly,

$$d(L_g)_{g'}(X_{g'}) = X_{g'g}$$

for all $g, g' \in G$.

Since L_g is a diffeomorphism, we can write this as $(L_g)_*X = X$ for all $g \in G$. Because $(L_g)_*$ is linear, the set of all left invariant vector fields over G is an $\mathbb{R}$-linear subspace of $\mathfrak{X}(G)$.

7.4.2 Proposition

If $X, Y \in \mathfrak{X}(G)$ are left-invariant, then so is $[X, Y]$.

Proof

For $g \in G$, we have
$$(L_g)_*[X, Y] = [(L_g)_*X, (L_g)_*Y] = [X, Y]. \qquad \blacksquare$$

7.4.3 Definition

An **Lie algebra** over a ring R is an R-module $\mathfrak{g}$ equipped with a **Lie bracket**: an R-bilinear map
$$[\bullet, \bullet]\colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g},$$
which is antisymmetric (meaning $[X, Y] = -[Y, X]$ for all $X, Y \in \mathfrak{g}$), and satisfying the Jacobi identity:
$$[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0.$$
We consider here only real Lie algebras (where $R = \mathbb{R}$).

Note:

The Jacobi identity and antisymmetry come together to give a slightly stronger property. Extend the Lie bracket of $\mathfrak{g}$ to finite words over $\mathfrak{g}$: $[\bullet]\colon \mathfrak{g}^* \to \mathfrak{g}$, by mapping a word $(X_1, \dots, X_n)$ to
$$[X_1, \dots, X_n] = [X_1, [X_2, \dots, [X_{n-1}, X_n]] \cdots].$$
Define the Lie bracket of the empty word to be 0 and $[X] = X$ for $X \in \mathfrak{g}$. Then for words $W \in \mathfrak{g}^*$ of length ≥ 2, we have that
$$\sum_\sigma [\sigma W] = 0,$$
where the sum is over all cycles in S_n, and σW is obtained by permuting the indices of the word accordingly.

7.4.4 Definition

If $\mathfrak{g}$ is a Lie algebra, a **Lie subalgebra** of $\mathfrak{g}$ is a submodule $\mathfrak{h} \subseteq \mathfrak{g}$ which is closed under the Lie bracket.

A **Lie algebra morphism** is a map between Lie algebras $A\colon \mathfrak{g} \to \mathfrak{h}$ which is a morphism of modules, and preserves the Lie bracket: $A[X, Y] = [AX, AY]$. A Lie algebra morphism with an inverse is a **Lie algebra isomorphism**. The set-theoretic inverse of a bijective Lie algebra morphism is a Lie algebra morphism, so this is the same as a bijective Lie algebra morphism.

7.4.5 Example

The space of vector fields over M, $\mathfrak{X}(M)$, is a Lie algebra. The space of left-invariant vector fields over a Lie group G is a Lie subalgebra of $\mathfrak{X}(G)$.

7.4.6 Example

If A is an associative algebra (an algebra in the usual sense), then it is also a Lie algebra whose bracket is

defined by
$$[X,Y] = XY - YX.$$
In particular $\mathrm{Mat}_n(\mathbb{R})$ is a Lie algebra. When considering $\mathrm{Mat}_n(\mathbb{R})$ as a Lie algebra with this bracket, we denote it by $\mathfrak{gl}_n(\mathbb{R})$ (to be explained). Similarly $\mathrm{Mat}_n(\mathbb{C})$ is a Lie algebra, which we denote $\mathfrak{gl}_n(\mathbb{C})$.

Every space can be made into a Lie algebra by making the Lie bracket identically zero. Such Lie algebras are called **Abelian**. If A is an associative commutative algebra, the Lie bracket defined above by $[X,Y] = XY - YX$ is Abelian.

7.4.7 Definition

Let G be a Lie group. Its **Lie algebra** is the space of all left-invariant vector fields in $\mathfrak{X}(G)$. We denote it by $\mathrm{Lie}(G)$.

7.4.8 Theorem

Let G be a Lie group with identity $e \in G$. Then the **evaluation map** $\varepsilon\colon \mathrm{Lie}(G) \to T_e(G)$ defined by $\varepsilon(X) = X_e$ is a vector space isomorphism. Thus $\mathrm{Lie}(G)$ is a finite-dimensional space of dimension $\dim G$.

Proof

Clearly the evaluation map is linear. By left-invariance, if $X_e = 0$ then $X_g = d(L_g)_g(X_e) = 0$ for all $g \in G$ so that ε is injective.

Now let $v \in T_eG$, and define the rough vector field v^L on G by
$$v^L|_g = d(L_g)_e(v).$$
That is, v^L's value at $g \in G$ is the derivation $d(L_g)_e(v)$, which maps $f \in C^\infty(G)$ to $v(f \circ L_g)$. Notice that this is left invariant:
$$d(L_g)_{g'}(v^L|_{g'}) = d(L_g)_{g'} d(L_{g'})_e(v) = d(L_{gg'})_e(v) = v^L_{gg'}.$$
And clearly $\varepsilon(v^L) = v$, so all that remains to show is that v^L is smooth.

It is sufficient to show that $v^L f$ is smooth for all $f \in C^\infty(G)$. Take a curve $\gamma\colon (-\delta,\delta) \to G$ such that $\gamma(0) = e$ and $\gamma'(0) = v$. Then for $g \in G$,
$$(v^L f)(g) = v^L|_g f = d(L_g)_e(v) f = v(f \circ L_g) = \gamma'(0)(f \circ L_g) = \left.\frac{d}{dt}\right|_{t=0} (f \circ L_g \circ \gamma)(t).$$
Now define $\varphi\colon (-\delta,\delta) \times G \to \mathbb{R}$ by $\varphi(t,g) = f \circ L_g \circ \gamma(t) = f(g\gamma(t))$. So we have $(v^L f)(g) = \partial\varphi/\partial t(0,g)$. Now, φ is the composition of smooth maps, and thus is smooth. Thus $v^L f$ is smooth, as desired. ■

We will continue to use the notation $\bullet^L$ for ε's inverse.

7.4.9 Corollary

Every left-invariant rough vector field on a Lie group is smooth.

Proof

If X is a rough vector field on G, then $v = X_e$ is in T_eG. X's left invariance implies that $X = v^L$, which is smooth. ■

This isomorphism induces a Lie algebra structure on T_eG:

$$[v,u] = \varepsilon[v^L, u^L] = [v^L, u^L]_e,$$

which maps

$$[v,u]f = [v^L, u^L]_e f = (v^L u^L - u^L v^L)_e f = v^L_e(u^L f) - u^L_e(v^L f) = v(u^L f) - u(v^L f).$$

Take for instance $G = \mathrm{GL}_n(\mathbb{R})$. We know that its tangent spaces are isomorphic to $\mathrm{Mat}_n(\mathbb{R})$, thus we have natural linear isomorphisms

$$\mathrm{Lie}(\mathrm{GL}_n(\mathbb{R})) \to T_{I_n}\mathrm{GL}_n(\mathbb{R}) \to \mathfrak{gl}_n(\mathbb{R}),$$

but is this also a Lie algebra morphism? Yes.

7.4.10 Proposition

The above natural isomorphism is an isomorphism of Lie algebras.

Proof

The natural isomorphism $T_{I_n}\mathrm{GL}_n(\mathbb{R}) \leftrightarrow \mathfrak{gl}_n(\mathbb{R})$ is the map

$$A^i_j \left.\frac{\partial}{\partial X^i_j}\right|_{I_n} \leftrightarrow (A^i_j).$$

We use Einstein summation here, viewing a lower index in a denominator as an upper index. Let $\mathfrak{g}$ be the Lie algebra of $\mathrm{GL}_n(\mathbb{R})$, then $A = (A^i_j) \in \mathfrak{gl}_n(\mathbb{R})$ maps through the isomorphisms to $A^L \in \mathfrak{g}$, defined by

$$A^L|_X = (dL_X)_{I_n}(A) = d(L_X)_{I_n}\left(A^i_j \left.\frac{\partial}{\partial X^i_j}\right|_{I_n} \right), \qquad X \in \mathrm{GL}_n(\mathbb{R}).$$

The map L_X is given by $A \mapsto XA$, whose differential is the $n^2 \times n^2$ matrix whose $(i,j),(i,j)$ coordinate is X^i_j, and zero elsewhere. Thus

$$A^L_X = X^i_j A^j_k \left.\frac{\partial}{\partial X^i_k}\right|_X .$$

Thus for $A, B \in \mathfrak{gl}_n(\mathbb{R})$, we have that (in local coordinates),

$$[A^L, B^L] = \left[X^i_j A^j_k \frac{\partial}{\partial X^i_k}, X^p_q B^q_r \frac{\partial}{\partial X^p_r} \right] = X^i_j A^j_k \frac{\partial}{\partial X^i_k}(X^p_q B^q_r)\frac{\partial}{\partial X^p_r} - X^p_q B^q_r \frac{\partial}{\partial X^p_r}(X^i_j A^j_k)\frac{\partial}{\partial X^i_k}.$$

Now, A and B are constants, and $\frac{\partial X^p_q}{\partial X^i_k} = \delta_{i=p,q=k}$ so that this is equal to

$$= X^i_j A^j_k B^k_r \frac{\partial}{\partial X^i_r} - X^p_q B^q_r \frac{\partial}{\partial X^p_k} = (X^i_j A^j_k B^k_r - X^i_j B^j_k A^k_r)\frac{\partial}{\partial X^i_r}.$$

Evaluating this at $X = I_n$ gives

$$[A^L, B^L]_{I_n} = (A^i_k B^k_r - B^i_k A^k_r) \left.\frac{\partial}{\partial X^i_r}\right|_{I_n},$$

which corresponds to the matrix commutator $[A,B]$. That is, $[A,B]^L_{I_n} = [A^L, B^L]_{I_n}$, and since left invariant vector fields are determined by their values at the identity, we see that $[A,B]^L = [A^L, B^L]$ so this is indeed a Lie algebra morphism. ■

For arbitrary Euclidean spaces V, we have $\mathrm{GL}(V)$ as the set of linear automorphisms, and $\mathfrak{gl}(V)$ as the Lie algebra of all linear endomorphisms. By choosing a basis for V, we see that there is an isomorphism of Lie algebras

$$\mathrm{Lie}(\mathrm{GL}(V)) \to T_{\mathrm{id}}\mathrm{GL}(V) \to \mathfrak{gl}(V).$$

7.4.11 Theorem

Let G, H be Lie groups with respective Lie algebras $\mathfrak{g}, \mathfrak{h}$. If $F\colon G \to H$ is a Lie group morphism, then for every $X \in \mathfrak{g}$ there is a unique vector field in $\mathfrak{h}$ that is F-related to X. Denoting this vector field by F_*X, $F_*\colon \mathfrak{g} \to \mathfrak{h}$ becomes a Lie algebra morphism.

Proof

If $Y \in \mathfrak{h}$ is F-related to X, it must satisfy $Y_e = dF_e(X_e)$ and thus is uniquely determined by

$$Y = (dF_e(X_e))^L.$$

We must now show that Y is indeed F-related to X. Since F is a Lie group morphism, $F(L_g g') = L_{Fg} F(g')$ i.e. $F \circ L_g = L_{Fg} \circ F$. Thus $dF \circ d(L_g) = d(L_{Fg}) \circ dF$, and so

$$dF(X_g) = dF(d(L_g)X_e) = d(L_{Fg})(dF(X_e)) = d(L_{Fg})(Y_e) = Y_g$$

as desired.

By naturality of Lie brackets, $F_*[X,Y] = [F_*X, F_*Y]$ (because F_*X is unique). Furthermore, F_* is clearly linear, by linearity of differentials and $\bullet^L$. ■

The map $F_*\colon \mathfrak{g} \to \mathfrak{h}$ is called an **induced Lie algebra morphism**.

7.4.12 Proposition

The correspondence $G \mapsto \mathrm{Lie}(G)$, $F \mapsto F_*$ is functorial. In particular isomorphic Lie groups have isomorphic Lie algebras.

Proof

By functorality of the differential, and since F_*X is determined by its value on the identity, we get the desired result. ■

Note that if $F\colon \mathfrak{g} \to \mathfrak{h}$ is a morphism of Lie algebras, then $\ker F \subseteq \mathfrak{g}$ and $\mathrm{im} F \subseteq \mathfrak{h}$ are Lie subalgebras. They are subspaces, and closed under brackets: for $X, Y \in \ker F$ then

$$F[X,Y] = [FX, FY] = [0,0] = 0,$$

and for $X, Y \in \mathfrak{g}$, $[FX, FY] = F[X,Y]$ so that both $\ker F, \mathrm{im} F$ are Lie subalgebras. Note further that $\ker F$ is an *ideal* in the sense that for $X \in \ker F$ and $Y \in \mathfrak{g}$,

$$F[X,Y] = [FX, FY] = [0, FY] = 0,$$

so that $[X,Y], [Y,X] \in \ker F$.

7.4.13 Theorem

Let $H \subseteq G$ be a Lie subgroup, and $\iota\colon H \hookrightarrow G$ the inclusion. Then the induced map $\iota_*\colon \mathrm{Lie}(H) \to \mathrm{Lie}(G)$ is an embedding of Lie algebras (meaning it is injective). So we can identify $\mathrm{Lie}(H)$ with its image in $\mathrm{Lie}(G)$,

$$\mathfrak{h} = \iota_*(\mathrm{Lie}(H)) = \{X \in \mathrm{Lie}(G) \mid X_e \in T_e H\}.$$

Proof

The differential of an immersion is injective, so ι_* must be: if $\iota_* X = \iota_* Y$ then they are equal at e and so

$$d\iota_e(X_e) = d\iota_e(Y_e) \Longrightarrow X_e = Y_e \Longrightarrow X = Y.$$

ι_*'s image are left invariant vector fields $X \in \mathrm{Lie}(G)$ whose values at the identity is of the form $d\iota_e(v)$ for $v \in T_e H$. We identify $T_e H$ in $T_e G$ via $d\iota_e$, so that the two characterizations of $\mathfrak{h}$ above are equal. ■

Note:
We make the following observation, that if $\Phi\colon M \to N$ is smooth of constant rank k, and $S = \Phi^{-1}n$ is a level set, then $T_p S$ is the kernel of $d\Phi_p\colon T_p M \to T_p N$ for all $p \in M$.
We know that S is an embedded submanifold of M of codimension k. First of all, $d\Phi_p \circ d\iota_p = d(\Phi \circ \iota)_p$, and $\Phi \circ \iota$ is constant and thus its differential is zero. Thus $T_p S \subseteq \mathrm{ker} d\Phi_p$, and they have equal dimension ($\mathrm{ker} d\Phi_p$ is $\dim M - k$ by rank-nullity), so they are equal.

7.4.14 Example

Recall that $\mathrm{O}(n)$ is the Lie subgroup of $\mathrm{GL}_n(\mathbb{R})$ given by orthogonal matrices. That is, it is the level set at I_n of $\Phi\colon \mathrm{GL}_n(\mathbb{R}) \to \mathrm{Mat}_n(\mathbb{R})$ given by $\Phi(A) = A^\top A$. Then $T_{I_n}\mathrm{O}(n)$ is the kernel of $d\Phi_{I_n}\colon T_{I_n}\mathrm{GL}_n(\mathbb{R}) \to T_{I_n}\mathrm{Mat}_n(\mathbb{R})$. The differential is given by $d\Phi_{I_n}(B) = B^\top + B$, so that

$$T_{I_n}\mathrm{O}(n) = \left\{B \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; B^\top + B = 0\right\} = \{\text{Skew-symmetric } n \times n \text{ matrices}\}.$$

Denote this subspace by $\mathfrak{o}(n)$; by the above proposition it is a Lie subalgebra of $\mathrm{GL}_n(\mathbb{R})$.

8 Integral Curves and Flows

8.1 Integral curves

8.1.1 Definition

Let V be a vector field on a smooth manifold M. An **integral curve** for V is a smooth curve $\gamma\colon J \to M$ (with J an interval in the real line) such that $\gamma'(t) = V_{\gamma(t)}$. If $0 \in J$, then $\gamma(0)$ is called the **starting point** of γ.

Recall that $\gamma'(t) \in T_{\gamma(t)}M$ is viewed as the derivation

$$\gamma'(t)f = (f \circ \gamma)'(t).$$

Equivalently, the differential $d\gamma_t$ is a map $\mathbb{R} \cong T_t J \to T_{\gamma(t)}M$, and then $\gamma'(t)f = d\gamma_t(1)(f)$, where $1 \in \mathbb{R}$ corresponds to the directional derivative in the direction of 1 (the usual 1-dimensional derivative).

Note:
In this section we will use the notation $\dot\gamma(t)$ for the derivative of a curve with respect to t. This is because often, the superscript is already being used as an index.

8.1.2 Example

Consider $\mathbb{R}^2$ and let $V = \partial/\partial x$ be the first-coordinate vector field. A curve $\gamma(t) = (\gamma^1(t), \gamma^2(t))$ has derivative $\gamma'(t) = \dot\gamma^1(t)\partial/\partial x + \dot\gamma^2(t)\partial/\partial y$. Then γ is an integral curve of V when it is of the form $\gamma(t) = (a + t, b)$.

Finding integral curves boils down to solving ODEs. For a vector field V, let $\gamma\colon J \to M$ be a smooth curve. On a coordinate domain $U \subseteq M$ we can write γ in local coordinates $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$ as well as V as $V^i(p)\, \frac{\partial}{\partial x^i}\big|_p$. Then the requirement that $\gamma'(t) = V_{\gamma(t)}$ is just

$$\dot{\gamma}^i(t)\, \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)} = V^i(\gamma(t))\, \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)},$$

which is just the system of ODEs:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \dots, \gamma^n(t)) \\ &\vdots \\ \dot{\gamma}^n(t) &= V^1(\gamma^1(t), \dots, \gamma^n(t)) \end{aligned}$$

Recall the existence and uniqueness of a solution to ODEs. Since everything here is smooth, so we can derive the following simple consequence.

8.1.3 Proposition

Let V be a smooth vector field on M. Then for every $p \in M$ there exists an $\varepsilon > 0$ and a unique smooth curve $\gamma\colon (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p.

8.1.4 Lemma

Let V be a smooth vector field on M, $J \subseteq \mathbb{R}$ an interval, and $\gamma\colon J \to \mathbb{R}$ an integral curve of V. For any $a \in \mathbb{R}$, the curve $\widetilde{\gamma}\colon a^{-1}J \to M$ defined by $\widetilde{\gamma}(t) = \gamma(at)$ is an integral curve of aV.

Similarly $\widehat{\gamma}\colon J - a \to M$ defined by $\widehat{\gamma}(t) = \gamma(t + a)$ is an integral curve of V.

Proof

We look at how $\widetilde{\gamma}$ acts on C^∞ functions defined in a neighborhood of t_0.

$$\widetilde{\gamma}'(t_0) f = \frac{d}{dt}\bigg|_{t_0} (f \circ \widetilde{\gamma})(t) = \frac{d}{dt}\bigg|_{t_0} (f \circ \gamma)(at),$$

which by the chain rule is just

$$= a(f \circ \gamma)'(at_0) = a\gamma'(at_0) f = aV_{\gamma(at_0)} f = aV_{\widetilde{\gamma}(t_0)} f.$$

The translation $\widehat{\gamma}$ is proven similarly.

8.1.5 Proposition

If $F\colon M \to N$ is smooth, then $X \in \mathfrak{X}(M), Y \in \mathfrak{X}(N)$ are F-related iff F takes integral curves of X to integral curves of Y. Meaning if γ is integral for X, then $F \circ \gamma$ is integral for Y.

Proof

If X and Y are F-related, let $\gamma\colon J \to M$ be integral for X, and define $\sigma\colon J \to N$ by $\sigma = F \circ \gamma$. Then

$$\sigma'(t) = (F \circ \gamma)'(t) = dF_{\gamma(t)}(\gamma'(t)) = dF_{\gamma(t)}(V_{\gamma(t)}) = Y_{F\circ\gamma(t)} = Y_{\sigma(t)},$$

as desired.

Conversely suppose F maps integral curves of X to integral curves of Y. Let $p \in M$ and let $\gamma\colon J \to M$ be an integral curve of X starting at p. Then $\sigma = F \circ \gamma$ is integral for Y so

$$Y_{F(p)} = (F \circ \gamma)'(0) = dF_{\gamma(0)}(\gamma'(0)) = dF_p(X_p),$$

meaning X and Y are F-related. ■

8.2 Flows

Let M be a smooth manifold and $V \in \mathfrak{X}(M)$. Now suppose that for every $p \in M$, V has a unique integral curve starting at p which is defined for all $t \in \mathbb{R}$ (this is a big assumption, but will suffice for the explanation), denoted by $\theta^{(p)}\colon \mathbb{R} \to M$. Now for each $t \in \mathbb{R}$ we define $\theta_t\colon M \to M$ by mapping $p \in M$ to $\theta^{(p)}(t)$. In other words we have

$$\theta\colon \mathbb{R} \times M \to M.$$

Translation implies that $t \mapsto \theta^{(p)}(s+t)$ is an integral curve for V starting at $q = \theta^{(p)}(s)$. Uniqueness implies that $\theta^{(q)}(t) = \theta^{(p)}(t+s)$, so that

$$\theta_t \circ \theta_s(p) = \theta_t(\theta^{(p)}(s)) = \theta^{(q)}(t) = \theta_{s+t}(p).$$

Since $\theta_0(p) = \theta^{(p)}(0) = p$, we see that $\theta\colon \mathbb{R} \times M \to M$ is an action of the additive group $\mathbb{R}$ on M.

This motivates the following definition.

8.2.1 Definition

A **global flow** on M is a continuous left action of $\mathbb{R}$ on M. In other words, a continuous map

$$\theta\colon \mathbb{R} \times M \to M$$

satisfying

$$\theta(0,p) = p, \qquad \theta(t+s,p) = \theta(t,\theta(s,p)).$$

Note:

Global flows are also called **one-parameter group actions**.

Given a global flow θ on M we define

$$\theta_t\colon M \to M,\ \theta_t(p) = \theta(t,p), \qquad \theta^{(p)}\colon \mathbb{R} \to M,\ \theta^{(p)}(t) = \theta(t,p),$$

as with Lie group actions. The image of the curve $\theta^{(p)}$ is the orbit of p under the action θ. And the requirement that θ be an action is equivalent to requiring

$$\theta_{t+s} = \theta_t \circ \theta_s.$$

If $\theta\colon \mathbb{R} \times M \to M$ is a global flow, define a tangent vector $V_p \in T_pM$ by

$$V_p = {\theta^{(p)}}'(0).$$

The assignment $p \mapsto V_p$ is a rough vector field on M, called the **infinitesimal generator** of θ.

8.2.2 Proposition

Let θ be a smooth global flow on M. Its infinitesimal generator V is then a smooth vector field on M, and each $\theta^{(p)}$ is an integral curve of V starting at p.

Proof

To show that V is smooth, it is sufficient to show that Vf is smooth for all $f \in C^\infty(M)$. Now, for $p \in M$,

$$Vf(p) = V_p f = \theta^{(p)\prime}(0) f = \left.\frac{d}{dt}\right|_0 f(\theta^{(p)}(t)) = \left.\frac{\partial}{\partial t}\right|_{(0,p)} f(\theta(t,p)).$$

Since $f \circ \theta \colon \mathbb{R} \times M \to \mathbb{R}$ is smooth by assumption, its partial derivatives are smooth as well. Thus Vf is smooth, as desired.

Now we need to show that $\theta^{(p)}$ is an integral curve of V. For $t \in \mathbb{R}$, let $q = \theta^{(p)}(t) = \theta(t,p)$ so that

$$V_{\theta^{(p)}(t)} = V_q = \theta^{(q)\prime}(0).$$

We want to show that this is equal to $\theta^{(p)\prime}(t)$. Now,

$$\theta^{(q)}(s) = \theta(s,q) = \theta(s,\theta(t,p)) = \theta(t+s,p) = \theta^{(p)}(t+s).$$

And so we have that

$$\theta^{(q)\prime}(0) = \theta^{(p)\prime}(t),$$

as desired. ■

While every global flow gives rise to a vector field, its infinitesimal generator, the converse is not true in general. The issue is the globalness: a vector field's integral curve does not in general need to be defined at every point on the real line.

8.2.3 Example

Let $M = \mathbb{R}^2 - 0$, and $V = \partial/\partial x$. The unique integral curve starting at $(-1,0)$ is $\gamma(t) = (-1+t, 0)$. But γ then cannot be defined at $t = 1$, as this is a removable singularity and defining it at $t = 1$ would require defining it to be equal to $0 \notin M$.

Another example is $M = \mathbb{R}^2$ and $V = x^2\partial/\partial x$. An integral curve $\gamma(t) = (\gamma^1(t), b)$ must satisfy $\dot{\gamma}^1(t) = \gamma^1(t)^2$ and thus has the solution $\gamma^1(t) = 1/(a-t)$. If we require it to start at $(1,0)$, then we have

$$\gamma(t) = \left(\frac{1}{1-t}, 0\right).$$

This also cannot be defined at $t = 1$.

Thus we must somehow localize the idea of a global flow.

8.2.4 Definition

Let M be a smooth manifold. A **flow domain** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for every $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} \mid (t,p) \in \mathcal{D}\}$ is an open interval containing 0.

A **(local) flow** on M is a continuous map $\theta \colon \mathcal{D} \to M$ where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, which satisfies the usual group action laws when defined. Explicitly, for all $p \in M$, $\theta(0,p) = p$ and

$$\theta(t, \theta(s,p)) = \theta(t+s, p),$$

for all $s \in \mathcal{D}^{(p)}$, and $t \in \mathcal{D}^{(\theta(s,p))}$ such that $s + t \in \mathcal{D}^{(p)}$.

As with global flows, we define $\theta_t(p) = \theta^{(p)}(t) = \theta(t,p)$ when $(t,p) \in \mathcal{D}$. For $t \in \mathbb{R}$ we define

$$M_t = \{p \in M \mid (t,p) \in \mathcal{D}\}$$

so that $p \in M_t$ iff $t \in \mathcal{D}^{(p)}$ iff $(t,p) \in \mathcal{D}$, and $\theta^{(p)} \colon \mathcal{D}^{(p)} \to M$, $\theta_t \colon M_t \to M$.

If θ is smooth, its **infinitesimal generator** is defined by $V_p = \theta^{(p)\prime}(0)$.

Note:
The term **local one-parameter group action** is also used for local flows.

8.2.5 Proposition

The infinitesimal generator of a flow is a smooth vector field, and $\theta^{(p)}$ is an integral curve for V for each $p \in M$.

Proof

The proof is the same as in the global case. Just verify that all terms are defined locally (in close enough neighborhoods). ■

8.2.6 Definition

A **maximal integral curve** is an integral curve which cannot be extended to a larger interval. A **maximal flow** is a flow that cannot be extended to a larger flow domain.

8.2.7 Theorem (Fundamental theorem on flows)

Let V be a smooth vector field on M. Then there is a unique maximal flow $\theta\colon \mathcal{D} \to M$ whose infinitesimal generator is V. The flow has the following properties:

(1) For each $p \in M$, the curve $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of V starting at p.

(2) If $s \in \mathcal{D}^{(p)}$ then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s \mid t \in \mathcal{D}^{(p)}\}$.

(3) For each $t \in \mathbb{R}$, M_t is open in M and $\theta_t\colon M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

Proof

Suppose that $\gamma, \widetilde{\gamma}\colon J \to M$ are two integral curves of V defined on the same interval such that they agree at some point $t_0 \in J$. Then let $\mathcal{S} = \{t \in J \mid \gamma(t) = \widetilde{\gamma}(t)\}$. This is nonempty since $t_0 \in \mathcal{S}$ by assumption, and it is closed by continuity (for $t \notin \mathcal{S}$ then by Hausdorff there are open sets around $\gamma(t), \widetilde{\gamma}(t)$ which don't intersect, their preimages under $\gamma, \widetilde{\gamma}$ are open and the intersection doesn't intersect $\mathcal{S}$). But $\mathcal{S}$ is also open: for $t_1 \in \mathcal{S}$, $\gamma, \widetilde{\gamma}$ are both solutions to the same ODE with the initial condition $\gamma(t_1) = \widetilde{\gamma}(t_1)$, and thus agree in a neighborhood of t_1. Since J is connected, $\mathcal{S}$ must be equal to J.
Thus any two integral curves defined on the same interval must be equal. So let us define $\mathcal{D}^{(p)}$ to be the union of all open intervals J with $0 \in J$ such that there is an integral curve of V starting at p defined on J. Then for $t \in \mathcal{D}^{(p)}$, define $\theta^{(p)}(t) = \gamma(t)$ where γ is an integral curve of V starting at p defined at t. This is well-defined by our above argument, and clearly the unique maximal integral curve starting at p.
Now let $\mathcal{D} = \{(t,p) \in \mathbb{R} \times M \mid t \in \mathcal{D}^{(p)}\}$, and define $\theta\colon \mathcal{D} \to M$ by $\theta(t,p) = \theta^{(p)}(t)$. θ satisfies the first point in our theorem statement. Now we need to verify that it satisfies the group laws. Clearly $\theta(0,p) = p$. For $p \in M$ and $s \in \mathcal{D}^{(p)}$ let $q = \theta(s,p)$, then the curve $\gamma\colon \mathcal{D}^{(p)} - s \to M$ defined by $\gamma(t) = \theta^{(p)}(t+s)$ starts at q and is also integral (by translation). By uniqueness, γ and $\theta^{(q)}$ agree on the intersection of their domains, which is equivalent to the second group law:

$$\theta^{(p)}(t+s) = \theta^{(q)}(t) \iff \theta(t+s,p) = \theta(t, \theta(s,p)).$$

The maximality of $\theta^{(q)}$ further implies that γ's domain is contained in $\mathcal{D}^{(q)}$. That is, $\mathcal{D}^{(p)} - s \subseteq \mathcal{D}^{(q)}$. Since $0 \in \mathcal{D}^{(p)}$ this implies $-s \in \mathcal{D}^{(q)}$ and $\theta^{(q)}(-s) = \theta(-s, \theta(s,p)) = p$. So if we repeat the above argument with $p = \theta(-s,q)$, we see that $\mathcal{D}^{(q)} + s \subseteq \mathcal{D}^{(p)}$ so that $\mathcal{D}^{(q)} - s = \mathcal{D}^{(p)}$, proving the second point.

Now we show that $\mathcal{D} \subseteq \mathbb{R} \times M$ is open, so that it is a flow domain, and that $\theta\colon \mathbb{R} \times M \to M$ is smooth. Define $W \subseteq \mathcal{D}$ to be the set of all $(t,p) \in \mathcal{D}$ where θ is smooth on a product neighborhood of (t,p) of the form $J \times U \subseteq \mathcal{D}$ where $0 \in J$ is a neighborhood and $U \subseteq M$ is a neighborhood of p. Then W is an open subset of $\mathbb{R} \times M$ and θ is smooth on W, so it is sufficient to show that $W = \mathcal{D}$.

Suppose not, so there is a $(\tau, p_0) \in \mathcal{D} - W$. We assume that $\tau > 0$, the proof for negative τ is similar. Define $t_0 = \inf\{t \in \mathbb{R} \mid (t, p_0) \notin W\}$. Taking a coordinate domain around p_0, we see that by uniqueness of ODE solutions, θ is smooth in a product neighborhood around $(0, p_0)$, so $t_0 > 0$. Since $t_0 \leq \tau$ and $\mathcal{D}^{(p_0)}$ is an open interval containing 0, we have that $t_0 \in \mathcal{D}^{(p_0)}$. Let $q_0 = \theta^{(p_0)}(t_0)$, and then again by ODE uniqueness, there is an $\varepsilon > 0$ and a neighborhood U_0 of q_0 such that $(-\varepsilon, \varepsilon) \times U_0 \subseteq W$. We will show that using the group action, we can extend θ smoothly to a neighborhood of (t_0, p_0). By maximality of θ, this implies $(t_0, p_0) \in W$, but W is open and thus this is a contradiction (since then a sequence of points outside of W converge to a point in W).

Choose $t_1 < t_0$ such that $t_1 + \varepsilon > t_0$ and $\theta^{(p_0)}(t_1) \in U_0$ (which exists because $\theta^{(p_0)}$ is continuous). Since $t_1 < t_0$, we have that $(t_1, p_0) \in W$ and so there is a product neighborhood $(t_1 - \delta, t_1 + \delta) \times U_1 \subseteq W$. This implies that θ is defined and smooth on $[0, t_1 + \delta) \times U_1$, by definition of W. Because $\theta(t_1, p_0) \in U_0$, we can choose U_1 small enough so that θ maps $\{t_1\} \times U_1$ into U_0. Define $\widetilde{\theta}\colon [0, t_1 + \varepsilon) \times U_1 \to M$ by

$$\widetilde{\theta}(t,p) = \begin{cases} \theta_t(p) & 0 \leq t < t_1 \\ \theta_{t-t_1} \circ \theta_{t_1}(p) & t_1 - \varepsilon < t < t_1 + \varepsilon \end{cases}.$$

The group action laws show that the two definitions agree where they overlap, and our choices of U_1, t_1, ε ensure that this is a smooth map.

Thus $t \mapsto \widetilde{\theta}(t,p)$ is an integral curve of V for each $p \in U_1$, so $\widetilde{\theta}$ is a smooth extension of θ to a neighborhood of (t_0, p_0). This is a contradiction, so $W = \mathcal{D}$. Thus θ is the unique maximal flow with infinitesimal generator V.

We now show that M_t is open. Recall that

$$M_t = \{p \in M \mid (t,p) \in \mathcal{D}\},$$

so clearly M_t is open since $\mathcal{D}$ is (it is the preimage of $\mathcal{D}$ under the map $p \mapsto (t,p)$). We also have

$$p \in M_t \Longrightarrow t \in \mathcal{D}^{(p)} \Longrightarrow \mathcal{D}^{(\theta_t(p))} = \mathcal{D}^{(p)} - t \Longrightarrow -t \in \mathcal{D}^{(\theta_t(p))} \Longrightarrow \theta_t(p) \in M_{-t}.$$

Thus θ_t is a map $M_t \to M_{-t}$, and smooth since θ is. The group laws show $\theta_t \circ \theta_{-t}$ is the identity, so that θ_t is then a diffeomorphism $M_t \to M_{-t}$ as desired. ■

8.2.8 Definition

The unique maximal flow with infinitesimal generator V, whose existence and uniqueness was proven above, is called the **flow generated by** V, or just the **flow of** V.

8.2.9 Proposition

Suppose $F\colon M \to N$ is smooth, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Let θ be the flow of X and η the flow of Y. If X and Y are F-related then for each $t \in \mathbb{R}$, $F(M_t) \subseteq N_t$ and $\eta_t \circ F = F \circ \theta_t$ on M_t:

$$\begin{tikzcd} M_t \arrow[r, "F"'] \arrow[d, "\theta_t"'] & N_t \arrow[d, "\eta_t"] \\ M_{-t} \arrow[r, "F"] & N_{-t} \end{tikzcd}$$

Proof

For any $p \in M$, $F \circ \theta^{(p)}$ is an integral curve of Y starting at $F(p)$. By uniqueness of integral curves, the maximal integral curve $\eta^{(F(p))}$ must be defined at least on $\mathcal{D}^{(p)}$ and $F \circ \theta^{(p)} = \eta^{(F(p)}$ on $\mathcal{D}^{(p)}$. This means that

$$p \in M_t \Longrightarrow t \in \mathcal{D}^{(p)} \Longrightarrow t \in \mathcal{D}^{(F(p))} \Longrightarrow F(p) \in N_t,$$

so $F(M_t) \subseteq N_t$. And also

$$F \circ \theta^{(p)}(t) = \eta^{(F(p))}(t) \Longrightarrow F \circ \theta_t(p) = \eta_t(F(p)) = \eta_t \circ F(p)$$

for all $p \in M_t$. ■

8.2.10 Corollary

Let $F\colon M \to N$ be a diffeomorphism. If $X \in \mathfrak{X}(M)$ has flow θ, then the flow of F_*X is $\eta_t = F \circ \theta_t \circ F^{-1}$ with domain $N_t = F(M_t)$.

Proof

By the above proposition, and going in reverse as F_*X and X are F^{-1}-related. ■

8.2.11 Definition

A vector field is **complete** if its flow is global. Equivalently if for every $p \in M$, it has an integral curve starting at p defined on all of $\mathbb{R}$.

8.2.12 Lemma (Uniform time lemma)

Let $V \in \mathfrak{X}(M)$ with flow θ. Suppose there is an $\varepsilon > 0$ such that for every $p \in M$, the domain of $\theta^{(p)}$ contains $(-\varepsilon, \varepsilon)$. Then V is complete.

Proof

Suppose V is not complete, so for some $p \in M$ we have that $\mathcal{D}^{(p)}$ is not all of $\mathbb{R}$. Without loss of generality, we assume that $\mathcal{D}^{(p)}$ is bounded above, let b be its supremum. Then let $t_0 \in \mathcal{D}^{(p)}$ such that $b - \varepsilon < t_0 < b$, and let $q \in \theta^{(p)}(t_0)$. They hypothesis implies that $\theta^{(q)}(t)$ is defined at least for $t \in (-\varepsilon, \varepsilon)$, so define $\gamma\colon (-\varepsilon, t + \varepsilon) \to M$ by

$$\gamma(t) = \begin{cases} \theta^{(p)}(t) & -\varepsilon < t < b \\ \theta^{(q)}(t - t_0) & t_0 - \varepsilon < t < t_0 + \varepsilon \end{cases}.$$

By the group action laws, the two definitions agree on their overlap: $\theta^{(q)}(t - t_0) = \theta(t - t_0, \theta(t_0, p)) = \theta(t, p)$. By the translation law, γ is an integral curve for V starting at p. But since $t_0 + \varepsilon > b$, this is a contradiction. ■

8.2.13 Theorem

Every compactly supported smooth vector field is complete.

Proof

Let V be compactly supported on M, and let $K = \operatorname{supp} V$ be its support. For $p \in K$ there is a neighborhood U_p of p and a positive number ε_p such that the flow of V is defined on at least $(-\varepsilon_p, \varepsilon_p) \times U_p$, since the flow domain is open. By compactness, finitely many U_p cover K: say $U_{p_1}, \ldots, U_{p_n}$ cover K. Let ε be the minimum of $\varepsilon_{p_1}, \ldots, \varepsilon_{p_n}$, so that the flow is defined on $(-\varepsilon, \varepsilon) \times K$. Since V is zero outside of K, every integral curve starting outside of K is constant and thus can be defined on all of $\mathbb{R}$. Thus by the above lemma, V is complete. ■

8.2.14 Corollary

Every vector field defined on a compact manifold is complete.

8.2.15 Theorem

Every left-invariant vector field on a Lie group is complete.

Proof

Let G be a Lie group and $X \in \operatorname{Lie}(G)$ with flow θ. There is some $\varepsilon > 0$ such that $\theta^{(e)}$ is defined on $(-\varepsilon, \varepsilon)$. Let $g \in G$ so that X is L_g-related to itself, so $L_g \circ \theta^{(e)}$ is an integral curve of g defined on $(-\varepsilon, \varepsilon)$. Thus $\theta^{(g)}$ is defined on at least $(-\varepsilon, \varepsilon)$ for all $g \in G$, thus X is complete by the uniform time lemma. ■

8.3 Flowouts

Recall that for an immersed submanifold $S \subseteq M$, $X \in \mathfrak{X}(M)$ is **tangent to** S if $X_p \in T_pS \subseteq T_pM$ for all $p \in S$. Equivalently, iff it is ι-related to a smooth vector field on U denoted $X|_S \in \mathfrak{X}(S)$.

8.3.1 Definition

For an immersed submanifold $S \subseteq M$ and $X \in \mathfrak{X}(M)$, we say that X is **nowhere tangent to** S if $X_p \notin T_pS$ for all $p \in S$.

8.3.2 Theorem (Flowout theorem)

Let M be a smooth manifold, $S \subseteq M$ be an embedded k-dimensional submanifold, and $V \in \mathfrak{X}(M)$ nowhere tangent to S. Let $\theta\colon \mathcal{D} \to M$ be V's flow, let $\mathcal{O} = (\mathbb{R} \times S) \cap \mathcal{D}$, and $\Phi = \theta|_{\mathcal{O}}$. Then:

(1) $\Phi\colon \mathcal{O} \to M$ is an immersion,

(2) $\partial/\partial t \in \mathfrak{X}(O)$ is Φ-related to V,

(3) There exists a smooth positive function $\delta\colon S \to \mathbb{R}$ such that the restriction of Φ to $\mathcal{O}_\delta$ is injective, where
$$\mathcal{O}_\delta = \{(t,p) \in \mathcal{O} \mid |t| < \delta(p)\} \subseteq \mathcal{O}$$
is a flow domain. Thus $\Phi(\mathcal{O}_\delta)$ is an immersed submanifold of M containing S, and V is tangent to this submanifold.

(4) If S has codimension 1, then $\Phi|_{\mathcal{O}_\delta}$ is a diffeomorphism onto an open submanifold of M.

Note:
The submanifold $\Phi(\mathcal{O}_\delta) \subseteq M$ is called a **flowout from S along V**.

Proof

First we prove (2). Fix $p \in S$ and let $\sigma\colon \mathcal{D}^{(p)} \to \mathbb{R} \times S$ be the curve $\sigma(t) = (t, p)$. Then $\Phi \circ \sigma(t) = \theta(t, p)$ is an integral curve of V, so for any $t_0 \in \mathcal{D}^{(p)}$ we have that

$$d\Phi_{(t_0,p)}\left(\left.\frac{\partial}{\partial t}\right|_{(t_0,p)}\right) = (\Phi \circ \sigma)'(t_0) = V_{(t_0,p)},$$

as desired.

Now we prove (1). The restriction of Φ to $\{0\} \times S$ is the restriction of the diffeomorphism $\{0\} \times S \cong S$ with the embedding $S \hookrightarrow M$ (since it is the map $\Phi(0,p) = \theta(0,p) = p$), and thus an embedding. So the restriction of $d\Phi_{(0,p)}$ to T_pS (viewed as a subspace of $T_{(0,p)}\mathcal{O} \cong T_0\mathbb{R} \oplus T_pS$) is the inclusion $T_pS \hookrightarrow T_pM$. Then $d\Phi_{(0,p)}$ maps the basis of $T_{(0,p)}\mathcal{O}$, $(\partial/\partial t|_{(0,p)}, E_1, \ldots, E_k)$ with $E_1, \ldots, E_k \in T_pS$, to $(V_p, E_1, \ldots, E_k)$. Since V is nowhere tangent to S, V_p is not in the span of $E_1, \ldots, E_k$ and thus this remains linearly independent. Thus $d\Phi_{(0,p)}$ is an injection.

To show that $d\Phi$ is injective elsewhere, let $(t_0, s_0) \in \mathcal{O}$ and let $\tau_{t_0}\colon \mathcal{O} \to \mathbb{R} \times S$ be the translation $\tau_{t_0}(t,p) = (t + t_0, p)$. By the group law for θ, the following diagram commutes (the horizontal maps may only be defined in a neighborhood of $(0, p_0)$ and p_0 respectively):

$$\begin{tikzcd} \mathcal{O} \arrow[r, "\tau_{t_0}"'] \arrow[d, "\Phi"'] & \mathcal{O} \arrow[d, "\Phi"'] \\ M \arrow[r, "\theta_{t_0}"] & M \end{tikzcd}$$

Indeed, $\theta_{t_0} \circ \Phi(t,p) = \theta_{t+t_0}(p) = \Phi \circ \tau_{t_0}(t,p)$. The horizontal maps are local diffeomorphisms (τ_{t_0} clearly is, θ_t is by the flow theorem), and thus the induced commutative square

$$\begin{tikzcd} T_{(0,p_0)}\mathcal{O} \arrow[r, "d(\tau_{t_0})_{(0,p_0)}"'] \arrow[d, "d\Phi_{(0,p_0)}"'] & T_{(t_0,p_0)}\mathcal{O} \arrow[d, "d\Phi_{(t_0,p_0)}"'] \\ T_{p_0}M \arrow[r, "d(\theta_{t_0})_{p_0}"] & T_{\Phi(t_0,p_0)}M \end{tikzcd}$$

has invertible horizontal maps. Thus the two vertical maps have the same rank, and so Φ has constant rank. Since it has full rank at $(0, p_0)$, it has full rank everywhere and is thus an immersion.

Now we prove (3). Let $p_0 \in S$ and let $(U, (x^i))$ be a chart for S centered at p_0, so that $U \cap S$ is the set $x^{k+1} = \cdots = x^n = 0$. Because V is not tangent to S, one of the $n-k$ last components of V_{p_0} is nonzero, say $V^j_{p_0}$. Shrinking U if necessary, we may assume that $\left|V^j(p)\right| \geq c$ for all $p \in U$.

Now $\Phi^{-1}U$ is open in $\mathbb{R} \times S$, so we may choose $\varepsilon_{p_0} > 0$ and a neighborhood W_{p_0} of p_0 such that $(-\varepsilon_{p_0}, \varepsilon_{p_0}) \times W_{p_0} \subseteq \mathcal{O}$ and $\Phi((-\varepsilon_{p_0}, \varepsilon_{p_0}) \times W_{p_0}) \subseteq U$. Let us write Φ in these local coordinates as

$$\Phi(t,p) = (\Phi^1(t,p), \ldots, \Phi^n(t,p)).$$

Because Φ is the restriction of a flow, we have that $\Phi'(t,p) = V(\Phi(t,p))$, so in particular

$$\frac{\partial \Phi^j}{\partial t}(t,p) = V^j(\Phi(t,p)),$$

and thus by the fundamental theorem of calculus, $\left|\Phi^j(t,p)\right| \geq c|t|$. So for $(t,p) \in (-\varepsilon_{p_0}, \varepsilon_{p_0}) \times W_{p_0}$ we have that $\Phi(t,p) \in S$ iff $t = 0$ (otherwise its jth component is nonzero).

Now let $\{\psi_p\}_{p \in S}$ be a partition of unity subordinate to $\{W_p\}_{p \in S}$ and define $f\colon S \to \mathbb{R}$ by

$$f(q) = \sum_{p \in S} \varepsilon_p \psi_p(q).$$

This is smooth and positive. For $q \in S$ there are finitely many p such that $\psi_p(q) > 0$, and if we let ε_{p_0} is

maximal among those ps then
$$f(q) \leq \varepsilon_{p_0} \sum_{p \in S} \psi_p(q) = \varepsilon_{p_0}.$$
Thus if $(t, q) \in \mathcal{O}$ and $|t| < f(q)$ then $(t, q) \in (-\varepsilon_{p_0}, \varepsilon_{p_0}) \times W_{p_0}$ so that $\Phi(t, q) \in S$ iff $t = 0$.

So define $\delta = \frac{1}{2}f$, and $\mathcal{O}_\delta$ as above:
$$\mathcal{O}_\delta = \{(t, p) \in \mathcal{O} \mid |t| < \delta(p)\}.$$
We show that Φ is injective on $\mathcal{O}_\delta$ (and thus its image is an immersed submanifold). Suppose $\Phi(t, q) = \Phi(t', q')$, and we can assume $f(q') \leq f(q)$. Our assumption means that $\theta_t(q) = \theta_{t'}(q')$, and thus $\theta_{t-t'}(q) = q' \in S$ by the group laws. Now we have
$$|t - t'| \leq |t| + |t'| \leq \frac{1}{2}f(q) + f(q') \leq f(q),$$
and thus as we showed this means $\theta_{t-t'}(q) = \Phi(t - t', q) \in S$ iff $t = t'$. It then follows that $q = q'$ (because we can recover the initial point from an integral flow).

Finally for (4), if S has codimension 1 then $\Phi|_{\mathcal{O}_\delta}$ is an injective smooth immersion between manifolds of the same dimension, and thus an embedding, and a diffeomorphism onto an open subset (the only submanifolds of codimension zero are open). ■

8.3.3 Definition

Let $V \in \mathfrak{X}(M)$, a **singular point** of V is a point $p \in M$ such that $V_p = 0$. Otherwise $p \in M$ is a **regular point**.

8.3.4 Proposition

Let $V \in \mathfrak{X}(M)$, and $\theta\colon \mathcal{D} \to M$ be its flow. If $p \in M$ is a singular point of V, then $\mathcal{D}^{(p)} = \mathbb{R}$ and $\theta^{(p)}$ is the constant curve $\theta^{(p)}(t) = p$. If p is a regular point, then $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ is a smooth immersion.

Proof

If $V_p = 0$, then the constant curve $\gamma\colon \mathbb{R} \to M$ given by $\gamma(t) = p$ is clearly integral. By uniqueness and maximality, it must be equal to $\theta^{(p)}$.

Suppose that $\theta^{(p)}$ is not an immersion; we will show that p is singular. Since it is not an immersion, ${\theta^{(p)}}'(s) = 0$ for some $s \in \mathcal{D}^{(p)}$. Let $q = \theta^{(p)}(s)$ so that $V_q = 0$ and q is singular. But then $\theta^{(q)}$ is constant, and
$$\theta^{(p)}(t) = \theta(t, p) = \theta(t - s, \theta(s, p)) = \theta(t - s, q) = q,$$
so that $p = \theta^{(p)}(0) = q$ and so p is singular. ■

8.3.5 Definition

If $\theta\colon \mathcal{D} \to M$ is a flow, a point $p \in M$ is an **equilibrium point** of θ if $\theta(t, p) = p$ for all $t \in \mathcal{D}^{(p)}$.

The above proposition shows that if θ is V's flow, then θ's equilibrium points are precisely V's singular points.

8.3.6 Theorem (Canonical form of flow)

Let $V \in \mathfrak{X}(M)$ and $p \in M$ a regular point of V. Then there exist smooth coordinates (s^i) in a neighborhood of p such that V has the coordinate representation $\partial/\partial s^1$. If $S \subseteq M$ is an embedded hypersurface (embedded submanifold of codimension 1) with $p \in S$ and $V_p \notin T_pS$, then the coordinates can be chosen so that s^1 is a local defining map for S.

Note:
If $S \subseteq M$ is embedded, a **local defining map** for S is a smooth map $\Phi\colon U \to N$ with $U \subseteq M$ open, such that $S \cap U$ is a regular level set of Φ.

Proof

If no S is given, let $(U, (x^i))$ be any smooth coordinates centered at p and take $S \subseteq U$ to be the hypersurface defined by $x^j = 0$ where j is any index where $V^j(p) \neq 0$.

Regardless, we now just assume we have an embedded hypersurface containing p where $V_p \notin T_pS$. We can potentially shrink S so that V is nowhere tangent to it. The flowout theorem gives us a flow domain $\mathcal{O}_\delta \subseteq \mathbb{R} \times S$ such that the flow of V restricts to a diffeomorphism from $\mathcal{O}_\delta$ to an open subset $W \subseteq M$ containing S. Take a product neighborhood $(0,p) \in (-\varepsilon, \varepsilon) \times W$ in $\mathcal{O}_\delta$, and choose a smooth local parameterization $X\colon \Omega \to S$ where Ω is an open subset of $\mathbb{R}^{n-1}$ with coordinates denoted by $s^2, \ldots, s^n$, and X's image is contained in W_0.

Then define $\Psi\colon (-\varepsilon, \varepsilon) \times \Omega \to M$ by

$$\Psi(t, s^2, \ldots, s^n) = \Phi(t, X(s^2, \ldots, s^n))$$

is a diffeomorphism onto a neighborhood of p in M. Because $(t, s^2, \ldots, s^n) \mapsto (t, X(s^2, \ldots, s^n))$ pushes $\partial/\partial t$ forward to itself, and $\Phi_*(\partial/\partial t) = V$ (since by the flowout theorem, $\partial/\partial t$ is Φ-related to V), it follows that $\Psi_*(\partial/\partial t) = V$.

Thus Ψ^{-1} is a smooth coordinate chart in which V has coordinate representation $\partial/\partial t$. Rename t to s^1 to complete the proof: $s^1 = 0$ gives W_0, so it is a local defining map for S. ■

8.4 Lie derivatives

Tangent spaces allow us to make sense of directional derivatives of real-valued functions on manifolds; an element of T_pM is viewed as a directional derivative at p. But what about the directional derivative of a vector field. In Euclidean space, if W is a vector field on $\mathbb{R}^n$ then we could define its directional derivative in the direction of $v \in T_p\mathbb{R}^n$ to be

$$D_vW(p) = \left.\frac{d}{dt}\right|_0 W_{p+tv} = \lim_{t \to 0} \frac{W_{p+tv} - W_p}{t}.$$

It is easy to see that

$$D_vW(p) = D_vW^i(p) \left.\frac{\partial}{\partial x^i}\right|_p,$$

where W^i are the components of W and D_vW^i is the usual directional derivative.

In smooth manifolds, W_{p+tv} makes no sense, but we could instead replace $p + tv$ with a curve γ starting at p with initial velocity v. The issue is then that $W_{\gamma(t)}$ and W_p lie in different tangent spaces, and thus there is no good way of combining them.

Instead we can replace $v \in T_pM$ with a vector field $V \in \mathfrak{X}(M)$, and so we can use the flow of V to push values of W back to p and then differentiate. This motivates the following.

8.4.1 Definition

Let M be smooth and without boundary, $V \in \mathfrak{X}(M)$ with flow θ. For any other smoot vector field W, define a rough vector field on M, denoted $\mathcal{L}_V W$ and called the **Lie derivative** of W with respect to V, to be

$$(\mathcal{L}_V W)_p = \frac{d}{dt}\bigg|_0 d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \lim_{t\to 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t},$$

provided the derivative exists. For small $t \neq 0$, θ_t is defined in a neighborhood of p, and θ_{-t} is its inverse, so both $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ and W_p are both elements of T_pM.

Note that we are differentiating a function $\mathbb{R} \to T_pM$ here. Since T_pM is Euclidean, this is well-defined as any linear isomorphism from $\mathbb{R}^n$ to it will give the same limit.

8.4.2 Lemma

The Lie derivative of W with respect to V exists for all $p \in M$ and $\mathcal{L}_V W$ forms a smooth vector field.

Proof

Let $p \in M$ and $(U, (x^i))$ be a smooth chart containing p. Choose an open interval $0 \in J_0$, and an open set $U_0 \subseteq U$ such that θ maps $J_0 \times U_0$ into U. For $(t, x) \in J_0 \times U_0$, write the components of θ as $(\theta^1(t, x), \ldots, \theta^n(t, x))$. Then for any $(t, x) \in J_0 \times U_0$, the matrix of $d(\theta_{-t})_{\theta_t(x)} \colon T_{\theta_t(x)}M \to T_xM$ is

$$\left(\frac{\partial \theta^i}{\partial x^j}(-t, \theta(t, x))\right).$$

Thus

$$d(\theta_{-t})_{\theta_t(x)}(W_{\theta_t(x)}) = \frac{\partial \theta^i}{\partial x^j}(-t, \theta(t, x)) W^j(\theta(t, x)) \left.\frac{\partial}{\partial x^i}\right|_x .$$

Because θ^i and W^j are smooth, the coefficient of $\partial/\partial x^i\big|_x$ depends smoothly on (t, x). It then follows that $(\mathcal{L}_V W)_x$, which is obtained by differentiating this and setting $t = 0$, exists for each $x \in U_0$ and depends smoothly on x. ■

8.4.3 Example

Let $v \in \mathbb{R}^n$ and W a smooth vector field on $\mathbb{R}^n$. Then $D_v W(p)$, which we defined above, is just $(\mathcal{L}_V W)_p$, where V is the vector field $v^i\, \partial/\partial x^i\big|$ with constant coefficients.

The definition of the Lie derivative is unwieldy for computations, but it turns out that it actually is just the Lie bracket in disguise.

8.4.4 Theorem

For $V, W \in \mathfrak{X}(M)$, $\mathcal{L}_V W = [V, W]$.

Proof

Let $\mathcal{R}(V) \subseteq M$ be the set of regular points of V (points p where $V_p \neq 0$). By continuity, $\mathcal{R}(V)$ is open in M, and its closure is V's support. We now show that $(\mathcal{L}_V W)_p = [V, W]_p$ for all $p \in M$ by considering three cases.

(1) If $p \in \mathcal{R}(V)$, then we can find a local chart $(U, (u^i))$ around p where V has the coordinate representation $V = \frac{\partial}{\partial u^1}$. In these coordinates, V's flow is then $\theta_t(u) = (u^1 + t, u^2, \ldots, u^n)$ (this boils down to the Euclidean case discussed in above examples). Then for each fixed t, the matrix $d(\theta_{-t})_{\theta_t(x)}$ in these coordinates is the identity. Thus

$$d(\theta_{-t})_{\theta_t(u)}(W_{\theta_t(u)}) = d(\theta_{-t})_{\theta_t(u)}\left(W^j(u^1+t,u^2,\dots,u^n)\left.\frac{\partial}{\partial u^j}\right|_{\theta_t(u)} \right) = W^j(u^1+t,u^2,\dots,u^n)\left.\frac{\partial}{\partial u^j}\right|_u .$$

And so by the definition of the Lie derivative,

$$(\mathcal{L}_V W)_u = \left.\frac{d}{dt}\right|_0 W^j(u^1+t,u^2,\dots,u^n)\left.\frac{\partial}{\partial u^j}\right|_u = \frac{\partial W^j}{\partial u^1}(u^1,\dots,u^n)\left.\frac{\partial}{\partial u^j}\right|_u .$$

Conversely,

$$[V,W] = \left[\frac{\partial}{\partial u^1}, W^j\frac{\partial}{\partial u^j}\right] = \left(V^i\frac{\partial}{\partial W^j}u^i - W^i\frac{\partial}{\partial V^j}u^i \right)\frac{\partial}{\partial u^j} = (W^j)u^1\frac{\partial}{\partial u^j},$$

since $V^1 = 1$ is constant and all other components are zero.

(2) If $p \in \operatorname{supp} V$ then by continuity we have the desired result.

(3) If $p \notin \operatorname{supp} V$ then $V = 0$ on a neighborhood of p, and so $[V,W]_p = 0$. Then θ_t is the identity map in a neighborhood of p, so $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = W_p$ and so $(\mathcal{L}_V W)_p = 0$ as well. ■

Thus we can extend the notion of a Lie derivative to manifolds with boundary simply by $\mathcal{L}_V W = [V,W]$. This is equivalent to embedding M into a smooth manifold without boundary $\widetilde{M}$ (e.g. its double), extending V and W smoothly, and computing the Lie derivative there. By the above theorem, this just gives $[V,W]$, so is independent on the extension.

This also gives a geometric interpretation of the Lie bracket as the Lie derivative, i.e. a directional derivative.

8.4.5 Corollary

Let $V, W, X \in \mathfrak{X}(M)$, then

(1) $\mathcal{L}_V W = -\mathcal{L}_W V$,

(2) $\mathcal{L}_V[W,X] = [\mathcal{L}_V W, X] + [W, \mathcal{L}_V X]$,

(3) $\mathcal{L}_{[V,W]}X = \mathcal{L}_V\mathcal{L}_W X - \mathcal{L}_W\mathcal{L}_V X$,

(4) If $g \in C^\infty(M)$, then $\mathcal{L}_V(gW) = (Vg)W + g\mathcal{L}_V W$,

(5) If $F\colon M \to N$ is a diffeomorphism, $F_*(\mathcal{L}_V W) = \mathcal{L}_{F_*V}F_*W$.

8.5 Commuting vector fields

8.5.1 Definition

Two vector fields $V, W \in \mathfrak{X}(M)$ **commute** if $VWf = WVf$ for all $f \in C^\infty(M)$, or equivalently if $[V,W] = 0$.

8.5.2 Definition

If θ is a smooth flow and $W \in \mathfrak{X}(M)$, then W is **invariant under** θ if W is θ_t-related to itself for all t.

More precisely, $W|_{M_t}$ is θ_t-related to $W_{M_{-t}}$:

$$d(\theta_t)_p(W_p) = W_{\theta(t,p)}$$

for all (t, p) in the flow domain.

8.5.3 Theorem

For $V, W \in \mathfrak{X}(M)$, the following are equivalent:

(1) V and W commute,

(2) V is invariant under W's flow,

(3) W is invariant under V's flow.

Proof

Let θ denote V's flow. If W is invariant under it, then we see that

$$d(\theta_{-t})_{\theta(t,p)}(W_{\theta(t,p)}) = W_p,$$

and by the definition of the Lie derivative $\mathcal{L}_V W = 0$ so $[V, W] = 0$.
Now if $\mathcal{L}_V(W) = [V, W] = 0$ then the derivative of $d(\theta_{-t}))_{\theta(t,p)}(W_{\theta(t,p)})$ is zero and thus it is constant over t. Plugging in $t = 0$ we see that it is equal to W_p, and thus we have the desired result. ∎

8.5.4 Corollary

Every vector field is invariant under its own flow.

We would like to discuss what it means for two flows to commute. For global flows, we could just require that $\psi_s \circ \theta_t = \theta_t \circ \psi_s$, but for local flows this may not always be defined. But even if we require that this only hold when both sides are defined, commuting vector fields may not have commuting flows. The issue is that if $\theta_t \circ \psi_s(p)$ is defined for $t = t_0, s = s_0$, then it must be defined for all t in some interval containing 0 and t_0, but ψ_s may not be defined all the way to $s = t_0$.

8.5.5 Definition

If θ, ψ are smooth flows on M, then we say they **commute** if whenever $p \in M$ and J, K are open intervals containing 0 such that one of $\theta_t \circ \psi_s(p), \psi_s \circ \theta_t(p)$ exist for all $(t, s) \in J \times K$, then both exist and are equal.

8.5.6 Theorem

Smooth vector fields commute iff their flows commute.

Proof

Let $V, W \in \mathfrak{X}(M)$ and θ, ψ be their respective flows. Suppose V and W commute and J, K are intervals such that $\psi_s \circ \theta_t(p)$ is defined for all $(t, s) \in J \times K$. Since V is invariant under ψ by assumption, let $s \in K$ and consider the curve $\gamma\colon J \to M$ given by $\gamma(t) = \psi_s \circ \theta_t(p)$. Then $\gamma(0) = \psi_s(p)$ and its velocity at $t \in J$ is

$$\gamma'(t) = d(\psi_s)_{\theta(t,p)}(\theta^{(p)\prime}(t)) = d(\psi_s)_{\theta(t,p)}(V_{\theta(t,p)}) = V_{\psi(s,\theta(t,p))} = V_{\gamma(t)},$$

so that $\gamma(t)$ is integral for V. Then

$$\gamma(t) = \theta^{(\psi(s,p))}(t) = \theta_t(\psi_s(p)),$$

so that θ, ψ commute.

Now if the flows commute, let $p \in M$ and $\varepsilon > 0$ small enough so that $\psi_s \circ \theta_t(p) = \theta_t \circ \psi_s(p)$ for $|t|, |s| < \varepsilon$. Now we have

$$W_{\theta_t(p)} = \left.\frac{d}{ds}\right|_{s=0} \psi^{(\theta_t(p))}(s) = \left.\frac{d}{ds}\right|_{s=0} \theta_t(\psi^{(p)}(s)) = d(\theta_t)_p(W_p).$$

Showing that W is θ_t-invariant, as desired. ∎

8.5.7 Definition

A **commuting frame** for a smooth manifold M is a local frame (E_i) such that each E_i, E_j commute.

Note that coordinate frames $\partial/\partial x|_p$ are commuting, as shown in our discussion of Lie brackets. Thus if a frame doesn't commut, it cannot be a coordinate frame.

8.5.8 Theorem

Let M be smooth and $(V_1, \ldots, V_k)$ be linearly independent commuting local vector fields, defined on an open subset W. Then for every $p \in W$ there exists a smooth coordinate chart $(U, (s^i))$ centered at p such that $V_i = \partial/\partial s^i$ for all $i = 1, \ldots, k$.

If $S \subseteq M$ is embedded and of codimension k and $p \in S$ is such that T_pS is complementary to the span of $V_1|_p, \ldots, V_k|_s$, then the coordinate chart can be chosen so that $S \cap U$ is the slice given by $s^1 = \cdots = s^k = 0$.

Proof

If no S is given, we can find one via an appropriate coordinate slice. Let $(U, (x^i))$ be the slice chart for S centered at p, with $U \subseteq W$ and $S \cap U$ being given by $x^1 = \cdots = x^k = 0$. Our assumptions show that $V_1, \ldots, V_k, \partial/\partial x^{k+1}, \ldots, \partial/\partial x^n$ form a local frame defined on U. Since this theorem is local, we can assume $U \subseteq \mathbb{R}^n$.

Let θ_i denote the flow of V_i. Then since the vector fields commute, there is an $\varepsilon > 0$ such that $(\theta_1)_{t_1} \circ \cdots \circ (\theta_k)_{t_k}$ is defined when $|t_i| < \varepsilon$ and on a neighborhood Y of p. Define $\Omega \subseteq \mathbb{R}^{n-k}$ by

$$\Omega = \left\{(s^{k+1}, \ldots, s^n) \;\middle|\; (0, \ldots, 0, s^{k+1}, \ldots, s^n) \in Y\right\}.$$

Define $\Phi\colon (-\varepsilon, \varepsilon)^k \times \Omega \to U$ by

$$\Phi(s^1, \ldots, s^k, s^{k+1}, \ldots, s^n) = (\theta_1)_{s^1} \circ \cdots \circ (\theta_k)_{s^k}(0, \ldots, 0, s^{k+1}, \ldots, s^n).$$

By construction $\Phi(\{0\} \times \Omega) = S \cap Y$. Now we will show that $\partial/\partial s^i$ is Φ-related to V_i for $i \le k$. For any $i \le k$ and $s_0 \in (-\varepsilon, \varepsilon)^k \times \Omega$, we have

$$d\Phi_{s_0}\left(\left.\frac{\partial}{\partial s^i}\right|_{s_0}\right) f = \left.\frac{\partial}{\partial s^i}\right|_{s_0} f(\Phi(s^1, \ldots, s^n)) = \left.\frac{\partial}{\partial s^i}\right|_{s_0} f((\theta_1)_{s^1} \circ \cdots \circ (\theta_k)_{s^k}(0, \ldots, 0, s^{k+1}, \ldots, s^n)).$$

Since the flows commute, we can reorder this composition to put θ_i first. And $t \mapsto (\theta_i)_t(q)$ is integral for V_i so this is equal to the derivative of f composed with this integral curve, equal to $V_i|_{\Phi(s_0)} f$ as desired.

Now we will show that $d\Phi_0$ is invertible. Since $d\Phi_0(\partial/\partial s^i_0) = V_i|_p$ by our above computation, and $\Phi(0,\overline{s}) = (0,\overline{s})$ we see that

$$d\Phi_0\left(\frac{\partial}{\partial s^i}_0\right) = \frac{\partial}{\partial s^i}_p, \qquad i > k.$$

Thus $d\Phi_0$'s image of the coordinate basis is itself a basis, so it is invertible.

So Φ is a diffeomorphism in a neighborhood of 0, and $\varphi = \Phi^{-1}$ is a smooth coordinate chart around p where V_i's coordinate representation is $\partial/\partial s^i$ and takes S to the slice $s^1 = \cdots = s^k = 0$. ■

9 Vector Bundles

9.1 Vector bundles

9.1.1 Definition

Let X be a topological space. A (real) **vector bundle** of rank k is a topological space E along with a surjective continuous map $\pi\colon E \to X$ satisfying the following conditions:

(1) for each $p \in M$, the fiber $E_p = \pi^{-1}p$ is endowed with a k-dimensional real vector space structure,

(2) for each $p \in M$ there is a neighborhood $p \in U \subseteq M$ and a homeomorphism $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$, called a **local trivialization** of E over U, satisfying

- **(i)** $\pi_U \circ \Phi = \pi$ where $\pi_U\colon U \times \mathbb{R}^k$ is the projection,
- **(ii)** for each $q \in U$, the restriction of Φ to E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

E is called the **total space** of the bundle, X is its **base**, and π the **projection**.

When $X = M$ and E are smooth manifolds, π is smooth, and the local trivializations are diffeomorphisms, then E is called a **smooth vector bundle**. The trivializations are called **smooth local trivializations**.

If there is a trivialization of E over all of M, then it is called a **global trivialization** and the bundle is called a **trivial bundle**. In such a case E is necessarily homeomorphic to $M \times \mathbb{R}^k$; when the bundle is smooth this is of course a diffeomorphism.

The requirement that $\pi_U \circ \Phi = \pi$ means that for $p \in U$ and $v \in E_p$, $\Phi(v) = (p, \bullet)$.

Note:

Unless specified otherwise, all vector bundles are assumed to be smooth.

9.1.2 Example

Define $E = M \times \mathbb{R}^k$, and $\pi\colon M \times \mathbb{R}^k \to M$ the obvious projection. Then $\pi^{-1}p = \{p\} \times \mathbb{R}^k$ has an obvious k-dimensional structure. This is a trivial bundle in the obvious way, when M is smooth it is a smooth trivial bundle.

The tangent bundle is the prototypical example of a vector bundle.

9.1.3 Proposition

Let M be a smooth n-manifold, and TM its tangent bundle. With the ordinary projection map, the usual vector space structure on each fiber (T_pM), and the usual topology and smooth structure, TM forms a smooth vector bundle of rank n over M.

Proof

For any smooth chart (U, φ) of M with coordinate functions (x^i), define $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^n$ by

$$\Phi\left(v^i \left. \frac{\partial}{\partial x^i} \right|_p \right) = (p, v^1, \dots, v^n).$$

Notice that the composite

$$\pi^{-1}U \xrightarrow{\Phi} U \times \mathbb{R}^n \xrightarrow{\varphi \times \mathrm{id}} \varphi(U) \times \mathbb{R}^n$$

is the smooth atlas of TM corresponding to the chart (U, φ). It and $\varphi \times \mathrm{id}$ are diffeomorphisms, thus Φ is smooth.

And clearly we have

$$\pi_U \circ \Phi = \pi,$$

completing the proof. ■

9.1.4 Proposition

Smooth vector bundles are smooth submersions.

Proof

Let $p \in M$, so there exists a neighborhood $p \in U$ with a diffeomorphism $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ such that $\pi = \pi_U \circ \Phi$. Now π_U and Φ are submersions, and thus so is $\pi|_{\pi^{-1}U}$. These $\pi^{-1}U$ cover E and so it is globally a smooth submersion. ■

9.1.5 Lemma

Let $\pi\colon E \to M$ be a smooth vector bundle of rank k. Suppose that $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi\colon \pi^{-1}(V) \to V \times \mathbb{R}^k$ are two smooth local trivializations with $U \cap V$ being nonempty. Then there is a smooth map $\tau\colon U \cap V \to \mathrm{GL}_k(\mathbb{R})$ such that the composition $\Phi \circ \Psi^{-1}\colon (U \cap V) \times \mathbb{R}^k \to (U \cap V) \times \mathbb{R}^k$ has the form

$$\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p)v).$$

Proof

Note that restricting Φ to $\pi^{-1}(U \cap V)$ maps into $(U \cap V) \times \mathbb{R}^k$: $\pi_1 \circ \Phi = \pi$, meaning Φ's first coordinate is in $\pi \circ \pi^{-1}(U \cap V) = U \cap V$.

The following diagram commutes

$$\begin{tikzcd}
(U \cap V) \times \mathbb{R}^k \arrow[dr, "\pi_1"'] & \pi^{-1}(U \cap V) \arrow[l, "\Psi"'] \arrow[r, "\Phi"] \arrow[d, "\pi"] & (U \cap V) \times \mathbb{R}^k \arrow[dl, "\pi_1"] \\
& U \cap V &
\end{tikzcd}$$

Meaning that $\pi_1 = \pi_1 \circ \Psi^{-1} \circ \Phi$, so $\Psi^{-1} \circ \Phi$ doesn't change the first coordinate. So there is a smooth map $\sigma\colon (U \cap V) \times \mathbb{R}^k \to \mathbb{R}^k$ such that

$$\Phi \circ \Psi^{-1}(p, v) = (p, \sigma(p, v)).$$

For a fixed $p \in U \cap V$, $\sigma(p, \bullet)$ is an invertible linear map $\mathbb{R}^k$ to $\mathbb{R}^k$. Thus there is an invertible matrix $\tau(p) \in \mathrm{GL}_k(\mathbb{R})$ such that $\sigma(p,v) = \tau(p)v$. The map τ is constant, as its columns are given by $C_i\tau(p) = \sigma(p, e_i)$ (e_i are the standard basis vectors), which are smooth. ∎

9.1.6 Definition

The map described in the above lemma, $\tau\colon (U \cap V) \to \mathrm{GL}_k(\mathbb{R})$, is called the **transition function** between the local trivializations Φ and Ψ.

9.1.7 Lemma

Let M be a smooth manifold, and suppose for each $p \in M$ we are given a real vector space E_p of some fixed dimension k. Let $E = \coprod_{p\in M} E_p$, and let $\pi\colon E \to M$ be the obvious projection (its component on E_p is the constant map to p). Suppose further that we have:

(1) an open cover $\{U_\alpha\}_\alpha$ of M;

(2) for each α, a bijective map $\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose restriction to each E_p for $p \in U_\alpha$ is a vector space isomorphism $E_p \to \{p\} \times \mathbb{R}^k$;

(3) for each α, β where $U_\alpha \neq U_\beta$ is nonempty, a smooth map $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}_k(\mathbb{R})$ such that the map $\Phi_\alpha \circ \Phi_\beta^{-1}$ on $(U_\alpha \cap U_\beta) \times \mathbb{R}^k$ is given by

$$\Phi_\alpha \circ \Phi_\beta^{-1}(p, v) = (p, \tau_{\alpha\beta}(p)v).$$

Then E has a unique topology and smooth structure making it into a smooth vector bundle over M of rank k, with π the projection, and $\{(U_\alpha, \Phi_\alpha)\}$ the trivializations.

Proof

For each $p \in M$, choose a U_α containing it, and a smooth chart (V_p, φ_p) for M with $p \in V_p \subseteq U_\alpha$. Then let $\widehat{V}_p = \varphi_p(V_p) \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, with $n = \dim M$. Define the map $\widetilde{\varphi}_p\colon \pi^{-1}V_p \to \widehat{V}_p \times \mathbb{R}^k$ by

$$\widetilde{\varphi}_p\colon \pi^{-1}\widehat{V}_p \xrightarrow{\Phi_\alpha} V_p \times \mathbb{R}^k \xrightarrow{\varphi_p \times \mathrm{id}} \widehat{V}_p \times \mathbb{R}^k.$$

We will now show that the collection of charts $\left\{(\pi^{-1}V_p, \widetilde{\varphi}_p)\right\}_{p\in M}$ make E into a vector bundle.

As a composition of bijective maps, $\widetilde{\varphi}_p$ is bijective, and its image $\widehat{V}_p \times \mathbb{R}^k$ is open in either $\mathbb{R}^n \times \mathbb{R}^k$ or $\mathbb{H}^n \times \mathbb{R}^k \cong \mathbb{H}^{n+k}$. For $p, q \in M$ we have

$$\widetilde{\varphi}_p(\pi^{-1}V_p \cap \pi^{-1}V_q) = \varphi_p(V_p \cap V_q) \times \mathbb{R}^k$$

because Φ_α maps $\pi^{-1}(X)$ to $X \times \mathbb{R}^k$ bijectively for $X \subseteq U_\alpha$. This is open, as φ_p is a homeomorphism. Furthermore, if two charts overlap then

$$\widetilde{\varphi}_p \circ \widetilde{\varphi}_q^{-1} = (\varphi_p \times \mathrm{id}) \circ \Phi_\alpha \circ \Phi_\beta^{-1} \circ (\varphi_q \times \mathrm{id})^{-1}.$$

The maps $\varphi_p \times \mathrm{id}$ and $\Phi_\alpha \circ \Phi_\beta^{-1}$ are diffeomorphisms, as desired.

The open cover $\{V_p\}_p$ of M has a countable subcover, so there is a countable subcovering of E by charts.

Finally we need to show that E is Hausdorff. If two points in E lie in the same E_p, then they lie in $\pi^{-1}V_p$. And if $\xi \in E_p$ and $\eta \in E_q$ for $p \neq q$, then taking disjoint $p \in V_p$ and $q \in V_q$, we have that $\xi \in \pi^{-1}V_p$ and $\eta \in \pi^{-1}V_q$ are disjoint neighborhoods. Thus E is a smooth manifold.

We now just need to verify that the data given forms a vector bundle. Φ_α is a diffeomorphism, because in terms of $(\pi^{-1}V_p, \widetilde{\varphi}_p)$ for E and $(V_p \times \mathbb{R}^k, \varphi_p \times \mathrm{id})$ for $U_\alpha \times \mathbb{R}^k$, the coordinate representation of Φ_α is just the identity. The coordinate representation of π relative to (V_p, φ_p) for M and its induced chart for E is

$$\varphi_p \circ \pi \circ \widetilde{\varphi}_p^{-1}(x, v) = \varphi_p \circ \pi \circ \Phi_\alpha^{-1}(\varphi_p^{-1}x, v) = \varphi_p(\varphi_p^{-1}x) = x,$$

which is smooth. By assumption, Φ_α maps E_p to $\{p\} \times \mathbb{R}^k$ so that $\pi_1 \circ \Phi_\alpha = \pi$.

This being the unique smooth structure follows from the requirement that Φ_α be diffeomorphisms. They are local diffeomorphisms whose domains cover E, and thus define the smooth structure on E. ■

9.1.8 Example

Suppose we have two vector bundles on M, $E' \to M$ and $E'' \to M$ of rank k', k'' respectively. Then we can define their **Whitney sum** to be the vector bundle $E \to M$ whose fiber at $p \in M$ is $E'_p \oplus E''_p$. The total space is $E = \coprod_{p \in M} E'_p \oplus E''_p$ and $\pi\colon E \to M$ is the obvious projection. For $p \in M$ choose a neighborhood U small enough so that there exist local trivializations (U, Φ') of E' and (U, Φ'') of E''. Then define $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^{k'} \times \mathbb{R}^{k''}$ by

$$\Phi(v', v'') = (\pi'(v'), \pi_{k'} \circ \Phi'(v'), \pi_{k''} \circ \Phi''(v'')),$$

where $\pi_{k'}\colon \mathbb{R}^{k'} \times \mathbb{R}^{k''} \to \mathbb{R}^{k'}$ is the projection. Note that Φ's restriction to $E_p = E'_p \oplus E''_p$ is a linear isomorphism to $\{p\} \times \mathbb{R}^{k'} \times \mathbb{R}^{k''}$ given by the linear isomorphisms of Φ', Φ''.

Now if we have another pair of local trivlializations $(\widetilde{U}, \widetilde{\Phi}'), (\widetilde{U}, \widetilde{\Phi}'')$ with transition maps $\tau'\colon U \cap \widetilde{U} \to \mathrm{GL}_{k'}(\mathbb{R})$ and $\tau''\colon U \cap \widetilde{U} \to \mathrm{GL}_{k''}(\mathbb{R})$. Then the transition function for $E' \oplus E''$ has the form

$$\widetilde{\Phi} \circ \Phi^{-1}(p, v', v'') = (p, \tau(p)(v', v'')), \qquad \tau(p) = \tau'(p) \oplus \tau''(p),$$

where the direct sum of matrices is given by the diagonal block matrix.

So by the above lemma, there is a unique vector bundle $E \to M$. This vector bundle $E = E' \oplus E''$ is called the **Whitney sum**.

9.1.9 Example

If $\pi\colon E \to M$ is a rank-k vector bundle, and $S \subseteq M$, then define the **restriction** of E to S to be the set $E|_S = \bigcup_{s \in S} E_s$, and the projection $E|_S \to S$ obtained by restricting π. If $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^k$ is a local trivialization, it restricts to a bijective map $\Phi|_S\colon (\pi|_S)^{-1}(U \cap S) \to (U \cap S) \times \mathbb{R}^k$. These form local trivializations for a continuous vector bundle structure on $E|_S$.

If $S \subseteq M$ is an immersed submanifold, then it can be shown by the above lemma that $E|_S$ has a smooth structure making it into a smooth vector bundle over S. When $E = TM$, $TM|_S$ is called the **ambient tangent bundle** over M.

9.2 Sections of vector bundles

9.2.1 Definition

Let $\pi\colon E \to M$ be a vector bundle. A **section** of E is a section of the map π, i.e. a map $\sigma\colon M \to E$ such that $\pi \circ \sigma = \mathrm{id}_M$ (i.e. $\sigma(p) \in E_p$). If we do not assume that σ is continuous, we call it a **rough section**, and if σ is smooth it is a **smooth section**.

For an open subset $U \subseteq M$, a (rough/continuous/smooth) **local section** is a (rough/continuous/smooth) map $\sigma\colon U \to E$ such that $\pi \circ \sigma = \mathrm{id}_M$. Sections on all of M are also called **global sections**.

The set of sections on E is denoted by $\Gamma(E)$.

The simplest example of a section is the **zero section**: $\zeta\colon M \to E$ which maps $p \in M$ to the zero vector of E_p.

This is a smooth section, as for every $p \in M$ we can take a local trivialization (U, Φ). Consider the composition

$$U \xrightarrow{\zeta} \pi^{-1}U \xrightarrow{\Phi} U \times \mathbb{R}^k,$$

which maps $p \in U$ to $(p, 0)$ (since $\zeta(p) = 0 \in E_p$ and Φ restricts to a linear isomorphism), and this is smooth. Since Φ is a diffeomorphism, ζ is smooth in U, and thus on all of M.

Note that sections of TM are the same as vector fields on M, so $\Gamma(TM) = \mathfrak{X}(M)$.

9.2.2 Example

A section of the ambient tangent bundle $TM|_S \to S$ is called a **vector field along** S. It is a map $X\colon S \to TM$ such that $X_p \in T_pM$ for $p \in S$. This is not a vector field *on* S which requires $X_p \in T_pS$.

9.2.3 Example

If $E = M \times \mathbb{R}^k$ is a trivial bundle, then there is a natural correspondence between sections of E and maps $M \to \mathbb{R}^k$. A map $F\colon M \to \mathbb{R}^k$ then defines the section $\widetilde{F}\colon M \to M \times \mathbb{R}^k$ by $\widetilde{F}(p) = (p, F(p))$, and vice versa.

In particular there is a natural correspondence between $C^\infty(M)$ and sections of the trivial line bundle $M \times \mathbb{R} \to M$.

9.2.4 Proposition

$\Gamma(E)$ is a $C^\infty(M)$-module under pointwise addition and multiplication: for $\sigma_1, \sigma_2 \in \Gamma(E)$ and $f \in C^\infty(M)$,

$$(\sigma_1 + \sigma_2)(p) = \sigma_1(p) + \sigma_2(p), \qquad (f\sigma)(p) = f(p)\sigma(p)$$

are sections of E.

Proof

Note that at least as maps, these are well-defined, since $\sigma_i(p) \in E_p$ which is a vector space. For $p \in M$ let (U, Φ) be a local trivialization with $p \in U$. Then since Φ is a linear space isomorphism on E_p,

$$\Phi \circ (\sigma_1 + \sigma_2)(p) = (p, \Phi \circ \sigma_1(p) + \Phi \circ \sigma_2(p)),$$

which is smooth, as desired. Similarly

$$\Phi \circ (f\sigma)(p) = (p, f(p)\Phi \circ \sigma),$$

which is again smooth. ∎

9.2.5 Lemma

Let $\pi\colon E \to M$ be a smooth vector bundle. Then for any $A \subseteq M$ closed contained in an open neighborhood U, and a section $\sigma\colon A \to E$ which is smooth (meaning it can be extended to a smooth map in a neighborhood of any $p \in A$), then there exists a global section $\widetilde{\sigma}\colon M \to E$ restricting to σ on A supported in U.

Proof

The proof is pretty much the same as the one for vector fields. ∎

9.2.6 Definition

Let $E \to M$ be a vector bundle, and $U \subseteq M$ open. A tuple of local sections of E over U, $(\sigma_1, \dots, \sigma_k)$, is **linearly independent** if $(\sigma_1(p), \dots, \sigma_k(p))$ is linearly independent in E_p for all $p \in U$. Similarly they **span** E if their values at p span E_p for all $p \in U$. A **local frame** for E over U is a linearly independent tuple of local sections over U which span E, i.e. their values form a basis for E_p for every $p \in U$.

9.2.7 Proposition

Let $E \to M$ be a smooth vector bundle of rank k.

(1) For any $p \in U$ and any linearly independent tuple of local sections of E over U, there is a neighborhood $p \in V \subseteq U$ such that the tuple can be extended to a local frame for E over V.

(2) If $(v_1, \dots, v_k)$ are linearly independent elements in E_p, there exists a neighborhood of p and a local frame over that neighborhood whose values at p are v_i.

(3) If $A \subseteq M$ is closed and $(\tau_1, \dots, \tau_k)$ are smooth linearly independent sections of $E|_A$, then there is a smooth local frame $(\sigma_1, \dots, \sigma_n)$ for E over some neighborhood of A such that $\sigma_i|_A = \tau_i$.

Proof

Suppose that $(\sigma_1, \dots, \sigma_k)$ is linearly independent but not a local frame. We show that we can extend it by another linearly independent section, and the result follows by inducting. Let (V, Φ) be a trivialization with $p \in V \subseteq U$, then $\Phi \circ \sigma_i(p) \in \{p\} \times \mathbb{R}^k$ are linearly independent and smooth. We want to construct a smooth map $\sigma\colon V \to \mathbb{R}^k$ which is linearly independent to these, and then $\Phi^{-1}\sigma$ gives the desired extension. Choose $v \in E_p$ to be linearly independent of $\Phi \circ \sigma_i(p)$, and then for a chart (V', φ) with $\varphi(p) = e_1$, define $\sigma(q) = \varphi(q)_1 v$ (where $\varphi(q)_1$ is the first coordinate of the real vector); this is clearly smooth. Then $\sigma(p) = v$ is linearly independent of $\Phi \circ \sigma_i(p)$, the matrix $\Phi \circ \sigma_i(p), \sigma(p)$ is full rank, and since the set of full rank matrices is open, there is a neighborhood of p where $\Phi \circ \sigma_i, \sigma$ is still full rank and thus linearly independent.

The construction for the second point is similar to the first.

For the last point, we can extend τ_i to a local frame of $E|_A$. The we can extend them to a neighborhood of A using a partition of unity. Then this extension is still linearly independent in a neighborhood of A, by virtue of being smooth and linear independence being an open property. ■

9.2.8 Example

If $E = M \times \mathbb{R}^k \to M$ is a trivial bundle, then the standard basis $e_1, \dots, e_k$ for $\mathbb{R}^k$ defines a global frame $(\widetilde{e}_i)$ for E by $\widetilde{e}_i(p) = (p, e_i)$.

9.2.9 Example

Suppose $\pi\colon E \to M$ is a smooth vector bundle and $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^k$ a smooth local trivialization to construct a local frame for E over U. Define $\sigma_1, \dots, \sigma_k\colon U \to E$ by $\sigma_i(p) = \Phi^{-1}(p, e_i) = \Phi^{-1} \circ \widetilde{e}_i(p)$. This is smooth as Φ is a diffeomorphism, and is a section since $\pi \circ \sigma_i(p) = \pi \circ \Phi^{-1}(p, e_i) = \pi_1(p, e_i) = p$. These form a basis over each E_p since $\Phi^{-1}(p, \bullet)$ is a linear isomorphism.

9.2.10 Proposition

Every local frame for a smooth vector bundle is associated with a smooth local trivialization as in the above example. That is, every local frame is of the form $\Phi^{-1} \circ \widetilde{e}_i$ for a local trivialization Φ.

Proof

If (σ_i) is a local frame for E over $U \subseteq M$, define $\Psi: U \times \mathbb{R}^k \to \pi^{-1}(U)$ by

$$\Psi(p, v^1, \dots, v^k) = v^i \sigma_i(p).$$

This is a bijection, since (σ_i) is a local frame. And $\sigma_i = \Psi \circ \widetilde{e}_i$, so if we can show Ψ^{-1} is a local trivialization, we are finished.

To show that Ψ is a diffeomorphism, it is sufficient to show it is a local diffeomorphism as it is bijective. For $q \in U$, choose a neighborhood $q \in V$ such that there exists a smooth trivialization $\Phi: \pi^{-1}V \to V \times \mathbb{R}^k$. Since Φ is a diffeomorphism, if we can show that $\Phi \circ \Psi$ is as well, then we will have that Ψ is a local diffeomorphism.

For each section σ_i, $\Phi \circ \sigma_i|_V: V \to V \times \mathbb{R}^k$ is smooth, and thus $\Phi \circ \sigma_i(p) = (p, \sigma_i^1(p), \dots, \sigma_i^k(p))$ for smooth σ_i^j. Thus

$$\Phi \circ \Psi(p, v^1, \dots, v^k) = \Phi(v^i \sigma_i(p)) = (p, v^i \sigma_i^1(p), \dots, v^i \sigma_i^k(p)),$$

which is clearly smooth.

To show that $(\Phi \circ \Psi)^{-1}$ is smooth, note that the matrix $(\sigma_i^j(p))$ is invertible, as $\sigma_i(p)$ form a basis. Taking the inverse of a matrix is smooth, and its inverse $(\tau_i^j(p))$ is smooth. We also have

$$(\Phi \circ \Psi)^{-1}(p, w^1, \dots, w^k) = (p, w^i \tau_i^1(p), \dots, w^i \tau_i^k(p))$$

which is also smooth. ■

9.2.11 Corollary

A smooth vector bundle is smoothly trivial iff it admits a global frame.

Proof

A global frame induces a global trivialization by the above proposition, and a global trivialization induces a global frame by the example above. ■

9.2.12 Definition

A smooth manifold M is **parallelizable** if TM admits a global frame.

The above corollary shows that M is parallelizable iff $TM \to M$ is trivial.

If (σ_i) is a local frame for E over $U \subseteq M$, and $\tau: M \to E$ is a rough section, then since $(\sigma_i(p))$ span E_p there are unique functions $\tau^i: M \to \mathbb{R}$ such that

$$\tau(p) = \tau^i(p)\sigma_i(p)$$

for all $p \in U$. These functions are called τ's **component functions** with respect to the local frame.

9.2.13 Proposition

A rough section is smooth on U iff its component functions with respect to a (equivalently, any) local frame over U are smooth.

Proof

Let $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^k$ be the local trivialization associated to the local frame (σ_i). Then for $\tau\colon U \to E$ rough, it is smooth iff $\Phi \circ \tau$ is, and

$$\Phi \circ \tau(p) = \Phi(\tau^i(p)\sigma_i(p)) = (p, \tau^1(p), \ldots, \tau^n(p)),$$

so that it is smooth iff each τ^i is. ■

9.2.14 Proposition

The topology and smooth structure on the tangent bundle TM is the unique one making it a vector bundle over M, with the given vector space structure on the fibers of the projection $\pi\colon TM \to M$, such that all coordinate vector fields are smooth local sections.

Proof

Suppose TM is endowed with some topology and smooth structure making it into a smooth vector bundle with the given properties. Let (U, φ) is any local chart, with the corresponding coordinate frame $(\partial/\partial x^i|)$ over U. Then there is a local trivialization $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^n$ associated with this local frame. By definition,

$$\Phi(v^i\, \partial/\partial x^i\big|_p) = (p, v^1, \ldots, v^n).$$

Then the map $(\varphi \times \mathrm{id}) \circ \Phi$ is the map $v^i \frac{\partial}{\partial x^i}\big|_p \mapsto (\varphi(p), v^1, \ldots, v^n)$. This is precisely $\widetilde{\varphi}$, the chart associated with φ. Thus TM must contain all such charts, which means it has the canonical smooth structure we gave it. ■

9.3 Bundle morphisms

9.3.1 Definition

If $\pi\colon E \to M, \pi'\colon E' \to M'$ are vector bundles, then a **bundle morphism** is a continuous map $F\colon E \to E'$ such that there exists a continuous $f\colon M \to M'$ making the below diagram commute

$$\begin{tikzcd} E \arrow[r, "F"'] \arrow[d, "\pi"'] & E' \arrow[d, "\pi'"'] \\ M \arrow[r, "f"] & M' \end{tikzcd}$$

And further that for each $p \in M$, $F|_{E_p}\colon E_p \to E'_{f(p)}$ is linear. We say that F **covers** f.

Let $\zeta\colon M \to E$ be the zero section, then if $f \circ \pi = \pi' \circ F$, we have $f = \pi' \circ F \circ \zeta$. Thus if the vector bundles and F are smooth, so is ζ, and thus so is f. So requiring that f be smooth for a smooth bundle morphism does not add any restriction.

Bundle morphisms, and smooth bundle morphisms, can obviously be composed, and thus form a category. So we can talk about bundle isomorphisms, and isomorphic (or smoothly isomorphic) vector bundles.

9.3.2 Definition

If $\pi\colon E \to M$ and $\pi'\colon E' \to M$ are vector bundles over the same base M, a **bundle morphism over** M is a bundle morphism $F: E \to E'$ over id_M. Equivalently it is a map $F: E \to E'$ such that $\pi = \pi' \circ F$ and $F|_{E_p}: E_p \to E_p$ is linear.

Clearly morphisms over a base M form a subcategory, so we can discuss **smooth bundle isomorphisms over** M.

9.3.3 Proposition

If $\pi: E \to M, \pi': E' \to M$ are smooth vector bundles over M, a bijective smooth bundle morphism over M, $F: E \to E'$, is an isomorphism over M.

Proof

We have that $\pi = \pi' \circ F$ and so $\pi \circ F^{-1} = \pi'$ is smooth. Since vector bundles are surjective submersions, this means that F^{-1} is smooth. It restricts linearly to the fibers because F does. ∎

9.3.4 Example

If $F: M \to N$ is smooth, $dF: TM \to TN$ is a smooth bundle morphism covering F.

Now if $E \to M, E' \to M$ are smooth bundles, let $\Gamma(E), \Gamma(E')$ be their space of smooth global sections. Then if $F: E \to E'$ is a smooth bundle morphism over M, it induces a map $\widetilde{F}: \Gamma(E) \to \Gamma(E')$ bu

$$\widetilde{F}(\sigma)(p) = (F \circ \sigma)(p).$$

This is indeed a section of π', since $\pi' \circ (F \circ \sigma) = \pi \circ \sigma = \mathrm{id}$.
This map is a morphism of $C^\infty(M)$-modules:

$$\widetilde{F}(u_1\sigma_1 + u_2\sigma_2) = F \circ (u_1\sigma_1) + F \circ (u_2\sigma_2) = u_1(F \circ \sigma_1) + u_2(F \circ \sigma_2),$$

since F is linear on fibers.

9.3.5 Lemma

All $C^\infty(M)$-morphisms $\Gamma(E) \to \Gamma(E')$ arise in this way.

Proof

Let $\mathcal{F}: \Gamma(E) \to \Gamma(E')$ be a $C^\infty(M)$-morphism. We claim that $\mathcal{F}$ acts locally, so that if $\sigma_1 = \sigma_2$ in an open $U \subseteq M$, then $\mathcal{F}(\sigma_1) = \mathcal{F}(\sigma_2)$ in U as well. By $\mathcal{F}$'s linearity it is sufficient to prove this for $\sigma = 0$ in U.
For $p \in U$ let $\psi \in C^\infty(M)$ be a smooth bump function supported in U and equal to 1 at p. Then $\psi\sigma$ is zero on M and thus

$$0 = \mathcal{F}(\psi\sigma) = \psi\mathcal{F}(\sigma),$$

and then plugging in p we see that $\mathcal{F}(\sigma)(p) = 0$ for all $p \in U$ as desired.
Now we show that if $\sigma_1(p) = \sigma_2(p)$ then $\mathcal{F}(\sigma_1)(p) = \mathcal{F}(\sigma_2)(p)$. Again it is sufficient to assume $\sigma(p) = 0$. Let $(\sigma_1, \ldots, \sigma_n)$ be a local frame for E in a neighborhood U of p, and let $\sigma = u^i\sigma_i$. Then $\sigma(p) = 0$ means that $u^i(p) = 0$ for all i. By our above point, since $\sigma = u^i\sigma_i$ locally,

$$\mathcal{F}(\sigma)(p) = (u^i\mathcal{F}(\sigma_i))(p) = 0$$

as desired.

Now define $F\colon E \to E'$ as follows. For $p \in M$ and $v \in E_p$, define $F(v) = \mathcal{F}(\widetilde{v})(p) \in E'_p$ where $\widetilde{v}$ is any global section of E such that $\widetilde{v}(p) = v$. This is independent on choice of $\widetilde{v}$, as we just showed. And it is linear in E_p since $\widetilde{v} + \widetilde{u}$ is a section mapping p to $v + u$, and $\lambda\widetilde{v}$ is one mapping p to λv. Also we have

$$\pi' \circ F(v) = \pi'(\mathcal{F}(\widetilde{v})(p)) = \pi' \circ \mathcal{F}(\widetilde{v})(p) = p = \pi(v),$$

since $\mathcal{F}(\widetilde{v})$ is a section of π'. For $p \in M$ and $\sigma \in \Gamma(E)$, by definition

$$(F \circ \sigma)(p) = \mathcal{F}(\widetilde{\sigma(p)})(p) = \mathcal{F}(\sigma)(p),$$

so that $F \circ \sigma = \mathcal{F}(\sigma)$ as desired.

We need only show that F is smooth. For $p \in M$, let (σ_i) be a local frame for E in a neighborhood of p. We can extend these to global sections $(\widetilde{\sigma}_i)$ in a smaller neighborhood of p. Potentially shrinking the neighborhood smaller, we can assume there is a local frame (σ'_j) for E' in that neighborhood, call it U. Then $\mathcal{F}(\widetilde{\sigma}_i)|_U = A_i^j \sigma'_j$ for $A_i^j \in C^\infty(U)$.

Then for any $q \in U$ and $v \in E_q$, we have $v = v^i \sigma_i(q)$ for some real numbers v^i and so

$$F(v^i \sigma_i(q)) = \mathcal{F}(v^i \widetilde{\sigma}_i)(q) = v^i \mathcal{F}(\widetilde{\sigma}_i)(q) = v^i A_i^j(q) \sigma'_j(q).$$

If Φ, Φ' are the local trivializations associated with σ_i, σ'_j then

$$\Phi' \circ F \circ \Phi^{-1}(q, v^1, \ldots, v^k) = \Phi' \circ F(v^i \sigma_i(q)) = \Phi'(v^i A_i^j(q) \sigma'_j(q)) = (q, v^i A_i^1(q), \ldots, v^i A_i^n(q)),$$

which is smooth. Since Φ', Φ^{-1} are diffeomorphisms, this shows that F is smooth on $\pi^{-1}U$. ■

9.4 Subbundles

9.4.1 Definition

Let $\pi_E\colon E \to M$ be a vector bundle. A **subbundle** is a topological subspace $D \subseteq E$ and a vector bundle $\pi_D\colon D \to M$ which is the restriction of π_E to D. Furthermore we require that for each $p \in M$, the subset $D_p = D \cap E_p$ is a linear subspace of E_p. A **smooth subbundle** is a continuous subbundle where $D \subseteq E$ is an embedded submanifold, and $\pi_D\colon D \to M$ is a smooth bundle.

9.4.2 Lemma

Let $\pi\colon E \to M$ be a smooth bundle, and suppose that for each $p \in M$ we have an m-dimensional subspace $D_p \subseteq E_p$. Then $D = \bigcup_p D_p \subseteq E$ is a smooth subbundle of E iff for every $p \in M$ there is a neighborhood $p \in U$ with smooth local sections $\sigma_i\colon U \to E$ such that $\sigma_i(q)$ form a basis for D_q at each $q \in U$.

Proof

If D is a smooth subbundle, then each $p \in M$ has a local trivialization over a neighborhood $p \in U$. Then there is a local frame (σ_i) over D. Then composing σ_i with the inclusion $D \hookrightarrow E$ gives the desired local sections for E.

Now suppose that π has rank k. To show that D is embedded, it is sufficient to show that for each $p \in M$ there is a neighborhood $p \in U$ such that $D \cap \pi^{-1}(U) \subseteq E$ is embedded. Let $\sigma_1, \ldots, \sigma_m$ be the local sections in the assumption of the lemma in a neighborhood $p \in U$. We can then complete these to a local frame $\sigma_1, \ldots, \sigma_k$ over E in a potentially smaller neighborhood. This is associated with a trivialization $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^k$ mapping

$$\Phi(s^i \sigma_i(q)) = (q, s^1, \ldots, s^k).$$

Then Φ takes $D \cap \pi^{-1}(U)$ to the subset $\{(q, s^1, \ldots, s^m, 0) \in U \times \mathbb{R}^k\}$, which is an embedded submanifold of

$U \times \mathbb{R}^k$. Thus $D \cap \pi^{-1}(U)$ is an embedded submanifold of E, as desired. Projecting this down to $\mathbb{R}^k$ gives the diffeomorphism $\pi_{\mathbb{R}^k} \circ \Phi \colon D \cap \pi^{-1}(U) \to U \times \mathbb{R}^m$, showing that D is also a smooth vector bundle itself. ■

9.4.3 Theorem

Let E, E' be smooth bundles over M, and $F \colon E \to E'$ be a smooth morphism over M. Define $\ker F \subseteq E, \operatorname{im} F \subseteq E'$ by

$$\ker F = \bigcup_{p \in M} \ker(F|_{E_p}), \qquad \operatorname{im} F = \bigcup_{p \in M} \operatorname{im}(F|_{E_p}).$$

Then $\ker F, \operatorname{im} F$ are smooth subbundles of E, E' respectively iff F has constant rank.

Proof

Note that $\ker(F|_{E_p})$ has codimension equal to the rank of F on $E|_p$, and similar for the image. Thus in order for $\ker F, \operatorname{im} F$ to be subbundles F must have constant rank.

Now suppose F has constant rank r, and suppose E, E' have respective ranks k, k'. Let $p \in M$ and choose a local frame σ_i for E over U. For each i, $F \circ \sigma_i \colon U \to E'$ is a smooth local section of E', and necessarily spans $(\operatorname{im} F)|_U$. Then we can assume $F \circ \sigma_1(p), \dots, F \circ \sigma_r(p)$ forms a basis for $\operatorname{im}(F|_{E_p})$, and then by continuity they remain linearly independent in a neighborhood of p. Since F has constant rank, these must form a basis for each $\operatorname{im}(F|_{E_q})$ for every q in the neighborhood. By the above lemma, we have finished.

For the kernel, let σ_i be as above, defined on U. Let $V \subseteq E|_U$ be the smooth subbundle spanned by σ_i, so that $F|_V \colon U \to (\operatorname{im} F)|_U$ is bijective, and thus a smooth bundle isomorphism. Define $\Psi \colon E|_U \to E|_U$ by $\Psi(v) = v - (F|_V)^{-1} \circ F(v)$. If $v \in V$ then $F(v) = F|_V(v)$ so that $F \circ \Psi(v) = F(v) - F \circ (F|_V)^{-1} \circ F(v) = F(v) - F(v) = 0$. And on the other hand, if $v \in \ker F$ then $\Psi(v) = v$, so again $F \circ \Psi(v) = 0$. Since V and $(\ker F)|_U$ span $E|_U$ together, we see that Ψ maps into $(\ker F)|_U$. It is the identity on the kernel, so it is actually a projection onto $(\ker F)|_U$. Thus Ψ has constant rank, and then $(\ker F)|_U = \operatorname{im}\Phi$ is a smooth subbundle of $E|_U$ by the above argument. This is true around each point, as desired. ■

10 The Cotangent Bundle

10.1 Covectors

Recall that for a finite-dimensional space V, its **dual** is the vector space $V^* = \hom(V, \mathbb{R})$ of all linear functionals. A basis (E_i) for V defines a **dual basis** (ε^i) for V^* by

$$\varepsilon^i(E_j) = \delta_{ij},$$

and extending by linearity. Then for $\omega \in V^*$, we have that $\omega = a_i \varepsilon^i$ where $\omega(E_i) = a_i$. There is a natural isomorphism between V and its double dual V^{**} by mapping $v \in V$ to the map $\widehat{v} \colon V^* \to \mathbb{R}$ given by $\widehat{v}(\sigma) = \sigma(v)$.

10.1.1 Definition

Let M be a smooth manifold, and $p \in M$. The **cotangent space** of M at p is defined to be $T_p^* M = (T_p M)^*$, the dual of the tangent space of M at p. Elements of the cotangent space are called **covectors**.

Given a coordinate chart (x^i) in an open subset $U \subseteq M$, for each $p \in U$ the local frame $\left(\partial/\partial x^i\big|_p\right)$ for $T_p M$ gives rise to its dual basis $(\lambda^i|_p)$ in $T_p^* M$ (we will give λ better notation soon). Then any covector $\omega \in T_p^* M$ can be write $\omega = \omega_i \lambda^i|_p$ where

$$\omega_i = \omega\left(\left.\frac{\partial}{\partial x^i}\right|_p\right).$$

If $(\widetilde{x}^i)$ is another coordinate chart whose domain contains p, then it defines the dual basis $(\widetilde{\lambda}^i|_p)$. We know that

$$\left.\frac{\partial}{\partial x^i}\right|_p = \frac{\partial \widetilde{x}^j}{\partial x^i}(p) \left.\frac{\partial}{\partial \widetilde{x}^j}\right|_p,$$

and so if $\omega = \omega_i \lambda^i|_p = \widetilde{\omega}_j \widetilde{\lambda}^j|_p$, we have

$$\omega_i = \omega\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \omega\left(\frac{\partial \widetilde{x}^j}{\partial x^i}(p) \left.\frac{\partial}{\partial \widetilde{x}^j}\right|_p\right) = \frac{\partial \widetilde{x}^j}{\partial x^i}(p)\widetilde{\omega}_j.$$

10.1.2 Definition

The **cotangent bundle** of a smooth manifold is the vector bundle $\pi: T^*M \to M$ given by mapping T_p^*M to p.

10.1.3 Proposition

There is a unique smooth structure on M making the cotangent bundle a vector bundle, where each fiber is given the linear structure of T_p^*M.

Proof

For a smooth chart (U, φ) for M, define $\Phi: \pi^{-1}(U) \to U \times \mathbb{R}^n$ by

$$\Phi(\xi_i \lambda^i|_p) = (p, \xi_1, \ldots, \xi_n),$$

where for $p \in \pi^{-1}(U)$, $\lambda^i|_p$ is the dual basis of T_p^*M. The restriction of Φ to T_p^*M is clearly a linear isomorphism.
If $(\widetilde{U}, \widetilde{\varphi})$ is another smooth chart with associated trivialization $\widetilde{\Phi}$, we see that

$$\Phi \circ \widetilde{\Phi}^{-1}(p, \widetilde{\xi}_1, \ldots, \widetilde{\xi}_n) = \left(p, \frac{\partial \widetilde{x}^j}{\partial x^1}(p)\widetilde{\xi}_j, \ldots, \frac{\partial \widetilde{x}^j}{\partial x^n}(p)\widetilde{\xi}_j\right),$$

which is smooth (the transition function is the Jacobian $\frac{\partial \widetilde{x}^j}{\partial x^i}$). Thus T^*M is a vector bundle. ■

The associated charts for these trivializations are $\widetilde{\varphi}: \pi^{-1}(U) \to \mathbb{R}^{2n}$ given by $\widetilde{\varphi} = (\varphi \times \mathrm{id}_{\mathbb{R}^n}) \circ \Phi$, i.e.

$$\widetilde{\varphi}(\xi_i \lambda^i|_p) = (\varphi(p), \xi_1, \ldots, \xi_n).$$

10.1.4 Definition

A section of T^*M is called a **covector field**.

As before we consider also rough and continuous covector fields. For a chart (U, φ) of M we have local covector fields $\lambda^i: U \to T^*M$ mapping $p \in U$ to $\lambda^i|_p$. These are local frames by construction, as the local trivializations in the above proposition are associated to them. If $\omega: M \to T^*M$ is a rough covector field, then it has a local representation $\omega = \omega_i \lambda^i|_p$ with $\omega_i: U \to \mathbb{R}$. As before, ω is smooth iff ω_i all are.
If X is a vector field of M, and ω a covector field, then we define $\omega(X): M \to \mathbb{R}$ by

$$\omega(X)(p) = \omega_p(X_p).$$

If we write $\omega = \omega_i \lambda^i|_p$ and $X = X^j\, \partial/\partial x^j\big|_p$ in terms of local coordinates from (U, φ), then $\omega(X)$'s coordinate representation is

$$\omega(X) = \omega_i X^i.$$

10.1.5 Proposition

Let M be a smooth manifold, and $\omega\colon M \to T^*M$ a rough covector field. The following are equivalent:

(1) ω is smooth,

(2) In every smooth coordinate chart, the components of ω are smooth,

(3) each point of M is contained in a coordinate chart in which ω has smooth components,

(4) for every $X \in \mathfrak{X}(M)$, $\omega(X)$ is smooth,

(5) for every open $U \subseteq M$ and every smooth local vector field $X \in \mathfrak{X}(U)$, $\omega(X)\colon U \to \mathbb{R}$ is smooth.

Proof

The first three are equivalent for general sections, and imply the fourth clearly, which implies the last. For the last point, notice that $\partial/\partial x^i\big|_p$ is a smooth local vector field and

$$\omega\left(\frac{\partial}{\partial x^i}\bigg|_p\right) = \omega_i,$$

its ith component. So the last point implies the third. ∎

10.2 Dual bundles

We can generalize the results from the previous section.

10.2.1 Definition

Let $\pi\colon E \to M$ be a vector bundle. Its **dual bundle** is the bundle $E^* = \coprod_{p\in M} E_p^*$, with the obvious projection onto M.

If we have a local frame (σ_i) for E, it defines its **dual frame** (Σ^i) for E^* where $\Sigma^i|_p \in E_p^*$ is defined to be the dual basis of $\sigma_i|_p$ for each p in the neighborhood of the local frame. If $(\widetilde{\sigma}_j)$ is another local frame, consider the associated trivializations $\Phi, \widetilde{\Phi}$, so that

$$\Phi(\xi^i\sigma_i|_p) = (p, \xi^1, \dots, xi^n).$$

So if $\sigma_i|_p = \widetilde{\xi}^j\widetilde{\sigma}_j$, then

$$\widetilde{\Phi}(\sigma_i|_p) = (p, \widetilde{\xi}).$$

Thus

$$\sigma_i|_p = \widetilde{\Phi}^j(\sigma_i|_p)\widetilde{\sigma}_j|_p.$$

And so if $\omega = \omega_i\Sigma^i|_p = \widetilde{\omega}_i\widetilde{\Sigma}^i|_p$ then

$$\omega_i = \omega(\sigma_i|_p) = \omega\Big(\widetilde{\Phi}^j(\sigma_i|_p)\widetilde{\sigma}_j|_p\Big) = \widetilde{\Phi}^j(\sigma_i|_p)\widetilde{\omega}_j|_p.$$

Recall that $\sigma_i|_p = \Phi^{-1}(p, e_i)$ so we can also write this as

$$\omega_i = \widetilde{\Phi}^j \circ \Phi^{-1}(p, e_i)\widetilde{\omega}_j|_p = (\widetilde{\Phi} \circ \Phi^{-1})^j(p, e_i)\widetilde{\omega}_j|_p.$$

10.2.2 Proposition

The dual bundle of a smooth vector bundle is a smooth vector bundle itself; it has a unique smooth structure making it into one. The double dual of a bundle is isomorphic to the original bundle.

Proof

Let $\pi: E \to M$ be a smooth vector bundle, and $\pi_*: E^* \to M$ be the projection. If (σ^i) is a local frame for E defined in U, then define the local trivialization $\Phi: \pi_*^{-1}(U) \to U \times \mathbb{R}^n$ by

$$\Phi(\xi_i \Sigma^i(p)) = (p, \xi_1, \ldots, \xi_n),$$

where $\Sigma^i: U \to E^*$ is defined at each point to be the dual basis for $\sigma^i(p)$. Then if we have another local frame $(\widetilde{\sigma}^i)$ and local trivialization $\widetilde{\Phi}$, we have

$$\Phi \circ \widetilde{\Phi}^{-1}(p, \xi_1, \ldots, \xi_n) = \Phi(\xi_i \widetilde{\Sigma}^i|_p) = \Phi((\widetilde{\Phi} \circ \Phi^{-1})^j(p, e_i)\xi_j \Sigma_i|_p) = (p, (\widetilde{\Phi} \circ \Phi^{-1})^j(p, e_i)\xi_j),$$

which is smooth.

The double dual is diffeomorphic to E since the smooth structure is unique. ■

If ω is a section of E^*, and (U, φ) is a chart for M with an associated section of E, (σ_i), and (Σ^i) its dual frame, then $\omega = \omega_i \Sigma^i$ with

$$\omega_i|_p = \omega_p(\sigma_i|_p).$$

And as before if ω is a section of E^* and X a section of E, then define $\omega(X): M \to \mathbb{R}$ by $\omega(X)(p) = \omega_p(X_p)$ and the results above apply.

10.2.3 Proposition

If (σ_i) is a rough local frame on $U \subseteq M$ for E, and (Σ_i) its dual frame for E^*, then σ_i is smooth iff Σ^i is.

Proof

Take a smooth local frame around a point in U, call it e_i, and its dual ε^i. These are smooth by assumption, and so we have

$$\sigma_i = a_i^j e_j, \qquad \Sigma^i = b_j^i \varepsilon^j.$$

Then σ_i is smooth iff a_i^j are, and Σ^i are smooth iff b_j^i are. Now,

$$\delta_{ij} = \Sigma^i(\sigma_j) = b_j^i a_i^j$$

implies that (a_i^j) and (b_j^i) are inverses matrices, and thus one is smooth iff the other. ■

Since E^* is a vector bundle, its space of sections $\Gamma(E)$ is a $C^\infty(M)$-module. In particular for the cotangent bundle T^*M, we denote its space of sections by $\mathfrak{X}^*(M)$.

10.3 Differential of a function

10.3.1 Definition

Let $f: M \to \mathbb{R}$ be real valued and smooth. We define a covector field $df: M \to T^*M$ by mapping $p \in M$ to the linear functional $df_p \in T_p^*M$ by

$$df_p(v) = vf, \qquad v \in T_pM.$$

10.3.2 Proposition

The differential of a smooth function is a smooth covector field.

Proof

df_p is indeed linear, so df is a rough covector field at least. Notice that for any vector field X on M, $df(X) = Xf$, which is smooth. ■

Let (x^i) be coordinates of $U \subseteq M$, and λ^i its corresponding coframe. Then $df_p = A_i(p)\lambda^i|_p$ for $A_i\colon U \to \mathbb{R}$. Then

$$A_i(p) = df_p\left(\frac{\partial}{\partial x^i}\bigg|_p\right) = \frac{\partial}{\partial x^i}\bigg|_p f = \frac{\partial f}{\partial x^i}(p).$$

Thus

$$df_p = \frac{\partial f}{\partial x^i}(p)\lambda^i|_p.$$

In the special case that f is a coordinate function $x^j\colon U \to \mathbb{R}$, we get that

$$dx^j_p = \frac{\partial x^j}{\partial x^i}(p)\lambda^i|_p = \lambda^j|_p.$$

So $\lambda^j|_p$ is equivalently viewed as $dx^j|_p$! So we can write

$$df = \frac{\partial f}{\partial x^i}dx^i.$$

10.3.3 Proposition

Let $f, g \in C^\infty(M)$.

(1) $d(af + bg) = a\,df + b\,dg$ for $a, b \in \mathbb{R}$;

(2) $d(fg) = f\,dg + g\,df$;

(3) $d(f/g) = (g\,df - f\,dg)/g^2$ when $g \neq 0$;

(4) if $J \subseteq \mathbb{R}$ is an interval containing f's image and $h\colon J \to \mathbb{R}$ is smooth, then $d(h \circ f) = (h' \circ f)df$;

(5) if f is constant, $df = 0$.

10.3.4 Proposition

If $f \in C^\infty(M)$, then $df = 0$ iff f is constant on M's components.

Proof

We assume that M is connected and show that $df = 0$ iff f is constant. One direct is easy, so assume $df = 0$. Let $p \in M$ and define $\mathcal{C} = \{q \in M \mid f(p) = f(q)\}$, a closed subset of M. If $q \in \mathcal{C}$, let U be a smooth coordinate ball centered at q. Then $\partial f/\partial x^i = 0$ on U (representing df via dx^i), so then f is constant on U. Thus $\mathcal{C}$ is open; it is then nonempty clopen so must be equal to all of M. ■

10.4 Pullbacks of covector fields

10.4.1 Definition

Let $F\colon M \to N$ smooth, then the differential at $p \in M$ $dF_p\colon T_pM \to T_{Fp}N$ dualizes to give the **pullback** of F at p, or the **cotangent map** of F at p: $dF_p^*\colon T_{Fp}^*N \to T_p^*M$.

The dual of a map $f\colon V \to W$ is the map $f^*\colon W^* \to V^*$ defined by $f^*(\varphi)(v) = \varphi(f(v))$. Thus the cotangent map is given by

$$dF_p^*(\omega)(v) = \omega(dF_p(v)), \qquad \omega \in T_{Fp}^*N,\ v \in T_pM.$$

Note:

The correspondence $(M, p) \mapsto T_p^*M$ is contravariant by this definition. Which is unfortunate since elements of T_p^*M are called *covectors* (covariant vectors).

Recall that the pushforward of vector fields are only defined in some cases, e.g. when the morphism is a diffeomorphism. But pullbacks of covector fields always exist.

10.4.2 Definition

Let $F\colon M \to N$ be smooth, and $\omega \in \mathfrak{X}^*(N)$ a covector field. Its **pullback** by F is $F^*\omega \in \mathfrak{X}^*(M)$ defined by

$$(F^*\omega) = dF_p^*(\omega_{Fp}),$$

or explicitly,

$$(F^*\omega)_p(v) = \omega_{Fp}(dF_p(v)).$$

10.4.3 Proposition

Let $F\colon M \to N$ be smooth, $u \in C^\infty(N)$, and $\omega \in \mathfrak{X}^*(N)$, then

$$F^*(u\omega) = (u \circ F)F^*\omega.$$

If u is in addition smooth, then

$$F^*du = d(u \circ F).$$

Proof

We compute,

$$(F^*(u\omega))_p = dF_p^*(u(F(p))\omega_{Fp}) = u(F(p))dF_p^*(\omega_{F(p)}) = ((u \circ F)dF^*\omega)_p,$$

as desired. And for $v \in T_pM$, we have

$$(F^*du)_p(v) = (dF_p^*(du_{Fp}))(v) = du_{Fp}(dF_p(v)) = dF_p(v)u$$

by the definition of du, and by the definition of $dF_p(v)$,

$$= v(u \circ F) = d(u \circ F)_p(v). \qquad \blacksquare$$

10.4.4 Proposition

If $F\colon M \to N$ is smooth and $\omega \in \mathfrak{X}^*(N)$, then $F^*\omega$ is smooth (if ω is continuous, then $F^*\omega$ is continuous).

Proof

Let $p \in M$ be arbitrary and choose coordinates (y^j) for a neighborhood V of $F(p)$. Let $U = F^{-1}(V)$, and write $\omega = \omega_j dy^j$ for continuous/smooth coordinates ω_j. Then we have, by the above proposition, (applied to $F|_U : U \to V$ smooth)

$$F^*\omega = F^*(\omega_j dy^j) = (\omega_j \circ F)F^* dy^j = (\omega_j \circ F)d(y^j \circ F).$$

This is continuous, and smooth if ω is. ■

We can write the formula in the above proof by

$$F^*\omega = (\omega_j \circ F)dF^j,$$

where F^j is the jth coordinate of F relative to the coordinates.

11 Constructions on Vector Bundles

11.1 Review of tensor products

This section will be rather short, assuming a basic familiarity with tensor products. The following definition is a brief overview of notation.

11.1.1 Definition

We denote the tensor product of modules by $A \otimes B$. We denote the dual of an R-module A by $A^* = \hom_R(A, R)$.

Note:

Primitive elements of the tensor product $A \otimes B$ are written $a \otimes b$; all other elements are sums of these.

The tensor product is associative (in a natural way), and we denote the tensor of n modules without parentheses. In an n-fold tensor product, we write the primitive elements without parentheses.

If $B_i = (b^i_1, \dots, b^i_{n_i})$ is a basis for each V_1, then

$$B = \left\{ b^1_{j_1} \otimes b^n_{j_n} \;\middle|\; b^i_{j_i} \in B_i \right\} \subseteq V_1 \otimes \cdots \otimes V_n$$

is a basis for the tensor product. In particular the dimension of a tensor product of vector spaces is the product of the dimensions of the vector spaces.

If A is an R-module, then $A \otimes R \cong A$ naturally. This natural isomorphism is given by mapping $a \otimes r \mapsto ra$.

The tensor product is functorial: if $F_i : A_i \to B_i$ are maps for $i = 1, 2$ then define $F_1 \otimes F_2 : A_1 \otimes A_2 \to B_1 \otimes B_2$ by

$$(F_1 \otimes F_2)(a_1 \otimes a_2) = (F_1(a_1)) \otimes (F_2(a_2))$$

and extending by linearity. This is well-defined.

In the case of linear functionals $\varphi_i : A_i \to \mathbb{R}$ we have that $\mathbb{R} \otimes \mathbb{R} \cong \mathbb{R}$, and this map is naturally isomorphic to

$$(\varphi_1 \otimes \varphi_2)(a_1 \otimes a_2) = \varphi_1(a_1)\varphi_2(a_2).$$

The universal property of the tensor product is that

$$\hom(V_1 \otimes \cdots \otimes V_n, A) \cong L(V_1, \dots, V_n; A)$$

naturally, where the right side is the space of multilinear maps $V_1 \otimes \cdots \otimes V_n \to A$.

Note:

For a field k and a finite-dimensional k-linear space V, we know that $\dim V^* = \dim V$. For a basis $B = (v_1, \dots, v_n)$ we define its **dual basis** $B^* = (v_1^*, \dots, v_n^*)$ in V^* as the linear functionals defined by

$$v_i^*(v_j) = \delta_{ij},$$

and extending by linearity. These are linearly independent, and thus must be a basis considering dimensions.

There is a natural map from a space to its double dual, $V \to V^{**}$. For $v \in V$ define $\widehat{v} \in V^{**}$ by $\widehat{v}(\varphi) = \varphi(v)$ for all $\varphi \in V^*$. This is injective, and considering dimensions is an isomorphism. It is natural, and thus for finite-dimensional spaces V and V^{**} are naturally isomorphic.

For infinite-dimensional spaces, $\dim V^*$ is strictly greater than $\dim V$, so in particular V is naturally a proper subspace of V^{**}, but they are not isomorphic.

11.1.2 Proposition

Let $V_1, \dots, V_n$ be vector spaces, then there is a natural isomorphism

$$(V_1 \otimes \cdots \otimes V_n)^* \cong V_1^* \otimes \cdots \otimes V_n^*.$$

Proof

They have the same dimension, and thus we need only find an injective map. We define a map $V_1^* \otimes \cdots \otimes V_n^*$ to the dual space on the left by mapping $\varphi_1 \otimes \cdots \otimes \varphi_n$ to the literal tensor product of maps:

$$(\varphi_1 \otimes \cdots \otimes \varphi_n)(a_1, \dots, a_n) = \varphi_1(a_1) \cdots \varphi_n(a_n).$$

If we choose a dual basis for each V_i^*, we see that it maps to a linearly independent set, and thus this is injective. ■

In particular, we then have the natural isomorphisms

$$V_1 \otimes \cdots \otimes V_n \cong (V_1 \otimes \cdots \otimes V_n)^{**} \cong (V_1^* \otimes \cdots \otimes V_n^*)^*.$$

Thus,

$$V_1 \otimes \cdots \otimes V_n \cong \hom(V_1^* \otimes \cdots \otimes V_n, \mathbb{R}) \cong L(V_1^*, \dots, V_n^*, \mathbb{R}).$$

So we can identify the tensor product $V_1 \otimes \cdots \otimes V_n$ with multilinear maps $V_1^* \times \cdots \times V_n^* \to \mathbb{R}$ naturally. Similarly, $V_1^* \otimes \cdots \otimes V_n^*$ can naturally be identified with multilinear maps $V_1 \times \cdots \times V_n \to \mathbb{R}$.

11.1.3 Definition

For a space A, denote by $T^k(A)$ the k-fold tensor product of A with itself:

$$T^k(A) = A \otimes \cdots \otimes A.$$

The empty tensor product is taken to be the ambient ring, so $T^0(A) = R$.

11.1.4 Definition

Let V be a finite-dimensional real vector space. A **covariant k-tensor** for $k > 0$ is an element of $V^* \otimes \cdots \otimes V^*$. Equivalently, it is a multilinear map $V \times \cdots \times V \to \mathbb{R}$. If α is a covariant k-tensor for some

k, it is called a **covariant tensor** and its **rank** is k. The space of covariant k-tensors is $T^k(V^*)$.

A **contravariant** k**-tensor** for $k > 0$ is an element of $V \otimes \cdots \otimes V = T^k(V)$. Equivalently, a multilinear map $V^* \times \cdots \times V^* \to \mathbb{R}$. The rank is defined similarly.

For $k, \ell \neq 0$ nonnegative, we define a **tensor of type** (k, ℓ) to be an element of

$$T^{(k,\ell)}(V) = V \otimes \cdots \otimes V \otimes V^* \otimes \cdots \otimes V^* = T^k(V) \otimes T^\ell(V^*).$$

Note that

$$T^{(0,k)}(V) = T^k(V^*), \qquad T^{(\ell,0)}(V) = T^\ell(V).$$

11.1.5 Definition

We define an action of S_k on $T^k(V)$, viewed as multilinear maps, by

$${}^\sigma\alpha(v_1, \ldots, v_n) = \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(n)})$$

for $\sigma \in S_k, \alpha \in T^k(V)$. This is indeed a group action, moreover it preserves the linear structure of $T^k(V)$ in that

$${}^\sigma(\alpha + \beta) = {}^\sigma\alpha + {}^\sigma\beta, \qquad {}^\sigma(c\alpha) = c^\sigma\alpha.$$

Viewing $T^k(V)$ as the k-fold tensor product of V^*, ${}^\sigma$ acts on primitive elements by

$$\sigma(v_1 \otimes \cdots \otimes v_k) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(k)},$$

and extending by linearity.

11.1.6 Definition

A covariant k-tensor $\alpha \in T^k(V^*)$ is **symmetric** if swapping its inputs does not change its value. Equivalently it has a trivial orbit under the above action: ${}^\sigma\alpha = \alpha$ for all $\sigma \in S_k$ (since transpositions generate S_k). Denote the space of symmetric covariant k-tensors by $\Sigma^k(V^*) \subseteq T^k(V^*)$.

Similarly a covariant k-tensor $\alpha \in T^k(V^*)$ is **antisymmetric** if swapping its inputs swaps the sign of its value. Equivalently if ${}^\sigma\alpha = \mathrm{sgn}\sigma\alpha$ for all $\sigma \in S_k$. Denote the space of alternating covariant k-tensors by $\Lambda^k(V^*) \subseteq T^k(V^*)$. Antisymmetric tensors are also called **exterior forms** or **multi-covectors**.

Define the map $\mathrm{Sym}\colon T^k(V^*) \to \Sigma^k(V^*)$ by

$$\mathrm{Sym}\alpha = \frac{1}{k!} \sum_{\sigma \in S_k} {}^\sigma\alpha.$$

This map is called **symmetrization**, and it is a projection onto $\Sigma^k(V^*)$. Indeed,

$${}^\tau\mathrm{Sym}\alpha = \frac{1}{k!} \sum_{\sigma \in S_k} {}^{\tau\sigma}\alpha = \mathrm{Sym}\alpha$$

so that Sym is well-defined, and $\mathrm{Sym}\alpha = \alpha$ iff α is symmetric. Clearly the equality implies α's symmetry since $\mathrm{Sym}\alpha$ is symmetric, and if α is symmetric then

$$\mathrm{Sym}\alpha = \frac{1}{k!} \sum_{\sigma \in S_k} \alpha = \alpha.$$

11.1.7 Definition

If $\alpha \in \Sigma^k(V^*)$ and $\beta \in \Sigma^\ell(V^*)$ are symmetric, then their tensor $\alpha \otimes \beta$ is not symmetric in general. Define their **symmetric product** to be the symmetrification of their tensor product:

$$\alpha\beta = \mathrm{Sym}(\alpha \otimes \beta).$$

Explicitly,

$$\alpha\beta(v_1, \ldots, v_{k+\ell}) = \frac{1}{(k+\ell)!} \sum_{\sigma \in S_{k+\ell}} \alpha(v_{\sigma 1}, \ldots, v_{\sigma k})\beta(v_{\sigma(k+1)}, \ldots, v_{\sigma \ell}).$$

11.1.8 Proposition

The symmetric product satisfies the following:

(1) it is bilinear,

(2) it is commutative,

(3) it is associative.

We will not prove this.
We can also define the **alternation** of a tensor by

$$\mathrm{Alt}\alpha = \frac{1}{k!} \sum_{\sigma \in S_k} \mathrm{sgn}\sigma^\sigma \alpha.$$

This is indeed antisymmetric,

$${}^\tau\mathrm{Alt}\alpha = \frac{1}{k!} \sum_{\sigma} \mathrm{sgn}\sigma^{\tau\sigma}\alpha = \mathrm{sgn}\tau \mathrm{Alt}\alpha.$$

If α is already antisymmetric then

$$\mathrm{Alt}\alpha = \frac{1}{k!} \sum_{\sigma} \mathrm{sgn}\sigma^\sigma \alpha = \frac{1}{k!} \sum_{\sigma} (\mathrm{sgn}\sigma)^2 \alpha = \alpha$$

so this is also a projection.
Notice that $\mathrm{Sym}\alpha = 0$ for antisymmetric tensors of dimension > 1 and similarly $\mathrm{Alt}\alpha = 0$ for symmetric tensors of dimension > 1.

11.2 Tensor fields

11.2.1 Definition
Let M be a smooth manifold. Define its **bundle of covariant k-tensors** by

$$T^k T^* M = \coprod_{p \in M} T^k(T_p^* M)$$

where $T_p^* M$ is the dual space of $T_p M$. Similarly the **bundle of contravariant k-tensors** by

$$T^k TM = \coprod_{p \in M} T^k(T_p M),$$

and in general the **bundle of (k, ℓ)-tensors** by

$$T^{(k,\ell)} TM = \coprod_{p \in M} T^{(k,\ell)}(T_p M).$$

To show that $T^{(k,\ell)}TM$ has the unique structure of a vector bundle over M, we prove that in general the tensor of two vector bundles is a vector bundle.

11.2.2 Definition

Let $\pi'\colon E' \to M$ and $\pi''\colon E'' \to M$ be vector bundles over M. Define their **tensor** to be $E = \coprod_{p\in M} E'_p \otimes E''_p$ with the obvious projection $\pi\colon E \to M$.

Let (σ_i) and $(\widetilde{\sigma}_i)$ be local frames defined in an open set U, and let $\Phi, \widetilde{\Phi}$ be their local trivializations. Meaning

$$\Phi(a^i\sigma_i|_p) = (p, \overline{a}),$$

and by assumption there is a smooth transition function $\tau\colon U \to \mathrm{GL}_n(\mathbb{R})$ such that

$$\widetilde{\Phi} \circ \Phi^{-1}(p, \overline{a}) = (p, \tau(p)\overline{a}).$$

Then if $X = a^i\sigma_i = \widetilde{a}^i\widetilde{\sigma}_i$, we have

$$\widetilde{\Phi} \circ \Phi^{-1}(p, \overline{a}) = \widetilde{\Phi}^{-1}(X_p) = (p, \widetilde{a}),$$

so that $\widetilde{a} = \tau(p)\overline{a}$, i.e. $\widetilde{a}^i = \tau(p)^i\overline{a}$ where $\bullet^i$ is the ith row.
In particular, $\sigma_i = \tau(p)_i^j\widetilde{\sigma}_j$.

11.2.3 Proposition

The tensor of two vector bundles has a unique smooth structure making it a vector bundle over M whose fibers are the tensors of the fibers of the original vector bundles.

Proof

Suppose E', E'' have rank k', k'' respectively. Let (σ_i) and (τ_j) be local frames of E', E'' respectively over an open subset $U \subseteq M$. Then define $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^{k'k''}$ by

$$\Phi(v^{ij}\sigma_i|_p \otimes \tau_j|_p) = (p, v^{ij}).$$

Then for another trivialization $\widetilde{\Phi}$ associated with $(\widetilde{\sigma}_i)$ and $(\widetilde{\tau}_j)$ we have that (T', T'' being the transition functions),

$$\sigma_i \otimes \tau_j = ((T')_i^k\widetilde{\sigma}_k) \otimes ((T'')_j^\ell\widetilde{\tau}_\ell) = (T')_i^k(T'')_j^\ell(\widetilde{\sigma}_k \otimes \widetilde{\tau}_\ell).$$

So the transition function is $T_j^i = \sum_{k,\ell}(T')_i^k \cdot (T'')_j^\ell$, which is smooth (as the sum and product of smooth functions) and invertible (since $\sigma_i \otimes \tau_j$ and $\widetilde{\sigma}_i \otimes \widetilde{\tau}_j$ form bases). ■

We have that

$$T^{(k,\ell)}TM = \coprod_{p\in M} T^{(k,\ell)}(T_pM)$$

is just the tensor of TM tensored with itself k times and T^*M with itself ℓ times, and is thus a vector bundle.
The local frames of $E \otimes E'$ are of the form $(\sigma_i \otimes \tau_j)$ for local frames σ_i of E and τ_j of E'. Thus the local frames of $T^{(k,\ell)}TM$ give representations of the form

$$A^{i_1\dots i_k}_{j_1\dots j_\ell}\frac{\partial}{\partial x^{i_1}} \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes dx^{j_\ell},$$

and $A^{i_1\dots i_k}_{j_1\dots j_\ell}$ are called the **components** of the local frame, we denote them also by A^I_J if the tuples I, J are understood.
We adopt the notation for the sections of the tensor bundles:

$$\mathcal{T}^k(M) = \Gamma(T^kT^*M), \qquad \mathcal{T}^{(k,\ell)}(M) = \Gamma(T^{(k,\ell)}TM).$$

We do not give special notation to $\Gamma(T^kTM)$, as it is not as important to us. Sections of T^kT^*M are called **covariant k-tensor fields**. Rough sections of $T^{(k,\ell)}TM$ are maps $A\colon M \to T^{(k,\ell)}TM$ such that for every $p \in M$,

$$A_p \in T^{(k,\ell)}T_pM = L(T_p^*M, \dots, T_p^*M, T_pM, \dots, T_pM;\ \mathbb{R}),$$

so A_p is a multlilinear functional from k-times T_p^*M and ℓ-times T_pM.

Note:

Note that a tensor of type (k, ℓ) is *contravariant* in its first k arguments and *covariant* in its last ℓ. This may seem unnatural, and indeed some authors think so! Some authors mean a tensor of type (k, ℓ) to mean it is *co*variant in the first k arguments and *contra*variant in its last ℓ; $T^{(k,\ell)}$ may have opposite meanings for different authors!

The convention adopted here is from Lee, but the course at Hebrew University (2026) utilizes the opposite convention.

11.2.4 Proposition

Let M be a manifold, and $A\colon M \to T^{(k,\ell)}TM$ be a rough section. Then the following are equivalent:

(1) A is smooth,

(2) In every coordinate chart A's components are smooth,

(3) Each point in M is contained in a chart where A's components are smooth,

(4) If $X_1, \dots, X_\ell \in \mathfrak{X}(M)$ and $\omega_1, \dots, \omega_k \in \mathfrak{X}^*(M)$, the function $A(\overline{\omega}, \overline{X})\colon M \to \mathbb{R}$ defined by

$$A(\overline{\omega}, \overline{X})(p) = A_p(\overline{\omega}|_p, \overline{X}|_p)$$

is smooth.

(5) Whenever $\overline{X}, \overline{\omega}$ are smooth vector and covector fields defined in some open subset $U \subseteq M$, $A(\overline{\omega}, \overline{X})$ is smooth on U.

Proof

The first three points are clearly equivalent, as they are for general vector bundles. We note that if $A = A^I_J \partial/\partial x^I \otimes dx^J$, then

$$A\left(dx^{j_1}, \dots, dx^{j_k}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_\ell}}\right) = A^I_J,$$

so that the last point implies the first. And if A is smooth, then in terms of local coordinates we have that if $X_i = X^j_i \partial/\partial x^j$ and $\omega^i = \omega^i_j dx^j$ then

$$A(\overline{\omega}, \overline{X}) = A^I_J \partial/\partial x^I \otimes dx^J(\overline{\omega}, \overline{X}) = A^I_J \omega_I X^J,$$

where $X^J = X^{j_1}_1 \cdots X^{j_k}_k$ and similar for ω_I. This is smooth, as desired. ∎

11.2.5 Proposition

Let M be a manifold, $A \in \mathcal{T}^{(k,\ell)}(M)$, $B \in \mathcal{T}^{(k',\ell')}(M)$, and $f \in C^\infty(M)$. Then $A \otimes B$ is a smooth tensor field in $\mathcal{T}^{(k+k',\ell+\ell')}$, and $fA \in \mathcal{T}^{(k,\ell)}$.

Proof

It is not hard to see that if A has local coordinates A^I_J, B has $B^{I'}_{J'}$ then $A \otimes B$ has coordinates

$$(A \otimes B)^{II'}_{JJ'} = A^I_J B^{I'}_{J'},$$

where we split the tuples II', JJ'. And similarly

$$(fA)^I_J = f \cdot A^I_J.$$

These are all smooth, as desired. ■

Our above proposition showed that a tensor field $A \in \mathcal{T}^{(k,\ell)}(M)$ induces a map

$$\mathfrak{X}^*(M) \times \cdots \times \mathfrak{X}^*(M) \times \mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M).$$

This is a multilinear $C^\infty(M)$-module morphism, which can be verified coordinate-wise.

11.2.6 Lemma

Multilinear $C^\infty(M)$ maps $\mathfrak{X}^*(M)^k \times \mathfrak{X}(M)^\ell \to C^\infty(M)$ are precisely those induced by (k,ℓ)-tensor fields.

Proof

We prove this for $(0,k)$-tensor fields, so $C^\infty(M)$-maps $\mathfrak{X}(M)^k \to C^\infty(M)$; the general case is similar, but messier. Let $\mathcal{A}$ be a $C^\infty(M)$-multilinear map. We first show that $\mathcal{A}$ acts locally, so if X_i vanishes in a neighborhood U of p then $0 = \mathcal{A}(\overline{X})$. Again, let ψ be a bump function supported in U such that $\psi(p) = 1$, so that $\psi X_i = 0$. Then

$$0 = \mathcal{A}(X_1, \ldots, \psi X_i, \ldots, X_n)(p) = \psi(p)\mathcal{A}(\overline{X})(p) = \mathcal{A}(\overline{X})(p)$$

meaning $\mathcal{A}(\overline{X})$ vanishes on U.

Now we show that if $X_i|_p = 0$ then $\mathcal{A}(\overline{X})(p) = 0$ so that $\mathcal{A}(\overline{X})(p)$ is dependent only on $\overline{X}$'s values at p. Let $X_i = X_i^j \partial/\partial x^j$ in a neighborhood of p, so the two vector fields agree in a neighborhood and thus we can replace X with this coordinate representation and not change $\mathcal{A}$'s value. Then

$$\mathcal{A}(X_1, \ldots, X_i, \ldots, X_n)(p) = X_i^j(p)\mathcal{A}(X_1, \ldots, \partial/\partial x^j, \ldots, X_n) = 0,$$

as desired. Note that we need to extend $X_i^j \partial/\partial x^j$ to a global tensor field, which can be done by the extension lemma.

Now define $A \colon M \to T^k T^* M$ by

$$A_p(v_1, \ldots, v_k)(p) = \mathcal{A}(V_1, \ldots, V_k)(p)$$

where V_i are any vector fields with $V_i(p) = v_i$. This is independent of choice of V_i, as we showed, and gives the desired tensor field (it is smooth since it induces $\mathcal{A}$ which is smooth). ■

Thus we can identify tensor fields with multilinear $C^\infty(M)$ morphisms.

There is a natural isomorphism $\hom(V, W) \cong V^* \otimes W$ given by mapping $\varphi \otimes w$ to the linear map $v \mapsto \varphi(v)w$. To show that this is an isomorphism, note that the dimensions are equal, and that it maps a basis to a linearly independent set. In particular, this gives a natural isomorphism

$$L(V, \ldots, V; V) \cong \hom(T^k(V), V) \cong T^k(V^*) \otimes V \cong T^{(1,k)}V.$$

This extends to tensor fields: there is a natural isomorphism of mutlilinear maps $\mathfrak{X}(M)^k \to \mathfrak{X}(M)$ and multilinear maps $\mathfrak{X}(M)^k \times \mathfrak{X}(M)^* \to C^\infty(M)$. This is because given $\mathcal{A} \colon \mathfrak{X}(M)^k \to \mathfrak{X}(M)$ we define $\widetilde{\mathcal{A}} \colon \mathfrak{X}(M)^k \times \mathfrak{X}(M)^* \to C^\infty(M)$ by

$$\widetilde{\mathcal{A}}(X_1, \ldots, X_n, \omega) = \omega(\mathcal{A}(X_1, \ldots, X_n)),$$

recalling that cotangent fields act on vector fields smoothly. Conversely given $\widetilde{\mathcal{A}}$ we define $\mathcal{A}$ by

$$\mathcal{A}(X_1, \ldots, X_n)\omega = \widetilde{\mathcal{A}}(X_1, \ldots, X_n, \omega)$$

recalling that by duality a vector field is just a mutlilinear map $X \colon \mathfrak{X}(M)^* \to C^\infty(M)$.

11.2.7 Proposition

$(1,k)$-tensor fields are naturally isomorphic to $C^\infty(M)$-multilinear maps

$$\mathfrak{X}(M)\times\cdots\times\mathfrak{X}(M)\to\mathfrak{X}(M).$$

In general $(\ell+1,k)$-tensor fields are naturally isomorphic to $C^\infty(M)$-multilinear maps

$$\mathfrak{X}^*(M)^\ell\times\mathfrak{X}(M)^k\to X(M).$$

This is an important basic result as many tensor fields are given in this manner.

11.2.8 Example

The Lie bracket $[\bullet,\bullet]$ is a map $\mathfrak{X}(M)\times\mathfrak{X}(M)\to\mathfrak{X}(M)$. But it is not multilinear over $C^\infty(M)$, and thus is *not* a tensor.

11.2.9 Definition

A **symmetric tensor field** is a covariant tensor field whose image consists of symmetric tensors.

11.2.10 Definition

Let $F\colon M\to N$ be smooth, then at any $p\in M$ and any k-tensor $\alpha\in T^k(T^*_{Fp}N)$ we define $dF^*_p(\alpha)\in T^k(T^*_pM)$ by

$$dF^*_p(\alpha)(v_1,\dots,v_n)=\alpha(dF_p(v_1),\dots,dF_p(v_n)).$$

This is the **pullback of** F **at** p. Then for a covariant k-tensor field A on N, we define the **pullback of** A **by** F to be the rough k-tensor field on M given by

$$(F^*A)_p=dF^*_p(A_{Fp}).$$

This acts on vectors $v_1,\dots,v_k\in T_pM$ by

$$(F^*A)_p(v_1,\dots,v_k)=A_{Fp}(dF_p(v_1),\dots,dF_p(v_k)).$$

11.2.11 Proposition

For $F\colon M\to N$ and $G\colon N\to P$ smooth, A,B covariant k-tensor fields on N, and $f\in C^\infty(N)$,

(1) $F^*(fB)=(f\circ F)F^*B$,

(2) $F^*(df)=d(f\circ F)$,

(3) $F^*(A\otimes B)=F^*A\otimes F^*B$,

(4) $F^*(A+B)=F^*A+F^*B$,

(5) F^*B is continuous, and smooth if B is smooth,

(6) $(G\circ F)^*B=F^*(G^*B)$,

(7) $(\mathrm{id}_N)^*B=B$.

Proof

Similar to before, by definitions. ■

11.2.12 Corollary

Let $F: M \to N$ be smooth, and $B \in \mathcal{T}^k(N)$. If $p \in M$ and (y^i) are smooth coordinates for N in a neighborhood of $F(p)$, then F^*B has the representation in a neighborhood of p:

$$F^*(B_I dy^I) = (B_I \circ F) d(y^{i_1} \circ F) \otimes \cdots \otimes d(y^{i_k} \circ F).$$

Proof

Immediate from the above proposition. ■

11.3 Lie derivatives of tensor fields

Suppose $V \in \mathfrak{X}(M)$ is a smooth vector field with flow θ with flow domain $\mathcal{D}$. Recall that for $t \in \mathbb{R}$, $\theta_t: M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}. For $p \in M$ and t small enough we have that $(t,p) \in \mathcal{D}$, and thus θ_t is a diffeomorphism from an open neighborhood of p to an open neighborhood of $\theta_t(p)$. So $d(\theta_t)_p^*$ pulls back tensors $T^k(T^*_{\theta(t,p)}M)$ to ones at p by

$$d(\theta_t)_p^*(A_{\theta_t(p)})(v_1, \ldots, v_k) = A_{\theta_t(p)}(d(\theta_t)_p v_1, \ldots, d(\theta_t)_p v_n).$$

This is the value of the pullback tensor field $\theta_t^* A$ at p.

11.3.1 Definition

If $V \in \mathfrak{X}(M)$ is a smooth vector field, and A a covariant tensor field on M, define its **Lie derivative** with respect to V by

$$(\mathcal{L}_V A)_p = \left.\frac{d}{dt}\right|_{t=0} (\theta_t^* A)_p = \lim_{t \to 0} \frac{d(\theta_t)_p^*(A_{\theta(t,p)}) - A_p}{t}.$$

11.3.2 Lemma

The Lie derivative exists for every p and defines a covariant tensor field on M.

Proof

For $p \in M$ let $(U, (x^i))$ be a smooth chart containing p. Choose an open interval $0 \in J_0$ and an open set $U_0 \subseteq U$ such that θ maps $J_0 \times U_0$ into U. For $(t,x) \in J_0 \times U_0$ we can write θ relative to the coordinates by

$$\theta(t,x) = (\theta^1(t,x), \ldots, \theta^n(t,x)).$$

Then $d(\theta_t)_x: T_x M \to T_{\theta(t,x)}$ has matrix representation

$$\left(\frac{\partial \theta^i}{\partial x^j}(t,x)\right).$$

If A has coordinate representation

$$A = A_J dx^J,$$

then

$$d(\theta_t)^*_x(A_{\theta_t(p)})\left(\left.\frac{\partial}{\partial x^I}\right|_x\right) = A_{\theta_t(p)}\left(\frac{\partial\theta^{i_k}}{\partial x^k}(t,x)\left.\frac{\partial}{\partial x^{i_k}}\right|_{\theta(t,x)}\right) = \frac{\partial\theta^{i_1}}{\partial x^1}(t,x)\cdots\frac{\partial\theta^{i_k}}{\partial x^k}(t,x)\cdot A_{i_1,\dots,i_k}.$$

This is a smooth coordinate representation and thus has a smooth derivative. ∎

11.3.3 Proposition

Let $V \in \mathfrak{X}(M)$ be smooth, $f \in C^\infty(M)$ (which can be regarded as T^0TM), and A, B are smooth covariant tensor fields on M (of potentially different type).

(1) $\mathcal{L}_V f = Vf$,

(2) $\mathcal{L}_V(fA) = (\mathcal{L}_V f)A + f(\mathcal{L}_V A)$,

(3) $\mathcal{L}_V(A \otimes B) = (\mathcal{L}_V A) \otimes B + A \otimes (\mathcal{L}_V B)$,

(4) if $X_1, \dots, X_k$ are smooth vector fields and A has type k, then

$$\mathcal{L}_V(A(X_1,\dots,X_k)) = (\mathcal{L}_V A)(X_1,\dots,X_k) + A(\mathcal{L}_V X_1,\dots,X_k) + \cdots + A(X_1,\dots,\mathcal{L}_V X_k).$$

Since $\mathcal{L}_V X = [V,X]$ for vector fields X, we have (by the first point the left-hand side above is just V on A's value),

$$(\mathcal{L}_V A)(X_1,\dots,X_n) = V(A(X_1,\dots,X_n)) - A([V,X_1],\dots,X_k) - \cdots - A(X_1,\dots,[V,X_k]).$$

Proof

Let θ be V's flow, then

$$\theta_t^*(f(p)) = f(\theta_t(p)) = f \circ \theta^{(p)}(t).$$

So the Lie derivative $\mathcal{L}_V f$ is the derivative of this:

$$(\mathcal{L}_V f)(p) = \left.\frac{d}{dt}\right|_{t=0} f \circ \theta^{(p)} = df_p(\theta^{(p)\prime}(0)) = df_p(V_p) = Vf(p).$$

For the other points, if p is a regular point then we choose a coordinate domain where V has representation $\partial/\partial u^1$. Then V's flow is just $\theta_t(u) = (u^1 + t, u^2, \dots, u^n)$ and then in these coordinates if $A = A_J dx^J$ we have that $\mathcal{L}_V A$'s coordinate representation is just A_J's partial derivatives with respect to u^1. Then all the other points follow by the product rule.

Then for points in Vs support it follows by continuity, and for points outside of the support everything becomes 0. ∎

For $f \in C^\infty(M)$ we have the covariant 1-tensor df given by $df(X) = Xf$. Notice then that

$$(\mathcal{L}_V df)(X) = V(df(X)) - df([V,X]) = V(Xf) - [V,X]f = VXf - VXf + XVf = XVf,$$

and this is just $df(XV)$ which is equal to $d(Vf)(X)$ and by above this is equal to $d(\mathcal{L}_V f)(X)$. Thus we have

$$\mathcal{L}_V df = d(\mathcal{L}_V f),$$

for all $f \in C^\infty(M)$.

11.3.4 Definition

If A is a covariant tensor field on M, and θ is a flow on M, then A is **invariant under** θ if for all (t,p) in θ's domain
$$d(\theta_t)_p^*(A_{\theta_t(p)}) = A_p.$$

11.3.5 Theorem

A is invariant under V's flow iff $\mathcal{L}_V A = 0$.

11.4 The Serre-Swan theorem

In this section we investigate the category of vector bundles over a smooth manifold M. Recall the strong connection between the $C^\infty(M)$-module of sections of a vector bundle and the vector bundle itself. In particular we recall the following result we proved earlier.

11.4.1 Theorem

Let $E \to M, F \to M$ be vector bundles over M. There is a natural correspondence between vector bundle morphisms $E \to F$ over M and $C^\infty(M)$-module morphisms $\Gamma(E) \to \Gamma(F)$.

Proof

We have already proven this, but we restate the construction of the bijections. Given a vector bundle morphism $\varphi\colon E \to F$, we define $\widetilde{\varphi}\colon \Gamma(E) \to \Gamma(F)$ by $s \mapsto \varphi \circ s$. Conversely given an module morphism $\psi\colon \Gamma(E) \to \Gamma(F)$ define $\widehat{\psi}\colon E \to F$ by $\widehat{\psi}(v_p) = \psi(\widehat{v})_p$ where $\widehat{v}$ is any section whose value at p is v.

Standard arguments are used to show that $\widehat{\psi}$ is well-defined: first one shows ψ acts locally, so sections agreeing on an open neighborhood have the same value under ψ in that neighborhood, then one shows that sections agreeing at a point agree at that point under ψ. Then $\widehat{\psi}$ is shown to be smooth by considering its representation under local trivializations.

It is clear that $\widetilde{\bullet}$ and $\widehat{\bullet}$ are inverses. ∎

Let us denote by VBun_M the category of vector bundles over M. Then we have a functor

$$\Gamma\colon \mathsf{VBun}_M \to \mathsf{Mod}_{C^\infty(M)}, \qquad E \mapsto \Gamma(E),$$

whose actions on vector bundle morphisms is described in the above theorem: $\Gamma(\varphi)(s) = \varphi \circ s$. The theorem says that Γ is fully faithful, and so we lose no information identifying φ and $\Gamma(\varphi)$.

Because Γ is fully faithful, it is an equivalence onto its essential image. As we will show later, its essential image is the collection of *projective* $C^\infty(M)$-modules.

11.4.2 Proposition

A vector bundle E is trivial iff E has a global frame iff $\Gamma(E)$ is free.

Conversely any free $C^\infty(M)$-module of finite rank is isomorphic to the sections of some trivial vector bundle.

Proof

The first equivalence was already shown, as the local trivialization related to a global frame is an isomorphism to a trivial bundle. And a basis for $\Gamma(E)$ is precisely a global frame for E.

Suppose F is a free $C^\infty(M)$-module, generated by $f_1, \dots, f_n$. Then $\Gamma(M \times \mathbb{R}^n)$ is free of rank n by the previous paragraph, and thus isomorphic to F. ■

For a bundle morphism $\varphi\colon E \to F$ and $p \in M$ let us denote by $\varphi_p = \varphi|_{E_p}\colon E_p \to F_p$, which is linear by definition. Now recall the following result we proved as well.

11.4.3 Proposition

A morphism of vector bundles over M is an isomorphism iff it is bijective.

Proof

Suppose $\pi\colon E \to M$ and $\pi'\colon F \to M$ are vector bundles and $\varphi\colon E \to F$ is a bijective vector bundle morphism. Since $\varphi_p\colon E_p \to F_p$ is then a linear isomorphism, E and F have the same rank and thus dimension. Since φ is by assumption bijective, it is sufficient to show that it is a local diffeomorphism, and for that it is sufficient to show that it is an immersion.

Now, we have

$$\pi = \pi' \circ \varphi,$$

so for all $v \in E$, say $v \in E_p$, we have

$$d\pi_v = d\pi'_{\varphi(v)} \circ d\varphi_v.$$

Let $x \in T_vE$ such that $d\varphi_v(x) = 0$ and thus $d\pi_v(x) = 0$.

Recall that for a submersion π, $v \in \pi^{-1}p$, we have that $\ker d\pi_v = T_v\pi^{-1}p$ (considering dimensions they are equal, and π is constant on $\pi^{-1}p$ so its differential is zero). Thus $x \in T_vE_p$, but φ_p is an isomorphism $E_p \to F_p$ and thus $d\varphi$ is an isomorphism on T_vE_p, so $x = 0$. Thus $d\varphi_v$ is an injection, so φ is a diffeomorphism. ■

Another way we could phrase this is that vector bundles are isomorphisms iff their actions on fibers are. Recall the following result which we proved.

11.4.4 Lemma

For a vector bundle morphism $\varphi\colon E \to F$, the following are equivalent:

(1) φ_p has constant rank for all $p \in M$,

(2) $\ker\varphi$ is a subbundle of E,

(3) $\operatorname{im}\varphi$ is a subbundle of F.

For a point $p \in M$ let us consider the map $\mathrm{ev}_p\colon C^\infty(M) \to \mathbb{R}$ which evaluates a smooth function at p. This map is clearly surjective (considering constant functions), and thus its kernel is a maximal ideal. We denote its kernel by

$$\mu_p = \ker \mathrm{ev}_p = \{f \in C^\infty(M) \mid f(p) = 0\}.$$

11.4.5 Lemma

For $s \in \Gamma(E)$, if $s(p) = 0$ then there are $f_i \in \mu_p$ and $s_i \in \Gamma(E)$ such that $s = \sum_i f_i s_i$. In other words, if $s(p) = 0$ then $s \in \mu_p\Gamma(E)$ (the product of a module and an ideal).

Therefore, for all $p \in M$ the following sequence splits

$$0 \to \mu_p\Gamma(E) \to \Gamma(E) \to E_p \to 0,$$

where the first map is inclusion, and the second is evaluating a section at p. Thus we have

$$E_p \cong \frac{\Gamma(E)}{\mu_p\Gamma(E)}.$$

Proof

Let U be a neighborhood of p over which we have a local frame $e_1, \dots, e_n$. So there are smooth functions $f_i \in C^\infty(U)$ such that

$$s = \sum_i f_i e_i$$

over U, and in particular we have $f_i(p) = 0$. Now let ψ be a bump function centered at p supported in U, so $\psi(p) = 1$ and $\operatorname{supp}\psi \subseteq U$. Then we have

$$\psi^2 s = \sum_i \psi^2 f_i e_i = \sum_i (\psi f_i)(\psi e_i).$$

Note that ψf_i and ψe_i can be extended smoothly over all of M by setting them to be zero outside of U. Then this equation holds over all of M, and we have

$$s = (1 - \psi^2)s + \sum_i (\psi f_i)\psi e_i.$$

And clearly $1 - \psi^2, \psi f_i \in \mu_p$ as desired.

The map $\Gamma(E) \to E_p$ is surjective as for every $v \in E_p$ there is a global section whose value at p is v (which we proved earlier). The kernel of the map $\Gamma(E) \to E_p$ is the set of all sections $s \in \Gamma(E)$ whose value at p is zero. We showed that this is equal to $\mu_p\Gamma(E)$, showing exactness. ■

11.4.6 Lemma

Let E, F be vector bundles over M. Then $\Gamma(E \oplus F) \cong \Gamma(E) \oplus \Gamma(F)$.

Proof

If $s \in \Gamma(E)$ and $t \in \Gamma(F)$ then $s \oplus t \in \Gamma(E \oplus F)$ where $(s \oplus t)|_p = s|_p \oplus t|_p$. It is easily seen via local frames that $s \oplus t$ is smooth, and that this map is an injective morphism. Similarly for $s \in \Gamma(E \oplus F)$ we map it to $s_0 \oplus s_1 \in \Gamma(E) \oplus \Gamma(F)$ where $s_0|_p$ is the projection of $s|_p$ onto E_p and similar for s_1. Again this is smooth, and these define inverse morphisms. ■

Now we state the following result, which will not be proven.

11.4.7 Lemma

For any vector bundle E, $\Gamma(E)$ is a finitely generated $C^\infty(M)$-module.

11.4.8 Lemma

For any vector bundle E, E is a direct summand of a trivial vector bundle.

Proof

Since $\Gamma(E)$ is fg, let $s_1, \ldots, s_n \in \Gamma(E)$ generate it. Let Q be the free rank-n $C^\infty(M)$-module, and define the surjection $\varphi\colon Q \to \Gamma(E)$ by mapping Q's basis to the s_i. Since Q is free, it is isomorphic to $\Gamma(F)$ where F is a free vector bundle, $F = M \times \mathbb{R}^n$. Thus $\varphi\colon \Gamma(F) \to \Gamma(E)$ is a morphism, and thus can be identified with a bundle morphism $\varphi\colon F \to E$.

Since $\Gamma(\varphi)$ is surjective, φ_p is surjective for all p. This is because we have the following commutative diagram

$$\begin{tikzcd}
\Gamma(E) \arrow[r] \arrow[d, "\Gamma(\varphi)"'] & E_p \arrow[d, "\varphi_p"'] \\
\Gamma(F) \arrow[r] & F_p
\end{tikzcd}$$

and all the arrows other than φ_p are surjective, implying φ_p is surjective. Thus φ has constant rank, and $\ker\varphi \subseteq F$ is a subbundle.

Now we claim that $\ker\varphi$ is a direct summand of F. Let $F = M \times \mathbb{R}^n$, then let $\ker\varphi^\perp$ be the subbundle of F whose fiber at $p \in M$ is the orthogonal complement of $\ker\varphi_p$ in $\mathbb{R}^n$ under the Euclidean product.

Now the Euclidean product of sections of F, which defines a function $\langle s, t\rangle_p = \langle s(p), t(p)\rangle$, is smooth. This is because the Euclidean product of the standard global frames e_i are constant, and the Euclidean product is tensorial, so smoothness follows from linearity. If we have a local frame for $\ker\varphi$ then extending it to a local frame for F and then applying Gram-Schmidt at each point gives us a local frame for $\ker\varphi^\perp$. This is because Gram-Schmidt is obtained via the Euclidean product which is smooth. Thus $\ker\varphi^\perp$ is a subbundle, and clearly $F = \ker\varphi \oplus \ker\varphi^\perp$.

Now we claim that the surjection $\varphi\colon F \to E$ induces an isomorphism $\varphi\colon \ker\varphi^\perp \to E$. This is because at each $p \in M$, φ_p's kernel is $\ker\varphi_p = (\ker\varphi)_p$ and thus forms an isomorphism $\ker\varphi_p^\perp \to E_p$. So $\varphi\colon \ker\varphi^\perp \to E$ is an isomorphism on the fibers, and thus an isomorphism of vector bundles, as desired. ■

11.4.9 Corollary

For a vector bundle E, $\Gamma(E)$ is a projective $C^\infty(M)$-module.

Proof

E is a direct summand of a trivial vector bundle, so $E \oplus F$ is trivial for some vector bundle F. Then we have that $\Gamma(E \oplus F) \cong \Gamma(E) \oplus \Gamma(F)$ is free, so $\Gamma(E)$ is projective. ■

11.4.10 Theorem (Serre-Swan)

Let M be a connected manifold. If P is a finitely-generated projective $C^\infty(M)$-module, it is isomorphic to the space of sections for some vector bundle E.

Thus Γ forms an equivalence of categories from VBun_M to the full subcategory of finitely-generated projective $C^\infty(M)$-modules.

Proof

Since P is projective, it is the direct summand of some finite-rank (since it is finitely-generated) free module. So we have $\Gamma(F) = P' \oplus Q$ where P', Q are modules and $P' \cong P$. We identify P with P', and thus we can view P' as a set of sections of F.
For $x \in M$ let us define

$$P_x = \{p(x) \mid p \in P\} \subseteq F_x.$$

This is a subspace of F_x, and similarly we define Q_x. Now we claim $F_x = P_x \oplus Q_x$.

For $v \in F_x$ there is a $s \in \Gamma(F)$ such that $s(x) = v$. Writing $s = p+q$ for $p \in P, q \in Q$, we have $v = p(x)+q(x)$, showing $P_x + Q_x = F_x$. Now if $p(x) = q(x)$ for $p \in P$ and $q \in Q$, then $(p-q)(x) = 0$ so $p - q \in \mu_x \Gamma(F)$ and thus

$$p - q = \sum_i f_i s_i = \sum_i f_i p_i + \sum_i f_i q_i,$$

where $f_i \in \mu_x$ and $s_i \in \Gamma(F)$ equal to $s_i = p_i + q_i$ for $p_i \in P, q_i \in Q$. But then $\sum_i f_i p_i \in P$ and so we have $p = \sum_i f_i p_i$ and $q = -\sum_i f_i q_i$. Thus $p(x) = \sum_i f_i(x) p_i(x) = 0$ and similarly $q(x) = 0$, so $v = 0$. So we have shown $F_x = P_x \oplus Q_x$ as desired.

Now we want to show that the union of P_x forms a subbundle of F, and that P gives its space of sections. Notice that $\dim P_x$ is independent of x. This is because if we take a basis for P_x of $p_1(x), \ldots, p_n(x)$, then by continuity $p_1(y), \ldots, p_n(y)$ remains linearly independent in a neighborhood of x. Thus $\dim P_y \geq \dim P_x$ in a neighborhood of x. Similarly we see $\dim Q_y \geq \dim Q_y$, and since $Q_x \oplus P_x = F_x$ whose dimension is constant, we see $\dim P_y$'s is constant in this neighborhood. Because M is connected, $\dim P_x$ must be constant over all of M.

If we choose any basis of P_x by $p_1(x), \ldots, p_n(x)$, then by continuity $p_1, \ldots, p_n$ are linearly independent in a neighborhood of x. Since we just showed that $\dim P_x$ is constant, this means $p_1, \ldots, p_n$ forms a local frame of P in a neighborhood of x. This means that $\overline{P} = \coprod_x P_x$ is indeed a subbundle of F.

Now we want to show that $P = \Gamma(\overline{P})$. Clearly $P \subseteq \Gamma(\overline{P})$ since P consists of sections of F falling into $\overline{P}$. And conversely for $s \in \Gamma(\overline{P}) \subseteq \Gamma(F)$ we have $s = p + q$ for $p \in P$ and $q \in Q$. Now $s(x) = p(x) + q(x)$ and since $s(x) \in P_x$, we must have $q(x) = 0$. So $s = p \in P$ as desired.

Since Γ is fully faithful, it is an equivalence onto its essential image. The previous corollary showed that Γ's essential image is contained in the subcategory of projective modules, and this one showed that it is in fact equal to it. ■

11.5 Functorial constructions on vector bundles

Thus far we have introduced some methods of constructing vector bundles from old ones. For example the Whitney sum, the dual bundle, and the tensor product of bundles. These are all liftings of constructions on vector spaces (direct sums, dual spaces, and tensor products respectively). In fact, we can generalize this as follows.

11.5.1 Definition

Let $\mathsf{Vec} = \mathsf{Vec}_{\mathbb{R}}^{\mathrm{fd}}$ denote the category of finite-dimensional vector spaces over $\mathbb{R}$. Then a single-variable covariant C^∞ **functor** is a functor $\mathcal{F} \colon \mathsf{Vec} \to \mathsf{Vec}$ such that the maps $\hom_{\mathbb{R}}(V, W) \to \hom_{\mathbb{R}}(\mathcal{F}(V), \mathcal{F}(W))$ are smooth (both sides are vector spaces). Contravariant C^∞ functors are defined similarly, and we can extend this to multivariate functors of mixed variance.

A multivariate C^∞ functor $\mathcal{F}(x_1, \ldots, x_n)$ is smooth iff $\mathcal{F}(V_1, \ldots, x_i, \ldots, V_n)$ is a single-variable C^∞ functor.

Examples of C^∞ functors are direct sums, taking the dual, and tensor products, as their actions on the hom-sets are linear and thus smooth. Then the construction on vector bundles done thus far is a special case of the following theorem.

Recall that a local trivialization is a diffeomorphism $\Phi \colon \pi^{-1} U \to U \times \mathbb{R}^n$ for an open subset $U \subseteq M$ such that projecting onto U gives π. So we can write Φ by (π, Φ_0), and Φ_0 restricts to linear isomorphisms $E_p \to \mathbb{R}^n$ on each E_p. In the following theorem we will identify a local trivialization with its second component (Φ_0), as no information is lost.

11.5.2 Theorem

Let $\mathcal{F}$ be a C^∞ functor. For vector bundles $E_1, \ldots, E_n \to M$ we define the vector bundle $\mathcal{F}(E_1, \ldots, E_n)$ by

$$\mathcal{F}(E_1, \ldots, E_n) = \coprod_{p \in M} \mathcal{F}(E_1|_p, \ldots, E_n|_p),$$

with the obvious projection, and this is indeed a vector bundle over M. Furthermore this construction is functorial.

Proof

Suppose $\mathcal{F}$ is covariant in the first n variables and contravariant in the last m. Let us denote by $E_1, \dots, E_n$ its first n inputs and $F_1, \dots, F_m$ its last m. If we have local trivializations Φ_i of the E_i and Ψ_i of the F_i all defined on the same open subset (by potentially shrinking it), we define a local trivialization Θ of $\mathcal{F}(\overline{E}, \overline{F})$ as

$$\Theta_p = \mathcal{F}(\Phi_1|_p, \dots, \Phi_n|_p, \Psi_1|_p^{-1}, \dots, \Psi_m|_p^{-1}).$$

That is, this is the definition of Θ on the fiber of p. Since $\mathcal{F}$ is contravariant in its last m arguments, this defines a map from $\mathcal{F}(\overline{E}_p, \overline{F}_p)$ to $V = \mathcal{F}(\mathbb{R}^{N_1}, \dots, \mathbb{R}^{N_n}, \mathbb{R}^{M_1}, \dots, \mathbb{R}^{M_m})$. Furthermore it is an isomorphism as functors preserve isomorphisms.

Now we must just show that different Θs are compatible. So suppose we have other local trivializations Φ_i' and Ψ_i' giving us Θ'. Then

$$\Theta' \circ \Theta^{-1}|_p = \Theta'|_p \circ \Theta|_p^{-1} = \mathcal{F}(\overline{\Phi}'|_p, (\overline{\Psi}')|_p^{-1}) \circ \mathcal{F}(\overline{\Phi}|_p, \overline{\Psi}|_p^{-1})^{-1} = \mathcal{F}(\overline{\Phi' \circ \Phi^{-1}}|_p, \overline{\Psi \circ (\Psi')^{-1}}|_p).$$

In the final equality we swap the order of Ψ and Ψ' due to contravariance. So if we let τ_i be the transition map from Φ_i to Φ_i', and σ_i the transition from Ψ_i' to Ψ_i, we see that the transition map θ from Θ to Θ' is given by

$$\theta(p) = \mathcal{F}(\overline{\tau}(p), \overline{\sigma}(p)).$$

Since $\mathcal{F}$ is smooth, we see that θ is smooth as desired.

Now suppose we have vector bundle morphisms $\varphi_i \colon E_i \to E_i'$ and $\psi_i \colon F_i' \to F_i$. Then define $\chi \colon \mathcal{F}(\overline{E}, \overline{F}) \to \mathcal{F}(\overline{E}', \overline{F}')$ defined on fibers by

$$\chi_p = \mathcal{F}(\overline{\varphi}_p, \overline{\psi}_p).$$

If we let Θ be a trivialization of $\mathcal{F}(\overline{E}, \overline{F})$ and Θ' a trivialization of $\mathcal{F}(\overline{E}', \overline{F}')$ we see that

$$\Theta' \circ \chi \Theta^{-1}|_p = \mathcal{F}(\overline{\Phi}' \circ \overline{\varphi} \circ \overline{\Phi}^{-1}|_p,\ \overline{\Psi} \circ \overline{\psi} \circ (\overline{\Psi}')^{-1}|_p).$$

Since φ_i, ψ_i are smooth, the arguments of $\mathcal{F}$ on the right are smooth in p and thus the left is smooth in p as well. Thus $\chi = \mathcal{F}(\overline{\varphi}, \overline{\psi})$ is smooth. This is clearly functorial. ∎

So for every C^∞ functor we associate a functor of vector bundles. This association itself is functorial: natural transformations of C^∞ functors map to natural transformations of functors of vector bundles functorally. In particular this means that isomorphic C^∞ functors give isomorphic vector bundle functors. This allows us to prove easily that the double dual of any vector bundle is naturally isomorphic to the original vector bundle for example.

Or consider the hom functor, $\hom_{\mathbb{R}} \colon \mathsf{Vec}^{\mathsf{op}} \times \mathsf{Vec} \to \mathsf{Vec}$. This is a linear functor (as in its actions on morphisms is linear) and thus smooth. Thus it defines a hom functor on vector bundles: for two vector bundles E, F we have a vector bundle $\hom(E, F)$ whose fibers are $\hom_{\mathbb{R}}(E_p, F_p)$. Since we have a natural isomorphism

$$\hom_{\mathbb{R}}(V, U) \cong V^* \otimes U,$$

this lifts to a natural isomorphism

$$\hom(E, F) \cong E^* \otimes F.$$

Now it is natural to ask what we can say about the space of sections of a functorial construction such as this. For example, we already know $\Gamma(E \oplus F) \cong \Gamma(E) \oplus \Gamma(F)$. It is not hard to see that $\Gamma(E^*) \cong \Gamma(E)^*$, where the latter dual is the dual over $C^\infty(M)$: $\Gamma(E)^* = \hom(\Gamma(E), C^\infty(M))$.

To do this we will first prove a few facts about projective modules.

11.5.3 Lemma

Let A be a commutative ring, and P, Q be finitely generated projective A-modules. Then

(1) $P \oplus Q$ is projective and $(P \oplus Q)^* \cong P^* \oplus Q^*$.

(2) P is **reflexive**: $P^{**} \cong P$.

(3) $P \otimes Q$ is projective.

(4) $P^* \otimes Q \cong \hom_A(P, Q)$ and $(P \otimes Q)^* \cong P^* \otimes Q^*$.

(5) $\hom_A(P, Q)$ is projective, in particular P^* is.

Proof

(1) $P \oplus Q$ is clearly projective. We have

$$(P \oplus Q)^* = \hom_A(P \oplus Q, A) \cong \hom_A(P, A) \oplus \hom_A(Q, A) = P^* \cong Q^*,$$

as desired and this holds for arbitrary modules.

(2) There is a natural map $P \to P^{**}$ which maps $x \in P$ to $\overline{x} \in P^{**}$ which maps $\varphi \in P^*$ to $\varphi(x)$. For fg free F this map is an isomorphism, as F^* has the same rank as F and it is easy to see that this map maps a basis of F to a linearly independent set of F^*. So if $P \oplus Q$ is free, then we have an isomorphism $P \oplus Q \to P^{**} \oplus Q^{**}$. It is easy to see that this isomorphism is the direct sum of the maps $P \to P^{**}$ and $Q \to Q^{**}$, so they must be isomorphisms.

(3) The tensor of two free modules is free, so if $P \oplus P', Q \oplus Q'$ are free then so is

$$(P \oplus P') \otimes (Q \oplus Q') \cong P \otimes Q \oplus \dots.$$

So $P \otimes Q$ is projective.

(4) There is a natural map $P^* \otimes Q \to \hom_A(P, Q)$ and this is an isomorphism when P is fg projective (independent of Q). The map is given by mapping $\varphi \otimes y$ to $x \mapsto \varphi(x)y$. For finite free P this is an isomorphism: take $x_1, \dots, x_n$ a basis then every $f\colon P \to Q$ is determined by $f(x_1), \dots, f(x_n)$ and we have a dual basis $\varphi_1, \dots, \varphi_n$ in P^* given by $\varphi_i(x_j) = \delta_{ij}$. Then $\sum_i \varphi_i \otimes f(x_i)$ maps to f.
Now if $P \oplus P'$ is free then we have an isomorphism

$$(P^* \otimes Q) \oplus ((P')^* \otimes Q) \to \hom_A(P, Q) \oplus \hom_A(P', Q)$$

and as before it is easy to see that this is the direct sum of the maps $P^* \otimes Q \to \hom_A(P, Q)$ so they must be isomorphisms.

(5) Suppose $P \oplus P'$ and $Q \oplus Q'$ are free of finite rank. Then

$$\hom_A(P \oplus P', Q \oplus Q') \cong \hom_A(P, Q) \oplus \hom_A(P, Q') \oplus \hom_A(P', Q) \oplus \hom_A(P', Q').$$

And the left-hand side is free, since finite biproducts commute with hom, so $\hom_A(P, Q)$ is a direct summand of a free module and thus projective.
Since $P^* = \hom_A(P, A)$ and A is free and thus projective, clearly P^* is projective. ∎

11.5.4 Proposition

For a vector bundle E, $\Gamma(E^*) \cong \Gamma(E)^*$.

Proof

Define $\varphi\colon \Gamma(E^*) \to \Gamma(E)^*$ by $\varphi(\overline{s})(t) = \overline{s}(t)$, where $\overline{s}(t) \in C^\infty(M)$ is given by $\overline{s}(t)(p) = \overline{s}_p(t_p)$. It is easily seen that $\overline{s}(t)$ is smooth by considering trivializations.

For trivial P, global sections of P yield global sections of P^*, so P^* is trivial as well and so we have an isomorphism $\Gamma(P^*) \cong \Gamma(P) \cong \Gamma(P)^*$ of free modules, obtained by mapping the dual global sections of $\Gamma(P^*)$ to $\Gamma(P)$ and then mapping to $\Gamma(P)^*$ (the last isomorphism is due to it being fg free). This is just φ, so in the case that P is free we have that φ is an isomorphism.

For general E, by Serre-Swan there is an F such that $E \oplus F \cong P$ trivial so we have $P^* \cong (E \oplus F)^* \cong E^* \oplus F^*$ where the last isomorphism is due to the lifting of the natural isomorphism of vector spaces. Thus we have an isomorphism (the first isomorphism is due to the above lemma)

$$\Gamma(E)^* \oplus \Gamma(F)^* \cong \Gamma(P)^* \cong \Gamma(P^*) \cong \Gamma(E^* \oplus F^*) \cong \Gamma(E^*) \oplus \Gamma(F^*)$$

Then we have a commutative diagram

$$\begin{tikzcd}
\Gamma(E^*) \arrow[r] \arrow[d, "\varphi"'] & \Gamma(E^*) \oplus (F^*) \arrow[d, "\cong"'] \\
\Gamma(E)^* \arrow[r] & \Gamma(E)^* \oplus \Gamma(F)^*
\end{tikzcd}$$

where the horizontal arrows are inclusions. This shows that φ is injective, and reversing the horizontal arrows to give projections shows φ is surjective. Note that this is the same as saying the map between free modules is the direct sum of the maps between projective modules. ■

We can also find φ's inverse explicitly. Let $\psi\colon \Gamma(E)^* \to \Gamma(E^*)$ by mapping $f\colon \Gamma(E) \to C^\infty(M)$ to $\psi(f) \in \Gamma(E^*)$ given by $\psi(f)_p \in E_p^*$ which maps $v \in E_p$ to $f(\overline{v})(p)$ where $\overline{v}$ is a global section extending v, then $\psi = \varphi^{-1}$. The above proof though generalizes nicely to prove a similar result for tensor products.

11.5.5 Proposition

For vector bundles E, F,

$$\Gamma(E \otimes F) \cong \Gamma(E) \otimes \Gamma(F),$$

where the latter tensor product is taken over $C^\infty(M)$.

Proof

There is a natural map $\Gamma(E) \otimes \Gamma(F) \to \Gamma(E \otimes F)$ given by mapping $s \otimes t$ to $s \otimes t \in \Gamma(E \otimes F)$ defined by $(s \otimes t)|_p = s|_p \otimes t|_p$. This is easily shown to be an isomorphism when E, F are trivial this is an isomorphism. Then in the general case we write E, F as direct summands of trivial bundles and use a similar argument as above to show that this map is an isomorphism in general. ■

11.5.6 Corollary

There is a natural isomorphism of vector bundles $(E \otimes F)^* \cong E^* \otimes F^*$.

Proof

There is an isomorphism

$$\Gamma((E \otimes F)^*) \cong \Gamma(E \otimes F)^* \cong \Gamma(E^*) \otimes \Gamma(F^*) \cong \Gamma(E^* \otimes F^*),$$

and by Serre-Swan since Γ is an equivalence we have $(E \otimes F)^* \cong E^* \otimes F^*$. ■

11.5.7 Corollary

For vector bundles E, F,

$$\Gamma(\hom(E, F)) \cong \hom_{C^\infty(M)}(\Gamma(E), \Gamma(F)) \cong \hom_{\mathsf{VBun}_M}(E, F).$$

Furthermore, for any vector bundle E, there is an adjunction $\bullet \otimes E \dashv \hom(E, \bullet)$.

Proof

There is a natural isomorphism $\hom(E, F) \cong E^* \otimes F$, thus

$$\Gamma(\hom(E, F)) \cong \Gamma(E^* \otimes F) \cong \Gamma(E)^* \otimes \Gamma(F) \cong \hom_{C^\infty(M)}(\Gamma(E), \Gamma(F)),$$

where the final isomorphism is due to the space of sections being fg projective. And since Γ is fully faithful we have the final isomorphism to vector bundle morphisms.
The tensor-hom adjunction for vector spaces gives an isomorphism

$$\hom(\bullet \otimes E, \bullet) \cong \hom(\bullet, \hom(E, \bullet)).$$

Then applying Γ and appealing to what we just proved, this gives an isomorphism

$$\hom_{\mathsf{VBun}_M}(\bullet \otimes E, \bullet) \cong \hom_{\mathsf{VBun}_M}(\bullet, \hom(E, \bullet)),$$

demonstrating the desired adjunction. ■

Recall that we categorized tensor fields as multilinear maps out of $\mathfrak{X}^*(M)^k \times \mathfrak{X}(M)^\ell$. We can generalize this easily.

11.5.8 Theorem

There is a natural isomorphism between sections of $E \otimes F$ and $C^\infty(M)$-bilinear maps $\Gamma(E)^* \times \Gamma(F)^* \to C^\infty(M)$. Inductively, there is a natural isomorphism

$$\Gamma(E_1 \otimes \cdots \otimes E_n) \cong \operatorname{MultLin}(\Gamma(E_1)^*, \ldots, \Gamma(E_n)^*;\ C^\infty(M)).$$

And in particular,

$$\Gamma(E^{\otimes k} \otimes (E^*)^{\otimes \ell}) \cong \operatorname{MultLin}(\Gamma(E^*)^k \times \Gamma(E)^\ell;\ C^\infty(M)).$$

Proof

Since fg projective modules are reflective, there is a natural isomorphism

$$\Gamma(E \otimes F) \cong \Gamma(E \otimes F)^{**} \cong \Gamma(E)^* \otimes \Gamma(F)^* = \hom_{C^\infty(M)}(\Gamma(E)^* \otimes \Gamma(F)^*, C^\infty(M)),$$

and by the universal property of tensor products this is isomorphic to the space of bilinear maps $\Gamma(E)^* \times \Gamma(F)^* \to C^\infty(M)$. The inductive case is then clear, and for the final isomorphism we just utilize the reflexivity of vector bundles: $E^{**} \cong E$. ■

12 Riemannian Manifolds

12.1 Riemannian manifolds

A symmetric 2-tensor $\alpha \in \mathrm{Sym}^2(V^*)$ is **positive definite** if $\alpha(v,v) > 0$ for all $v \neq 0$. A symmetric covariant 2-tensor field is positive definite if it is positive definite at every point.

12.1.1 Definition

Let M be a smooth manifold, a **Riemannian metric** on M is a smooth symmetric covariant 2-tensor field on M. A **Riemannian manifold** is a smooth manifold M equipped with a Riemannian metric g.

Since g_p is an positive semidefinite and symmetric, it forms an inner product on T_pM. We denote this by $\langle \bullet, \bullet \rangle_g$.

> **Note:**
> Equivalently, we can characterize a Riemannian metric to be an association to each $p \in M$ an inner product $\langle \bullet, \bullet \rangle_g$ on T_pM which is smooth in the sense that $\langle X, Y \rangle_g$ is in $C^\infty(M)$ for every $X, Y \in \mathfrak{X}(M)$.

In any smooth coordinate chart (x^i), a Riemannian metric has components

$$g = g_{ij} dx^i \otimes dx^j.$$

Note then that by symmetry, $g_{ij} = g_{ji}$ and so

$$g = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ij} dx^i \otimes dx^j) = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ji} dx^i \otimes dx^j) = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ij} dx^j \otimes dx^i) = g_{ij} dx^i dx^j,$$

where $dx^i dx^j$ is the symmetric product.

12.1.2 Example

The **Euclidean metric** on $\mathbb{R}^n$ is $\overline{g}$ given by gloabl coordinates

$$\overline{g} = \delta_{ij} dx^i dx^j.$$

Writing α^2 for the symmetric product of α with itself, we have that

$$\overline{g} = (dx^1)^2 + \cdots + (dx^n)^2.$$

Then for $v, w \in T_p\mathbb{R}^n$ we have

$$\overline{g}_p(v, w) = \delta_{ij} v^i w^j = \sum_i v^i w^i = v \cdot w,$$

so that the Euclidean metric is the Euclidean inner product on $T_p\mathbb{R}^n$.

12.1.3 Example

If (M_1, g^1) and (M_2, g^2) are Riemannian manifolds, we can define a Riemannian metric $g = g^1 \oplus g^2$ on $M_1 \times M_2$ by

$$g((v_1, w_1), (v_2, w_2)) = g^1(v_1, v_2) + g^2(w_1, w_2),$$

for $(v_i, w_i) \in T_pM_1 \oplus T_qM_2 \cong T_{(p,q)}(M_1 \times M_2)$. This is the sum of two smooth tensor fields, and thus smooth itself, and symmetric. It is positive definite at each point because g^1, g^2 are.
Given local coordinates (x^i) for M_1 and (y^j) for M_2, if g^1 has components $g^1_{ij} dx^i dx^j$ and g^2 has components $g^2_{ij} dy^i dy^j$, then g's coordinate representation is

$$(g_{ij}) = \begin{pmatrix} g^1_{ij} & \\ & g^2_{ij} \end{pmatrix}.$$

12.1.4 Proposition

Every smooth manifold admits a Riemannian metric.

Proof

Let $(U_\alpha, \varphi_\alpha)$ cover M by countably many coordinate charts. Then $\varphi_\alpha\colon U_\alpha \to \mathbb{R}^m$ is smooth and thus we can pull back the Euclidean metric to U_α by $g_\alpha = \varphi_\alpha^* \overline{g}$. Its coordinate representation is $\delta_{ij} dx^i dx^j$ relative to φ_α. Now let $\{\psi_\alpha\}$ be a partition of unity subordinate to U_α, and define

$$g = \sum_\alpha \psi_\alpha g_\alpha.$$

This is smooth, by local finiteness. It is also obviously symmetric, so only positivity needs to be checked. All the functions in question are nonnegative, so we need to show that $g(v,v) \neq 0$ when $v \neq 0$. Indeed,

$$g(v,v) = \sum_\alpha \psi_\alpha(p) g_\alpha(v,v),$$

and if $0 \neq v \in T_pM$ then some $\psi_\alpha(p)$ must be positive (since they sum to 1), and thus $g(v,v) \geq \psi_\alpha(p) g_\alpha(v,v) > 0$. ■

Note that such a choice of Riemannian metric is not natural.

12.1.5 Definition

If (M,g) is a Riemannian metric, then the **norm** of $v \in T_pM$ is defined to be $|v| = \langle v,v\rangle_g^{1/2} = g_p(v,v)^{1/2}$. $v, w \in T_pM$ are **orthogonal** if $\langle v,w\rangle_g = 0$.

12.1.6 Proposition

Let (M,g) be a Riemannian manifold, and (X_j) a smooth local frame for M over an open subset $U \subseteq M$. Then there is a smooth local frame (E_j) over U which is orthonormal at every point, and $\operatorname{span}(E_1|_p, \ldots, E_j|_p) = \operatorname{span}(X_1|_p, \ldots, X_j|_p)$ for all $j = 1, \ldots, n$.

In particular for every $p \in M$ there is an orthonormal frame in a neighborhood around p.

Proof

Apply Gram-Schmidt, and note that the resulting fields are smooth. ■

12.1.7 Definition

If $F\colon M \to N$ is smooth, and g is a Riemannian metric on N, then the pullback F^*g is a smooth 2-tensor field on M. If it is positive definite, it is a Riemannian metric on M, called the **pullback metric**.

12.1.8 Proposition

The pullback F^*g is Riemannian iff F is a smooth immersion.

Proof

The pullback is defined by
$$F^*g_p(v,w) = g(dF_pv, dF_pw),$$
and in general $\langle F\bullet, F\bullet\rangle$ is an inner product only when F is injective. ■

12.1.9 Definition

If (M,g) and $(N,\widetilde{g})$ are Riemannian manifolds, a smooth map $F\colon M \to N$ is a **(Riemannian) isometry** if it is a diffeomorphism which satisfies $F^*\widetilde{g} = g$. A **local isometry** is one where for every $p \in M$ there is a neighborhood $p \in U$ such that $F|_U\colon U \to N$ is an isometry onto an open subset of N. Equivalently, if it satisfies $F^*\widetilde{g} = g$ and F is a local diffeomorphism.

If there exists an isometry between Riemannian manifolds, we say they are **isometric**. A Riemmanian n-manifold is a **flat Riemannian manifold** (and the metric is a **flat metric**) if it is locally isometric to $(\mathbb{R}^n, \overline{g})$.

12.1.10 Proposition

Let (M,g) be a Riemannian manifold. The following are equivalent:

(1) g is flat;

(2) Each point of M is contained in the domain of a smooth coordinate chart in which g has the coordinate representation $g = \delta_{ij}dx^idx^j$;

(3) Each point of M is contained in the domain of a smooth coordinate chart in which the coordinate frame is orthonormal;

(4) Each point of M is contained in the domain of a commuting orthnormal frame.

Proof

Clearly (2) and (3) are equivalent, and if g is flat then every point lies in an open subset $U \subseteq M$ which is isometric to an open subset of $\mathbb{R}^n$. The isometry forms a local chart $(U,(x^i))$ where $\partial/\partial x^i$ are orthonormal. And conversely if $(U,(x^i))$ is a local chart where $\partial/\partial x^i$ are orthonormal, then the chart is an isometry as it maps the local coordinate frame $\partial/\partial x^i$ to the Euclidean coordinate frame which is orthonormal. It is sufficient for a map to preserve a single orthonormal frame for it to be an isometry.

Clearly (3) implies (4) as coordinate frames are commuting. Now (4) implies (3) since by the canonical form for commuting frames, if (E_i) is a commuting local orthonormal frame then there is a coordinate frame $(U,(x^i))$ such that $E_i = \partial/\partial x^i$. ■

12.1.11 Definition

If $S \subseteq M$ is an immersed submanifold with (M,g) Riemannian, then S inherits the pullback metric ι^*g where $\iota\colon S \hookrightarrow M$ is the inclusion. This is called the **induced metric**.

The induced metric is indeed a metric since ι is by definition an immersion. Under this metric S is called a **Riemannian submanifold**.

12.1.12 Example

The induced metric $\iota^* g$ induced on S^n by the inclusion $S^n \hookrightarrow \mathbb{R}^{n+1}$ is called the **round** or **standard metric** on the sphere.

12.1.13 Definition

Let (M, g) be a Riemannian n-manifold, and $S \subseteq M$ a k-dimensional submanifold. We say that for $p \in S$, a vector $v \in T_pM$ is **normal to** S if it is orthogonal to T_pS with respect to $\langle \bullet, \bullet \rangle_g$. The **normal space to** S at p is the subspace $N_pS \subseteq T_pM$ of all vectors orthogonal to T_pS (i.e. $N_pS = (T_pS)^\perp$).
Let $NS = \coprod_{p \in S} N_pS$, and $\pi_{NS} \colon NS \to S$ to be the obvious projection.

12.1.14 Proposition

$NS \to S$ forms a smooth $(n-k)$-rank vector bundle over S, with a unique smooth structure where the linear structures on the fibers are given by N_pS. For each $p \in S$ there is a smooth frame for NS on a neighborhood of p that is orthonormal with respect to g.

Proof

Take a coordinate frame (X_i) for T_pS, and then we can extend it and apply Gram-Schmidt to get a coordinate frame (X_i, Y_j) where Y_j are in N_pS. These form a basis, and thus satisfy the local frame criterion for subbundles. ■

12.2 The Riemannian distance function

12.2.1 Definition

Suppose (M, g) is a Riemannian manifold, and $\gamma \colon [a, b] \to M$ a piecewise smooth curve segment. γ's **length** is

$$L_g(\gamma) = \int_a^b |\gamma'(t)|_g \, dt.$$

This is well-defined since $\gamma'(t)$ is defined at all but finitely many values of t.

Note that if $F \colon M \to N$ is a local isometry, then since $F^* g_N = g_M$ we have that F preserves lengths: if $\gamma \colon [a, b] \to M$ is piecewise smooth,

$$L_{g_N}(F \circ \gamma) = \int_a^b |(F \circ \gamma)'(t)| = \int_a^b \left| dF_{\gamma(t)} \circ \gamma'(t) \right| = \int_a^b |\gamma'(t)| = L_{g_M}(\gamma).$$

12.2.2 Definition

A **reparametrization** of a curve $\gamma \colon [a, b] \to M$ is a curve of the form $\widetilde{\gamma} = \gamma \circ \varphi$ where $\varphi \colon [c, d] \to [a, b]$ is a diffeomorphism.

12.2.3 Proposition

Curve length is invariant under reparametrization.

Proof

Let $\varphi\colon [c,d] \to [a,b]$ be a diffeomorphism and $\gamma\colon [a,b] \to M$ a piecewise smooth curve. Since φ is a diffeomorphism, either $\varphi' > 0$ or $\varphi' < 0$ everywhere. Assume φ' is always positive, then

$$L_g(\gamma \circ \varphi) = \int_c^d |(\gamma \circ \varphi)'(t)| = \int_c^d |\varphi'(t)\gamma'(\varphi(t))| = \int_c^d |\gamma'(\varphi(t))|\,\varphi'(t) = \int_a^b |\gamma'(t)|\,,$$

where the final equality is due to change of variables. If $\varphi' < 0$ then we'd have to multiply the second to last term by -1, then change of variables swaps the sign again. ■

12.2.4 Definition

Let (M,g) be a connected Riemannian manifold. For $p,q \in M$ define their **Riemannian distance**, denoted $d_g(p,q)$, to be the infimum over $L_g(\gamma)$ over all piecewise smooth curves γ from p to q.

12.2.5 Example

Consider $\mathbb{R}^n$ with the Euclidean metric g. Then $d_g(p,q) = |p-q|$, i.e. the Riemannian distance is the same as Euclidean distance.
Indeed, $\gamma\colon [0,1] \to \mathbb{R}^n$ with $\gamma(t) = tp + (1-t)q$ has $\gamma'(t) = p - q$ and thus

$$L_g(\gamma) = \int_0^1 |p-q| = |p-q|\,.$$

So $d_g(p,q) \leq |p-q|$.
Now if γ is any other piecewise smooth curve, we need to show that $L_g(\gamma) \geq |p-q|$. We have that

$$L_g\gamma = \int_a^b |\gamma'(t)|\; dt \geq \left|\int_a^b \gamma'(t)\right| = |\gamma(b) - \gamma(a)| = |p-q|\,,$$

as desired.

12.2.6 Lemma

Let g be a Riemannian metric on an open subset $U \subseteq \mathbb{R}^n$. Given a compact subset $K \subseteq U$, there exist positive constants c, C such that for all $x \in K$ and $v \in T_x\mathbb{R}^n$,

$$c\,|v|_{\overline{g}} \leq |v|_g \leq C\,|v|_{\overline{g}}\,.$$

Proof

Define $L \subseteq T\mathbb{R}^n$ by

$$L = \left\{ (x,v) \in T\mathbb{R}^n \;\middle|\; x \in K, |v|_{\overline{g}} = 1 \right\}.$$

Under the canonical identification of $T\mathbb{R}^n$ with $\mathbb{R}^n \times \mathbb{R}^n$, this is just $L = K \times S^{n-1}$, which is compact. Since $|v|_g$ is continuous and strictly positive on L, there are positive constants c, C such that $c \leq |v|_g \leq C$ ($|v|_g$ obtains its maximum and minimum on a compact set) for all $(x,v) \in L$. If $x \in K$ and $v \in T_x\mathbb{R}^n$ is nonzero, let $\lambda = |v|_{\overline{g}}$ so that $(x, \lambda^{-1}v) \in L$ so

$$|v|_g = \lambda\left|\lambda^{-1}v\right|_g \leq \lambda C = C\,|v|_{\overline{g}}\,,$$

and similar for c. ■

12.2.7 Theorem

The Riemannian distance function on a connected Riemannian manifold (M, g) is a metric in the usual sense. It induces the same topology on M as its manifold topology.

Proof

All of the conditions for d_g being a metric are clear, other than positivity. Note that the triangle inequality is because if γ connects a to b, and $\widetilde{\gamma}$ connects b to c, then we can concatenate the curves to get a curve from a to c whose length is the sum of the curves.

Now we need to show that $d_g(p, q) \neq 0$ when $p \neq q$. Let U be a chart containing p but not q. We use the coordinate map to identify U with a an open subset of $\mathbb{R}^n$, and let $\overline{g}$ be the Euclidean metric in these coordinates. Let V be a regular coordinate ball of radius ε centered at p such that $\overline{V} \subseteq U$. Then the above lemma shows that there are constants $c, C > 0$ such that $c\,|v|_{\overline{g}} \leq |v|_g \leq C\,|v|_{\overline{g}}$ for all $v \in T_xM$ when $x \in \overline{V}$. Then for any smooth curve γ lying in $\overline{V}$, $cL_{\overline{g}}(\gamma) \leq L_g(\gamma) \leq CL_{\overline{g}}(\gamma)$.

Now suppose $\gamma\colon [a, b] \to M$ is piecewise smooth from p to q. Let t_0 be the infimum of all t such that $\gamma(t) \notin \overline{V}$. By continuity, $\gamma(t_0) \in \partial V$ and so $\gamma(t) \in \overline{V}$ for $a \leq t \leq t_0$. Thus

$$L_g(\gamma) \geq L_g(\gamma|_{[a,t_0]}) \geq cL_{\overline{g}}(\gamma|_{[a,t_0]}) \geq cd_{\overline{g}}(p, \gamma(t_0)) = c\varepsilon.$$

The final inequality is because $d_{\overline{g}}$ agrees with normal Euclidean distance, and $|p - \gamma(t_0)| = \varepsilon$ since $\gamma(t_0) \in \partial V$. Thus we conclude that $d_g(p, q) \geq c\varepsilon > 0$ as desired.

Finally to show that the two topologies coincide, we show that open sets in one are open in the other and vice versa. Suppose that $U \subseteq M$ is open, and let $p \in U$, and let V be a regular coordinate ball of radius ε around p such that $\overline{V} \subseteq U$. We have that $d_g(p, q) \geq c\varepsilon$ when $q \notin \overline{V}$, and thus if $d_g(p, q) < c\varepsilon$ then $q \in \overline{V} \subseteq U$. So the open ball of radius $c\varepsilon$ centered around p is contained in U, since p was arbitrary this means U is open.

Conversely if W is open in the metric topology, let $p \in W$. Let V be a regular coordinate ball around p of radius r, and let $\overline{g}$ be the Euclidean metric on $\overline{V}$. Then we have positive constants $c, C > 0$ bounding the metric of d_g by $d_{\overline{g}}$ as before. Let $\varepsilon < r$ be small enough such that the metric ball around p of radius $C\varepsilon$ is contained in W, and let V_ε be the set of points of $\overline{V}$ whose Euclidean distance from p is less than ε. If $q \in V_\varepsilon$ let γ be the straight line segment from p to q, then

$$d_g(p, q) \leq L_g(\gamma) \leq CL_{\overline{g}}(\gamma) < C\varepsilon.$$

Thus V_ε is contained in the metric ball of radius $C\varepsilon$ around p, and thus in W. V_ε is open in the manifold topology (since it is an open set under the coordinate chart of $\overline{V}$), and $p \in V_\varepsilon$, we see that W is open in the manifold topology. ■

12.2.8 Definition

A Riemannian manifold (M, g) is **complete**, and g is a **complete Riemannian metric**, if (M, d_g) is a complete metric space. A subset $B \subseteq M$ is **bounded** if it is bounded in the metric space.

12.2.9 Corollary

Every smooth manifold is metrizable.

Proof

If M is a manifold without boundary, then choose a Riemannian metric g on M. If M is connected, then the above proposition shows that it is metrizable by d_g. Otherwise, let $\{M_i\}$ be the components of M, each of which is metrizable by $d_g|_{M_i}$. The disjoint union of metrizable spaces is metrizable: let $p_i \in M_i$ and define

$$d_g(x, y) = d_g(x, p_i) + 1 + d_g(p_j, y),$$

when x, y are not in the same component.

If M has boundary, it can be embedded into a manifold without boundary (its double), which is metrizable. Subspaces of metrizable spaces are metrizable. ■

12.3 The musical isomorphisms

If V is a finite-dimensional inner product space, there is a natural isomorphism $V \to V^*$ given by $v \mapsto \langle v, \bullet \rangle$. This is clearly linear and injective, since if $\langle v, \bullet \rangle = 0$ then $\langle v, v \rangle = 0$ so $v = 0$. We can also compute its inverse: let e_i be an orthonormal basis of V, then for $\varphi \in V^*$ define $v = \sum_i \varphi(e_i) e_i$.

Thus have a bundle isomorphism

$$\widehat{g}: TM \to T^*M, \qquad \widehat{g}(v)(w) = g(v, w).$$

This is a bundle isomorphism since it corresponds to the morphism of sections $\widehat{g}: \mathfrak{X}(M) \to \mathfrak{X}^*(M)$, $\widehat{g}(X)(Y) = g(X, Y)$. Indeed, $\widehat{g}(X)$ is a smooth covector field because its action on vector fields is smooth. And clearly $\widehat{g}$ is $C^\infty(M)$-linear. Furthermore, $\widehat{g}$ is bijective on the fibers, so that $\widehat{g}$ is a bijective bundle morphism over M and thus an isomorphism.

If (E_i) is a local frame with dual coframe (ε^i), let $g = g_{ij}\varepsilon^i\varepsilon^j$ be the representation of the metric with respect to this frame. Suppose $X \in \mathfrak{X}(M)$ is a vector field with representation $X = X^i E_i$, then

$$\widehat{g}(X)(E_j) = g(X^i E_i, E_j) = X^i g_{ij}$$

so that

$$\widehat{g}(X) = (X^i g_{ij})\varepsilon^j.$$

It is customary to write the components of $\widehat{g}(X)$ as X_j, so that $X_j = X^i g_{ij}$. Due to this notational convention, the action of $\widehat{g}$ on vector fields is called "lowering an index", and we write $X^\flat$ for $\widehat{g}(X)$. The symbol $\flat$ is a musical symbol called the **flat** which denotes the lowering of a tone, and thus $X^\flat$ is called X-flat.

The inverse $\widehat{g}^{-1}$ acts on covector fields $\omega = \omega_j \varepsilon^j$ and maps them to vector fields

$$\widehat{g}^{-1}(\omega) = \omega^i E_i, \qquad \omega^i = g^{ij}\omega_j,$$

where (g^{ij}) is the inverse matrix of (g_{ij}). This is called "raising an index", and we write $\omega^\sharp$ for $\widehat{g}^{-1}(\omega)$; borrowing the musical symbol for raising a tone.

Thus we have bijective correspondences

$$\begin{array}{ccll}
\mathfrak{X}(M) & \underset{\bullet^\sharp}{\overset{\bullet^\flat}{\rightleftarrows}} & \mathfrak{X}^*(M), & \\
X^i E_i & \longmapsto & X_i \varepsilon^i, & X_i = g_{ij} X^j \\
\omega^i E_i & \longleftarrow & \omega_i \varepsilon^i, & \omega^i = g^{ij}\omega_j.
\end{array}$$

These correspondences are called the **musical isomorphisms**. Note that we can raise and lower the indices of non-smooth vector and covector fields in the same way.

The gradient

One application of the musical isomorphisms is to recover the notion of the gradient of a real-valued function. Recall that if $f: \mathbb{R}^n \to \mathbb{R}$ is a smooth function, its gradient $\operatorname{grad} f$ is the vector of its partial derivatives. It is the vector with the unique property that

$$\operatorname{grad} f|_p \cdot v = D_v f,$$

where $D_v f$ is the directional derivative of f in the direction of v.

In the case of Riemannian manifolds, $D_v f$ is replaced with $df(v)$ for $v \in T_pM$ so that this becomes

$$\langle \operatorname{grad} f, v \rangle_p = df_p(v),$$

which is to say that $\operatorname{grad} f = (df)^\sharp$. Thus in any local frame (E_i) with dual frame (ε^i), we have that $df = (E_i f)\varepsilon^i$ and so

$$\operatorname{grad} f = (g^{ij} E_i f) E_j.$$

So if (E_i) is an orthonormal frame, the components of $\operatorname{grad} f$ are the same as the components of df; but this is not true in general. In the case of $\mathbb{R}^n$ we see that $\operatorname{grad} f = \partial f/\partial x_i \partial/\partial x_i$.

The musical isomorphisms on tensors

Now we can generalize the musical isomorphisms to any tensor. If F is a (k, ℓ)-tensor, then it corresponds to a map

$$F \colon \mathfrak{X}^*(M)^{\otimes k} \otimes \mathfrak{X}(M)^{\otimes \ell} \to C^\infty(M),$$

and we can use the musical isomorphisms on any index $i \in \{1, \dots, k + \ell\}$ to change a $\mathfrak{X}(M)$ to a $\mathfrak{X}^*(M)$ or vice versa. That is, if i is a covariant index, meaning its input is a vector, then we obtain $F^\sharp$ by making that index contravariant. Explicitly, $F^\sharp$ is a $(k + 1, \ell - 1)$-tensor obtained by

$$F^\sharp(\alpha_1, \dots, \alpha_{k+\ell}) = F(\alpha_1, \dots, \alpha_i^\sharp, \dots, \alpha_{k+\ell}).$$

Similarly if i is a contravariant index, then $F^\flat$ is obtained by making that index covariant:

$$F^\flat(\alpha_1, \dots, \alpha_{k+\ell}) = F(\alpha_1, \dots, \alpha_i^\flat, \dots, \alpha_{k+\ell}).$$

In this way $F^\flat$ is a $(k - 1, \ell + 1)$-tensor.

Note that $X \in \mathfrak{X}(M)$ is a $(1, 0)$-tensor and this definition makes $X^\flat$ a $(0, 1)$-tensor (covector field) by

$$X^\flat(Y) = X(Y^\flat) = Y^\flat(X) = \widehat{g}(Y)(X) = g(X, Y)$$

which is the same as our previous definition of $X^\flat$.

In terms of a local frame, $F^\sharp$ is obtained by multiplying the components of F with g^{pq} where we set either p or q to be the index of F which we raise. Similar for $F^\flat$. For example, if we have a mixed 3-tensor

$$A = A_i{}^j{}_k \varepsilon^i \otimes E_j \otimes \varepsilon^k,$$

and we want to lower the middle index to obtain a covariant 3-tensor $A^\flat$, we get

$$A^\flat = A_{ijk} \varepsilon^i \otimes \varepsilon^j \otimes \varepsilon^k, \qquad A_{ijk} = g_{j\ell} A_i{}^\ell{}_k.$$

While just writing $F^\sharp, F^\flat$ is ambiguous, as we have not specified which index to raise/lower, we will often omit the index from our notation and instead specify it in words when necessary.

The trace of a tensor

We turn to another application of the musical isomorphisms. Recall that $T^{(1,1)}V = V \otimes V^* \cong \hom(V, V)$, so that we can unambiguously take the trace of a $(1, 1)$-tensor. Explicitly, if $F = F^i_j E_i \otimes \varepsilon^j$ is a $(1, 1)$-tensor, then its trace is the sum

$$\operatorname{tr}(F) = F^i_i.$$

We can extend this to mixed tensors: if F is a mixed $(k + 1, \ell + 1)$-tensor, for any choice of a covariant index and contravariant index we can form F's trace. We can assume that the choice of indices are the last ones; then for covectors $\omega^1, \dots, \omega^k$ and vectors $v_1, \dots, v_\ell$, we have a $(1, 1)$-tensor

$$F(\omega^1, \dots, \omega^k, \bullet, v_1, \dots, v_\ell, \bullet) \in T^{(1,1)}V.$$

Thus we can define $\operatorname{tr}(F)(\overline{\omega}, \overline{v})$ to be the trace of this tensor. This defines a (k, ℓ)-tensor. If F has components $F^{I,i}_{J,j}$, then $\operatorname{tr}(F)$'s components are the sums $F^{I,m}_{J,m}$.

So tr defines a bundle morphism

$$\operatorname{tr} \colon T^{(k+1,\ell+1)}TM \to T^{(k,\ell)}TM.$$

Thus far we have not had to use the musical morphisms. But if we have a covariant k-tensor h with $k \geq 2$, then $h^\sharp$ is a $(1, k-1)$-tensor for any choice of index to raise. We can then take its trace, resulting in a covariant $(k-2)$-tensor,

$$\mathrm{tr}_g h = \mathrm{tr}(h^\sharp).$$

If h's components are $h = h_I \varepsilon^I$, then $h^\sharp = g^{ij} h_{i_1,\dots,i_{k-1},i} E_j \otimes \varepsilon^{i_1} \otimes \cdots \otimes \varepsilon^{i_{k-1}}$. Thus $\mathrm{tr}_g h$ has components

$$(\mathrm{tr}_g h)_{i_1,\dots,i_{k-2}} = (h^\sharp)^j_{i_1,\dots,i_{k-2},j} = g^{ij} h_{i_1,\dots,i_{k-2},j}.$$

For example if h is a 2-covector field, so $h = h_{i,j}\varepsilon^i \otimes \varepsilon^j$ then $h^\sharp = g^{i\ell} h_{\ell,j} E_i \otimes \varepsilon^j$ so that

$$\mathrm{tr}_g h = g^{i\ell} h_{\ell i}$$

is a smooth function.

Note that we can take the trace of all sorts of tensors. For example, a tensor valued in another manifold. We will revisit this when necessary, e.g. with mean curvature.

Metrics on tensors

We can also use the musical isomorphisms to define the inner product of tensors. If M is Riemannian, then we can define an inner product on the cotangent space $T_p^* M$ for $p \in M$ by

$$\langle \omega, \eta \rangle_g = \langle \omega^\sharp, \eta^\sharp \rangle_g.$$

Note that applying this inner product on smooth covector fields results in a smooth real-valued function (by smoothness of g).

In terms of a local frame (E_i) with dual coframe (ε^j), if $\omega = \omega_i \varepsilon^i$ and $\eta = \eta_j \varepsilon^j$, then

$$\langle \omega, \eta \rangle_g = \langle g^{ik}\omega_k E_i, g^{j\ell}\eta_\ell E_j \rangle = g^{ik} g^{j\ell} \omega_k \eta_\ell g_{ij} = g^{ij} \omega_i \eta_j$$

where the last equality is because g^{ik}, g_{ij} are inverses. Using our convention for raising indices, where $\omega^i = g^{ij}\omega_j$, we see that

$$\langle \omega, \eta \rangle = \omega^i \eta_i = \omega_i \eta^i.$$

Note that it is not the case in general that $(\varepsilon^i)^\sharp = E_i$. This is true iff (E_i), equivalently (ε^i), is orthonormal. Indeed, $(\varepsilon^i)^\sharp = g^{ij} E_j$ is equal to E_i iff g^{ij} is the identity, iff g_{ij} is, which means (E_i) is orthonormal.

We can naturally define a metric on a vector bundle $E \to M$ to be a metric on each fiber E_p which varies smoothly as p does, in the sense that for any local sections σ, τ, $\langle \sigma, \tau \rangle$ is a smooth function. Equivalently, this is just a section of $\Gamma(E^* \otimes E^*)$ which is symmetric and positive-definite on each fiber.

If $E_1, E_2 \to M$ are vector bundles over M with smooth metrics, we can define a smooth metric on $E_1 \otimes E_2$ by

$$\langle \alpha \otimes \beta, \alpha' \otimes \beta' \rangle = \langle \alpha, \alpha' \rangle \cdot \langle \beta, \beta' \rangle, \qquad \alpha, \alpha' \in \Gamma(E_1), \quad \beta, \beta' \in \Gamma(E_2).$$

This can be viewed as being obtained via the isomorphism

$$\Gamma((E_1 \otimes E_2)^* \otimes (E_1 \otimes E_2)^*) \cong \Gamma(E_1^* \otimes E_1^*) \otimes \Gamma(E_2^* \otimes E_2^*).$$

Since we have metrics on TM, T^*M, by this result we can define a unique inner product on the tensor bundles $T^{(k,\ell)}M$. It is given uniquely by

$$\langle \alpha_1 \otimes \cdots \otimes \alpha_{k+\ell}, \beta_1 \otimes \cdots \otimes \beta_{k+\ell} \rangle = \langle \alpha_1, \beta_1 \rangle \cdots \langle \alpha_{k+\ell}, \beta_{k+\ell} \rangle$$

where α_i, β_i are vector or covector fields, where appropriate. We see that if F, G are (k,ℓ)-tensors with components F^I_J, G^I_J, then

$$\langle F, G \rangle = g_{i_1,r_1} \cdots g_{i_k,r_k} g^{j_1,s_1} \cdots g^{j_\ell,s_\ell} F^{i_1,\dots,i_k}_{j_1,\dots,j_\ell} G^{r_1,\dots,r_k}_{s_1,\dots,s_\ell}.$$

13 The Exponential Map

13.1 The exponential map

13.1.1 Definition

Let G be a Lie group, a **one-parameter subgroup** is a Lie group morphism $\gamma\colon \mathbb{R} \to G$.

Note that a one-parameter subgroup is not a subgroup of G, it is a morphism, but its image is a subgroup (but we care about γ too).

13.1.2 Theorem

Let G be a Lie group, then the one-parameter subgroups of G are precisely the maximal integral curves of left-invariant vector fields starting at $e \in G$.

Proof

If γ is a maximal integral curve of $X \in \mathrm{Lie}(G)$ starting at e, then since left-invariant vector fields are complete, γ is defined on all of $\mathbb{R}$. Since X is L_g-related to itself for all $g \in G$, this means that $L_g \circ \gamma$ is integral for X starting at g. In particular $L_{\gamma(s)} \circ \gamma$ is integral for X starting at $\gamma(s)$. But $t \mapsto \gamma(t+s)$ is as well, so $\gamma(t+s) = \gamma(s) \circ \gamma(t)$, meaning γ is a Lie group morphism.

Conversely if γ is a one-parameter subgroup, let $X = \gamma_*(d/dt)$ be the pushforward of $d/dt \in \mathfrak{X}(\mathbb{R})$. Since d/dt is left-invariant on $\mathbb{R}$, X is left-invariant on G. We just need to show that γ is integral for X. Now,

$$\gamma'(t_0) = d\gamma_{t_0}\left(\left.\frac{d}{dt}\right|_{t_0}\right) = X_{\gamma(t_0)}$$

as desired. ■

Given $X \in \mathrm{Lie}(G)$, the one-parameter subgroup of G defined as above is said to be **generated** by it. Because X is determined by its value at the identity, a one-parameter subgroup is determined by its velocity at e. So we have correspondences

$$\{\text{one-parameter subgroups of } G\} \leftrightarrow \mathrm{Lie}(G) \leftrightarrow T_eG.$$

13.1.3 Proposition

For any $A \in \mathfrak{gl}_n(\mathbb{R})$ define

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

This converges to an invertible matrix in $\mathrm{GL}_n(\mathbb{R})$, and the one-parameter subgroup of $\mathrm{GL}_n(\mathbb{R})$ generated by A is $\gamma(t) = e^{tA}$.

Proof

See homework. ■

13.1.4 Proposition

Let $H \subseteq G$ be a Lie subgroup. Then the one-parameter subgroups of H are precisely the one-parameter subgroups of G whose initial velocities lie in T_eH.

Proof

If $\gamma\colon \mathbb{R} \to H$ is a one-parameter subgroup, then composing with the inclusion $H \hookrightarrow G$ gives a one-parameter subgroup of G whose initial velocity is $d\iota_e(\gamma'(0)) \in d\iota_e(T_eH)$ which we identify with $T_eH \subseteq T_eG$.

Conversely if $\gamma\colon \mathbb{R} \to G$ is a one-parameter subgroup whose initial velocity lies in T_eH, let $\widetilde{\gamma}\colon \mathbb{R} \to H$ be the unique one-parameter subgroup whose initial velocity is the same as γ's. By the previous paragraph, composing $\iota \circ \widetilde{\gamma}$ gives a one-parameter subgroup of G with the same initial velocity as γ. By uniqueness, $\iota \circ \widetilde{\gamma} = \gamma$. ∎

13.1.5 Example

For a Lie subgroup H of $\mathrm{GL}_n(\mathbb{R})$, its one-parameter subgroups are curves of the form e^{tA} where $A \in \mathrm{Lie}(H)$ (since the initial velocity of e^{tA} is A).

13.1.6 Definition

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The **exponential map** of G is the map $\exp\colon \mathfrak{g} \to G$ defined by

$$\exp X = \gamma(1)$$

where $X \in \mathfrak{g}$ is left-invariant and γ is the one-parameter subgroup generated by X.

13.1.7 Proposition

Let G be a Lie group, then for any $X \in \mathrm{Lie}(G)$, $\gamma(t) = \exp(tX)$ is the one-parameter subgroup of G generated by X.

Proof

Let $\gamma\colon \mathbb{R} \to G$ be the one-parameter subgroup of G generated by X, so by the rescaling lemma $\widetilde{\gamma}\colon t \mapsto \gamma(ts)$ is the integral curve of sX starting at e. Thus

$$\exp(sX) = \widetilde{\gamma}(1) = \gamma(s),$$

as desired. ∎

13.1.8 Example

For $G = \mathrm{GL}_n(\mathbb{R})$ this shows that the exponential map is $\exp(A) = e^A$, explaining the name.

13.1.9 Proposition

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Because $\mathfrak{g} \cong T_eG$ is Euclidean, it has a unique smooth structure.

(1) The exponential map is smooth.

(2) For any $X \in \mathfrak{g}$ and $s, t \in \mathbb{R}$, $\exp(s+t)X = \exp(sX)\exp(tX)$.

(3) For any $X \in \mathfrak{g}$, $(\exp X)^{-1} = \exp(-X)$.

(4) For any $X \in \mathfrak{g}$ and $n \in \mathbb{Z}$, $(\exp X)^n = \exp(nX)$.

(5) The differential $(d\exp)_0 \colon T_0\mathfrak{g} \to T_eG$ is the identity map, under the canonical identification of $T_0\mathfrak{g}$ and T_eG with $\mathfrak{g}$.

(6) The exponential restricts to a diffeomorphism from a neighborhood of $0 \in \mathfrak{g}$ to a neighborhood of $e \in G$.

(7) If H is another Lie group with Lie algebra $\mathfrak{h}$, and $\Phi \colon G \to H$ a Lie group morphism, the diagram commutes

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "\Phi_*"'] \arrow[d, "\exp"'] & \mathfrak{h} \arrow[d, "\exp"'] \\ G \arrow[r, "\Phi"] & H \end{tikzcd}$$

In other words, exp is a natural map from the Lie algebra functor (mapping $G \mapsto \mathfrak{g}$) to the identity functor of Lie groups.

(8) The flow θ of $X \in \mathfrak{g}$ is given by $\theta_t = R_{\exp(tX)}$ (right multiplication by $\exp(tX)$).

Proof

For $X \in \mathfrak{g}$ denote its flow by $\theta_{(X)}$. For the first point, we need to show that $\theta_{(X)}^{(e)}(1)$ depends smoothly on X. Define a vector field Ξ on $G \times \mathfrak{g}$ by

$$\Xi_{(g,X)} = (X_g, 0) \in T_gG \oplus T_X\mathfrak{g} \cong T_{(g,X)}(G \times \mathfrak{g}).$$

This is smooth, as we show now. Let $X_1, \dots, X_n$ be a basis for $\mathfrak{g}$ and let (x^i) be the corresponding global coordinates of $\mathfrak{g}$ (the coordinates represented by this basis). Let (w^i) be any local coordinates for G, then for $f \in C^\infty(G \times \mathfrak{g})$ we have

$$\Xi f(w^i, x^i) = \Xi_{(w^i, X^i)} f = ((x^i X_i)_{w^i}, 0) f = x^i X_i f(w^j).$$

This is clearly a smooth function, so Ξ is smooth.

Let Θ be Ξ's flow, we can see that

$$\Theta_t(g, X) = (\theta_{(X)}(t, g), X),$$

since differentiating this with respect to t gives $((\theta_{(X)}^{(g)})'(t_0), 0) = (X_{\theta_{(X)}^{(g)}(t_0)}, 0)$ which is equal to Ξ at $\Theta_{t_0}(g, X)$.

This is smooth as flow is smooth, and so $\Theta_1(e, X) = (\theta_{(X)}^{(e)}(1), X)$ is smooth, as desired.

The next three points are clear, since $t \mapsto \exp(tX)$ is a one-parameter subgroup, so a group morphism.

To prove (5), let $X \in \mathfrak{g}$ be arbitrary and $\sigma \colon \mathbb{R} \to \mathfrak{g}$ be given by $\sigma(t) = tX$. Then $\sigma'(0) = X$ (under the identification $T_0\mathfrak{g} \cong \mathfrak{g}$, σ and X are identified),

$$(d\exp)_0(X) = (d\exp)_0(\sigma'(0)) = (\exp \circ \sigma)'(0) = \left.\frac{d}{dt}\right|_{t=0} \exp(tX) = X,$$

since $\exp(tX)$ is the one-parameter subgroup of X.

Part (6) is due to the inverse function theorem.

To prove naturality we need to show $\exp(\Phi_* X) = \Phi(\exp X)$ for all $X \in \mathfrak{g}$. We show the more general

$$\exp(t\Phi_* X) = \Phi(\exp(tX))$$

for all $t \in \mathbb{R}$. The left-hand side is the one-parameter subgroup generated by $\Phi_* X$. Let $\sigma(t) = \Phi(\exp(tX))$; we want to show that it is also the one-parameter subgroup of $\Phi_* X$, i.e. it is a Lie group morphism satisfying $\sigma'(0) = (\Phi_* X)_e$. As the composition of Lie group morphisms, it itself is a Lie group morphism. And

$$\sigma'(0) = d\Phi_e\left(\left.\frac{d}{dt}\right|_{t=0} (\exp(tX) \right) = d\Phi_e(X_e) = (\Phi_* X)_e.$$

For the final point we want to show $(\theta_{(X)})_t = R_{\exp(tX)}$. We use the fact that L_g maps integral curves of X to integral curves of X, so $t \mapsto L_g(\exp(tX))$ is an integral curve of X starting at g, so it is equal to $\theta^{(g)}_{(X)}$. Thus

$$R_{\exp(tX)}(g) = g\exp(tX) = L_g(\exp(tX)) = \theta^{(g)}_{(X)}(t) = (\theta_{(X)})_t(g)$$

as desired. ■

Note:
Importantly, the above proposition does not imply $\exp(X+Y) = \exp(X)\exp(Y)$. As it turns out, when G is connected this is only true if it is Abelian.

13.2 Normal subgroups

13.2.1 Lemma

Let G be a connected Lie group, and $H \subseteq G$ a connected Lie subgroup. Let $\mathfrak{g}, \mathfrak{h}$ be their respective Lie algebras, then H is normal in G iff for all $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$,

$$\exp(X)\exp(Y)\exp(-X) \in H.$$

Proof

Since $\exp(-X) = \exp(X)^{-1}$, if H is normal this holds. Conversely, let $V \subseteq \mathfrak{g}$ be a neighborhood of 0 and $W \subseteq G$ such that $\exp\colon V \to W$ is a diffeomorphism. Since the exponential of H is the restriction of that of G, shrinking V if necessary, the restriction of exp to $V \cap \mathfrak{h}$ is a diffeomorphism to a neighborhood $U_0 \subseteq H$ of e. We can further assume that V is symmetric (if $X \in V$ then $-X \in V$) by potentially further shrinking V.

This implies that $ghg^{-1} \in H$ whenever $g \in U$ and $h \in U_0$ (since they are in the image of exp). Recall that we showed that U_0 and U generate H, G respectively we get the desired result. ■

13.2.2 Definition

If $\mathfrak{g}$ is a Lie algebra, a **kernel** is a subspace $\mathfrak{h} \subseteq \mathfrak{g}$ which is closed under Lie bracketing from $\mathfrak{g}$: for $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$, $[X, Y] \in \mathfrak{h}$.

For a Lie group G and $g \in G$, the map $C_g\colon G \to G$ given by conjugation with g: $C_g(h) = ghg^{-1}$ is a Lie group isomorphism, and so if we define its induced map $\mathrm{Ad}(g) = (C_g)_*\colon \mathfrak{g} \to \mathfrak{g}$. This defines a map

$$\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$$

called the **Adjoint representation** of G.

13.2.3 Proposition

The adjoint representation is indeed a representation.

Proof

It is well-defined since C_g are Lie group isomorphisms so their induced maps are linear automorphisms of

$\mathfrak{g}$. Since $C_{g_1 g_2} = C_{g_1} \circ C_{g_2}$ we see that $\mathrm{Ad}(g_1 g_2) = \mathrm{Ad}(g_1) \circ \mathrm{Ad}(g_2)$, so it is a morphism. We need to show only that Ad is smooth.

Let $C \colon G \times G \to G$ be given by $C(g,h) = ghg^{-1}$, which is smooth. Let $X \in \mathfrak{g}$ and $g \in G$, so $\mathrm{Ad}(g)X$ is the left-invariant vector field whose value at $e \in G$ is

$$(\mathrm{Ad}(g)X)_e = ((C_g)_* X)_e = \left.\frac{d}{dt}\right|_{t=0} C_g(\exp tX) = \left.\frac{d}{dt}\right|_{t=0} C(g, \exp tX) = dC_{(g,e)}(0, X_e).$$

Because dC is smooth, this is smoothly dependent on X and g. If we choose a basis (E_i) for $\mathfrak{g}$, then $\mathrm{GL}(\mathfrak{g})$ has a basis with respect to this. Let (ε^j) be the dual basis of (E_i), then the matrix entries of $\mathrm{Ad}(g)$ is $(\mathrm{Ad}(g))_i^j = \varepsilon^j(\mathrm{Ad}(g)E_i)$. Plugging in $X = E_i$ to the above we see that these coordinates are smooth.

■

13.2.4 Definition

If $\mathfrak{g}$ is a Lie algebra, define its **adjoint representation** $\mathrm{ad} \colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ by

$$\mathrm{ad}(X)Y = [X, Y].$$

Note that its codomain is $\mathfrak{gl}(\mathfrak{g})$, not $\mathrm{GL}(\mathfrak{g})$.

This is a smooth map, as it is a linear map between Euclidean spaces.

13.2.5 Theorem

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Let $\mathrm{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ be its adjoint representation, then its induced Lie algebra representation $\mathrm{Ad}_* \colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ is just $\mathrm{Ad}_* = \mathrm{ad}$.

Proof

Let $X \in \mathfrak{g}$, then $\mathrm{Ad}_* X$ is determined by its value at the identity, an element of $\mathfrak{gl}(\mathfrak{g})$. Now, we have

$$(\mathrm{Ad}_* X)Y = \left(\left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX) \right) Y = \left.\frac{d}{dt}\right|_{t=0} (\mathrm{Ad}(\exp tX)Y).$$

Now $C_g = R_{g^{-1}} \circ L_g$ so that $\mathrm{Ad}(g) = (R_{g^{-1}})_* \circ (L_g)_*$, and thus

$$(\mathrm{Ad}(\exp(tX))Y)_e = dR_{\exp(-tX)} \circ dL_{\exp(tX)}(Y_e) = dR_{\exp(-tX)}(Y_{\exp(tX)}),$$

since Y is left-invariant. Recall that the flow of X is given by $\theta_t(g) = R_{\exp(tX)}(g)$ so this is equal to

$$= d(\theta_{-t})(Y_{\theta_t(e)}).$$

Thus if we take the derivative of this with respect to t and setting $t = 0$, we get

$$((\mathrm{Ad}_* X)Y)_e = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})(Y_{\theta_t(e)}) = (\mathcal{L}_X Y)_e = [X, Y]_e.$$

Since these vector fields are determined by their values at the identity, we see that $(\mathrm{Ad}_* X)Y = \mathrm{ad}(X)Y$ as desired. ■

13.2.6 Theorem

Let $H \subseteq G$ be a connected Lie subgroup, and denote the respective Lie algebras by $\mathfrak{h}, \mathfrak{g}$. Then H is a normal subgroup of G iff $\mathfrak{h}$ is an ideal of $\mathfrak{g}$.

Proof

Consider the commutative diagram

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "\mathrm{Ad}(g)"'] \arrow[d, "\exp"'] & \mathfrak{g} \arrow[d, "\exp"'] \\ G \arrow[r, "C_g"] & G \end{tikzcd}$$

for all $g \in G$. If $\mathfrak{h}$ is an ideal then for $Y \in \mathfrak{h}$ and $g = \exp X$ we see that

$$\exp(\mathrm{Ad}(\exp X)Y) = C_{\exp X}(\exp Y) = (\exp X)(\exp Y)(\exp(-X)).$$

Conversely a similar commutative diagram for $\mathrm{Ad}_* = \mathrm{ad}$ yields

$$\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}X)$$

so that

$$\mathrm{Ad}(\exp X)Y = \exp(\mathrm{ad}X)Y = \sum_{k=0}^{\infty} \frac{1}{k!}(\mathrm{ad}X)^k Y.$$

Since $\mathfrak{h}$ is an ideal, $(\mathrm{ad}X)Y = [X, Y] \in \mathfrak{h}$, and by induction $(\mathrm{ad}X)^k Y \in \mathfrak{h}$. Furthermore $\mathfrak{h}$ is closed in $\mathfrak{g}$ so that $\mathrm{Ad}(\exp X)Y \in \mathfrak{h}$, and thus its exponent is in $\mathfrak{h}$. So we have $(\exp X)(\exp Y)(\exp(-X)) \in \mathfrak{h}$, so H is normal.
Conversely if H is normal, then for $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$ then by the commutative diagram

$$\exp(\mathrm{Ad}(\exp(tX))sY) = (\exp tX)(\exp sY)(\exp tX)^{-1} \in H.$$

Since $\mathrm{Ad}(\exp(tX))$ is linear over $\mathbb{R}$, it follows that

$$\exp(s\mathrm{Ad}(\exp(tX))Y) = \exp(\mathrm{Ad}(\exp(tX))sY),$$

which is in H for all s, and thus $\mathrm{Ad}(\exp(tX))Y \in \mathfrak{h}$ (since its integral curve lies in H). By above,

$$\left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp(tX)Y) = [X, Y],$$

so that $[X, Y] \in \mathfrak{h}$ and thus $\mathfrak{h}$ is an ideal. ■

13.3 Bi-invariant metrics

In this section we discuss metrics on Lie groups. Since Riemannian metrics are typically denoted by g, we will in this section denote elements of a Lie group by φ, ψ, etc.

13.3.1 Definition

Let G be a Lie group. A Riemmanian metric g on G is **left-invariant** if $L_\varphi^* g = g$ for all $\varphi \in G$. Similarly we define **right-invariant** Riemannian metrics, and **bi-invariant metrics** are metrics which are both left- and right-invariant.

13.3.2 Lemma

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Then a Riemannian metric g on G is left-invariant iff for all $X, Y \in \mathfrak{g}$ the function $g(X, Y) \in C^\infty(G)$ is constant.

Proof

Recall that $g(X, Y)(\varphi) = g_\varphi(X_\varphi, Y_\varphi)$; we claim that if g is left-invariant this is equal to $g_e(X_e, Y_e)$. Indeed, by left-invariance of g, X, Y,

$$g_e(X_e, Y_e) = (L_\varphi^* g)_e(X_e, Y_e) = g_\varphi(d(L_\varphi)_e X_e, d(L_\varphi)_e Y_e) = g_\varphi(X_\varphi, Y_\varphi),$$

as desired. The same computation shows that $L_\varphi^* g = g$ when $g(X, Y)$ is constant. ■

From this lemma we deduce that every left-invariant Riemannian metric arises uniquely from an inner product on $\mathfrak{g}$. That is, $g \mapsto g_e$ is a bijection.

13.3.3 Definition

Let V be a Euclidean space, $\mathrm{GL}(V)$ be its group of linear automorphisms. Let $H \subseteq \mathrm{GL}(V)$ be a subgroup, then we say an inner product $\langle \bullet, \bullet \rangle$ is ***H*-invariant** if $\langle h\bullet, h\bullet \rangle = \langle \bullet, \bullet \rangle$ for all $h \in H$.

13.3.4 Proposition

Let G be a Lie group with Lie algebra $\mathfrak{g}$, and g a left-invariant Riemannian metric on G. Let $\langle \bullet, \bullet \rangle$ denote its associated inner product on $\mathfrak{g}$ (i.e. g_e). Then g is bi-invariant iff $\langle \bullet, \bullet \rangle$ is $\mathrm{Ad}(G) \subseteq \mathrm{GL}(\mathfrak{g})$-invariant.

Proof

For $\varphi \in G$ recall that $C_\varphi = R_{\varphi^{-1}} \circ L_\varphi$. So for $X \in \mathfrak{g}$ we have that

$$\mathrm{Ad}(\varphi)X = (R_{\varphi^{-1}})_*(L_\varphi)_* X = (R_{\varphi^{-1}})_* X,$$

by left-invariance of X. Thus for all $\psi \in G$ and $X, Y \in \mathfrak{g}$,

$$(R_{\varphi^{-1}}^* g)_\psi(X_\psi, Y_\psi) = g_{\psi\varphi^{-1}}((R_{\varphi^{-1}})_* X_{\psi\varphi^{-1}}, (R_{\varphi^{-1}})_* Y_{\psi\varphi^{-1}}) = g_{\psi\varphi^{-1}}(\mathrm{Ad}(\varphi)X_{\psi\varphi^{-1}}, \mathrm{Ad}(\varphi)Y_{\varphi\psi^{-1}}).$$

By left-invariance of g this is just equal to

$$\langle \mathrm{Ad}(\varphi)X, \mathrm{Ad}(\varphi)Y \rangle.$$

So if the inner product is $\mathrm{Ad}(G)$-invariant we see that $(R_{\varphi^{-1}})^* g = g$ for all φ so that g is right-invariant as well, so bi-invariant.

Conversely if g is bi-invariant then we have that $(R_{\varphi^{-1}})^* g = g$. So setting $\psi = e$ we see that $g_e = \langle \mathrm{Ad}(\varphi)\bullet, \mathrm{Ad}(\varphi)\bullet \rangle$ so that g_e is $\mathrm{Ad}(G)$-invariant. ■

13.3.5 Lemma

Let V be a Euclidean space, and H a subgroup of $\mathrm{GL}(V)$. Then there is an H-invariant inner product on V iff H has compact closure in $\mathrm{GL}(V)$.

In the following proof we integrate differential forms, see the later section on that for details.

Proof

First if there is an H-invariant inner product $\langle \bullet, \bullet \rangle$ on V, then H is contained in the subgroup $\mathrm{O}(V) \subseteq \mathrm{GL}(V)$ of linear automorphisms of V which are orthogonal with respect to this inner product. Choosing an orthonormal basis of V yields a Lie group isomorphism $\mathrm{O}(V) \cong \mathrm{O}(n) \subseteq \mathrm{GL}_n(\mathbb{R})$. Now $\mathrm{O}(n)$ is compact in $\mathrm{GL}_n(\mathbb{R})$, because it is closed and bounded (the norm of an orthogonal matrix is $\mathrm{tr}(A^\top A) = \mathrm{tr}(I) = n$) and closed bounded subspaces of Euclidean space are compact. Thus $\mathrm{O}(V)$ is compact, and so H's closure is a closed subset of $\mathrm{O}(V)$ and thus compact as well.

Conversely suppose H has compact closure K in $\mathrm{GL}(V)$. Then K is a subgroup of $\mathrm{GL}(V)$, as a simple limiting argument will show. As it is a closed subset of $\mathrm{GL}(V)$ it is an embedded Lie subgroup. Now let $\langle \bullet, \bullet \rangle_0$ be an arbitrary inner product on V, and let ω be a right-invariant non-vanishing top-form on K.

Such ω exists, as we can take (E_i) to be a global frame of right-invariant vector fields on K, then the dual coframe (ε^i) are right-invariant as well so that

$$\omega = \varepsilon^1 \wedge \cdots \wedge \varepsilon^n$$

is right-invariant and non-vanishing. So take ω to also be the orientation form of K.

Now for $x, y \in V$ define $f_{x,y} \colon K \to \mathbb{R}$ by $f_{x,y}(k) = \langle kx, ky \rangle_0$. Then define a new inner product on V by

$$\langle x, y \rangle = \int_K f_{x,y} \omega.$$

Note that this is well-defined since K is compact. Symmetry and bilinearity follow from definition. For nonzero $x \in V$, $f_{x,x} > 0$ on K so that $\langle x, x \rangle > 0$. Thus this is indeed an inner product.

It remains to show that this inner product is H-invariant. Because $H \subseteq K$, we will show it is K-invariant. For $k_0 \in K$ and $x, y \in V$ and $k \in K$ we have

$$f_{k_0 x, k_0 y}(k) = \langle k k_0 x, k k_0 y \rangle = f_{x,y} \circ R_{k_0}(k)$$

where $R_{k_0} \colon K \to K$ is right-multiplication.

Now,

$$\langle k_0 x, k_0 y \rangle = \int_K f_{k_0 x, k_0 y} \omega = \int_K (f_{x,y} \circ R_{k_0}) \omega$$

and because ω is right-invariant,

$$R_{k_0}^*(f_{x,y}\omega) = R_{k_0}^*(f_{x,y}) R_{k_0}^*(\omega) = (f_{x,y} \circ R_{k_0})\omega$$

so that

$$\langle k_0 x, k_0 y \rangle = \int_K R_{k_0}^*(f_{x,y}\omega) = \int_K f_{x,y}\omega = \langle x, y \rangle.$$

The final equality is because R_{k_0} is an orientation-preserving diffeomorphism: it is orientation-preserving because $R_{k_0}^* \omega = \omega$ as it is right-invariant, and ω is the orientation form. ∎

The following is then an immediate result of this lemma and the above proposition.

13.3.6 Theorem

A Lie group G has a bi-invariant Riemannian metric iff $\mathrm{Ad}(G)$ has compact closure in $\mathrm{GL}(\mathfrak{g})$.

Proof

By the above proposition, G has a bi-invariant metric iff there is an $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$. The above lemma shows that this is iff $\mathrm{Ad}(G)$ has compact closure in $\mathrm{GL}(\mathfrak{g})$, as desired. ∎

13.3.7 Corollary

Every compact Lie group has a bi-invariant Riemannian metric.

Proof

$\mathrm{Ad}(G)$ is the continuous image of G under Ad, and thus compact. ∎

13.3.8 Example

If G is Abelian, then $C_g = \mathrm{id}$ and thus Ad is trivial and so every left-invariant metric is bi-invariant.

14 Connections

14.1 Connections

14.1.1 Definition

Let $E \to M$ be a smooth vector bundle over M, and denote its space of sections by $\Gamma(E)$. A **connection** in E is a map

$$\nabla\colon \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$$

written $(X, Y) \mapsto \nabla_X Y$ which has the following properties:

(1) $\nabla_X Y$ is $C^\infty(M)$-linear in X:

$$\nabla_{f_1 X_2 + f_2 X_2} Y = f_1 \nabla_{X_1} Y + f_2 \nabla_{X_2}, \qquad f_1, f_2 \in C^\infty(M).$$

(2) $\nabla_X Y$ is $\mathbb{R}$-linear in Y:

$$\nabla_X (a_1 Y_1 + a_2 Y_2) = a_1 \nabla_X Y_1 + a_2 \nabla_X Y_2, \qquad a_1, a_2 \in \mathbb{R}.$$

(3) ∇ satisfies the product rule: for $f \in C^\infty(M)$:

$$\nabla_X (fY) = f \nabla_X Y + (Xf) Y.$$

Note:
Alternatively, we can define a connection to be an $\mathbb{R}$-linear map

$$\widetilde{\nabla}\colon \Gamma(E) \to \Gamma(T^* M \otimes E)$$

which satisfies the product rule:

$$\widetilde{\nabla}(fY) = f \widetilde{\nabla} Y + df \otimes Y.$$

This is because

$$\Gamma(T^* M \otimes E) \cong \Gamma(TM)^* \otimes \Gamma(E) \cong \hom(\mathfrak{X}(M), \Gamma(E))$$

naturally.

14.1.2 Lemma

If ∇ is a connection over E, and $X, \widetilde{X} \in \mathfrak{X}(M)$ and $Y, \widetilde{Y} \in \Gamma(E)$ are sections which agree on a neighborhood of $p \in M$, then

$$\nabla_X Y|_p = \nabla_{\widetilde{X}} \widetilde{Y}|_p.$$

Proof

By linearity it is sufficient to assume $X = 0$ or $Y = 0$ in a neighborhood of p and show that in each case $\nabla_X Y = 0$. For Y, suppose Y is zero in a neighborhood U of p. Let ψ a bump function which is 1 at p and supported in U. Thus $\varphi Y = 0$, and so by the axioms

$$0 = \nabla_X(\varphi Y) = \varphi \nabla_X Y + (X\varphi)Y.$$

Since Y is zero on φ's support, the second term is zero. So evaluating the first term at p gives

$$0 = \varphi(p)\nabla_X Y|_p = \nabla_X Y|_p,$$

as desired.

Choosing a similar bump function for X gives

$$0 = \nabla_{\varphi X} Y = \varphi \nabla_X Y$$

and evaluating at p gives the desired result again. ■

14.1.3 Lemma

Let ∇ be a connection in $E \to M$. For an open subset $U \subseteq M$ there is a unique connection ∇^U on the restricted bundle $E|_U$ such that for all $X \in \mathfrak{X}(M)$ and $Y \in \Gamma(E)$,

$$\nabla^U_{X|_U}(Y|_U) = (\nabla_X Y)|_U.$$

Proof

First we show uniqueness. For $X \in \mathfrak{X}(U)$ and $Y \in \Gamma(E|_U)$ arbitrary and $p \in U$ we use a bump function to define sections $\widetilde{X} \in \mathfrak{X}(U)$ and $\widetilde{Y} \in \Gamma(E|_U)$ such that $\widetilde{X}$ and $\widetilde{Y}$ agree with X, Y on a neighborhood of p so that

$$\nabla^U_X Y|_p = \nabla^U_{\widetilde{X}|_U}(\widetilde{Y}|_U)|_p = (\nabla_{\widetilde{X}} \widetilde{Y})|_p,$$

showing uniqueness. This also demonstrates how to construct ∇^U. By the above lemma $\nabla_{\widetilde{X}} \widetilde{Y}|_p$ is independent on choice of $\widetilde{X}, \widetilde{Y}$, and it is easy to see that the above formula defines a connection. ■

Note:
The restricted connection is usually denoted ∇ as well instead of ∇^U. This lemma shows there can be no confusion.

14.1.4 Proposition

$\nabla_X Y$ depends only on X's value at p and Y's values in any neighborhood of p.

Proof

The lemma before this showed that the connection only depends on X and Y in any neighborhood of p. To show that it depends only on X_p, it is sufficient to assume $X_p = 0$. Let U be a coordinate neighborhood of p and write $X = X^i \partial/\partial x^i$ so that $X^i(p) = 0$. Then at p by the above lemma on restricting the connection,

$$\nabla_X Y|_p = \nabla_{X^i \partial/\partial x^i} Y|_p = (X^i \nabla_{\partial/\partial x^i} Y)|_p = 0$$

since $X^i(p) = 0$. ∎

Our main concern are connections in the tangent bundle, called just **connections on** M. These are connections

$$\nabla \colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M).$$

Let (E_i) be a smooth local frame for TM on an open subset $U \subseteq M$. Then for every i, j we can represent $\nabla_{E_i} E_j$ in U over this coordinate frame. So there are smooth functions $\Gamma^k_{ij} \in C^\infty(U)$ such that

$$\nabla_{E_i} E_j = \Gamma^k_{ij} E_k.$$

These are n^3 smooth functions ($n = \dim M$) called the **connection coefficients** of ∇ in the given frame.

14.1.5 Proposition

Let ∇ be a connection on M, E_i a local frame over U, and Γ^k_{ij} be ∇'s connection coefficients in this frame. Then for $X, Y \in \mathfrak{X}(U)$ with representations $X = X^i E_i, Y = Y^j E_j$, one has

$$\nabla_X Y = (X(Y^k) + X^i Y^j \Gamma^k_{ij}) E_k.$$

Proof

Shut up and compute.

$$\begin{aligned} \nabla_X Y = \nabla_X (Y^j E_j) &= X(Y^j) E_j + Y^j \nabla_X E_j = X(Y^j) E_j + Y^j \nabla_{X^i E_i} E_j \\ &= X(Y^j) E_j + X^i Y^j \nabla_{E_i} E_j \\ &= X(Y^j) E_j + X^i Y^j \Gamma^k_{ij} E_k. \end{aligned}$$

∎

14.1.6 Proposition

Let ∇ be a connection on M, and (E_i) and $(\widetilde{E}_j)$ are two local frames over U related by $\widetilde{E}_i = A^j_i E_j$ for a matrix $A = (A^j_i)$. Let $\Gamma^k_{ij}, \widetilde{\Gamma}^k_{ij}$ be the connection coefficients relative to these frames, then

$$\widetilde{\Gamma}^k_{ij} = (A^{-1})^k_p A^q_i A^r_j \Gamma^p_{qr} + (A^{-1})^k_p A^q_i E_q(E^p_j).$$

Proof

Tedious computations. ■

14.1.7 Proposition

If TM has a global frame (E_i) then any choice of smooth functions $\Gamma_{ij}^k \in C^\infty(M)$ defines a connection on M with these as the connection coefficients relative to (E_i).

Proof

Write $X = X^i E_i$ and $Y = Y^j E_j$, and show that the formula above

$$\nabla_X Y = (X(Y^k) + X^i Y^j \Gamma_{ij}^k) E_k$$

defines a connection on M. This is clear, or an easy set of computations. ■

14.1.8 Example

$T\mathbb{R}^n$ has a global frame $\partial/\partial x^i$. So define $\Gamma_{ij}^k = 0$ for all i, j, k, this gives us the **Euclidean connection**:

$$\overline{\nabla}_X Y = X(Y^i) \frac{\partial}{\partial x^i}.$$

Note:
Linear combinations of connections need not be connections. They are obviously still linear, but need not satisfy the product rule. For example if ∇ is a connection then

$$2\nabla_X(fY) = 2(XfY + f\nabla_X Y).$$

But the product rule requires

$$(2\nabla_X(fY)) = XfY + f(2\nabla_X)Y = XfY + 2f\nabla_X Y,$$

which are clearly not equal. Similarly for sums.

14.1.9 Proposition

Every manifold admits a connection on it.

Proof

Let M be a smooth manifold and cover it with coordinate charts $\{U_\alpha\}$. These are all diffeomorphic to Euclidean space and thus can be given the (restriction of the) Euclidean connection ∇^α. Let $\{\varphi_\alpha\}$ be a partition of unity subordinate to $\{U_\alpha\}$ and define

$$\nabla_X Y = \sum_\alpha \varphi_\alpha \nabla_X^\alpha Y.$$

Since partitions of unity are locally finite, this defines a smooth vector field on M. This is clearly still linear, but we need to show the product rule, as explained in the note above.

We have

$$\nabla_X(fY) = \sum_\alpha \varphi_\alpha(XfY + f\nabla_X^\alpha Y) = \left(\sum_\alpha \varphi_\alpha\right)XfY + f\sum_\alpha \varphi_\alpha \nabla_X^\alpha Y = XfY + f\nabla_X Y$$

because the sum of a partition of unity is one. ■

14.1.10 Proposition

Let ∇^0, ∇^1 be two connections on M, then the map $D\colon \mathfrak{X}(M)\times\mathfrak{X}(M)\to\mathfrak{X}(M)$ given by

$$D(X,Y) = \nabla_X^1 Y - \nabla_X^0 Y$$

is bilinear over $C^\infty(M)$ and thus defines a $(1,2)$-tensor field called the **difference tensor**.

Proof

Additivity is clear. Multiplicity is by multiplicity in the lower input and the product rule. ■

14.1.11 Theorem

Let ∇^0 be a connection on M, then the set $\mathcal{A}(TM)$ of all connections on M is equal to

$$\mathcal{A}(TM) = \left\{\nabla^0 + D \;\middle|\; D \in \mathcal{T}^{(1,2)}TM\right\},$$

where $\nabla^0 + D$ is the connection given by $(\nabla^0 + D)_X Y = \nabla_X^0 Y + D(X,Y)$.

Proof

The proposition above shows that $\mathcal{A}(TM)$ is contained in this affine space, and it is routine to show that elements of the affine space are all connections. ■

14.2 Covariant derivatives of tensor fields

Recall that a $(1,1)$-tensor over V is just a linear map $V\to V$. Thus we can naturally define the trace of such a tensor to be the trace of its corresponding linear map. In general we define the **trace operator**

$$\mathrm{tr}\colon T^{(k+1,\ell+1)}(V)\to T^{(k,\ell)}(V)$$

by mapping $F\in T^{(k+1,\ell+1)}(V)$ to $\mathrm{tr}(F)\in T^{(k,\ell)}(V)$ where $\mathrm{tr}(F)(\omega^1,\dots,\omega^k,v_1,\dots,v_\ell)$ is defined to be the trace of the $(1,1)$-tensor

$$F(\omega^1,\dots,\omega^k,\bullet,v_1,\dots,v_\ell,\bullet)\in T^{(1,1)}(V).$$

This is clearly a tensor by linearity of the trace. In terms of a basis

$$(\mathrm{tr}F)_J^I = F_{J,m}^{I,m}$$

summing over m.

Note that we can take the trace of a tensor on different indices, not just the last ones, as long as one is covariant and the other contravariant.

14.2.1 Proposition

Let M be a smooth manifold with connection ∇. Then ∇ uniquely determines a connection on the tensor bundle $T^{(k,\ell)}TM$ also denoted ∇, such that

(1) In $T^{(1,0)}TM = TM$, ∇ agrees with the given connection.

(2) In $T^{(0,0)}TM = M \times \mathbb{R}$ (whose sections are identified with real-valued functions), ∇ is ordinary differentiatio of functions,
$$\nabla_X f = Xf.$$

(3) ∇ satisfies the product rule with respect to tensor products,
$$\nabla_X(F \otimes G) = (\nabla_X F) \otimes G + F \otimes (\nabla_X G).$$

(4) ∇ commutes with trace (contractions): if tr is the trace of a tensor on any pair of indices, one covariant and one contravariant, then
$$\nabla_X(\mathrm{tr}F) = \mathrm{tr}(\nabla_X F).$$

Furthermore the connection satisfies

(1) For a vector field Y and covector field ω,
$$\nabla_X(\omega(Y)) = (\nabla_X\omega)(Y) + \omega(\nabla_X Y).$$

(2) For $F \in \mathcal{T}^{(k,\ell)}TM$, covector fields $\omega^1, \ldots, \omega^k$, and vector fields $Y^1, \ldots, Y^\ell$,
$$\begin{aligned}(\nabla_X F)(\omega^1, \ldots, \omega^k, Y_1, \ldots, Y_\ell) &= X(F(\overline{\omega}, \overline{Y})) \\ &\quad - \sum_{i=1}^{k} F(\omega^1, \ldots, \nabla_X\omega^i, \ldots, \omega^k, \overline{Y}) - \sum_{j=1}^{\ell} F(\overline{\omega}, Y_1, \ldots, \nabla_X Y_j, \ldots, Y_\ell).\end{aligned}$$

Proof

First we show that the conditions imply the two properties at the end. Note that $\omega(Y) = \omega_i Y^i = \mathrm{tr}(\omega \otimes Y)$. Thus
$$\nabla_X\omega(Y) = \nabla_X(\mathrm{tr}(\omega \otimes Y)) = \mathrm{tr}\nabla_X(\omega \otimes Y) = \mathrm{tr}(\nabla_X\omega \otimes Y + \omega \otimes \nabla_X Y) = (\nabla_X\omega)(Y) + \omega(\nabla_X Y).$$
The second property is proven by induction, noting that
$$F(\overline{\omega}, \overline{Y}) = \mathrm{tr}\cdots\mathrm{tr}(F \otimes \overline{\omega} \otimes \overline{Y})$$
where we take $k+\ell$ traces, each operating on an upper index of F and the lower index of the corresponding covector field, or a lower index of F and the upper index of the corresponding vector field.
For uniqueness, notice that by the conditions and properties,
$$(\nabla_X\omega)(Y) = \langle \nabla_X\omega, Y\rangle = \nabla_X(\omega(Y)) - \omega(\nabla_X Y).$$
Thus we have a unique method of computing the covariant derivatives of covector fields, and by the second property this gives us an expression for $\nabla_X F$ for any tensor, showing uniqueness.
Now to show existence, define $\nabla_X\omega$ for covector fields as in the previous equation, and extend this to all tensors. Then $\nabla_X F$ is $C^\infty(M)$-multilinear (as can be checked), thus defining a smooth tensor field. Checking that ∇_X actually defines a connection is similarly routine. ∎

The expressions for the covariant derivative given in the proof/proposition above are not useful for computations.

14.2.2 Proposition

Let ∇ be a connection on M, (E_i) a local frame, (ε^j) its dual coframe, and Γ_{ij}^k the connection coefficients of ∇ with respect to (E_i). Let $X = X^i E_i$ be a local vector field.

(1) The covariant derivative of $\omega = \omega_i \varepsilon^i$ is given locally by

$$\nabla_X \omega = (X(\omega_k) - X^j \omega_i \Gamma_{jk}^i)\varepsilon^k.$$

(2) If $F \in \Gamma(T^{(k,\ell)}TM)$ is a smooth tensor field with local representation

$$F = F_J^I E_I \otimes \varepsilon^J,$$

then its covariant derivative is

$$\nabla_X F = \left(X(F_J^I) + \sum_{s=1}^{k} X^m F_J^{I_{\hat{s}}^p} \Gamma_{mp}^{i_s} - \sum_{s=1}^{\ell} X^m F_{J_{\hat{s}}^p}^{I} \Gamma_{mj_s}^{p} \right) E_I \otimes \varepsilon^J,$$

where $I_{\hat{s}}^p = (i_1, \ldots, i_{s-1}, p, i_{s+1}, \ldots, i_k)$.

Proof

We know that

$$(\nabla_X \omega)(Y) = \nabla_X(\omega(Y)) - \omega(\nabla_X Y).$$

And

$$\nabla_X \omega = (\nabla_X \omega)(E_i)\varepsilon^i = (\nabla_X(\omega(E_i)) - \omega(\nabla_X E_i))\varepsilon^i.$$

Since $\omega(E_i) = \omega_i$ and ∇_X of a map is just its derivative relative to X, and by expanding ∇_X by linearity, we get that this is equal to

$$= \big(\nabla_X(\omega_i) - \omega(X^j \Gamma_{ij}^k E_k)\big)\varepsilon^i = \big(X(\omega_i) - X^j \omega_k \Gamma_{ij}^k\big)\varepsilon^i.$$

The second formula is by considering $(\nabla_X F)(\varepsilon^I, E_J)$ via the formula in the theorem above. ∎

Note that $\nabla_\bullet F$ is $C^\infty(M)$-linear for any F, and thus defines a tensor:

14.2.3 Proposition

Let ∇ be a connection on M, and $F \in \Gamma(T^{(k,\ell)}TM)$ a tensor field. Define

$$\nabla F \colon \mathfrak{X}^*(M)^k \times \mathfrak{X}(M)^\ell \times \mathfrak{X}(M) \to C^\infty(M)$$

by

$$\nabla F(\omega^I, Y_J, X) = (\nabla_X F)(\omega^I, Y_J).$$

This defines a smooth $(k, \ell+1)$-tensor field on M called the **total covariant derivative** of F.

Proof

$\nabla_X F$ is a tensor field and thus is $C^\infty(M)$-multilinear. Since ∇ is a connection, $\nabla_\bullet F$ is $C^\infty(M)$-linear. Thus in total ∇F is $C^\infty(M)$-multilinear and thus defines a tensor field. ∎

When representing the total covariant derivative in a local coordinate frame, we have

$$\nabla F = \nabla F(\varepsilon^I, E_J, E_j) E_I \otimes \varepsilon^J \otimes \varepsilon^j = (\nabla_{E_j} F)(\varepsilon^I, E_J) E_I \otimes \varepsilon^J \otimes \varepsilon_j.$$

It is customary to denote the components by

$$F^I_{J;j} = \nabla F(\varepsilon^I, E_J, E_j),$$

since the final index (j) has different significance than the rest.
So for example, if Y is a vector field $Y = Y^i E_i$ then

$$\nabla Y = Y^i_{;j} E_i \otimes \varepsilon^j, \qquad Y^i_{;j} = \nabla_{E_j} Y(\varepsilon^i) = E_j Y^i + Y^k \Gamma^i_{jk}.$$

And similarly for covector field $\omega = \omega_i \varepsilon^i$,

$$\nabla \omega = \omega_{i;j} \varepsilon^i \otimes \varepsilon^j, \qquad \omega_{i;j} = E_j \omega_i - \omega_k \Gamma^k_{ij}.$$

In general we have

$$F^I_{J;m} = \nabla_{E_m} F(\varepsilon^I, E_J)$$

and we know the coordinate representation of $\nabla_{E_j} F$, giving us

$$F^I_{J;j} = E_m(E^I_J) + \sum_{s=1}^{k} F^{I^p_{\hat{s}}}_J \Gamma^{i_s}_{mp} - \sum_{s=1}^{\ell} F^I_{J^p_{\hat{s}}} \Gamma^p_{mj_s}.$$

Since ∇F is itself a tensor field, we can take its own total covariant derivative: $\nabla^2 F = \nabla\nabla F$, which is a $(k, \ell+2)$-tensor field. Define for $X, Y \in \mathfrak{X}(M)$,

$$\nabla^2_{X,Y} F(\bullet) = \nabla^2 F(\bullet, Y, X).$$

The reason for swapping order of X, Y is because $\nabla^2_{X,Y}$ should be read inside-out: first we differentiate with respect to Y and then X.
Note that $\nabla^2_{X,Y} F$ does not equal $\nabla_X(\nabla_Y F)$. This can be seen because the former is $C^\infty(M)$-linear in Y but the latter is not.

14.2.4 Proposition

$$\nabla^2_{X,Y} F = \nabla_X(\nabla_Y F) - \nabla_{\nabla_X Y} F.$$

Proof

Note that

$$\nabla_Y F = \mathrm{tr}(\nabla F \otimes Y),$$

where the trace is on its final two indices. This can be checked directly by considering the representations of each by the coordinate bases (both give $F^I_{J;m} Y^m$). Similarly one can check that

$$\nabla^2_{X,Y} F = \mathrm{tr}(\mathrm{tr}(\nabla^2 F \otimes X) \otimes Y),$$

the inner trace being on the last index of $\nabla^2 F$ with X's, and the outer trace the last free index (the second-to-last in $\nabla^2 F$) with Y's.
Thus

$$\nabla_X \nabla_Y F = \nabla_X(\mathrm{tr}(\nabla F \otimes Y)) = \mathrm{tr}\nabla_X(\nabla F \otimes Y) = \mathrm{tr}(\nabla_X(\nabla F) \otimes Y + \nabla F \otimes \nabla_X Y),$$

applying the same idea again gives

$$= \mathrm{tr}(\mathrm{tr}(\nabla^2 F \otimes X)) + \mathrm{tr}(\nabla F \otimes \nabla_X Y)).$$

The first term is $\nabla^2_{X,Y} F$, the second is $\nabla_{\nabla_X Y} F$, as desired. ■

14.2.5 Example

Let $u \in C^\infty(M)$ be a smooth function. Then $\nabla u \in \mathfrak{X}^*(M)$ is just du; they both act on vector fields the same way: $\nabla_X u = Xu = du(X)$.

The 2-tensor $\nabla^2 u = \nabla(du)$ is called the **covariant Hessian** of u. By the above proposition:

$$\nabla^2 u(Y,X) = \nabla^2_{X,Y} u = \nabla_X(\nabla_Y u) - \nabla_{\nabla_X Y} u = X(Yu) - (\nabla_X Y)u.$$

It is not uncommon to have tensors of type $(1,k)$, as they arise from maps $\mathfrak{X}(M)^k \to \mathfrak{X}(M)$. One example is the curvature tensor, which we will discuss. Given a map $F\colon \mathfrak{X}(M)^k \to \mathfrak{X}(M)$ we identify it with the $(1,k)$-tensor $F(\omega, \overline{X}) = \omega(F(\overline{X}))$. Its covariant derivative $\nabla_Y F$ is also a tensor of type $(1,k)$ and thus a map $\mathfrak{X}(M)^k \to \mathfrak{X}(M)$, which we describe here.

By the above formulas,

$$\begin{aligned}
(\nabla_Y F)(\omega, \overline{X}) &= Y(F(\omega,\overline{X})) - F(\nabla_Y \omega, \overline{X}) - \sum_i F(\omega, \dots, \nabla_Y X_i, \dots) \\
&= Y(\omega(F\overline{X})) - (\nabla_Y \omega)(F\overline{X}) - \sum_i \omega(F(\dots, \nabla_Y X_i, \dots)) \\
&= Y(\omega(F\overline{X})) - Y(\omega(F\overline{X})) + \omega(\nabla_Y(F\overline{X})) - \sum_i \omega(F(\dots, \nabla_Y X_i, \dots)) \\
&= \omega\left(\nabla_Y(F\overline{X}) - \sum_i F(\dots, \nabla_Y X_i, \dots)\right).
\end{aligned}$$

Thus $\nabla_Y F$ corresponds to the map $\mathfrak{X}(M)^k \to \mathfrak{X}(M)$ given by

$$(\nabla_Y F)(\overline{X}) = \nabla_Y(F\overline{X}) - \sum_i F(\dots, \nabla_Y X_i, \dots).$$

14.3 Pullbacks of vector bundles and connections

14.3.1 Definition

If $F\colon M \to N$ is a smooth map, and $\pi\colon E \to N$ a vector bundle on N, we define its **pullback** to be the vector bundle $\pi^*\colon F^*E \to M$ defined by

$$F^*E = \{(p,e) \in M \times E \mid F(p) = \pi(e)\} \subseteq M \times E,$$

where π^* is projection onto the first component.

Projecting onto the second component gives a map $f\colon F^*E \to E$ so that the following diagram commutes

$$\begin{tikzcd}
F^*E \arrow[r, "\pi^*"] \arrow[d, "f"'] & M \arrow[d, "F"] \\
E \arrow[r, "\pi"] & N
\end{tikzcd}$$

To show that this is indeed a vector bundle we need to give local trivializations. Let (U,Φ) be a local trivialization of E, where $\Phi\colon \pi^{-1}U \to U \times \mathbb{R}^k$, then $(F^{-1}U, \Phi^*)$ is a local trivialization of F^*E where

$$\Phi^*\colon (\pi^*)^{-1}(F^{-1}U) \to F^{-1}U \times \mathbb{R}^k, \qquad \Phi^*(p,e) = (p, \mathrm{proj}_2(\Phi(e))).$$

Note that this is well-defined, since $(\pi^*)^{-1}(F^{-1}U) = f^{-1}(\pi^{-1}U)$ so if (p,e) is in it then $e \in \pi^{-1}U$ and $Fp = \pi e$. Thus $\Phi(e)$ is defined, and $p \in F^{-1}U$.

Now Φ^* is also bijective. It is injective since if $\Phi^*(p, e) = \Phi^*(q, e')$ then clearly $p = q$, and so $Fp = \pi e = \pi e'$ so $e, e' \in E_{Fp}$. Then $\operatorname{proj}_2 \Phi|_{E_{Fp}}$ is a bijection $E_{Fp} \to \mathbb{R}^k$ so $e = e'$. It is surjective since if $(p, v) \in F^{-1}U \times \mathbb{R}^k$ then consider the bijection $\Phi|_{E_{Fp}}: E_{Fp} \to \mathbb{R}^k$, there is an $e \in E_{Fp}$ such that $\Phi(e) = v$. And $e \in E_{Fp}$ means that $\pi e = Fp$ so (p, e) is the preimage of (p, v).

And Φ^*'s restriction to $(\pi^*)^{-1}\{p\} = \{(p, e) \mid \pi e = Fp\}$ is

$$\Phi^*(p, e) = (p, \operatorname{proj}_2 \Phi(e))$$

whose second component is $\operatorname{proj}_2 \Phi|_{E_{Fp}}$ which is a linear isomorphism to $\mathbb{R}^k$.

If Ψ is another local trivialization, then

$$\Phi^* \circ (\Psi^*)^{-1}(p, v) = \Phi^*(p, \Psi^{-1}(Fp, v)) = (p, \operatorname{proj}_2 \Phi \Psi^{-1}(Fp, v)),$$

and the second coordinate is linear, it is equal to $\tau(Fp)v$ where τ is the transition function between Φ, Ψ.

Thus $\pi^*: F^*E \to M$ is a vector bundle over M. Moreover, f is a bundle morphism over F. Indeed,

$$f|_{F^*E_p}: F^*E_p \to E_{Fp}$$

is the map from $F^*E_p = \{(p, e) \in F^*E\}$ to E_{Fp} by mapping $(p, e) \mapsto e$. This is clearly linear.

The pullback bundle is precisely a categorical pullback. That is, it satisfies a certain universal property. Specifically, if $\tau: X \to M$ is any other vector bundle over M and $g: X \to E$ is a vector bundle morphism over F then there is a unique vector bundle morphism $h: X \to F^*E$ over M commuting everything. This is summarized in the following diagram:

$$\begin{tikzcd}
X \arrow[dr, dashed, "\exists! h"] \arrow[drr, bend left, "g"] \arrow[ddr, bend right, "\tau"'] & & \\
& F^*E \arrow[r, "f"] \arrow[d, "\pi^*"'] & E \arrow[d, "\pi"] \\
& M \arrow[r, "F"'] & N
\end{tikzcd}$$

The map h is given explicitly by $h(x) = (\tau(x), g(x))$.

Taking the pullback is functorial. To formalize this we construct a category whose objects are pairs $(F: M \to N, \pi: E \to N)$ where F is smooth and π is a vector bundle. A morphism $(F: M \to N, \pi: E \to N) \to (F': M' \to N', \pi': E' \to N')$ is then a triplet of maps $(f_M: M \to M', f_N: N \to N', f_E: E \to E')$ such that the obvious diagram commutes:

$$f_N \circ F = F' \circ f_M, \qquad \pi' \circ f_E = f_N \circ \pi,$$

and furthermore that f_E forms a vector bundle morphism, so is linear over π's fibers. This is just a subcategory of a functor category (of shape $\bullet \to \bullet \leftarrow \bullet$). Then the pullback is then just the restriction of the pullback (as a limit) to this subcategory.

Since taking limits is functorial, given such a morphism in this category there is a unique map $G: F^*E \to (F')^*E'$ which commutes with everything. Explicitly it is given by

$$G(p, e) = (f_M p, f_E e).$$

Then G forms a bundle morphism over f_M.

More useful is another form of functorality. Suppose we have the following diagram:

$$\begin{tikzcd}
& & E \arrow[d, "\pi"] \\
M \arrow[r, "F"] & N \arrow[r, "G"] & P
\end{tikzcd}$$

We can fill this in by taking the pullback twice:

$$
\begin{tikzcd}
F^*(G^*E) \arrow[r, "f"'] \arrow[d, "\pi^{**}"] & G^*E \arrow[r, "g"'] \arrow[d, "\pi^*"] & E \arrow[d, "\pi"] \\
M \arrow[r, "F"] & N \arrow[r, "G"] & P
\end{tikzcd}
$$

But we can also pull back E through $G \circ F$ directly:

$$
\begin{tikzcd}
(G \circ F)^*E \arrow[drrr, bend left, "h"] \arrow[ddr, bend right, "\tau"'] & & & \\
& F^*(G^*E) \arrow[r, "f"'] \arrow[d, "\pi^{**}"] & G^*E \arrow[r, "g"'] \arrow[d, "\pi^*"] & E \arrow[d, "\pi"] \\
& M \arrow[r, "F"] & N \arrow[r, "G"] & P
\end{tikzcd}
$$

By the universal property of the pullback, we see that there are unique maps between $(G \circ F)^*E$ and $F^*(G^*E)$, and their composition must then by the identity. So we see that $(G \circ F)^*E$ and F^*G^*E are naturally isomorphic. Since id^*E is naturally isomorphic to E (showing this explicitly is easy, but we can also appeal to the universal property). Thus the pullback is functorial in the sense that

$$(G \circ F)^*E = F^*G^*E, \qquad \mathrm{id}^*E = E$$

up to natural isomorphism.

In particular we see that if F is a diffeomorphism, F^*E is naturally isomorphic to E.

14.3.2 Proposition

If $f\colon E \to E'$ is a vector bundle isomorphism over M, then we can push forward a connection on E to E'. Explicitly, let ∇ be a connection on E, then there is a unique connection $\widetilde{\nabla}$ on E' such that

$$\widetilde{\nabla}_X Y = f \circ \nabla_X f^{-1} Y$$

for all $X \in \mathfrak{X}(M)$ and $Y \in \Gamma(E')$.

Proof

Since f is a bundle isomorphism, $f^{-1}Y$ is a section of E for $Y \in \Gamma(E')$. Then $\nabla_X f^{-1}Y$ is a section of E, and $f \circ \nabla_X f^{-1} Y$ is a section of E'. Seeing that this is a connection is simple. ∎

14.3.3 Proposition

Sections of the pullback bundle F^*E are equivalent to maps $\sigma\colon M \to E$ where $\sigma(p) \in E_{Fp}$. Viewing these as $C^\infty(M)$-modules, they are isomorphic.

Proof

We want to give a natural correspondence

$$\Gamma(F^*E) = \{\sigma\colon M \to F^*E \mid \pi^* \circ \sigma = \mathrm{id}\} \leftrightarrow \{\sigma\colon M \to E \mid \sigma(p) \in E_{Fp}\}.$$

If $\sigma \in \Gamma(F^*E)$ is a section, then define $\widetilde{\sigma} = f \circ \sigma$. This is a smooth map and $f \circ \sigma(p) \in E_{Fp}$ since

$\pi \circ f \circ \sigma(p) = F \circ \pi^* \circ \sigma(p) = F(p)$. Conversely if $\sigma\colon M \to E$ is smooth with $\sigma(p) \in E_{Fp}$ then define $\sigma^*(p) = (p, \sigma(p))$, which is well-defined since $\pi\sigma(p) = Fp$ by assumption. Then $\pi^* \circ \sigma^*(p) = \pi^*(p, \sigma(p)) = p$ so σ^* is a section.

These are inverses, since

$$(\widetilde{\sigma})^*(p) = (p, \widetilde{\sigma}(p)) = (p, f \circ \sigma(p)),$$

and this is equal to $\sigma(p)$ by definition ($\sigma(p) = (p, e)$ and $e = f(p, e)$). And

$$\widetilde{\sigma^*}(p) = f \circ \sigma^*(p) = f(p, \sigma(p)) = \sigma(p).$$

■

Note that this gives us a map $\Gamma(E) \to \Gamma(F^*E)$ mapping $\sigma \in \Gamma(E)$ to $\sigma \circ F$. In particular if $E_i\colon U \to E$ is a local frame for E, then $E_i^* = E_i \circ F$ is a local frame for F^*E in $F^{-1}U$. This is because $E_i^*|_p = E_i|_{Fp}$ are linearly independent and the rank of E and F^*E are the same.

Now consider connections on pullback bundles, so maps

$$\nabla\colon \mathfrak{X}(M) \times \Gamma(F^*E) \to \Gamma(F^*E)$$

satisfying the conditions of a connection.

14.3.4 Proposition

Let $F\colon M \to N$ be smooth, $\pi\colon E \to N$ a vector bundle on N, and $\pi^*\colon F^*E \to M$ its pullback bundle. Further let ∇ be a connection in E, then there is a unique connection ∇^* in F^*E such that

$$\nabla^*_X(Y \circ F)|_p = (\nabla_{dF_p(X_p)} Y)|_{Fp}$$

for all $X \in \mathfrak{X}(M)$, $Y \in \Gamma(E)$, and $p \in M$.

Note:

Vector fields along F of the form $Y \circ F$ where Y is a vector field on a neighborhood of $\mathrm{im} F$ are called **extendible**.

Proof

To show uniqueness, let $p \in M$ and let (E_i) be a local frame of $F(p) \in U \subseteq N$ and consider its pullback frame (E_i^*) of F^*E. Then ∇^*_X is determined by its values on $E_i^* = E_i \circ F$, which are given by the above formula. Thus it is unique, and it is easy to check that ∇^* is a connection. ■

If F is not clear from context, we also may denote the pullback of a connection ∇ by ∇^F.

While in general we cannot push a connection *forward*, we can pull back a connection via a diffeomorphism. In particular if $F\colon M \to N$ is a diffeomorphism and ∇ is a connection on $E \to M$, then we can pull back ∇ via $F^{-1}\colon N \to M$. That is, we have a connection $\nabla^{F^{-1}}$ on $(F^{-1})^*E$ given by the above proposition.

A specific case of interest are pullbacks of the tangent bundle, i.e. F^*TN.

14.3.5 Definition

Let $F\colon M \to N$ be smooth, then sections of the pullback F^*TN are called **vector fields along** F, and the space of sections is denoted

$$\Gamma(F^*TN) = \mathfrak{X}(F).$$

More generally the sections of the pullback $F^*T^{(k,\ell)}TN$ are called **tensor fields along** F.

As we saw, a vector field along F amounts to a map $X\colon M \to TN$ such that $X_p \in T_{Fp}N$ for all $p \in M$. Similarly a tensor field along F amounts to a map $X\colon M \to T^{(k,\ell)}TN$ such that $X_p \in T^{(k,\ell)}T_{Fp}N$ for all $p \in M$. The case of most importance is when $F = \gamma$ is a curve.

When $F\colon M \to N$ is a diffeomorphism, we know that TM and TN are isomorphic vector bundles, and thus TM is isomorphic to the pullback F^*TN. Explicitly the isomorphism is $h\colon F^*TN \to TM$ by

$$h(p,u) = (p, (dF^{-1})_{Fp}u), \qquad p \in M, u \in T_{Fp}N.$$

If ∇ is a connection on N, then pulling it back gives a connection on F^*TN. Since F^*TN and TM are isomorphic vector bundles over M, this defines a connection on TM via h. Explicitly this connection is given by

$$\widetilde{\nabla}_X Y = h \circ \nabla^F_X h^{-1} Y.$$

Evaluating at p gives

$$\widetilde{\nabla}_X Y|_p = h\nabla_{dF_p(X_p)}(h^{-1} \circ Y \circ F^{-1}).$$

Note that $h^{-1}YF^{-1}$ is just $dF_{F^{-1}p}Y_{F^{-1}p}$, i.e. $(F_*Y)_p$. So this is equal to

$$\widetilde{\nabla}_X Y|_p = (dF^{-1})_{Fp}\nabla_{dF_p(X_p)}(F_*Y)_p,$$

and thus

$$\widetilde{\nabla}_X Y = (F^{-1})_*\nabla_{F_*X}(F_*Y).$$

14.3.6 Proposition

If $F\colon M \to N$ is a diffeomorphism, and ∇ a connection on N, then we define a connection $F^*\nabla$ on M by

$$(F^*\nabla)_X Y = (F^{-1})_*\nabla_{F_*X}(F_*Y).$$

This is called the **pullback** of ∇.

Note then that there are two notions of a pullback. The one just introduced is a special case of the one obtained by pulling back to a pullback bundle.

14.4 Parallel transport

14.4.1 Definition

Let ∇ be a connection on M, and $F\colon A \to M$ a smooth map from an open subset $A \subseteq \mathbb{R}^n$ to M. We define the operator

$$\frac{D}{\partial t_i}\colon \Gamma(F^*T^{(k,\ell)}TM) \to \Gamma(F^*T^{(k,\ell)}TM)$$

by

$$\frac{D}{\partial t_i}X = \nabla^F_{\partial/\partial t_i}X,$$

the **covariant derivative along** F.

The case we are most interested is when $n = 1$ and $F = \gamma$ is a curve. In such a case we write D_t for $D/\partial t$.

14.4.2 Theorem (Existence and uniqueness of parallel transport)

Let $\gamma\colon (a,b) = I \to M$ be a curve, $t_0 \in (a,b)$, and $v \in T^{(k,\ell)}T_{\gamma(t_0)}M$. Then there is a unique tensor field X along γ such that $X(t_0) = v$ and $D_t X = 0$.

Proof

We prove this for when X is a vector field; the proof for tensor fields is similar but more index-heavy.

We assume that $\alpha(a,b)$ is contained in a coordinate domain (U,φ). Otherwise can cover (a,b) with subintervals which γ maps into coordinate domains of M. By uniqueness, the vector fields in the subintervals agree on their intersections, and thus we can glue them together to get the desired vector field X.

For $X \in \mathfrak{X}(\gamma)$, consider its representation relative to local the local frame $\partial_i \circ \gamma$ ($\partial_i = \partial/\partial x^i$), $X = X^i(\partial_i \circ \gamma)$. Then

$$D_t X = \nabla^\gamma_{dt}(X^i(\partial_i \circ \gamma)) = dX^i\left(\frac{d}{dt}\right)\partial_i + X^i \nabla_{d\gamma(d/dt)}\partial_i.$$

If γ's local representation is $(\gamma^1(t), \dots, \gamma^n(t))$ then

$$d\gamma(d/dt) = \dot\gamma(t) = \dot\gamma^i(t)(\partial_i \circ \gamma).$$

Thus

$$D_t X = \frac{\partial X^i}{\partial t}\partial_i + X^i\dot\gamma^j \nabla_{\partial_j}\partial_i = \frac{\partial X^i}{\partial t}\partial_i + X^i\dot\gamma^j \Gamma^k_{ij}(\alpha(t))\partial_k.$$

So $D_t X = 0$ iff

$$\frac{\partial X^i}{\partial t}(t) + X^i(t)\dot\gamma^j(t)\Gamma^k_{ij}(\alpha(t)) = 0, \qquad \text{for } k = 1, \dots, n.$$

This is a linear ODE in X, and thus has a unique solution given any initial value. ■

14.4.3 Definition

A vector field $X \in \mathfrak{X}(\gamma)$ is called **parallel along** γ if $D_t X = 0$. By the above theorem, for any initial value v, there is a unique vector field parallel along γ starting at v.

For $t_0, t_1 \in I$ define

$$P^\gamma_{t_0,t_1}\colon T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$$

to map $v \in T_{\gamma(t_0)}M$ to $X(t_1)$ where X is the unique vector field parallel along γ with $X(t_0) = v$. This is called the **parallel transport map**, X is also called the **parallel transport** of v along γ.

14.4.4 Proposition

Parallel transport is a linear isomorphism $P^\gamma_{t_0,t_1}\colon T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$.

Proof

It is easy to verify that if X, Y are the parallel transports of $v, u \in T_{\gamma(t_0)}M$ along γ, then $X + Y$ is the parallel transport of $v + u$. Similarly cX is the parallel transport of cv, proving linearity.

Furthermore if X is the parallel transport of v, then X is also the parallel transport of $u = X(t_1)$. So the parallel transport map maps u to $X(t_0) = v$, showing that $P^\gamma_{t_0,t_1}$ and $P^\gamma_{t_1,t_0}$ are inverses. ■

14.4.5 Corollary

For any curve $\gamma\colon I \to M$ there exists a global frame of γ^*TM consisting of parallel vector fields along γ.

Proof

Choose any $t_0 \in I$ and a basis $v_1, \dots, v_n$ of $T_{\gamma(t_0)}M$. Then the above proposition shows that parallel transporting v_i keeps it a basis (as it is an isomorphism). That is if we let X_i be the parallel transport of v_i, then X_i forms a global frame (in the pullback), and by definition each X_i is parallel. ■

14.5 Geodesics

14.5.1 Definition

Let ∇ be a connection on M, and $\gamma\colon I \to M$ a smooth curve. Its **acceleration** is defined to be the vector field $D_t\gamma'(t)$. γ is a **geodesic** if its acceleration is zero: $D_t\gamma'(t) = 0$.

Note that γ is a geodesic iff its derivative is parallel along γ.
In local coordinates if $\gamma(t) = (x^1(t), \dots, x^n(t))$ then

$$D_t\gamma' = D_t\big(\dot{x}^i(t)\partial_i\big) = \ddot{x}^i\partial_i + \dot{x}^i\dot{x}^j\Gamma_{ij}^k(\gamma(t))\partial_k,$$

so γ is a geodesic iff this is zero.

14.5.2 Theorem (Existence and uniqueness of geodesics)

Let ∇ be a connection on M. Then for every $p \in M$, $w \in T_pM$, and $t_0 \in \mathbb{R}$, there is an open interval $t_0 \in I \subseteq \mathbb{R}$ and a geodesic $\gamma\colon I \to M$ where $\gamma(t_0) = p$ and $\gamma'(t_0) = w$. Any two such geodesics are equal on their common domain.

Proof

Our ODE above is second-order, so we introduce auxillary variables $v^i = \dot{x}^i$. Then our geodesic equation becomes

$$\dot{x}^k(t) = v^k(t), \qquad \dot{v}^k(t) = -v^i(t)v^j(t)\Gamma_{ij}^k(x(t)).$$

Consider $(x^1, \dots, x^n, v^1, \dots, v^n)$ as coordinates on $U \times \mathbb{R}^n$, then the equation above is just the flow equation of the vector field $G \in \mathfrak{X}(U \times \mathbb{R}^n)$ given by

$$G_{(x,v)} = v^k \left.\frac{\partial}{\partial x^k}\right|_{(x,v)} - v^iv^j\Gamma_{ij}^k \left.\frac{\partial}{\partial v^k}\right|_{(x,v)}.$$

By the fundamental theorem on flows, for every $(p, w) \in U \times \mathbb{R}^n$, G has a unique integral curve $\zeta\colon I \to U \times \mathbb{R}^n$ where $\zeta(t_0) = (p, w)$. If we write $\zeta(t) = (x^1(t), \dots, x^n(t), v^1(t), \dots, v^n(t))$ then we see $\gamma(t) = (x^1(t), \dots, x^n(t))$ satisfies the existence.

Uniqueness follows because any other geodesic $\widetilde{\gamma}$ satisfying the properties defines an integral curve of G by $\widetilde{\zeta}(t) = (\gamma(t), \dot{\gamma}(t))$, and thus must be equal to ζ on their common domain by uniqueness of flows. ■

14.5.3 Definition

A **maximal** geodesic is one whose domain cannot be extended.

14.5.4 Corollary

For any initial condition $\gamma(0) = p, \gamma'(0) = w \in T_pM$, there is a unique maximal geodesic satisfying these conditions.

Proof

Let I be the union of all the open intervals containing 0 on which the initial conditions are satisfied (there is a geodesic). All the geodesics on these intervals agree on their intersections, and thus gluing them together yields a geodesic from I which is clearly maximal. ■

14.5.5 Definition

The unique maximal geodesic γ with $\gamma(0) = p, \gamma'(0) = v$ is called simpy the **geodesic with initial point** p **and initial velocity** v, and is denoted simply by γ_v (p can be recovered by $p = \pi(v)$).

Note our vector field G defined in our proof above. We claim that this vector field actually extends to a global vector field on the tangent bundle: $G \in \mathfrak{X}(TM)$. For any coordinate domain $U \subseteq M$ with coordinates (x^i) then let (x^i, v^i) denote the local coordinates for $\pi^{-1}U \subseteq TM$ so that in these coordinates G is given by the formula above:

$$G_{(x,v)} = v^k \left.\frac{\partial}{\partial x^k}\right|_{(x,v)} - v^i v^j \Gamma^k_{ij} \left.\frac{\partial}{\partial v^k}\right|_{(x,v)}.$$

To show that G defines a global vector field, we need to show that the definition above agrees on the intersection of coordinate domains. Instead of doing this forcefully, we instead will show that G acts on a smooth function $f \in C^\infty(TM)$ by

$$Gf(p, v) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma_v(t), \gamma_v'(t)).$$

Since this is independent on the coordinates, we see that G is well-defined on all of TM.

Let $\gamma_v = (x^1, \dots, x^n)$ and $\gamma_v' = (v^1, \dots, v^n)$, so $\dot{x}^i = v^i$. Then the right-hand side is

$$\left.\frac{d}{dt}\right|_{t=0} f(x^1(t), \dots, x^n(t), v^1(t), \dots, v^n(t)) = \left(\frac{\partial f}{\partial x^k}(x(t), v(t))\dot{x}^k(t) + \frac{\partial f}{\partial v^k}(x(t), v(t))\dot{v}^k\right)_{t=0}.$$

By the geodesic equations we know $\dot{x}^k = v^k$ and $\dot{v}^k = -v^i v^j \Gamma^k_{ij}(x)$, and $x(0) = p, v(0) = v$, so this is equal to

$$= \frac{\partial f}{\partial x^k}(p, v)v^k - \frac{\partial f}{\partial v^k}(p, v)v^i v^j \Gamma^k_{ij}(p).$$

Which is precisely $Gf(p, v)$, as desired.

14.5.6 Definition

The vector field $G \in \mathfrak{X}(TM)$ defined above is called the **geodesic vector field**. γ is a geodesic iff (γ, γ') is an integral curve of G iff there is an integral curve ζ of G whose projection onto M is γ.

14.5.7 Proposition

Let $M, \widetilde{M}$ be smooth manifolds, $\widetilde{\nabla}$ a connection on $\widetilde{M}$, and $\varphi\colon M \to \widetilde{M}$ a diffeomorphism. Let $\nabla = \varphi^*\widetilde{\nabla}$ be the pullback connection on M, and let $\gamma\colon I \to M$ be a smooth curve.

(1) φ commutes with covariant derivatives:

$$d\varphi \circ D_t V = \widetilde{D}_t(d\varphi \circ V),$$

where D_t is the covariant derivative along γ with respect to ∇, $\widetilde{D}_t$ along $\varphi \circ \gamma$ with respect to $\widetilde{\nabla}$, and $V \in \mathfrak{X}(\gamma)$.

(2) φ maps geodesics to geodesics: if γ is a ∇-geodesic, then $\varphi \circ \gamma$ is a $\widetilde{\nabla}$-geodesic.

(3) φ commutes with parallel transport:

$$d\varphi_{\gamma(t_1)} \circ P^{\gamma}_{t_0 t_1} = P^{\varphi\circ\gamma}_{t_0 t_1} \circ d\varphi_{\gamma(t_0)}.$$

Proof

Recall that

$$\nabla_X Y = \varphi_*^{-1}\widetilde{\nabla}_{\varphi_* X}(\varphi_* Y).$$

Thus for extendible $V \circ \gamma$,

$$D_t(V \circ \gamma) = \nabla_{\partial/\partial t} V = \varphi_*^{-1}\widetilde{\nabla}_{\varphi_*\partial/\partial t}(\varphi_* V).$$

And from this we get the desired result, recalling that

$$\varphi_* X|_p = d\varphi_{\varphi^{-1}p}(X_{\varphi^{-1}p}).$$

The second point follows from the first, since if $D_t\gamma' = 0$ then

$$\widetilde{D}_t((\varphi \circ \gamma)') = \widetilde{D}_t(d\varphi \circ \gamma') = d\varphi \circ D_t\gamma' = 0.$$

So $\varphi \circ \gamma$ is a geodesic.

And if X is the parallel transport of $v \in T_{\gamma(t_0)}M$ then $d\varphi \circ X$ is the parallel transport of $d\varphi_{\gamma(t_0)}v$ since $\widetilde{D}_t(d\varphi \circ X) = 0$ by the first point. The result follows. ■

14.6 The exponential map

14.6.1 Lemma (Rescaling lemma)

For every $p \in M, v \in T_pM, c, t \in \mathbb{R}$,

$$\gamma_{cv}(t) = \gamma_v(ct),$$

whenever either side is defined (i.e. they are defined on the same interval and are equal).

Proof

If $c = 0$ then both sides are just p for all $t \in \mathbb{R}$. Assume $c \neq 0$, then it is sufficient to assume $\gamma_v(ct)$ is defined and to show $\gamma_{cv}(t)$ is and is equal to it. The other direction is due to setting $v = v/c$ and $t = t/c$.

Suppose $\gamma = \gamma_v$'s domain is $I \subseteq \mathbb{R}$, and define $\widetilde{\gamma}\colon c^{-1}I \to M$ by $\widetilde{\gamma}(t) = \gamma(ct)$. We show that $\widetilde{\gamma}$ is a geodesic with initial point p and initial velocity cv, then by uniqueness and maximality $\widetilde{\gamma} = \gamma_{cv}$. It is immediate that $\widetilde{\gamma}(0) = \gamma(0) = p$. Furthermore,

$$\widetilde{\gamma}'(t) = c\gamma'(ct) \Longrightarrow \widetilde{\gamma}'(0) = c\gamma'(0) = cv$$

as desired. Now we need to show that $\widetilde{\gamma}$ is a geodesic.

Let D_t and $\widetilde{D}_t$ denote the differentiation operators along $\gamma, \widetilde{\gamma}$ respectively. Then in local coordinates

$$\widetilde{D}_t\widetilde{\gamma}'(t) = \left(\frac{d}{dt}\dot{\widetilde{\gamma}}^k + \Gamma^k_{ij}(\widetilde{\gamma}(t))\dot{\widetilde{\gamma}}^i\dot{\widetilde{\gamma}}^j\right)\partial_k,$$

by the chain rule,

$$= \big(c^2\ddot{\gamma}^k(ct) + c^2\Gamma^k_{ij}(\gamma(ct))\dot{\gamma}^i(ct)\dot{\gamma}^j(ct)\big)\partial_k = c^2 D_t\gamma'(ct) = 0,$$

so $\widetilde{\gamma}$ is indeed a geodesic, as desired. ■

Let $\mathcal{E} \subseteq TM$ be the subset

$$\mathcal{E} = \{v \in TM \mid \gamma_v \text{ is defined on an interval containing } [0,1]\}.$$

Then define the **exponential map**

$$\exp\colon \mathcal{E} \to M, \qquad \exp(v) = \gamma_v(1).$$

We also define, for $p \in M$, $\mathcal{E}_p = \mathcal{E} \cap T_pM$ and the **restricted exponential map** $\exp_p\colon \mathcal{E}_p \to M$ to be just the restriction of the exponential to $\mathcal{E}_p$.

Note:
The exponential map here is not in general related to the exponential of a Lie group. They are related closely for bi-invariant metrics, but otherwise they may be different. We will thus make it clear which exponential map we are dealing with.

14.6.2 Definition

A subset S of a vector space V is **star-shaped** with respect to $x \in S$ if for every $y \in S$, the line connecting x and y is contained in S.

14.6.3 Proposition

(1) For every $p \in M$, $\mathcal{E}$ is an open subset of TM containing every zero vector (i.e. for $0 \in \mathcal{E}_p$ for all p), and each $\mathcal{E}_p$ is star-shaped with respect to 0.

(2) The exponential map is smooth.

(3) For each $v \in TM$ the geodesic γ_v is given by

$$\gamma_v(t) = \exp(tv),$$

when either side is defined.

(4) For each $p \in M$, $d(\exp_p)_0 \colon T_pM \cong T_0(T_pM) \to T_pM$ is the identity map of T_pM, under the usual identification of $T_0(T_pM)$ with T_pM. Thus there is a neighborhood V of $0 \in T_pM$ and U of p such that $\exp_p \colon V \to U$ is a diffeomorphism.

Proof

Let $n = \dim M$. The rescaling lemma says that

$$\exp(cv) = \gamma_{cv}(1) = \gamma_v(c)$$

whenever either side is defined, giving us (3).
When $v \in \mathcal{E}_p$ then γ_v is defined on at least $[0, 1]$, thus for $t \in [0, 1]$ the rescaling lemma gives us

$$\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t),$$

so γ_{tv} is defined on at least $[0, 1]$ as well; $tv \in \mathcal{E}_p$, so $\mathcal{E}_p$ is star shaped with respect to 0.
Now we show that $\mathcal{E}$ is open and exp is smooth. Let G be the geodesic vector field, so by the fundamental theorem of flows there is an open subset $\mathcal{D} \subseteq \mathbb{R} \times TM$ containing $\{0\} \times TM$ and a smooth map $\theta \colon \mathcal{D} \to TM$ such that $\theta^{(p,v)}(t) = \theta(t, (p, v))$ is the unique maximal integral curve of G starting at (p, v).
If $(p, v) \in \mathcal{E}$ then γ_v is defined on at least $[0, 1]$, and thus so is the integral curve of G starting at $(p, v) \in TM$. So $(1, (p, v)) \in \mathcal{D}$, and thus there is a neighborhood of $(1, (p, v))$ in $\mathbb{R} \times TM$ contained in $\mathcal{D}$. In particular there is an open neighborhood U of (p, v) such that $\{1\} \times U \subseteq \mathcal{D}$. This means that $U \subseteq \mathcal{E}$ and so $\mathcal{E}$ is open.
Geodesics are the projections of integral curves of G, so

$$\exp(v) = \gamma_v(1) = \pi \circ \theta(1, (p, v)),$$

which is smooth.
To compute $d(\exp_p)_0(v)$, let τ be a curve in T_pM starting at 0 whose initial velocity is v. Take $\tau(t) = tv$, and thus

$$d(\exp_p)_0(v) = \left.\frac{d}{dt}\right|_{t=0} (\exp_p \circ \tau)(t) = \left.\frac{d}{dt}\right|_{t=0} \exp_p(tv) = \left.\frac{d}{dt}\right|_{t=0} \gamma_v(t) = v,$$

as desired. ■

15 The Levi Civita Connection

15.1 Metric connections

In this section we are concerned with connections on Riemannian manifolds. Recall that the Euclidean connection on $\mathbb{R}^n$ is defined by

$$\overline{\nabla}_X Y = X(Y^i)\frac{\partial}{\partial x^i}, \qquad \Gamma_{ij}^k = 0.$$

This interacts nicely with the euclidean metric:

$$\overline{\nabla}_X \langle Y, Z\rangle = \langle \overline{\nabla}_X Y, Z\rangle + \langle Y, \overline{\nabla}_X Z\rangle.$$

This can be verified easily:

$$\overline{\nabla}_X \langle Y, Z\rangle = \overline{\nabla}_X (\sum_i Y^i Z^i) = \sum_i X(Y^i Z^i) = \sum_i X(Y^i)Z^i + Y^i X(Z^i),$$

while

$$\langle \overline{\nabla}_X Y, Z\rangle = \left\langle X(Y^i)\frac{\partial}{\partial x^i}, Z\right\rangle = \sum_i X(Y^i)Z^i$$

so the identity is easily seen to be true.

15.1.1 Definition

Let (M, g) be a Riemannian manifold, and ∇ a connection on M. Then ∇ is said to be **compatible** with g, or a **metric connection** if it satisfies

$$\nabla_X \langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

for all vector fields $X, Y, Z \in \mathfrak{X}(M)$.

15.1.2 Proposition

Let (M, g) be a Riemannian manifold, and ∇ a connection on M. The following are equivalent:

(1) ∇ is compatible with g.

(2) g is parallel with respect to ∇: $\nabla g = 0$.

(3) If (E_i) is a smooth local frame, then the connection coefficients of ∇ satisfy

$$\Gamma_{ki}^\ell g_{\ell j} + \Gamma_{kj}^\ell g_{i\ell} = E_k(g_{ij}).$$

(4) If V and W are smooth vector fields along a curve γ, then

$$\frac{d}{dt}\langle V, W\rangle = \langle D_t V, W\rangle + \langle V, D_t W\rangle.$$

(5) If V, W are parallel vector fields along a curve γ, then $\langle V, W\rangle$ is constant along γ.

(6) Given any smooth curve γ, every parallel transport map along γ is a linear isometry.

(7) Given any smooth curve γ, every orthonormal basis at a point of γ can be extended to a parallel orthonormal frame along γ.

Proof

First we prove the equivalence of the first two points. Recall that

$$(\nabla g)(Y,Z,X) = (\nabla_X g)(Y,Z) = X(g(Y,Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z) = \nabla_X \langle Y, Z\rangle - \langle \nabla_X Y, Z\rangle - \langle Y, \nabla_X Z\rangle.$$

Thus ∇g is zero iff ∇ is compatible with g, as desired.

Now we prove the equivalence of (2) and (3). Recall that ∇g's components are given by

$$g_{ij;k} = E_k(g_{ij}) - \Gamma^{\ell}_{ki} g_{ij} - \Gamma^{\ell}_{kj} g_{i\ell},$$

so ∇g is zero iff the identity holds.

We prove now the equivalence of (1) and (4). If ∇ is compatible with g, let V, W be smooth vector fields along $\gamma\colon I \to M$. Let $t_0 \in I$ and choose coordinates (x^i) of a neighborhood of $\gamma(t_0)$, with $V = V^i\partial_i, W = W^j\partial_j$ in a neighborhood of t_0. Then

$$\frac{d}{dt}\langle V, W\rangle = \frac{d}{dt}(V^iW^j\langle \partial_i, \partial_j\rangle) = (\dot{V}^iW^j + V^i\dot{W}^j)\langle \partial_i, \partial_j\rangle + V^iW^j\frac{d}{dt}\langle \partial_i, \partial_j\rangle.$$

Note that for extendible $X\circ\gamma, Y\circ\gamma$,

$$\frac{d}{dt}\langle X\circ\gamma, Y\circ\gamma\rangle = \frac{d}{dt}(\langle X, Y\rangle\circ\gamma) = d_{\gamma(t)}\langle X, Y\rangle\circ\gamma'(t).$$

And since ∇ is g-compatible:

$$d_{\gamma(t)}\langle X, Y\rangle = \nabla\langle X, Y\rangle|_{\gamma(t)} = (\langle \nabla X, Y\rangle + \langle X, \nabla Y\rangle)|_{\gamma(t)},$$

since $df = \nabla f$ for all real-valued f, so this is equal to

$$\frac{d}{dt}\langle X\circ\gamma, Y\circ\gamma\rangle = \left\langle \nabla_{\gamma'(t)}X|_{\gamma(t)}, Y|_{\gamma(t)}\right\rangle + \left\langle X|_{\gamma(t)}, \nabla_{\gamma'(t)}Y|_{\gamma(t)}\right\rangle.$$

Taking $X = \partial_i$ and $Y = \nabla_j$ we have our desired result.

Conversely if (4) holds, then it holds for extendible vector fields and so if V, W are vector fields on M, we get the result by considering $V\circ\gamma, W\circ\gamma$ and recalling that

$$D_t(V\circ\gamma)(t) = \nabla_{\gamma'(t)}V.$$

Now we prove (4) implies (5) implies (6) implies (7) implies (4). If V, W are parallel along γ then $D_tV = D_tW = 0$ so $\langle V, W\rangle$ has zero derivative and is thus constant. So (4) implies (5).

Now assume (5), let $v_0, w_0 \in T_{\gamma(t_0)}M$ and let V, W be their parallel transports along γ, so $V(t_0) = v_0$ and $W(t_0) = w_0$, $P^{\gamma}_{t_0t_1}v_0 = V(t_1), P^{\gamma}_{t_0t_1}w_0 = W(t_1)$. Because V, W are parallel along γ, $\langle V, W\rangle$ is constant so

$$\left\langle P^{\gamma}_{t_0t_1}v_0, P^{\gamma}_{t_0t_1}w_0\right\rangle = \langle V(t_1), W(t_1)\rangle = \langle V(t_0), W(t_0)\rangle = \langle v_0, w_0\rangle,$$

so parallel transport is indeed an isometry.

Now if (b_i) is an orthonormal basis of $T_{\gamma(t)}M$, then extend b_i via its parallel transport. Since parallel transport is an isometry, this remains an orthonormal basis.

Finally assume (7), we will show (4). Let V, W be smooth vector fields along γ, and let (E_i) be an orthonormal frame parallel along γ which exists by (7). Write $V = V^iE_i, W = W^jE_j$. Now $D_tV = \dot{V}^iE_i$ since E_i is parallel, and thus

$$\langle D_tV, W\rangle = \dot{V}^iW^j\langle E_i, E_j\rangle = \sum_i \dot{V}^iW^i.$$

And

$$\langle V, W\rangle = \sum_i V^iW^i \Longrightarrow \frac{d}{dt}\langle V, W\rangle = \sum_i \dot{V}^iW^i + V^i\dot{W}^i = \langle D_tV, W\rangle + \langle V, D_tW\rangle$$

as desired. ∎

15.1.3 Corollary

Let (M, g) be a Riemannian manifold, ∇ a metric connection, and $\gamma\colon I \to M$ a curve.

(1) $|\gamma'(t)|$ is constant iff $D_t\gamma'(t)$ is orthogonal to $\gamma'(t)$ for all $t \in I$.

(2) If γ is a geodesic, then $|\gamma'(t)|$ is constant.

Proof

By the above proposition

$$\frac{d}{dt}\langle\gamma'(t), \gamma'(t)\rangle = 2\langle\gamma'(t), D_t\gamma'(t)\rangle,$$

and the result follows. ■

We now turn to another nice property of the Euclidean connection. Notice that for two vector fields X, Y on Euclidean space

$$[X, Y] = X(Y^i)\frac{\partial}{\partial x^i} - Y(X^i)\frac{\partial}{\partial x^i} = \overline{\nabla}_X Y - \overline{\nabla}_Y X.$$

This is a nice property, so we give it a name.

15.1.4 Definition

A connection ∇ on a manifold M is **symmetric** if

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

for all vector fields $X, Y \in \mathfrak{X}(M)$.

Note that for a connection ∇ we can define a tensor, the **torsion tensor**, $\tau\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ by

$$\tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

This is easily seen to be $C^\infty(M)$-multilinear, and thus a $(1, 2)$-tensor field. So ∇ is symmetric iff its torsion is identically zero.

Notice relative to a coordinate frame ∂_i,

$$\tau(\partial_i, \partial_j) = \Gamma_{ij}^k E_k - \Gamma_{ji}^k E_k$$

and thus τ vanishes iff $\Gamma_{ij}^k = \Gamma_{ji}^k$ for all i, j, k. So ∇ is symmetric iff $\Gamma_{ij}^k = \Gamma_{ji}^k$ relative to every coordinate frame (not necessarily every local frame).

15.1.5 Theorem (Fundamental theorem of Riemannian geometry)

On a Riemannian manifold (M, g) there exists a unique symmetric metric connection ∇. This connection is called the **Levi-Civita connection** (or **Riemannian connection**).

Proof

We show uniqueness first. If ∇ is symmetric and metric, then

$$\begin{aligned} X\langle Y, Z\rangle &= \langle\nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle, \\ Y\langle Z, X\rangle &= \langle\nabla_Y Z, X\rangle + \langle Z, \nabla_Y X\rangle, \\ Z\langle X, Y\rangle &= \langle\nabla_Z X, Y\rangle + \langle X, \nabla_Z Y\rangle. \end{aligned}$$

By symmetry we get

$$X\langle Y,Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_Z X\rangle - \langle Y, [X,Z]\rangle,$$
$$Y\langle Z,X\rangle = \langle \nabla_Y Z, X\rangle + \langle Z, \nabla_X Y\rangle - \langle Z, [Y,X]\rangle,$$
$$Z\langle X,Y\rangle = \langle \nabla_Z X, Y\rangle + \langle X, \nabla_Y Z\rangle - \langle X, [Z,Y]\rangle.$$

Adding the first two and subtracting the third yields

$$X\langle Y,Z\rangle + Y\langle Z,X\rangle - Z\langle X,Y\rangle = 2\langle \nabla_X Y, Z\rangle + \langle Y,[X,Z]\rangle + \langle Z,[Y,X]\rangle - \langle X,[Z,Y]\rangle.$$

Solving for $\langle \nabla_X Y, Z\rangle$ yields

$$\langle \nabla_X Y, Z\rangle = \frac{1}{2}\big(X\langle Y,Z\rangle + Y\langle Z,X\rangle - Z\langle X,Y\rangle - \langle Y,[X,Z]\rangle - \langle Z,[Y,X]\rangle + \langle X,[Z,Y]\rangle\big).$$

This uniquely defines ∇.

We can then use this formula to prove that the resulting connection is indeed a connection. By the formula above, if $(U,(x^i))$ is a coordinate chart then

$$\langle \nabla_{\partial_i}\partial_j, \partial_\ell\rangle = \frac{1}{2}(\partial_i\langle \partial_j,\partial_\ell\rangle + \partial_j\langle \partial_\ell,\partial_i\rangle - \partial_\ell\langle \partial_i,\partial_j\rangle).$$

Recall that $g_{ij} = \langle \partial_i, \partial_j\rangle$ and $\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k \partial_k$. So

$$\Gamma_{ij}^k g_{k\ell} = \frac{1}{2}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - g_\ell g_{ij}).$$

Multiplying by g's inverse, we see

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_i g_{i\ell} - g_\ell g_{ij}).$$

This defines a connection in each chart, and by uniqueness the connections from different charts agree on their overlaps. Thus ∇ is a well-defined connection.

By symmetry of g_{ij}, we see $\Gamma_{ij}^k = \Gamma_{ji}^k$ in every coordinate domain, so ∇ is symmetric.

To show compatibility, we just show $\Gamma_{ki}^\ell g_{\ell j} + \Gamma_{kj}^\ell g_{i\ell} = \partial_k(g_{ij})$. This is just a direct computation. ∎

The proof of the theorem gave us an explicit formula for the Levi-Civita connection, as well as explicit formulas for its connection coefficients. The connection coefficients of the Levi-Civita connection of (M,g) are called the **Christoffel symbols** of g.

15.1.6 Proposition (Naturality of the Levi-Civita connection)

The Levi-Civita connection is natural. Suppose $\varphi\colon (M,g) \to (\widetilde{M},\widetilde{g})$ is an isometry of Riemannian manifolds, and let $\widetilde{\nabla}$ be the Levi-Civita connection on $\widetilde{M}$. Then $\nabla = \varphi^*\widetilde{\nabla}$ is the Levi-Civita connection on M.

Proof

It is sufficient to show that the pullback is symetric and metric. φ being an isometry means

$$\langle Y_p, Z_p\rangle = \langle d\varphi_p Y_p, d\varphi_p Z_p\rangle = \left\langle (\varphi_* Y)_{\varphi(p)}, (\varphi_* Z)_{\varphi(p)}\right\rangle.$$

Meaning $\langle Y, Z\rangle = \langle \varphi_* Y, \varphi_* Z\rangle \circ \varphi$. Thus

$$\begin{aligned} X\langle Y, Z\rangle &= X(\langle \varphi_* Y, \varphi_* Z\rangle \circ \varphi) \\ &= (\varphi_* X\langle \varphi_* Y, \varphi_* Z\rangle) \circ \varphi \\ &= \left(\left\langle \widetilde{\nabla}_{\varphi_* X}(\varphi_* Y), \varphi_* Z\right\rangle + \left\langle \varphi_* Y, \widetilde{\nabla}_{\varphi_* X}\varphi_* Z\right\rangle\right) \circ \varphi. \end{aligned}$$

Since φ^{-1} is an isometry, by the same reason we have

$$\left\langle \widetilde{\nabla}_{\varphi_* X}(\varphi_* Y), \varphi_* Z\right\rangle \circ \varphi = \left\langle d\varphi_*^{-1}\widetilde{\nabla}_{\varphi_* X}(\varphi_* Y), Z\right\rangle$$

which is just $\langle \nabla_X Y, Z\rangle$ by definition, so we have the desired equality for a metric connection. Symmetry is due to pullbacks commuting with Lie brackets:

$$\begin{aligned} \nabla_X Y - \nabla_Y X &= \varphi_*^{-1}(\widetilde{\nabla}_{\varphi_* X}(\varphi_* Y) - \widetilde{\nabla}_{\varphi_* Y}(\varphi_* X)) \\ &= \varphi_*^{-1}[\varphi_* X, \varphi_* Y] \\ &= [X, Y]. \end{aligned}$$

As desired. ■

15.1.7 Corollary

An isometry maps geodesics to geodesics.

Proof

If φ is an isometry, recall that as a diffeomorphism it maps $\varphi^*\nabla$-geodesics to ∇-geodesics. Since these are the Levi-Civita connections of their respective manifolds, the result follows.

15.1.8 Corollary (Naturality of the exponential map)

If $\varphi\colon (M, g) \to (\widetilde{M}, \widetilde{g})$ is an isometry of Riemannian manifolds with Levi-Civita connections $\nabla, \widetilde{\nabla}$, then for every $p \in M$ the following diagram commutes:

$$\begin{tikzcd} \mathcal{E}_p \arrow[r, "d\varphi_p"'] \arrow[d, "\exp_p"'] & \widetilde{E}_{\varphi p} \arrow[d, "\exp_{\varphi(p)}"'] \\ M \arrow[r, "\varphi"] & \widetilde{M} \end{tikzcd}$$

Where $\mathcal{E}_p$ is the domain of the restricted exponential $\exp_p$.

Proof

φ maps geodesics of M to geodesics of $\widetilde{M}$. So if γ is a geodesic starting at p with initial velocity $v \in T_pM$, then $\varphi \circ \gamma$ is a geodesic starting at $\varphi(p)$ with initial velocity $d\varphi_p v$. Thus

$$\exp_{\varphi p}(d\varphi_p v) = \varphi \circ \gamma(1) = \varphi \circ \exp_p(v)$$

as desired. ■

15.1.9 Proposition

Let (M, g) be Riemannian. Then the Levi-Civita connection ∇ induces a connection on the tensor bundles compatible with the metric in the sense that

$$X\langle F, G\rangle = \langle \nabla_X F, G\rangle + \langle F, \nabla_X G\rangle$$

for every $X \in \mathfrak{X}(M)$ and $F, G \in \Gamma(T^{(k,\ell)}TM)$.

Proof

It is sufficient to prove this for tensor fields which are the tensors of vector and covector fields, $\alpha_1 \otimes \cdots \alpha_{k+\ell}$. In such a case the result follows from the formula for the inner product of tensors and the connection on tensors. ■

15.1.10 Proposition

The musical isomorphisms commute with the Levi-Civita connection. That is, if F is a tensor field with contravariant index i and $\flat$ represents lowering the index, then

$$\nabla(F^\flat) = (\nabla F)^\flat,$$

and if G has covariant index i and $\sharp$ denotes raising the ith index then

$$\nabla(G^\sharp) = (\nabla G)^\sharp.$$

Proof

Recall that $F^\flat = \operatorname{tr}(F \otimes g)$ where the trace is taken on the ith and last index of $F \otimes g$. Since g is parallel $\nabla(F \otimes g) = \nabla(F) \otimes g + F \otimes \nabla(g) = \nabla(F) \otimes g$. Thus

$$\nabla(F^\flat) = \nabla(\operatorname{tr}(F \otimes g)) = \operatorname{tr}(\nabla(F \otimes g)) = \operatorname{tr}(\nabla F \otimes g) = (\nabla F)^\flat.$$

Similar for sharp. ■

Note:
For a real-valued smooth function $f \in C^\infty(M)$, recall we defined its covariant Hessian to be

$$\operatorname{hess} f(Y, X) = \nabla^2 f(Y, X) = \nabla^2_{Y,X} f = X(Yf) - (\nabla_X Y)f.$$

Note that this is a symmetric tensor for all f iff the connection is symmetric:

$$\nabla^2 f(Y, X) - \nabla^2 f(X, Y) = X(Yf) - Y(Xf) - (\nabla_X Y)f + (\nabla_Y X)f = [X, Y]f - (\nabla_X Y - \nabla_Y X)f.$$

Now if M is Riemannian and the connection is metric, then

$$\operatorname{hess} f(Y, X) = \langle \nabla_X \operatorname{grad} f, Y\rangle,$$

where $\operatorname{grad} f = (df)^\sharp$, i.e. it is the unique vector field where $\langle \operatorname{grad} f, Y\rangle = df(Y) = Yf$. Indeed, the right-hand side is

$$\langle \nabla_X \operatorname{grad} f, Y\rangle = X\langle \operatorname{grad} f, Y\rangle - \langle \operatorname{grad} f, \nabla_X Y\rangle = X(Yf) - (\nabla_X Y)f.$$

It is easy to see that in the case of $M = \mathbb{R}^n$,

$$\operatorname{hess} f(\partial_i, \partial_j) = f_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

So this is a generalization of the Hessian matrix, justifying the name.

15.2 Normal neighborhoods

Let (M, g) be a Riemannian manifold. Recall that for $p \in M$ the restricted exponential map $\exp_p\colon \mathcal{E}_p \to M$ is defined on an open subset $\mathcal{E}_p \subseteq T_pM$, and its differential at zero is invertible. Thus there is an open neighborhood V of the origin of T_pM such that $U = \exp_p(V)$ is open and $\exp_p\colon V \to U$ is a diffeomorphism.

15.2.1 Definition

If U is a neighborhood of $p \in M$ which is the diffeomorphic image under $\exp_p$ of a *star-shaped* neighborhood of $0 \in T_pM$ then U is called a **normal neighborhood** of p.

If U is a neighborhood of $p \in M$ diffeomorphic under $\exp_p$ to an open $0 \in V \subseteq T_pM$, then consider an open ball $B_\varepsilon(0) \subseteq V$ (with respect to the Riemannian metric). Then $\exp_p(B_\varepsilon(0))$ is a normal neighborhood of p, called a **geodesic ball**. If $\overline{B}_\varepsilon(0) \subseteq V$, then $\exp_p(S_\varepsilon(0))$, where $S_\varepsilon(0) = \partial B_\varepsilon(0)$, is called the **geodesic sphere** around p.

Now if we have an orthonormal basis (b_i) for T_pM, consider the basis isomorphism $B\colon T_pM \to \mathbb{R}^n$ which maps $x^i b_i \mapsto (x^1, \dots, x^n)$. If $U = \exp_p(V)$ is a normal neighborhood of p, consider the smooth coordinate map $\varphi = B^{-1} \circ (\exp_p|_V)^{-1}\colon U \to \mathbb{R}^n$. This coordinate chart (U, φ) is called the **Riemannian normal coordinate chart** centered at p, its coordinates are the **Riemannian normal coordinates**.

Riemannian normal coordinates are unique in the following sense.

15.2.2 Proposition

Let (M, g) be a Riemannian manifold, $p \in M$, and U a normal neighborhood of p. For every normal coordinate chart on U centered at p, the coordinate basis is orthonormal at p; and for every orthonormal basis (b_i) for T_pM there is a unique normal coordinate chart (x^i) on U such that $\partial_i|_p = b_i$ for all i.

Any two normal coordinate charts $(x^i), (\widetilde{x}^j)$ are related by

$$\widetilde{x}^j = A_i^j x^i,$$

for some constant matrix $(A_i^j) \in \mathrm{O}(n)$. (This is not true for pseudo-Riemannian manifolds.)

Proof

Let φ be a normal coordinate chart on U centered at p with coordinates (x^i). This means that $\varphi = B^{-1} \circ \exp_p^{-1}$ where $B\colon \mathbb{R}^n \to T_pM$ is the basis isomorphism determined by an orthonormal basis (b_i) for T_pM.

Note that $d\varphi_p^{-1} = d(\exp_p)_0 \circ dB_0 = B$ since $d(\exp_p)_0$ is the identity and B is linear. Thus

$$\partial_i|_p = d\varphi_p^{-1} \partial_i|_0 = B(\partial_i|_0) = b_i$$

showing that $\partial_i|_p = b_i$ is indeed orthonormal.

Conversely any orthonormal basis (b_i) yields a normal coordinate chart φ given by $B^{-1} \circ \exp_p^{-1}$ where B is (b_i)'s basis isomorphism. The computation shows that $\partial_i|_p = b_i$.

If $\widetilde{\varphi} = \widetilde{B}^{-1} \circ \exp_p^{-1}$ is another such chart then

$$\widetilde{\varphi} \circ \varphi^{-1} = \widetilde{B}^{-1} \circ \exp_p^{-1} \circ \exp_p \circ B = \widetilde{B}^{-1} \circ B,$$

which is a linear isometry of $\mathbb{R}^n$ (since B maps an orthonormal basis to an orthonormal basis). Letting $A = \widetilde{B}^{-1} \circ B$ we have the desired result.

Notice then that two normal coordinates (x^i) and $(\widetilde{x}^i)$ are equal iff A is the identity which is iff $\widetilde{\varphi} = \varphi$. Thus a normal coordinate chart and normal coordinates uniquely determine one another. ■

15.2.3 Proposition

Let (M, g) be Riemannian, and $(U, (x^i))$ a normal coordinate chart centered at $p \in M$. Then,

(1) The coordinates of p are $(0, \ldots, 0)$,

(2) The components of the metric at p are $g_{ij} = \delta_{ij}$ (or $\pm\delta_{ij}$ in the pseudo-Riemannian case),

(3) For every $v = v^i \partial_i|_p \in T_pM$, the geodesic γ_v starting at p with initial velocity v is represented in normal coordinates by the line

$$\gamma_v(t) = (tv^1, \ldots, tv^n),$$

as long as t is in some interval such that $\gamma_v(I) \subseteq U$.

(4) The Christoffel symbols in these coordinates vanish at p.

(5) All the first partial derivatives of g_{ij} in these coordinates vanish at p.

Proof

Point 1 is clear, point 2 is because $\partial_i|_p$ are orthonormal as per the previous proposition. Point 3 is because $\gamma_v(t) = \exp_p(tv)$ which in these coordinates is

$$\varphi \circ \gamma_v(t) = \varphi \circ \exp_p(tv) = B^{-1}(tv) = (tv^1, \ldots, tv^n)$$

as desired.

Now for point 4, let $v = v^i \partial_i|_p \in T_pM$. Then the geodesic equation for $\gamma_v(t) = (tv^1, \ldots, tv^n)$ simplifies to

$$\Gamma^k_{ij}(tv)v^i v^j = 0.$$

Evaluating at zero shows that $\Gamma^k_{ij}(0)v^i v^j = 0$ for every k and every $v \in T_pM$. In particular if we choose $v = \partial_\ell$ for some ℓ we see that $\Gamma^k_{\ell\ell}(0) = 0$ for each k, ℓ. Then choosing $v = \partial_a + \partial_b$ and $v = \partial_a - \partial_b$ we get the desired result.

And for point 5, we have that

$$\Gamma^\ell_{ki} g_{\ell j} + \Gamma^\ell_{kj} g_{i\ell} = \partial_k(g_{ij}),$$

and the left-hand side is zero. ■

15.2.4 Definition

Geodesics starting at p and lying in a normal neighborhood of p are called **radial geodesics**, and the above proposition shows that in terms of normal coordinates such geodesics have the representation

$$\gamma_v(t) = (tv^1, \ldots, tv^n).$$

16 Differential Forms

16.1 Alternating covectors

Recall that a covariant k-tensor $\alpha \in T^k(V^*)$ is **alternating** if ${}^\sigma\alpha = \mathrm{sgn}\sigma\alpha$ for all $\sigma \in S_k$. This is equivalent to α being zero when any of its inputs are repeated. We denote the subspace of alternating covariant tensors by $\Lambda^k(V^*)$. We have a projection $\mathrm{Alt}\colon T^k(V^*) \to \Lambda^k(V^*)$ by

$$\mathrm{Alt}\alpha = \frac{1}{k!}\sum_{\sigma\in S_k}(\mathrm{sgn}\sigma)({}^\sigma\alpha).$$

16.1.1 Definition

A covariant k-tensor $\alpha \in \Lambda^k(V^*)$ is called a **k-covector**.

16.1.2 Definition

Let $I = (i_1, \ldots, i_k)$ be a multi-index, so a tuple of length k of indices. For $\sigma \in S_k$ define

$$I_\sigma = (i_{\sigma(1)}, \ldots, i_{\sigma(k)}),$$

so that $I_{\sigma\tau} = (I_\sigma)_\tau$.

For a multi-index I and a list of vectors $(v_1, \ldots, v_k)$ we write $v_I = (v_{i_1}, \ldots, v_{i_k})$. For a dual basis $(\varepsilon^1, \ldots, \varepsilon^n)$ for V^*, define ε^I by

$$\varepsilon^I(v_1, \ldots, v_k) = \det(\varepsilon^{i_k}(v_j))_j^k.$$

This is called an **elementary k-covector**.

Indeed, ε^I is a covector since it is multilinear by multilinearity of the determinant and the ε^is, and if any of its inputs are repeated then the determinant is zero.

For multi-indices I, J of length k we define

$$\delta^I_J = \det(\delta^{i_k}_{j_\ell})^k_\ell = \begin{cases} \mathrm{sgn}\sigma & \text{if neither } I \text{ nor } J \text{ have repeated indices and } J = I_\sigma \\ 0 & \text{else} \end{cases}.$$

16.1.3 Proposition

Let (E_i) be a basis for V and (ε^i) its dual basis in V^*, and let I be a multi-index of length k.

(1) If I has a repeated index $\varepsilon^I = 0$,

(2) If $J = I_\sigma$ then $\varepsilon^I = (\mathrm{sgn}\sigma)\varepsilon^J$,

(3) We have the identity $\varepsilon^I(E_J) = \delta^I_J$.

Proof

Everything here is a standard computation. ■

Call a multi-index $I = (i_1, \ldots, i_k)$ **increasing** if $i_1 < \cdots < i_k$.

16.1.4 Proposition

Let V be an n-dimensional vector space, then for any basis (ε^i) for V^* and any $k \le n$,

$$\mathcal{E} = \left\{ \varepsilon^I \;\middle|\; I \text{ is an increasing multi-index of length } k \right\}$$

forms a basis for $\Lambda^k(V^*)$ and thus

$$\dim \Lambda^k(V^*) = \binom{n}{k}.$$

If $k > n$ then $\Lambda^k(V^*) = 0$.

Proof

Clearly if $k > n$ then any covector is zero, since any of its inputs must be linearly dependent and thus give a value of zero. Let (E_i) be dual to (ε^i), and let $\alpha \in \Lambda^k(V^*)$ then write $\alpha_I = \alpha(E_I)$. Now, for the following sum over increasing indices and an arbitrary multi-index J we have

$$\sum_I \alpha_I \varepsilon^I(E_J) = \sum_I \alpha_I \delta^I_J.$$

If J has a repeated index this is zero, as $\delta^I_J = 0$ for all I, and this is equal to α_J. Otherwise there is a σ such that $J_\sigma = I$ is increasing (order by increasing value), then this is equal to

$$= (\mathrm{sgn}\sigma)\alpha_I = \alpha_J.$$

So we see that

$$\sum_I \alpha_I \varepsilon^I = \alpha$$

and thus $\mathcal{E}$ spans $\Lambda^k(V^*)$.
Now suppose there are α_I such that $\sum_I \alpha_I \varepsilon^I = 0$. Then for an increasing index J, we see that $\varepsilon^I(E_J) = \delta^I_J$ and thus

$$0 = \sum_I \alpha_I \delta^I_J = \alpha_J$$

for all J so that $\mathcal{E}$ is linearly independent. ■

16.1.5 Proposition

Let V be n-dimensional, $T\colon V \to V$ a linear operator, and $\omega \in \Lambda^n(V^*)$. Then for all $v_1, \ldots, v_n \in V$ we have

$$\omega(Tv_1, \ldots, Tv_n) = \det(T)\omega(v_1, \ldots, v_n).$$

Proof

Choose a basis (E_i) for V with dual basis (ε^i). Let (T^j_i) represent T with respect to this basis, and let $T_i = TE_i = T^j_i E_j$. By the above proposition $\omega = c\varepsilon^{1\ldots n}$ for some number c.
Since both sides of the identity are multilinear and alternating, it is sufficient to check this when $v_i = E_i$. Then

$$\begin{aligned}\omega(TE_1, \ldots, TE_n) = \omega(T_1, \ldots, T_n) = c\varepsilon^{1\ldots n}(T_1, \ldots, T_n) &= c\det(\varepsilon^i(T_j)) \\ &= c\det(T^i_j) = c\det T = \det T\omega(E_1, \ldots, E_n).\end{aligned}$$

As desired. ■

16.1.6 Definition

For $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^\ell(V^*)$ define their **wedge product** to be the $(k+\ell)$-covector:

$$\omega \wedge \eta = \frac{(k+\ell)!}{k!\ell!}\mathrm{Alt}(\omega \otimes \eta).$$

The coefficient in the above definition is to normalize it so that it behaves nicely.

16.1.7 Lemma

For a basis (ε^i) of V^* and multi-indices I, J, we have

$$\varepsilon^I \wedge \varepsilon^J = \varepsilon^{IJ},$$

where IJ denotes their concatenation.

Proof

Some tedious computations and a case analysis. ■

16.1.8 Proposition

For multicovectors $\omega, \omega', \eta, \eta', \xi$ over V, then the wedge product has the following properties:

(1) Bilinearity,

(2) Associativity,

(3) Anticommutativity: if $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^\ell(V^*)$ then

$$\omega \wedge \eta = (-1)^{k\ell}\eta \wedge \omega.$$

(4) If (ε^i) is any basis for V^* and I is any multi-index,

$$\varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k} = \varepsilon^I.$$

(5) For any 1-covectors $\omega^1, \ldots, \omega^k$ and $v_1, \ldots, v_k$,

$$\omega^1 \wedge \cdots \wedge \omega^k(v_1, \ldots, v_n) = \det(\omega^j(v_i)).$$

Proof

Bilinearity follows from the definition since the tensor and Alt are bilinear. The rest of the cases follow from analysis on elementary multicovectors by bilinearity.

Associativity follows from the above lemma, as does anticommutativity since $IJ = (JI)_\sigma$ where σ swaps the first ℓ and last k indices. This can be written as $k\ell$ transposition so it has sign $(-1)^{k\ell}$. (4) is a direct result of the above lemma via induction. Then (5) is clear by reducing it to the case of $\omega^j = \varepsilon^{i_j}$. ■

A k-covector η is **decomposable** if it can be written $\eta = \omega^1 \wedge \cdots \wedge \omega^k$ for 1-covectors ω^i. By the above proposition, ε^I are decomposable and thus every k-covector can be written as the linear combination of decomposable

k-covectors. It is easy to show that the wedge product is the unique map satisfying the above properties.

16.1.9 Definition

For an n-dimensional vector space V, define its **exterior algebra**

$$\Lambda(V^*) = \bigoplus_{k=0}^{n} \Lambda^k(V^*).$$

This is a real vector space of dimension 2^n, and has a graded algebra structure given by the wedge product.

16.1.10 Definition

For a vector $v \in V$ define a map $i_v\colon \Lambda^k(V^*) \to \Lambda^{k-1}(V^*)$ called **interior multiplication** by v as

$$i_v\omega(w_1, \ldots, w_{n-1}) = \omega(v, w_1, \ldots, w_{m-1}), \qquad \omega \in \Lambda^k(V^*),\ w_i \in V.$$

We also write

$$v \lrcorner\, \omega = i_v\omega.$$

16.1.11 Lemma

For $v \in V$, $i_v \circ i_v = 0$ and

$$i_v(\omega \wedge \eta) = (i_v\omega) \wedge \eta + (-1)^k \omega \wedge (i_v\eta).$$

Proof

Clearly $i_v \circ i_v = 0$ since this inserts v twice to an alternating tensor. For the other claim, it is sufficient to prove this for decomposable covectors. In general we can show

$$v \lrcorner\, (\omega^1 \wedge \cdots \wedge \omega^k) = \sum_{i=1}^{k} (-1)^{i-1} \omega^i(v) \omega^1 \wedge \cdots \wedge \widehat{\omega}^i \wedge \cdots \wedge \omega^k.$$

Indeed, writing $v = v_1$ and applying $v_2, \ldots, v_k$ to both sides gives

$$\omega^1 \wedge \cdots \wedge \omega^k(v_1, \ldots, v_k) = \sum_{i=1}^{k} (-1)^{i-1} \omega^i(v_1)(\omega^1 \wedge \cdots \wedge \widehat{\omega}^i \wedge \cdots \wedge \omega^k(v_2, \ldots, v_k).$$

It is easy to see that the right-hand side is the expansion of the determinant of $\omega^i(v_j)$ along the ith row, and the left-hand side is this determinant as well. ■

16.2 Differential forms

It is easy to see that $V \to \Lambda^k V^*$ is smoothly tensorial, and as discussed this means that $E \mapsto \Lambda^K E^*$ is tensorial for vector bundles over a smooth manifold M. This is the vector bundle with fibers $\Lambda^k E_p^*$. By considering the trivializations we constructed over smooth tensorial constructions, we can see that for any local frame (s_i) of E with dual frame (ε^i) in E^*, then

$$\left(\varepsilon^I \;\middle|\; I \text{ is a multi-index of length } k\right)$$

forms a local frame of $\Lambda^k E^*$; where ε^I on the fiber of p is given by $\varepsilon^I_p = \varepsilon^{i_1}_p \wedge \cdots \wedge \varepsilon^{i_k}_p$.

Similarly we have the vector bundle $\Lambda(E^*) = \bigoplus_{k=0}^m \Lambda^k(E^*)$ where m is E's rank. Then we have the isomorphism of $C^\infty(M)$-modules

$$\Gamma(\Lambda(E^*)) \cong \bigoplus_{k=0}^{m} \Gamma(\Lambda^k(E^*)).$$

This is then a graded $C^\infty(M)$-algebra with multiplication given by the wedge product. Explicitly for $\omega \in \Gamma(\Lambda^k(E^*))$ and $\eta \in \Gamma(\Lambda^\ell(E^*))$, then we define $\omega \wedge \eta \in \Gamma(\Lambda^{k+\ell}(E^*))$ pointwise by

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p.$$

Then the ε^I above is just $\varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k}$.

Our main concern is when $E = TM$.

16.2.1 Definition

A **differential k-form** of a smooth manifold is a section of $\Lambda^k T^*M$. Define

$$\Omega^k(M) = \Gamma(\Lambda^k T^*M)$$

to be the space of differential k-forms. And we define the graded algebra of differential forms

$$\Omega^*(M) = \Gamma(\Lambda(T^*M)) \cong \bigoplus_{k=0}^{n} \Omega^k(M).$$

So for any chart (x^i) of M there is a local frame of $\Omega^k(M)$ given by $(dx^I)_I$ where $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ ranges over increasing multi-indices I.

Recall that any smooth map $F\colon M \to N$ pulls back covariant tensor fields: for $A \in \mathcal{T}^k(T^*N)$ we define $F^*A \in \mathcal{T}^k(T^*M)$ by

$$(F^*A)_p(v_1, \ldots, v_k) = A_{Fp}(dF_p(v_1), \ldots, dF_p(v_n)).$$

Clearly we see that if $A = \omega$ is a differential k-form, then $F^*\omega$ is as well. So F defines a map $F^*\colon \Omega^k(N) \to \Omega^k(M)$ for all k.

16.2.2 Lemma

Let $F\colon M \to N$ be smooth.

(1) $F^*\colon \Omega^k(N) \to \Omega^k(M)$ is $\mathbb{R}$-linear.

(2) $F^*(\omega \wedge \eta) = (F^*\omega) \wedge (F^*\eta)$.

(3) In any smooth chart,

$$F^*\left(\sum_I \omega_I dy^I\right) = \sum_I (\omega_I \circ F) d(y^{i_1} \circ F) \wedge \cdots \wedge d(y^{i_k} \circ F).$$

Proof

(1) is clear as it is true for general covariant vector fields that F^* is additive and $F^*(fA) = (f \circ F)F^*(A)$. (2) is due to linearity and a computation showing it is true for $\omega = dx^I$ and $\eta = dx^J$. Then (3) follows from (1) and (2) since $F^*(df) = d(f \circ F)$. ■

16.2.3 Lemma

Let $F\colon M \to N$ be smooth mapping between n-manifolds which maps a local chart $(U, (x^i))$ into $(V, (y^j))$.

Let u be a continuous real-valued function on V. Then

$$F^*(udy^1 \wedge \cdots \wedge dy^n) = (u \circ F)(\det DF)dx^1 \wedge \cdots \wedge dx^n,$$

where DF is the Jacobian of F in these coordinates.

Proof

Since the fiber of $\Lambda^n T^* M$ is spanned by $dx^1 \wedge \cdots \wedge dx^n$ at each point, we just need to evaluate both sides at $\partial/\partial x^i$. By the previous lemma,

$$F^*(udy^1 \wedge \cdots \wedge dy^n) = (u \circ F)d(F^1) \wedge \cdots \wedge d(F^n),$$

where F^i is $y_i \circ F$, the projection of F relative to the chart. Evaluating at $\partial/\partial x^1, \ldots, \partial/\partial x^n$ shows

$$dF^1 \wedge \cdots \wedge dF^n\left(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n}\right) = \det\left(dF^j\left(\frac{\partial}{\partial x^i}\right)\right) = \det\left(\frac{\partial F^j}{\partial x^i}\right) = \det(DF).$$

This gives us the desired result. ∎

16.2.4 Corollary

If $(U, (x^i))$ and $(\widetilde{U}, (\widetilde{x}^j))$ are overlapping coordinate charts, then

$$d\widetilde{x}^1 \wedge \cdots \wedge d\widetilde{x}^n = \det\left(\frac{\partial \widetilde{x}^j}{\partial x^i}\right)dx^1 \wedge \cdots \wedge dx^n.$$

Proof

Apply the previous lemma to $F = \mathrm{id}$. ∎

Interior multiplication acts on differential forms.

16.2.5 Definition

Let $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$ be open. For $X \in \mathfrak{X}(U)$ and $\omega \in \Omega^k(U)$, define the $k-1$-differential form $i_X\omega = X \lrcorner \omega \in \Omega^{k-1}(M)$ by

$$(X \lrcorner \omega)_p = X_p \lrcorner \omega_p.$$

By considering local charts, this is smooth.

16.3 Exterior derivatives

16.3.1 Definition

We define the **exterior derivative** on $\mathbb{R}^n$ to be the operator d which maps k-forms to $(k+1)$-forms in the following way. For $\omega = \sum_I \omega_I dx^I$ (sum over I increasing) define

$$d\omega = d\sum_I \omega_I dx^I = \sum_I d\omega_I \wedge dx^I,$$

where $dx^I \in \Omega^1(\mathbb{R}^n)$ is the differential of the smooth real-valued ω_I, and the sum is over increasing I.

Since

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i,$$

we see that we can write the exterior derivative as

$$d\omega = \sum_I \sum_i \frac{\partial \omega_I}{\partial dx^i} dx^i \wedge dx^I.$$

For example, for a 1-form this is

$$d(\omega_j dx^j) = d\omega_j \wedge dx^j = \sum_{i,j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j = \sum_{i<j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j + \sum_{i>j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j,$$

by swapping i and j in the second sum and by anticommutativity, this is equal to

$$d(\omega_j dx^j) = \sum_{i<j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} \right) dx^i \wedge dx^j.$$

16.3.2 Proposition

(1) d is $\mathbb{R}$-linear.

(2) If ω is a smooth k-form and η a smooth ℓ-form over an open subset $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$, then

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^k \omega \wedge (d\eta).$$

(3) $d \circ d = 0$.

(4) d commutes with pullbacks: if $F\colon U \to V$ is a smooth map between open subsets of Euclidean or Euclidean half-space, then

$$F^*(d\omega) = d(F^*\omega)$$

for all $\omega \in \Omega^k(V)$.

Proof

(1) is clear, and (2) follows from the case $\omega = udx^I$ and $\eta = vdx^J$. But first we claim $d(udx^I) = du \wedge dx^I$ even if I is not increasing. Let $J = I_\sigma$ be increasing (if I has repeated indices then both sides are zero), then $udx^I = (\operatorname{sgn}\sigma) udx^J$ then

$$d(udx^I) = (\operatorname{sgn}\sigma) du \wedge dx^J = du \wedge dx^I$$

as desired. Thus

$$\begin{aligned} d(u\,dx^I \wedge v\,dx^J) = d(uv\,dx^I \wedge dx^J) = d(uv) \wedge dx^I \wedge dx^J &= (v\,du + u\,dv) \wedge dx^I \wedge dx^J \\ &= v\,du \wedge dx^I \wedge dx^J + u\,dv \wedge dx^I \wedge dx^J. \end{aligned}$$

The first term is equal to $d(u\,dx^I) \wedge (v\,dx^J)$, and the second is equal to $(-1)^k (u\,dx^I) \wedge d(v\,dx^J)$ by anticommutativity, as desired.

First we show (3) for a 0-form, i.e. a smooth function $u \in C^\infty(M)$. Here,

$$d(du) = d\left(\frac{\partial u}{\partial x^j}\, dx^j \right) = \frac{\partial^2 u}{\partial x^i x^j}\, dx^i \wedge dx^j = \sum_{i<j} \left(\frac{\partial^2 u}{\partial x^i x^j} - \frac{\partial^2 u}{\partial x^j x^i} \right) dx^i \wedge dx^j = 0.$$

Now, in the general case we use (2) and the case of $k = 0$,

$$d(d\omega) = d \sum_I d\omega_I \wedge dx^I = \sum_I d(d\omega_I) \wedge dx^I - \sum_I d\omega_I \wedge d(dx^I).$$

The first sum is zero since $d(d\omega_I) = 0$ by the $k = 0$ case. The second sum is zero as well, as again by (2) we can write it as the sum of terms of the form $d\omega_I \wedge dx^{i_1} \wedge \cdots \wedge d(dx^{i_j}) \wedge \cdots \wedge dx^{i_k} = 0$.
For (4) we need only consider $\omega = u\, dx^I$. The left-hand side is

$$F^*(d(u\, dx^I)) = F^*(du \wedge dx^I) = F^*(du) \wedge F^*(dx^I),$$

we know that $F^*(du) = d(u \circ F)$ and it is easy to see $F^*(dx^I) = d(x^I \circ F)$, so this is equal to

$$F^*(d(u\, dx^I)) = d(u \circ F) \wedge d(x^I \circ F).$$

While

$$d(F^*(u\, dx^I)) = d((u \circ F)\, d(x^I \circ F)) = d(u \circ F) \wedge (x^I \circ F)$$

as well. ■

This allows us to define the exterior derivative on arbitrary smooth manifolds.

16.3.3 Theorem

Let M be a smooth manifold with or without boundary. Then there are unique operators $d\colon \Omega^k(M) \to \Omega^{k+1}(M)$ called **exterior differentiation**, such that

(1) d is linear over $\mathbb{R}$.

(2) For $\omega \in \Omega^k(M)$ and $\eta \in \Omega^\ell(M)$, then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

(3) $d \circ d = 0$.

(4) For $f \in \Omega^0(M) = C^\infty(M)$, df is the differential of f.

Proof

First we prove existence. For $\omega \in \Omega^k(M)$ we define $d\omega$ using the exterior derivative of Euclidean space. That is, for a chart (U, φ) define

$$d\omega = \varphi^*\, d((\varphi^{-1})^*\omega).$$

We want to show that this is well-defined: for any other chart (V, ψ) then $\varphi \circ \psi^{-1}$ is a diffeomorphism of Euclidean space, so by above

$$(\varphi \circ \psi^{-1})^*\, d((\varphi^{-1})^*\omega) = d((\varphi \circ \psi^{-1})^*(\varphi^{-1})^*\omega) = d((\psi^{-1})^*\omega).$$

Thus

$$\varphi^*\, d((\varphi^{-1})^*\omega) = \psi^* \circ (\varphi \circ \psi^{-1})^*\, d((\varphi^{-1})^*\omega) = \psi^*\, d((\psi^{-1})^*\omega).$$

Then the properties (1) through (4) are satisfied as they are satisfied in Euclidean space.
Now we show uniqueness. If d is an operator satisfying the conditions, we will show that it is the exterior derivative already given. For a k-form ω, we show that $d\omega$ is determined locally. It is sufficient to show that if ω is zero in a neighborhood of p, then $d\omega$ is zero in that neighborhood, say U. Let $\psi \in C^\infty(M)$ be a bump function equal to 1 in a neighborhood of p and supported in U. So $\psi\omega$ is identically zero, and by linearity $d(\psi\omega) = 0$. But $\psi\omega = \psi \wedge \omega$, and so by the properties,

$$0 = d(\psi\omega) = d\psi \wedge \omega + \psi d\omega.$$

Then at p we have $d\psi_p = 0$ (since it is constant in a neighborhood of p) and $\psi(p) = 1$ so

$$0 = d\omega|_p,$$

as desired.

So now for $\omega \in \Omega^k(M)$, let (U, φ) be a coordinate chart of M. Write $\omega = \sum_I \omega_I dx^I$ in this coordinate chart. We can extend ω_I and dx^i globally by means of a bump function identically 1 in U without loss of generality, and the exterior derivative will be independent of such an extension. Then by the properties,

$$d\omega = \sum_I d\omega_I \wedge dx^I,$$

showing uniqueness. ■

Note:

For a graded algebra $A = \bigoplus_k A^k$, an endomorphism $T\colon A \to A$ is said to have **degree** m if it maps A^k into A^{k+m} for all k. And it is an **antiderivation** if

$$T(xy) = (Tx)y + (-1)^k x(Ty)$$

for all $x \in A^k, y \in A^\ell$. The previous theorem can be summarized as saying there is a unique extension of the differential $\Omega^0(M) \to \Omega^0(M)$ to an antiderivation of $\Omega^*(M)$ of degree $+1$ whose square is zero.

16.3.4 Proposition

If $F\colon M \to N$ is smooth, then the pullback $F^*\colon \Omega^k(N) \to \Omega^k(M)$ commutes with exterior differentiation for all k:

$$F^*(d\omega) = d(F^*\omega), \qquad \omega \in \Omega^k(N).$$

Proof

Let (U, φ) and (V, ψ) be coordinate charts for M, N respectively. Then locally

$$F^*(d\omega) = F^*\psi^*\, d((\psi^{-1})^*\omega) = \varphi^* \circ (\psi \circ F \circ \varphi^{-1})^*\, d((\psi^{-1})^*\omega),$$

and since $\psi \circ F \circ \varphi^{-1}$ is a smooth Euclidean map and the Euclidean exterior derivative commutes with smooth maps,

$$F^*(d\omega) = \varphi^*\, d((\psi \circ F \circ \varphi^{-1})^*(\psi^{-1})^*\omega) = \varphi^*\, d((\varphi^{-1})^* F^*\omega) = d(F^*\omega).$$ ■

16.3.5 Definition

A smooth differential form $\omega \in \Omega^k(M)$ is **closed** if $d\omega = 0$ and **exact** if there is a smooth $(k-1)$-differential form $\eta \in \Omega^{k-1}(M)$ such that $\omega = d\eta$. Since $d \circ d = 0$ all exact forms are closed.

Note:

By $\mathbb{R}$-linearity of d and $d \circ d = 0$, the $\Omega^k(M)$ form a **cochain complex**:

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{n-1}(M) \xrightarrow{d} \Omega^n(M) \to 0.$$

This is called the **De Rham complex** of M. The cycles of this complex are by definition the closed forms, and the boundaries are the exact forms. The cohomology of this complex (the **De Rham cohomology**) is of much importance, and we will discuss it later.

16.3.6 Proposition

For a one-form $\omega \in \Omega^1(M)$ and $X, Y \in \mathfrak{X}(M)$,

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]).$$

Proof

Both sides are additive, so it is sufficient to consider the case $\omega = u\,dv$ for u, v smooth. Then by the formula for wedge products of 1-forms,

$$d(u\,dv)(X, Y) = du \wedge dv(X, Y) = du(X)\,dv(Y) - dv(X)\,du(Y) = X(u)Y(v) - X(v)Y(u).$$

The right-hand side is

$$\begin{aligned} X(u\,dv(Y)) - Y(u\,dv(X)) - u\,dv([X, Y]) &= X(uY(v)) - Y(uX(v)) - u[X, Y]v \\ &= X(u)Y(v) + uXY(v) - Y(u)X(v) - uYX(v) - u[X, Y]v \\ &= X(u)Y(v) - Y(u)X(v). \end{aligned}$$

So we have equality. ■

16.3.7 Corollary

Let M be a smooth n-manifold, with a local frame (E_i) and dual frame (ε^i). Let b^i_{jk} be the components of the exterior derivative of ε^i in this frame, and let c^i_{jk} be the components of $[E_j, E_k]$:

$$d\varepsilon^i = \sum_{j<k} b^i_{jk}\varepsilon^j \wedge \varepsilon^k, \qquad [E_j, E_k] = c^i_{jk}E_i.$$

Then $b^i_{jk} = -c^i_{jk}$.

Proof

By the previous proposition, for $j < k$,

$$d\varepsilon^i(E_j, E_k) = E_j(\varepsilon^i(E_k)) - E_k(\varepsilon^i(E_j)) - \varepsilon^i([E_j, E_k]).$$

Since $\varepsilon^i(E_j)$ is constant, the first two terms are zero. The last term is equal to $\varepsilon^i(c^\ell_{jk}E_\ell) = c^i_{jk}$. So $d\varepsilon^i(E_j, E_k) = -c^i_{jk}$, and we also see it is equal to b^i_{jk} since $\varepsilon^\ell \wedge \varepsilon^s(E_j, E_k) = 1$ iff $\ell = j$ and $s = k$. ■

16.3.8 Proposition

Let $\omega \in \Omega^k(M)$ and $X_1, \dots, X_{k+1} \in \mathfrak{X}(M)$. Then

$$d\omega(X_1, \dots, X_{k+1}) = \sum_i (-1)^{i-1} X_i(\omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1})) + \sum_{i<j} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_{k+1}).$$

Proof

Let the right-hand side be $D\omega(X_1, \dots, X_{k+1})$, and the two sums by $A(X_1, \dots, X_{k+1})$ and $B(X_1, \dots, X_{k+1})$. $D\omega$ is clearly multilinear over $\mathbb{R}$; and we show it is multilinear over $C^\infty(M)$. Let $f \in C^\infty(M)$, we consider $A(X_1, \dots, fX_p, \dots, X_{k+1})$ and $B(X_1, \dots, fX_p, \dots, X_{k+1})$.
For A, clearly f factors out of the pth term in the sum. For the other terms,

$$\sum_{i \neq p} (-1)^{i-1} X_i(f\omega(X_1, \dots, \widehat{X}_i, \dots, X_{k+1})) = \sum_{i \neq p} (-1)^{i-1} \big(fX_i(\omega(\dots, \widehat{X}_i, \dots)) + (X_i f)\omega(\dots, \widehat{X}_i, \dots)\big).$$

Thus

$$A(\dots fX_p \dots) = fA(\dots X_p \dots) + \sum_{i \neq p} (-1)^{i-1} (X_i f)\omega(\dots, \widehat{X}_i, \dots).$$

Now consider B. When $i \neq p$ and $j \neq p$ clearly f factors out. We know

$$[fX_p, X_j] = f[X_p, X_j] - (X_i f)X_p, \qquad [X_i, fX_p] = f[X_i, X_p] + (X_i f)X_p,$$

and so

$$B(\dots fX_p \dots) = fB(\dots X_p \dots) + \sum_{p<j} (-1)^{p+j+1} (X_j f)\omega(X_p, \dots, \widehat{X}_p, \dots, \widehat{X}_j, \dots) + \sum_{i<p} (-1)^{i+p} (X_i f)\omega(X_p, \dots, \widehat{X}_i, \dots, \widehat{X}_p, \dots).$$

Rearranging the order of X_p in each of the last two sums to place X_p in the pth index gives the sums of $(-1)^{j+1}(X_j f)\omega(\dots \widehat{X}_j \dots)$ and $(-1)^{i-1}(X_i f)\omega(\dots \widehat{X}_i \dots)$. These cancel out the additional sum from A, so we see that $D\omega(\dots fX_p \dots) = fD\omega(\dots X_p \dots)$ as desired.
Now that we know both $D\omega, d\omega$ are multilinear, we can restrict our attention to vector fields from any local frame. So take a chart $(U, (x^i))$, and by the obvious $\mathbb{R}$-linearity of D and d in ω, we can assume $\omega = u\, dx^I$ for I increasing. Then

$$d\omega = du \wedge dx^I = \sum_m \frac{\partial u}{\partial x^m}\, dx^m \wedge dx^I.$$

For any multi-index $J = (j_1, \dots, j_{k+1})$ we have

$$d\omega(\partial_J) = \sum_m \frac{\partial u}{\partial x^m} \delta_J^{mI}.$$

The only non-zero terms are when m is equal to one of the indices from J, say $m = j_p$. In such a case $\delta_J^{mI} = (-1)^{p-1}\delta_{\widehat{J}_p}^I$ where $\widehat{J}_p = (j_1, \dots, \widehat{j}_p, \dots, j_{k+1})$. Thus

$$d\omega(\partial_J) = \sum_p (-1)^{p-1} \frac{\partial u}{\partial x^{j_p}} \delta_{\widehat{J}_p}^I.$$

On the other hand, the Lie brackets are zero so

$$D\omega(\partial_J) = \sum_p (-1)^{p-1} \frac{\partial}{\partial x^{j_p}} \left(u\, dx^I \left(\frac{\partial}{\partial x^{j_1}}, \dots, \widehat{\frac{\partial}{\partial x^{j_p}}}, \dots, \frac{\partial}{\partial x^{j_{k+1}}} \right) \right).$$

The innermost application of dx^I is just $\delta_{\widehat{J}_p}^I$, and so this is equal to

$$= \sum_p (-1)^{p-1} \frac{\partial u}{\partial x^{j_p}} \delta_{\widehat{J}_p}^I = d\omega(\partial_J)$$

as desired. ∎

Since k-forms are k-tensors, we can take their Lie derivatives. By the formula

$$(\mathcal{L}_V\omega)(X_1,\dots,X_k) = V(\omega(X_1,\dots,X_k)) - \sum_i \omega(X_1,\dots,[V,X_i],\dots,X_k),$$

we see that the Lie derivative of a k-form is a k-form.

16.3.9 Proposition

The Lie derivative commutes with wedge products as follows: for $\omega,\eta \in \Omega^*(M)$ and $V \in \mathfrak{X}(M)$,

$$\mathcal{L}_V(\omega\wedge\eta) = (\mathcal{L}_V\omega)\wedge\eta + \omega\wedge(\mathcal{L}_V\eta).$$

Proof

This follows from the general properties of lie derivatives: it is $\mathbb{R}$-linear, $\mathcal{L}_V(A\otimes B) = (\mathcal{L}_V A)\otimes B + A\otimes(\mathcal{L}_V B)$, and ${}^\sigma\mathcal{L}_V(A) = \mathcal{L}_V({}^\sigma A)$. ■

16.3.10 Theorem (Cartan's magic formula)

Let $\omega \in \Omega^k(M)$ and $V \in \mathfrak{X}(M)$, then

$$\mathcal{L}_V\omega = V \lrcorner (d\omega) + d(V \lrcorner \omega).$$

Proof

We induct on k. For $f \in \Omega^0(M)$, we have

$$V \lrcorner (df) + d(V \lrcorner f) = V \lrcorner df = df(V) = V(f) = \mathcal{L}_V f.$$

Now let $k \geq 1$ and write $\omega = \omega_I\, dx^I$ relative to a chart. Writing $u = x^{i_1}$ and $\beta = \omega_i\, dx^{i_2}\wedge\cdots\wedge dx^{i_k}$, every term in the sum can be written as $du\wedge\beta$ for $u \in C^\infty(M)$ and $\beta \in \Omega^{k-1}(M)$. By linearity of $\mathcal{L}_V$ it is sufficient to consider only this case.
We know that $\mathcal{L}_V(du) = d(\mathcal{L}_V u) = d(Vu)$. Then by the above proposition and induction,

$$\mathcal{L}_V(du\wedge\beta) = \mathcal{L}_V(du)\wedge\beta + du\wedge\mathcal{L}_V(\beta) = d(Vu)\wedge\beta + du(V\lrcorner d\beta + d(V\lrcorner\beta)).$$

Since both d and interior multiplication by V are antiderivations, and $V \lrcorner du = du(V) = Vu$, we see

$$\begin{aligned} V \lrcorner d(du\wedge\beta) + d(V\lrcorner(du\wedge\beta)) &= V\lrcorner(-\,du\wedge d\beta) + d((Vu)\beta - du\wedge(V\lrcorner\beta)) \\ &= -(Vu)\,d\beta + du\wedge(V\lrcorner d\beta) + d(Vu)\wedge\beta + (Vu)\,d\beta + du\wedge d(V\lrcorner\beta). \end{aligned}$$

Which gives us the expression computed for $\mathcal{L}_V(du\wedge\beta)$. ■

16.3.11 Corollary

The Lie derivative commutes with exterior derivatives:

$$\mathcal{L}_V(d\omega) = d(\mathcal{L}_V\omega)$$

for all $V \in \mathfrak{X}(M)$ and $\omega \in \Omega^*(M)$.

Proof

From Cartan's formula,

$$\mathcal{L}_V(d\omega) = V \lrcorner\, d(d\omega) + d(V \lrcorner\, d\omega) = d(V \lrcorner\, d\omega)$$
$$d\mathcal{L}_V(\omega) = d(V \lrcorner\, d\omega + d(V \lrcorner\, \omega)) = d(V \lrcorner\, d\omega).$$

These are equal. ■

16.4 Orientations

16.4.1 Definition

Two bases $(E_1, \ldots, E_n)$ and $(\widetilde{E}_1, \ldots, \widetilde{E}_n)$ of a vector space V are **consistently oriented** if their transformation matrix has positive determinant. That is, the matrix (B_i^j) given by $E_i = B_i^j \widetilde{E}_j$ has positive determinant. This is clearly an equivalence relation with two equivalence classes.

An **orientation** for V is a choice of one of these equivalence classes. For a basis $(E_1, \ldots, E_n)$, denote by $[E_1, \ldots, E_n]$ its equivalence class and $-[E_1, \ldots, E_n] = [-E_1, \ldots, E_n]$ the other equivalence class. Any basis in this orientation is said to be **positively oriented**, or just **oriented**, and all others are **negatively oriented**.

An orientation of the zero-dimensional space is a choice of one of ± 1.

16.4.2 Example

The **standard orientiation** of $\mathbb{R}^n$ is $[e_1, \ldots, e_n]$.

16.4.3 Proposition

Let V be an n-dimensional vector space. Then every nonzero $\omega \in \Lambda^n(V^*)$ determines an orientation $\mathcal{O}_\omega$ of V, where $(E_1, \ldots, E_n)$ is oriented iff $\omega(E_1, \ldots, E_n) > 0$. Two nonzero n-covectors determine the same orientation iff they are positive multiples of one another.

Proof

The zero-dimensional case is immediate since $\Lambda^0(V^*) \cong \mathbb{R}$. Otherwise let $\omega \in \Lambda^n(V^*)$, and (E_i) and $(\widetilde{E}_i)$ be bases. We claim that they are consistently oriented iff $\omega(E_1, \ldots, E_n)$ and $\omega(\widetilde{E}_1, \ldots, \widetilde{E}_n)$ have the same sign. Let $B = (B_i^j)$ be their transition matrix so

$$\omega(\widetilde{E}_1, \ldots, \widetilde{E}_n) = \omega(BE_1, \ldots, BE_n) = \det(B)\omega(E_1, \ldots, E_n).$$

So they have the same sign iff $\det(B) > 0$, meaning the bases are consistently ordered as desired. The last claim is clear. ■

We say that a covector $\omega \in \Lambda^n(V^*)$ which determines the orientation of V is an **(positively) oriented covector**. For example, $e_1 \wedge \cdots \wedge e_n$ is a positively oriented covector for the standard orientation of $\mathbb{R}^n$.

Now we can discuss orientations of manifolds.

16.4.4 Definition

A **(pointwise) orientation** of a smooth manifold is a choice of orientation of T_pM for each $p \in M$. If M is endowed with a pointwise orientation, a local frame (E_i) is **(positively) oriented** if $(E_i|_p)$ is a positively oriented basis for each p in the frame's domain. We similarly define a negatively oriented frame.

A pointwise orientation is **continuous** if every point of M is in the domain of an oriented local frame. An **orientation** of M is a continuous pointwise orientation. A manifold is said to be **orientable** if it possesses a continuous pointwise orientation, otherwise it is **nonorientable**.

An **oriented manifold** is a pair $(M, \mathcal{O})$ where $\mathcal{O}$ is an orientation on M. Denote the orientation of T_pM by $\mathcal{O}_p$.

This definition is valid for manifolds with boundary. For a zero-dimensional manifold, a pointwise orientation is a choice of ± 1 for every point $p \in M$. In the zero-dimensional case every pointwise orientation is continuous.

Note that if M is oriented, any local frame defined on a connected domain is either positively or negatively oriented. Let (E_i) be a local frame on a connected domain U, then define $U \to \mathbb{R}$ by mapping $p \in U$ to $+1$ if $(E_i|_p)$ is positively oriented and -1 otherwise. Then this is locally constant: for $p \in U$ take a positively oriented frame (F_i) defined in a neighborhood of p, which we can restrict to a connected open subset of U. Let B be the transition map between E and F, then E_p is positively oriented iff $\det(B_p) > 0$, so in this open subset the map is equal to $\operatorname{sgn}\det(B)$. Since $\det(B)$ is nonzero, continuous, and the domain is connected, we have that its sign is constant. So the map is locally constant, and since the domain is connected it must be constant.

16.4.5 Proposition

Let M be a smooth n-manifold, then any nonvanishing n-form ω on M determines a unique orientation for which ω is positively oriented at each point. Conversely if M is given an orientation, there is a smooth nonvanishing n-form on M that is positively oriented at each point.

Proof

Let $\omega \in \Omega^n(M)$ be nonvanishing. Then ω_p defines an orientation on T_pM for all $p \in M$, we check that this orientation is continuous. This is trivial for $n = 0$, so we assume $n \geq 1$.

By the above discussion, every local frame on a connected neighborhood should be positively or negatively oriented. So let $p \in M$ and take a local frame on a connected neighborhood U of p, call it (E_i) with dual coframe (ε^j). Write $\omega = f\varepsilon^1 \wedge \cdots \wedge \varepsilon^n$, and since ω is nonvanishing f is nonzero over all of U so

$$\omega(E_1, \ldots, E_n) = f \neq 0.$$

Since U is connected, f is either always positive or always negative. If f is always positive then (E_i) is positively oriented and we have finished. Otherwise f is always negative so $\omega(-E_1, E_2, \ldots, E_n) = -f$ is always positive and we take the local frame $(-E_1, E_2, \ldots, E_n)$ and we have finished.

For $p \in M$ we take a connected neighborhood U_p with positively oriented frame (E_i^p). Let (ε_p^j) be its dual coframe and define the local n-form $\omega^p = \varepsilon_p^1 \wedge \cdots \varepsilon_p^n$. Then let ψ_p be a partition of unity subordinate to $\{U_p\}_p$ and define

$$\omega = \sum_p \psi_p \omega^p.$$

This is nonvanishing since for any (E_i^p), $\omega^q(E_i^p) \geq 0$ for all q since it is positively oriented. Since $\psi_p(p) = 1$ this implies $\omega(E_i^p) > 0$. And since ω is a positive linear combination of positively oriented n-forms at each point it is positively oriented at each point. ■

16.4.6 Definition

A nowhere-vanishing n-form on M is called an **orientation form**. If M is oriented, an n-form giving its orientation is called **(positively) oriented**.

Two orientation forms $\omega, \widetilde{\omega}$ are easily seen to give the same orientation iff $\omega = f\widetilde{\omega}$ for some positive smooth real-valued f.

16.4.7 Definition

A smooth coordinate chart is **(positively) oriented** if its coordinate frame $(\partial/\partial x^i)$ is positively oriented, and **negatively oriented** if it the coordinate frame is negatively oriented.

An atlas $\{(U_i, \varphi_i)\}_i$ is **consistently oriented** iff the transition maps $\varphi_j \circ \varphi_i^{-1}$ have positive determinant for all i, j.

16.4.8 Proposition

Let M be a smooth positive-dimensional manifold. Any consistently oriented smooth atlas for M uniquely determines an orientation for which the charts are all positively oriented.

Conversely if M is oriented, and either $\partial M = \emptyset$ or $\dim M > 1$, then the collection of all oriented smooth charts is a consistently oriented atlas for M.

Proof

Let (U_i, φ_i) be a consistently oriented smooth atlas. Then each chart determines a pointwise orientation of its domain, and consistency proves that these pointwise orientations agree on their intersections. Since the chart covers the manifold, this is a continuous orientation.

If M is oriented and $\partial M = \emptyset$ or $\dim M > 1$, each point in M is in the domain of a connected smooth chart. If the domain is negatively oriented, we swap x^1 with $-x^1$ to obtain a positively oriented chart. As they are all positively oriented, their transition functions must have positive Jacobian.

For a boundary chart, it maps into $\mathbb{H}^n$ so the last coordinate must be positive. For $\dim M = 1$ this is the only coordinate so we cannot swap its sign. ∎

The following proposition is easily seen.

16.4.9 Proposition

If $M_1, \dots, M_k$ are orientable smooth manifolds, then there is a unique orientation on $M = \prod_i M_i$ called the **product orientation**, such that if ω_i is an orientation form for M_i then $\pi_1^*\omega_1 \wedge \cdots \wedge \pi_n^*\omega_k$ is an orientation form for the product orientation.

16.4.10 Definition

Let M, N be oriented smooth manifolds with positive dimension. A local diffeomorphism $F\colon M \to N$ is **orientation preserving** if for each $p \in M$ its differential dF_p maps oriented bases of T_pM to oriented bases of $T_{Fp}N$. Similarly it is **orientation reversing** if it maps negatively oriented bases to oriented bases.

For zero-dimensional manifolds, F is orientation preserving if for every $p \in M$, both p and $F(p)$ have the same orientation (choice of ± 1).

It is easy to see that F is orientation preserving iff its Jacobian with respect to any two oriented bases has positive determinant, iff it pulls back positively oriented n-forms to positively oriented n-forms.

16.4.11 Proposition

Let $F\colon M \to N$ be a local diffeomorphism with N oriented. Then there is a unique orientation on M making F orientation preserving. This is called the **pullback orientation**.

Proof

Let ω be a positively oriented n-form on N. Then for F to be orientation preserving, $F^*\omega$ must be positively-oriented. So we just need to show that $F^*\omega$ is nonvanishing and thus defines a continuous orientation on M. And indeed, if (E_i) is a local frame of M then

$$F^*\omega(E)_p = \omega_{Fp}(dF_pE),$$

and since dF_pE is a basis, ω does not vanish on it. Thus $F^*\omega$ is non-vanishing. ■

Pulling back orientations is functorial. If we denote the pullback orientation by $F^*\mathcal{O}$ then $(G \circ F)^*\mathcal{O} = F^*(G^*\mathcal{O})$. This is because pulling back the orientation n-form is functorial.

16.4.12 Proposition

Every parallelizable smooth manifold is orientable.

Proof

If M is parallelizable, let (E_i) be a global frame. Then define a pointwise orientation on M by making $(E_i|_p)$ positively oriented for all $p \in M$. This is then trivially continuous. ■

16.4.13 Example

S^1 is parallelizable, as we can define a global vector field V to be the unit tangent (say in the clockwise direction). Thus S^1 is orientable. Then as the product of S^1s, T^n is also parallelizable and thus orientable.

S^n is parallelizable for $n = 1, 3, 7$, and thus also orientable in those cases. We will see that S^n is orientable for all n.

Note that Lie groups are parallelizable too, as any basis for their identity's tangent space extends to a global frame.

16.4.14 Proposition

Suppose M is an oriented n-manifold, and S is an immersed hypersurface in M (i.e. of codimension one). Suppose further there is a vector field along S that is nowhere tangent to S, call it N.

Then S has a unique orientation such that for every $p \in S$ if $E_1, \ldots, E_{n-1}$ is an oriented basis for T_pS iff $N_p, E_1, \ldots, E_{n-1}$ is an oriented basis for T_pM. If ω is an orientation form for M, then $\iota_S^*(N \lrcorner\, \omega)$ is an orientation form for S relative to this orientation.

Proof

Define $\sigma = \iota_S^*(N \lrcorner\, \omega)$, we claim this is non-vanishing. Let $E_1, \ldots, E_{n-1}$ be a basis for T_pS then since N is nowhere tangent to S means that it is not in the span of the E_i so $N_p, E_1, \ldots, E_{n-1}$ is a basis for T_pM. Then

$$\sigma_p(E_1, \ldots, E_{n-1}) = \omega_p(N_p, E_1, \ldots, E_{n-1}) \neq 0,$$

as ω is non-vanishing. Since $\sigma_p(E_1, \ldots, E_{n-1}) > 0$ iff $\omega_p(N_p, E_1, \ldots, E_{n-1}) > 0$, this is the desired orientation. It is continuous as it is given by a smooth n-form. Uniqueness is obvious since the condition uniquely defines the pointwise orientation. ■

16.4.15 Example

S^n is an embedded hypersurface of $\mathbb{R}^{n+1}$, and $N = x^i\partial_i$ is nowhere tangent to it. Thus it induces an orientation on S^n, this is the **standard orientation** on S^n.

16.4.16 Corollary

Every regular level set of a smooth real-valued function is orientable.

Proof

Let $f\colon M \to \mathbb{R}$ be smooth, and S be a regular level set of f. Endow M with a Riemannian metric, and let $N = \operatorname{grad} f|_S$. Then N is nowhere tangent to S as it is nonzero for points on S (as it is regular there), and $\langle \operatorname{grad} f_p, v\rangle = df_p(v) = 0$ for $v \in T_pS$, so $\operatorname{grad} f_p \notin T_pS$. ■

16.4.17 Definition

Let M be a smooth n-manifold with boundary. For a point $p \in \partial M$, a tangent vector $v \in T_pM - T_p\partial M$ is **inward pointing** if there is a smooth curve $\gamma\colon [0, \varepsilon) \to M$ with $\gamma(0) = p$ and $\gamma'(0) = v$. If there is a smooth curve $\gamma\colon (-\varepsilon, 0] \to M$ similarly, v is **outward pointing**.

16.4.18 Proposition

Let $p \in \partial M$ and $(U, (x^i))$ a boundary chart centered at p, and let $v \in T_pM$. Write $v = a^i\partial_i$, then

(1) $v \in T_p\partial M$ iff $a^n = 0$,

(2) v is inward-pointing iff $a^n > 0$.

(3) v is outward-pointing iff $a^n < 0$.

In particular v is inward-pointing iff $-v$ is outward-pointing, and all tangent vectors at a boundary point are either in the tangent space of the boundary, or inward or outward-pointing.

Proof

The boundary points of M lying in U are precisely those where $x^n = 0$. So $\partial M \cap U$ is the level set of 0 under x^n. So $T_p\partial M$ are those $v \in T_pM$ where $d(x^n)_pv = 0$, which occurs iff $a^n = 0$.

If v is inward-pointing, let $\gamma\colon [0, \varepsilon) \to M$ be smooth with $\gamma'(0) = v$. Let $(U, (x^i))$ be any chart centered at p, write $\gamma = (\gamma^1, \dots, \gamma^n)$. We then have $\gamma^n \geq 0$ and so it is locally non-decreasing (since it cannot go below 0). Then

$$\gamma'(0) = \dot\gamma^i(0)\partial_i = v,$$

so $a^n = \dot\gamma^n(0) \geq 0$ as it is locally non-decreasing, and since $v \notin T_p\partial M$ we see that $a^n > 0$ as desired.

Conversely if $a^n > 0$ in $(U, (x^i))$ then we want $\dot\gamma^i(0) = a^i$ and $\gamma^i(0) = 0$. This ODE has a solution γ defined in some $(-\varepsilon, \varepsilon)$. As $\dot\gamma^n(0) > 0$, it is increasing and thus restricted to $[0, \delta)$ for some $\delta \leq \varepsilon$ we have $\gamma^n(t) \geq 0$. This then defines a smooth curve $\gamma\colon [0, \delta) \to M$ with $\gamma'(0) = v$. ■

16.4.19 Lemma

Let M be smooth with boundary, then there exists a smooth vector field on M whose restriction to ∂M

is everywhere inward-pointing. Similarly we can make it outward pointing.

Proof

If X is inward-pointing, then $-X$ is outward pointing, so it is sufficient to prove it for inward-pointing. Cover ∂M with boundary charts, and for each boundary chart U_i define X_i to be ∂_n. Define X_0 to be the zero vector field on the interior of M. Let ψ_i be a partition of unity subordinate to $U_i, \mathrm{Int} M$ and define

$$X = \sum_i \psi_i X_i.$$

Since $X_i(p) = \partial_n$ has positive coefficient, it is inward pointing and thus has positive coefficient relative to all coordinate charts. Then relative to U_j we see that the coefficients of $\psi_(p) X_i(p)$ are all positive and thus $X(p)$ is positive. It is smooth, and thus an inward pointing vector field. ■

Notice that by definition an inward/outward pointing vector field is nowhere tangent to the boundary of the manifold and since ∂M is embedded it induces an orientation.

16.4.20 Proposition

Let M be an oriented n-manifold with $n \geq 1$. Then ∂M is orientable, and all outward pointing vector fields induce the same orientation on it.

Proof

Let ω be an orientation form for M, and let N be a smooth outward-pointing vector field on M. Then as we showed $\sigma = \iota^*_{\partial M}(N \lrcorner \omega)$ is an orientation form on the boundary.

If $\widetilde{N}$ is another outward-pointing vector field, then it is sufficient to show that for any boundary chart $(U, (x^i))$, $N_p, \partial_1, \dots, \partial_{n-1}$ is oriented iff $\widetilde{N}_p, \partial_1, \dots, \partial_{n-1}$ is. Since $N_p, \widetilde{N}_p$ have coefficient in ∂_n both negative, the transition matrix has determinant $N^n_p / \widetilde{N}^n_p > 0$, so one is oriented iff the other is. ■

16.4.21 Example

S^n is the boundary ∂B^{n+1}, showing that it is oriented again.

16.4.22 Definition

The induced orientation on ∂M by an outward-pointing vector field is called the **induced orientation**, or the **Stokes orientation** of the boundary.

16.4.23 Proposition

Let (M, g) be an oriented Riemannian n-manifold, with $n \geq 1$. Then there is a unique smooth orientation form $\omega_g \in \Omega^n(M)$ satisfying both of the equivalent properties:

(1) $\omega_g(E_1, \dots, E_n) = 1$ for every local oriented orthonormal frame (E_i).

(2) $\omega_g = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$ for every oriented local chart $(U, (x^i))$.

ω_g is called the **Riemannian volume form**.

Proof

Either point clearly uniquely determines ω_g, showing equivalence is done by representing ∂_i with respect to E_j and recalling that the determinant of the transition must be positive. Then (1) defines ω_g locally as $\varepsilon^1 \wedge \cdots \wedge \varepsilon^n$ where ε^i is the dual coframe of E_i, and this is well-defined as is readily shown. ■

17 Integration

17.1 Integration of differential forms

When we think of integration, usually our theories are built on the integration of real-valued functions. This is not suitable in the context of manifolds in general. At least not in a way that would behave nicely. We would want our integrals to be independent of choice of coordinates, but this is easily seen to be impossible.

Take for example a closed ball $C \subseteq \mathbb{R}^n$ and the constant function $f: C \to \mathbb{R}^n$ given by $f(x) = 1$. Any sensible theory of integration would mandate that the integral of f is equal to the volume of C:

$$\int_C f = \mathrm{Vol}(C).$$

But this is clearly not coordinate-independent, as C is diffeomorphic to a larger closed ball with a larger volume say.

As we will see, we can solve this via differential forms.

17.1.1 Definition

A **domain of integration** in $\mathbb{R}^n$ is a bounded subset with measure zero boundary. For a domain of integration $D \subseteq \mathbb{R}^n$, let ω be a continuous n-form on $\overline{D}$. Such an n-form can be written $\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ for a continuous $f: \overline{D} \to \mathbb{R}$.

Then we define ω's **integral** over D as

$$\int_D \omega = \int_D f\, dV,$$

where dV denotes integration relative to the Lebesgue measure of $\mathbb{R}^n$. We can write this more clearly as

$$\int_D f\, dx^1 \wedge \cdots \wedge dx^n = \int_D f\, dx^1 \cdots dx^n.$$

So we can view integration as "removing the wedges".

More generally, let U be an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ and ω is a compactly supported n-form on U, then define

$$\int_U \omega = \int_D \omega$$

where $D \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$ is any domain of integration (e.g. a rectangle) containing $\mathrm{supp}\omega$ where we potentially extend ω to be zero outside of U.

This definition does not depend on choice of domain of integration. This is because for any two domains of integration, ω is zero outside of their intersection.

17.1.2 Proposition

Suppose D, E are open domains of integration in $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G: \overline{D} \to \overline{E}$ is a smooth map which restricts to an orientation-preserving or orientation-reversing diffeomorphism from D to E. Then for an n-form ω on $\overline{E}$, we have

$$\int_D G^*\omega = \pm \int_E \omega,$$

where the sign is flipped when G is orientation-reversing.

Proof

Denote the standard coordinates of E by (y^i) and those of D by (x^i). If G is orientation preserving, let $\omega = f\,dy^1 \wedge \cdots \wedge dy^n$, then we have

$$\int_E \omega = \int_E f\,dV = \int_D (f \circ G)|\det DG|\,dV = \int_D (f \circ G)\det(DG)\,dV$$
$$= \int_D (f \circ G)\det(DG)\,dx^1 \wedge \cdots \wedge dx^n = \int_D G^*\omega.$$

If G is orientation-reversing $|\det DG| = -\det(DG)$ and the sign flips. ■

17.1.3 Lemma

Suppose U is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, and K is a compact subset of U. Then there is an open domain of integration D such that $K \subseteq D \subseteq \overline{D} \subseteq U$.

Proof

For each $p \in K$ take a open ball or half-open ball whose closure is contained in U. As K is compact, this has a finite subcovering of balls; let D be their union. D is then bound, and its boundary is contained in the union of the boundaries of the balls. As a finite union of measure zero sets, the boundary of D has measure zero. ■

So we can now generalize our above proposition to compactly supported differential n-forms defined on any open subset, not just a domain of integration.

17.1.4 Proposition

If U, V are open subsets of $\mathbb{R}^n$ or $\mathbb{H}^n$ and $G\colon U \to V$ is an orientation-preserving or reversing diffeomorphism, for any compactly supported n-form ω on V,

$$\int_V \omega = \pm \int_U G^*\omega.$$

The sign depends on if G preserves or reverses orientation (positive and negative respectively).

Proof

Take an open domain of integration $E \subseteq V$ such that $\operatorname{supp}\omega \subseteq E$. Then $D = G^{-1}E$ is open, and since diffeomorphisms preserve measure-zero-ness and boundaries, we see that D is a domain of integration. Then we have by the previous proposition,

$$\int_V \omega = \int_E \omega = \pm \int_D G^*\omega = \pm \int_U G^*\omega.$$ ■

We now extend these definitions to arbitrary oriented smooth manifolds.

Let M be an oriented smooth n-manifold, and ω an n-form on M. First suppose that ω is compactly supported in the domain of a single smooth chart (U, φ). Such a chart is either positively or negatively oriented (if connected, otherwise require it to have a consistent orientation), then we define

$$\int_M \omega = \pm \int_{\varphi(U)} (\varphi^{-1})^* \omega,$$

where the right integral is done in Euclidean (half) space, and the sign depends on the orientation of the chart (positive when oriented, negative when negatively oriented).

17.1.5 Lemma

The above definition is independent on choice of smooth chart containing suppω.

Proof

If (U, φ) and $(\widetilde{U}, \widetilde{\varphi})$ both contain suppω then $\widetilde{\varphi} \circ \varphi^{-1}$ is a diffeomorphism $\varphi(U \cap \widetilde{U})$ to $\widetilde{\varphi}(U \cap \widetilde{U})$. It is orientation-preserving when both are consistently oriented, and negatively oriented otherwise.

If they are consistently oriented, then

$$\int_{\widetilde{\varphi}(U)} (\widetilde{\varphi}^{-1})^* \omega = \int_{\widetilde{\varphi}(U \cap \widetilde{U})} (\widetilde{\varphi}^{-1})^* \omega = \int_{\varphi(U \cap \widetilde{U})} (\widetilde{\varphi} \circ \varphi^{-1})^* \circ (\widetilde{\varphi}^{-1})^* \omega$$
$$= \int_{\varphi(U \cap \widetilde{U})} (\varphi^{-1})^* \omega = \int_{\varphi(U)} (\varphi^{-1})^* \omega.$$

If they have different orientations then the above computation swaps signs, which is what we need by definition. ∎

Now suppose that ω is a compactly supported n-form on all of M. Let U_i be a finite cover of suppω, and let ψ_i be a partition of unity subordinate to it. Define ω's integral as

$$\int_M \omega = \sum_i \int_M \psi_i \omega.$$

Since $\psi_i \omega$ is compactly supported (as both ω and ψ_i are), this is well-defined, but it is not immediately clear why this would be independent on choice of partition of unity.

Note:

One may ask why we even allow for negatively oriented charts, and instead just use positively oriented charts. The reason is that for a one-dimensional manifold with boundary, we cannot ensure the existence of positively oriented boundary charts.

17.1.6 Proposition

The definition of $\int_M \omega$ does not depend on choice of open cover or partition of unity.

Proof

Suppose $\widetilde{U}_i$ is another finite open cover of suppω with $\widetilde{\psi}_j$ a subordinate partition of unity. Then we have for each i,

$$\int_M \psi_i \omega = \int_M \left(\sum_j \widetilde{\psi}_j \right) \psi_i \omega = \sum_j \int_M \widetilde{\psi}_j \psi_i \omega.$$

Summing over i we see

$$\sum_i \int_M \psi_i \omega = \sum_{i,j} \int_M \widetilde{\psi}_j \psi_i \omega.$$

Each term in the sum is compactly supported in a single smooth chart (e.g. U_i), so each term is uniquely defined. The same process shows

$$\sum_j \int_M \widetilde{\psi}_j \omega = \sum_{i,j} \int_M \widetilde{\psi}_j \psi_i \omega,$$

proving equality. ■

For the zero-dimensional case, a compactly supported 0-form is just a real valued function $f \colon M \to \mathbb{R}$ which is nonzero on finitely many points. So we define

$$\int_M f = \sum_{p \in M} \pm f(p),$$

where the sign is determined by choice of orientation at p.

If $S \subseteq M$ is an immersed oriented k-submanifold (with induced orientation), and ω is a k-form on M whose restriction to S is compactly supported, we define

$$\int_S \omega = \int_S \iota_S^* \omega,$$

where ι_S^* is inclusion. In particular for compact oriented M, and an $(n-1)$-form ω on M,

$$\int_{\partial M} \omega = \int_{\partial M} \iota_{\partial M}^* \omega,$$

where ∂M has the induced (Stokes) orientation.

Note:

It is possible to extend integration to non-compactly-supported forms, but this introduces issues of convergence. Such definitions are useful, but unnecessary for our purposes.

17.1.7 Proposition

Suppose M, N are non-empty oriented smooth n-manifolds, ω, η are compactly supported n-forms on M. The integral has the following properties.

(1) **Linearity**: for $a, b \in \mathbb{R}$:

$$\int_M a\omega + b\eta = a \int_M \omega + b \int_M \eta.$$

(2) **Orientation reversal**: if $-M$ denotes M with the opposite orientation, then

$$\int_{-M} \omega = -\int_M \omega.$$

(3) **Positivity**: if ω is a positively oriented orientation form, $\int_M \omega > 0$.

(4) **Diffeomorphism invariance**: if $F \colon N \to M$ is a orientation-preserving or reversing diffeomorphism,

$$\int_M \omega = \pm \int_N F^* \omega,$$

with positive sign when F preserves orientation, and negative sign when it reverses it.

Proof

(1) is clear at each step in the definition (in Euclidean space, then this implies it for forms supported in a single chart, and this implies it in general). (2) is an immediate consequence of (4) via $F = \mathrm{id}_M$ as it reverses orientation.

For (3) if ω is positively oriented and (U, φ) is a positively oriented smooth chart, $(\varphi^{-1})^*\omega$ is a positive form on $\varphi(U)$, so $f\, dx^1 \wedge \cdots \wedge dx^n$ for f positive. Then its integral over U is positive. For negatively oriented charts, f is negative, but we swap signs so the integral remains positive. Then for a partition of unity, each term in the sum is nonnegative, and there must be at least one strictly positive term.

For (4) it is sufficient to prove this for ω supported in a single chart, since a partition of unity in M maps to a partition of unity of N via F. So suppose (U, φ) is positively oriented with $\mathrm{supp}\omega \subseteq U$. If F is positively oriented, $(F^{-1}U, \varphi \circ F)$ is an oriented smooth atlas for N whose domain contains $\mathrm{supp}F^*\omega$. The result then follows from diffeomorphism invariance of the Euclidean integral. The other cases are handled similarly. ■

17.2 Stoke's theorem

17.2.1 Theorem

Let M be an oriented n-manifold with boundary, and ω a compactly supported smooth $(n-1)$-form on M. Then

$$\int_M d\omega = \int_{\partial M} \omega,$$

where ∂M is understood to have the Stokes orientation.

Proof

First we consider the case that $M = \mathbb{H}^n$. Since ω is compactly supported, its support is contained within a rectangle $A = [-R, R]^{n-1} \times [0, R]$. Writing ω in terms of the standard coordinates,

$$\omega = \sum_{i=1}^n \omega_i\, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n.$$

Thus,

$$d\omega = \sum_{i=1}^n d\omega_i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = \sum_{i,j} \frac{\partial \omega_i}{\partial x^j}\, dx^j \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n =$$
$$\sum_i (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i}\, dx^1 \wedge \cdots \wedge dx^i \wedge \cdots \wedge dx^n.$$

Then we compute,

$$\int_{\mathbb{H}^n} d\omega = \sum_{i=1}^n (-1)^{i-1} \int_A \frac{\partial \omega_i}{\partial x^i}\, dx^1 \wedge \cdots \wedge dx^i \wedge \cdots \wedge dx^n = \sum_{i=1}^n (-1)^{i-1} \int_0^R \int_0^R \cdots \int_0^R \frac{\partial \omega_i}{\partial x^i}\, dx^1 \cdots dx^n.$$

By Fubini we can change the order of integration as to integrate the ith term first. When $i \neq n$ this sum becomes, by the fundamental theorem of calculus,

$$\sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R \frac{\partial \omega_i}{\partial x^i}\, dx^1 \cdots dx^n = \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \cdots \int_{-R}^R \left[\omega_i(x)\right]_{x^i=-R}^{x^i=R} dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n.$$

Each term here is zero, since we chose R so that $\omega = 0$ when $x^i = \pm R$. Thus

$$\int_{\mathbb{H}^n} d\omega = (-1)^{n-1} \int_{-R}^R \cdots \int_{-R}^R \int_0^R \frac{\partial \omega_n}{\partial x^n}\, dx^n\, dx^1 \cdots dx^{n-1} = (-1)^n \int_{-R}^R \cdots \int_{-R}^R \omega_n(x^1, \ldots, x^{n-1}, 0)\, dx^1 \cdots dx^{n-1}.$$

Now we compute the other term, keeping in mind that x^n vanishes on $\partial\mathbb{H}^n$,

$$\int_{\partial\mathbb{H}^n} \omega = \sum_i \int_{A\cap\partial\mathbb{H}^n} \omega_i(x^1, \dots, x^{n-1}, 0)\, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n.$$

Because x^n vanishes on $\partial\mathbb{H}^n$, the pullback of dx^n to the boundary is zero. Thus all terms for $i \neq n$ are zero, so

$$\int_{\partial\mathbb{H}^n} \omega = \int_{A\cap\partial\mathbb{H}^n} \omega_i(x^1, \dots, x^{n-1}, 0)\, dx^1 \wedge \cdots \wedge dx^{n-1}.$$

Since $(x^1, \dots, x^{n-1})$ is positively oriented for $\partial\mathbb{H}^n$ when n is even and negatively oriented when n is odd, this is equal to the previous term. So

$$\int_{\mathbb{H}^n} d\omega = \int_{\partial\mathbb{H}^n} \omega$$

as desired.

For $M = \mathbb{R}^n$, the support of ω is contained in $A = [-R, R]^n$. In this case, the computations show that all the terms in the integral of ω over A are zero (the same computations as above). Since $\partial\mathbb{R}^n = \emptyset$, both integrals are equal to zero.

Now for M be a smooth manifold with boundary, and ω an $(n-1)$-form compactly supported in the domain of a single positively or negatively oriented chart (U, φ). If φ is positively oriented,

$$\int_M d\omega = \int_{\mathbb{H}^n} (\varphi^{-1})^*\, d\omega = \int_{\mathbb{H}^n} d((\varphi^{-1})^*\omega),$$

which is then equal to, as we showed,

$$= \int_{\partial\mathbb{H}^n} (\varphi^{-1})^*\omega.$$

Since $d\varphi$ takes outward-pointing vectors to outward-pointing vectors, it follows that $\varphi|_{U\cap\partial M}$ is orientation-preserving, and so this is equal to

$$= \int_{\partial M} \omega$$

as desired. For φ negatively oriented, we get a similar result. For an interior chart, the same result follows with $\mathbb{R}^n$ instead of $\mathbb{H}^n$.

Finally for ω compactly supported, let ψ_i be a partition of unity subordinate to an open cover U_i of suppω. Then

$$\int_{\partial M} \omega = \sum_i \int_{\partial M} \psi_i\omega = \sum_i \int_M d(\psi_i\omega) = \sum_i \int_M d\psi_i \wedge \omega + \psi\, d\omega =$$
$$\int_M d\left(\sum_i \psi_i\right) \wedge \omega + \int_M \left(\sum_i \psi_i\right)\omega = 0 + \int_M \omega.$$

The final equality is because $\sum_i \psi_i = 1$ is constant and thus its derivative is zero. ■

17.2.2 Example

Let $\gamma\colon [a, b] \to M$ be a smooth embedding, so that $S = \gamma([a, b])$ is an embedded 1-submanifold with boundary in M. If we orient S in such a way as to make S orientation-preserving, then for any $f \in C^\infty(M)$,

$$\int_{[a,b]} \gamma^*\, df = \int_S df = \int_{\partial S} f = f(\gamma(b)) - f(\gamma(a)).$$

We define the **line integral** of a one-form $\omega \in \Omega^1(M)$ along γ to be

$$\int_\gamma \omega = \int_{[a,b]} \gamma^*\omega.$$

This shows that

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a)).$$

This is called the **fundamental theorem of line integrals**.

Note that for $\gamma\colon [a,b] \hookrightarrow \mathbb{R}$ the inclusion, this is just the fundamental theorem of calculus. So Stoke's is a proper strengthening of the fundamental theorem of calculus.

The following corollaries are immediate from Stoke's, but important enough to state explicitly.

17.2.3 Corollary

If M is a compact oriented smooth manifold without boundary, the integral of every exact form over M is zero:

$$\int_M d\omega = 0, \qquad \text{if } \partial M = \emptyset.$$

17.2.4 Corollary

Suppose M is a compact oriented smooth manifold with boundary. The integral of any closed form over ∂M is zero:

$$\int_{\partial M} \omega = 0, \qquad \text{if } d\omega = 0.$$

17.2.5 Corollary

Suppose M is a smooth manifold with or without boundary, and $S \subseteq M$ is an oriented compact smooth k-dimensional submanifold without boundary. Let ω be a closed k-form on M. If $\int_S \omega \neq 0$ then the following are both true:

(1) ω is not exact,

(2) S is not the boundary of an oriented compact smooth submanifold with boundary in M.

17.2.6 Example

Consider the 1-form $\omega = (x\,dy - y\,dx)/(x^2+y^2)$. This is a closed 1-form, and if we let $\gamma\colon [0,2\pi] \to M$ be given by $\gamma(t) = (\cos t, \sin t)$ then

$$\int_{S^1} \omega = \int_{[0,2\pi]} \gamma^*\omega = \int_0^{2\pi} \frac{\cos t(\cos t\,dt) - \sin t(-\sin t\,dt)}{\sin^2 t + \cos^2 t} = \int_0^{2\pi} dt = 2\pi.$$

Thus ω is not exact, and furthermore S^1 is not the boundary of a compact submanifold of $\mathbb{R}^2 - 0$.

17.2.7 Theorem (Green's theorem)

Suppose D is a compact regular domain in $\mathbb{R}^2$, and P, Q are smooth real-valued functions on D. Then

$$\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{\partial D} P\,dx + Q\,dy.$$

Proof

This is just Stoke's on $P\,dx + Q\,dy$. ■

18 Curvature

18.1 Families of curves

18.1.1 Definition

Given intervals $I, J \subseteq \mathbb{R}$, a continuous map $\Gamma\colon J \times I \to M$ is called a **one-parameter family of curves**. This defines two collections of curves:

(1) the **main curves**: for each $s \in J$ the curve $\Gamma_s\colon I \to M$ by $\Gamma_s(t) = \Gamma(s,t)$.

(2) the **transverse curves**: for $t \in I$, the curve $\Gamma^{(t)}\colon J \to M$ by $\Gamma^{(t)}(s) = \Gamma(s,t)$.

If Γ is smooth, then denote the velocity vectors of the main and transverse curves by

$$\partial_t\Gamma(s,t) = (\Gamma_s)'(t), \qquad \partial_s\Gamma(s,t) = (\Gamma^{(t)})'(s),$$

both of which lie in $T_{\Gamma(s,t)}M$. These are vector fields along Γ (smooth when Γ is smooth).

18.1.2 Definition

A curve $\gamma\colon I \to M$ is **regular** if γ' is nonzero. A curve $\gamma\colon [a,b] \to M$ is **admissible** if there is a partition $a_0 < \cdots < a_k$ of $[a,b]$ such that $\gamma|_{[a_i,a_{i+1}]}$ is regular for all i.

A one-parameter family of curves Γ is called **admissible** if its domain is of the form $J \times [a,b]$ for some open interval J, and there is a partition $a_0 < \cdots < a_k$ of $[a,b]$ such that Γ is smooth on each rectangle $J \times [a_i, a_{i+1}]$, and finally Γ_s is an admissible curve.

For an admissible family Γ, the transverse curves $\Gamma^{(t)}$ are smooth (as Γ is smooth on $J \times \{t\}$), but the main curves are only piecewise regular. Thus $\partial_s\Gamma, \partial_t\Gamma$ are smooth on the rectangles $J \times [a_i, a_{i+1}]$ but not necessarily on the whole domain.

If Γ is an admissible family, then $\partial_s\Gamma$ is a piecewise smooth vector field along Γ. This means it is continuous, and its restriction to each $J \times [a_i, a_{i+1}]$ is smooth. As explained its restrictions are smooth, and we explain why it is continuous.

Note that for each i, $\partial_s\Gamma$'s values on $J \times \{a_i\}$ depend only on the values of Γ on that set, since the derivative is taken with respect to s. And $\partial_s\Gamma$ is smooth on $J \times [a_{i-1}, a_i]$ and $J \times [a_i, a_{i+1}]$, and their values must agree on their overlap (taking the limit as t goes to a_i).

Conversely, $\partial_t\Gamma$ is not in general continuous at $t = a_i$.

18.1.3 Definition

If $\gamma\colon [a,b] \to M$ is an admissible curve, a **variation** of γ is an admissible family of curves $\Gamma\colon J \times [a,b] \to M$ such that J is an open interval containing 0 and $\Gamma_0 = \gamma$.

It is a **proper variation** if in addition all the main curves have the same endpoints: $\Gamma_s(a) = \gamma(a)$ and $\Gamma_s(b) = \gamma(b)$ for all $s \in J$.

If Γ is a variation of γ, then the **variation field** of Γ is the smooth vector field $V(t) = \partial_s\Gamma(0,t)$ along γ. We say that a vector field V along γ is **proper** if $V(a)$ and $V(b)$ are both zero.

18.1.4 Lemma

If γ is an admissible curve and V a piecewise smooth vector field along γ, then V is the variation field of some variation of γ. If V is proper, the variation can be taken to be proper as well.

Proof

Let $\Gamma(s,t) = \exp_{\gamma(t)}(sV(t))$. By compactness of $[a,b]$ there is some $\varepsilon > 0$ such that Γ is defined on $(-\varepsilon,\varepsilon)\times[a,b]$. Then Γ is smooth on each $(-\varepsilon,\varepsilon)\times[a_i,a_{i+1}]$ when V is smooth on $[a_i,a_{i+1}]$, and it is continuous on its whole domain.

The variation field of Γ is V:

$$\partial_s\Gamma(0,t) = \partial_s|_{s=0}\exp_{\gamma(t)}(sV(t)) = V(t)$$

because $\exp_p(sv)$ is the geodesic with initial velocity v.

Furthermore if V is proper, so $V(a), V(b)$ are zero, then

$$\Gamma(s,a) = \exp_{\gamma(a)}(0) = \gamma(a)$$

and similar for b, so Γ is proper. ■

18.1.5 Lemma

Let $\Gamma\colon J\times[a,b]\to M$ be an admissible family of curves. Then on every rectangle $J\times[a_i,a_{i+1}]$ where Γ is smooth,

$$D_s\partial_t\Gamma = D_t\partial_s\Gamma.$$

Proof

This is local so we compute in local coordinates (x^i) around a point $\Gamma(s_0,t_0)$. Writing

$$\Gamma(s,t) = (x^1(s,t),\ldots,x^n(s,t)),$$

we have

$$\partial_t\Gamma = \frac{\partial x^k}{\partial t}\partial_k, \qquad \partial_s\Gamma = \frac{\partial x^k}{\partial s}\partial_k.$$

Then using the formula for the covariant derivative,

$$\begin{aligned} D_s\partial_t\Gamma &= \left(\frac{\partial^2 x^k}{\partial st} + \frac{\partial x^i}{\partial t}\frac{\partial x^j}{\partial s}\Gamma^k_{ji}\right)\partial_k, \\ D_t\partial_s\Gamma &= \left(\frac{\partial^2 x^k}{\partial ts} + \frac{\partial x^i}{\partial s}\frac{\partial x^j}{\partial t}\Gamma^k_{ji}\right)\partial_k. \end{aligned}$$

By symmetry of the second partial derivative and the connection, we see that these are equal. ■

18.2 The curvature tensor

Let $\overline{\nabla}$ be the standard connection on $\mathbb{R}^n$, then a direct computation shows

$$\overline{\nabla}_X\overline{\nabla}_Y Z - \overline{\nabla}_Y\overline{\nabla}_X Z = \overline{\nabla}_{[X,Y]}Z.$$

This can be shown to be tensorial (which we will do), and thus we need only verify this for coordinate frames. And since $\overline{\nabla}_{\partial_i}\partial_j = 0$ this is clear.

This means that any Riemannian manifold which is flat, meaning locally isometric to $\mathbb{R}^n$, must satisfy the above identity with its own Levi-Civita connection.

18.2.1 Definition

Let (M,g) be Riemannian, then define a map $R\colon \mathfrak{X}(M)\times\mathfrak{X}(M)\times\mathfrak{X}(M)\to\mathfrak{X}(M)$ by

$$R(X,Y)Z=\nabla_Y\nabla_X Z-\nabla_X\nabla_Y Z+\nabla_{[X,Y]}Z.$$

Note:

Conventions vary on the sign of R. Some authors (e.g. Lee) define the curvature tensor with the opposite sign to ours:

$$R(X,Y)Z=\nabla_X\nabla_Y Z-\nabla_Y\nabla_X Z-\nabla_{[X,Y]}Z.$$

In this course R was defined as in the previous definition.

18.2.2 Proposition

R is multilinear over $C^\infty(M)$ and thus defines a $(1,3)$-tensor field on M.

Proof

Linearity in Z is clear, and multilinearity over $\mathbb{R}$ is as well. We show linearity in Y:

$$\begin{aligned} R(X,fY)Z &= -\nabla_X\nabla_{fY}Z+\nabla_{fY}\nabla_X Z+\nabla_{[X,fY]}Z \\ &= -\nabla_X(f\nabla_Y Z)+f\nabla_Y\nabla_X Z+\nabla_{f[X,Y]+(Xf)Y}Z \\ &= -(Xf)\nabla_Y Z-f\nabla_X\nabla_Y Z+f\nabla_Y\nabla_X Z+f\nabla_{[X,Y]}Z+(Xf)\nabla_Y Z \\ &= fR(X,Y)Z. \end{aligned}$$

Linearity in X is done similarly. ■

We adopt the convention that the last index in R is its contravariant (upper) one. So in terms of a coordinate frame,

$$R=R_{ijk}^{\ell}dx^i\otimes dx^j\otimes dx^k\otimes\partial_\ell,\qquad R(\partial_i,\partial_j)\partial_k=R_{ijk}^{\ell}\partial_\ell.$$

18.2.3 Proposition

The components of R in any coordinate frame are given by

$$R_{ijk}^{\ell}=-\partial_i\Gamma_{jk}^{\ell}+\partial_j\Gamma_{ik}^{\ell}-\Gamma_{jk}^{m}\Gamma_{im}^{\ell}+\Gamma_{ik}^{m}\Gamma_{jm}^{\ell}.$$

Proof

A standard computation. ■

Note:

Curvature in the one-dimensional case is uninteresting. This is because $R(X,Y)=-R(Y,X)$, and in the one-dimensional case this clearly implies $R=0$. In particular if we consider the image of a curve to be a one-dimensional manifold (when the curve is not self-intersecting), then its curvature is zero.

18.2.4 Proposition

Let Γ be a smooth one-parameter family of curves $J \times I \to M$. Then for every vector field V along Γ,

$$D_s D_t V - D_t D_s V = R(\partial_t \Gamma, \partial_s \Gamma) V.$$

Proof

Locally write

$$\Gamma(s,t) = (\gamma^1(s,t), \ldots, \gamma^n(s,t)), \qquad V(s,t) = V^j(s,t)\partial_j|_{\Gamma(s,t)}.$$

Then

$$D_t V = \frac{\partial V^i}{\partial t}\partial_i + V^i D_t \partial_i,$$

and so

$$D_s D_t V = \frac{\partial^2 V^i}{\partial st}\partial_i + \frac{\partial V^i}{\partial t} D_s \partial_i + \frac{\partial V^i}{\partial s} D_t \partial_i + V^i D_s D_t \partial_i.$$

Thus we get

$$D_s D_t V - D_t D_s V = V^i (D_s D_t \partial_i - D_t D_s \partial_i).$$

Now we need to compute the commutator. Let us write

$$S = \partial_s \Gamma = \frac{\partial \gamma^k}{\partial s}\partial_k, \qquad T = \partial_t \Gamma = \frac{\partial \gamma^j}{\partial t}\partial_j.$$

Because ∂_i is extendible,

$$D_t \partial_i = \nabla_T \partial_i = \frac{\partial \gamma^j}{\partial \partial t}\nabla_{\partial_j}\partial_i.$$

Since $\nabla_{\partial_j}\partial_i$ is also extendible,

$$\begin{aligned} D_s D_t \partial_i &= \frac{\partial^2 \gamma^j}{\partial st}\nabla_{\partial_j}\partial_i + \frac{\partial \gamma^j}{\partial t}\nabla_S(\nabla_{\partial_j}\partial_i) \\ &= \frac{\partial^2 \gamma^j}{\partial st}\nabla_{\partial_j}\partial_i + \frac{\partial \gamma^j}{\partial t}\frac{\partial \gamma^k}{\partial s}\nabla_{\partial_k}\nabla_{\partial_j}\partial_i. \end{aligned}$$

We then see that

$$D_s D_t \partial_i - D_t D_s \partial_i = \frac{\partial \gamma^j}{\partial t}\frac{\partial \gamma^k}{\partial s}(\nabla_{\partial_k}\nabla_{\partial_j}\partial_i - \nabla_{\partial_j}\nabla_{\partial_k}\partial_i) = \frac{\partial \gamma^j}{\partial t}\frac{\partial \gamma^k}{\partial s}R(\partial_j, \partial_k) = R(T,S)\partial_i.$$

Thus

$$D_s D_t V - D_t D_s V = V^i R(T,S)\partial_i = R(T,S)V$$

as desired. ■

18.2.5 Definition

Let (M,g) be a Riemannian manifold. Define its **Riemannian curvature tensor** to be the tensor obtained by lowering R's index: $\mathrm{Rm} = R^\flat$. Explicitly, it is a $(0,4)$-tensor field given by

$$\mathrm{Rm}(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle.$$

In terms of local coordinates,

$$\mathrm{Rm} = R_{ijk\ell}\, dx^i \otimes dx^j \otimes dx^k \otimes dx^\ell, \qquad R_{ijk\ell} = g_{\ell m} R_{ijk}^m.$$

Thus

$$R_{ijk\ell} = g_{\ell m}(-\partial_i \Gamma^m_{jk} + \partial_j \Gamma^m_{ik} - \Gamma^p_{jk}\Gamma^m_{ip} + \Gamma^p_{ik}\Gamma^m_{jp}).$$

Riemannian curvature is preserved by local isometries.

18.2.6 Proposition

If $\varphi\colon M \to \widetilde{M}$ is a local isometry, let $\mathrm{Rm}, \widetilde{\mathrm{Rm}}$ be the Riemannian curvature tensors of $M, \widetilde{M}$ respectively. Then $\varphi^* \widetilde{\mathrm{Rm}} = \mathrm{Rm}$.

Proof

We have

$$\varphi^* \widetilde{\mathrm{Rm}}(X, Y, Z, W)_p = \widetilde{\mathrm{Rm}}_{\varphi p}(d\varphi_p X, d\varphi_p Y, d\varphi_p Z, d\varphi_p W) = \langle R_{\varphi p}(d\varphi_p X, d\varphi_p Y) d\varphi_p Z, d\varphi_p W\rangle.$$

Then using the definition of the curvature tensor and the naturality of Levi-Civita, we get the desired result. ■

18.2.7 Proposition

The following identities hold for the Riemannian curvature tensor:

(1) $\mathrm{Rm}(W, X, Y, Z) = -\mathrm{Rm}(X, W, Y, Z)$,

(2) $\mathrm{Rm}(W, X, Y, Z) = -\mathrm{Rm}(W, X, Z, Y)$,

(3) $\mathrm{Rm}(W, X, Y, Z) = \mathrm{Rm}(Y, Z, W, X)$,

(4) $\mathrm{Rm}(W, X, Y, Z) + \mathrm{Rm}(X, Y, W, Z) + \mathrm{Rm}(Y, W, X, Z) = 0$.

Proof

The first is immediate because $R(W, X)Y = -R(X, W)Y$. To prove (2) it is sufficient to show that $\mathrm{Rm}(W, X, Y, Y) = 0$ for all Y (equivalence of antisymmetric multilinear maps). This follows from expanding $WX\left|Y\right|^2, XW\left|Y\right|^2, [W, X]\left|Y\right|^2$ and equating them.
To prove the fourth, it is sufficient to show

$$R(W, X)Y + R(X, Y)W + R(Y, W)X = 0,$$

which can be shown via a tedious computation.
Then the third follows from the others, by a tedious expansion. ■

18.3 Minimizing curves

18.3.1 Definition

If $I, K \subseteq \mathbb{R}$ are intervals, $\gamma\colon I \to M$ a geodesic, and $\Gamma\colon K \times I \to M$ a variation of γ. It is called a **variation through geodesics** if its main curves $\Gamma_s(t) = \Gamma(s, t)$ are all geodesics (requiring Γ be smooth).

18.3.2 Theorem (Jacobi equation)

Let γ be a geodesic in M and J a vector field along γ. If J is the variation field of a variation through geodesics, then J satisfies the **Jacobi equation**:

$$D_t^2 J + R(\gamma', J)\gamma' = 0.$$

Proof

Let Γ be a variation through geodesics of γ such that J is its variation field, meaning $J(t) = S(0,t)$. Let $T = \partial_t \Gamma(s,t)$ and $S = \partial_s \Gamma(s,t)$ so T is a geodesic. Then $D_t T = 0$ identically.
Then if we take the covariant derivative with respect to s we get $D_s D_t T = 0$. Recall curvature measures the non-commutativity of covariant derivatives, so

$$0 = D_s D_t T = D_t D_s T + R(T,S)T = D_t D_t S + R(T,S)T,$$

where the last equality is due to symmetry. Evaluating at $s = 0$ gives us

$$D_t^2 J + R(\gamma', J)\gamma'$$

since $\gamma' = T(0,t)$ and $S(0,t) = J(t)$. ■

18.3.3 Definition
A smooth vector field along a geodesic satisfying the Jacobi equation is called a **Jacobi field**.

18.3.4 Lemma
If $\gamma\colon I \to M$ is geodesic, starting at v and $J \in \mathfrak{X}(\gamma)$ a Jacobi field, then

$$\langle \gamma', J \rangle = \langle v, J(0) \rangle + \langle v, J'(0) \rangle t.$$

Proof

Differentiating

$$\frac{d}{dt}\langle \gamma', J \rangle = \langle D_t\gamma', J \rangle + \langle \gamma', D_t J \rangle = \langle \gamma', J' \rangle.$$

And

$$\frac{d}{dt}\langle \gamma', J' \rangle = \langle \gamma', J'' \rangle = -\langle \gamma', R(\gamma', J)\gamma' \rangle,$$

where last equality is due to Jacobi equation.

$$= -\mathrm{Rm}(\gamma', J, \gamma', \gamma') = 0,$$

by symmetry of the Riemannian curvature tensor. Thus $\langle \gamma', J \rangle = a + bt$. Evaluating the function and its derivative at 0 gives us

$$a = \langle \gamma'(0), J(0) \rangle, \qquad b = \langle \gamma'(0), J'(0) \rangle,$$

as desired. ■

18.3.5 Lemma (Gauss)

Let $p \in (M, g)$, and $w, v \in T_pM$, then

$$\big\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\big\rangle_{\exp_p(v)} = \langle v, w\rangle_p.$$

Where $d(\exp_p)_v \colon T_vT_pM \cong T_pM \to T_{\exp_p(v)}M$ under the natural identification $T_vT_pM \cong T_pM$.

Proof

Define

$$f\colon (-\varepsilon, \varepsilon) \times [0,1] \to M, \qquad (s,t) \mapsto \exp_p(t(v + sw)).$$

So $f(s,1) = \exp_p(v + sw)$ and $t \mapsto f(t,s)$ is a geodesic for every s, and thus it is a variation through geodesics. Thus $J = \partial_s f(0,t)$ is a Jacobi field along $\gamma(t) = \exp_p(tv)$. Then

$$J(1) = \frac{\partial f}{\partial s}(0,1) = \frac{d}{ds}\exp_p(v + sw)|_{s=0} = d(\exp_p)_v(w),$$

and

$$\gamma'(1) = d(\exp_p)_v(v).$$

Thus by the above lemma

$$\big\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\big\rangle = \langle \gamma', J\rangle(1) = \langle v, J(0)\rangle + \langle v, J'(0)\rangle.$$

So we compute J, J' at zero.

$$J(0) = \frac{\partial f}{\partial s}(0,0) = \frac{d}{ds}\exp_p(0)|_{s=0} = 0,$$

since $\exp_p(0) = p$ is constant. And

$$J'(0) = \frac{D}{\partial t}\frac{\partial f}{\partial s}(0,0) = \frac{D}{\partial s}\bigg|_{s=0}\Big(\frac{\partial}{\partial t}\bigg|_{t=0} f\Big).$$

$\exp_p(t(v + sw))$ is a geodesic starting at $v + sw$, so its derivative with respect to t at zero is $v + sw$, so this is equal to

$$J'(0) = \frac{D}{\partial s}\bigg|_{s=0}(v + sw) = w.$$

Thus

$$\big\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\big\rangle = \langle v, w\rangle$$

as desired. ∎

18.3.6 Theorem

Let $B \subseteq M$ be a geodesic ball around $p \in M$, and let $\gamma\colon I \to B$ be a geodesic starting at p. Let $c\colon I \to M$ be a piecewise smooth curve such that $c(0) = p, c(1) = \gamma(1)$.

Then $L_g(\gamma) \leq L_g(c)$, and if there is equality then c is a reparameterization of γ, i.e. there is a non-decreasing $\varphi\colon I \to I$ such that $c = \gamma \circ \varphi$.

Proof

We assume $c(t) \in B$ for all t, and $c(t) \neq p$ for all $t \in (0,1]$. If $c(t) = p$, then we can consider $c|_{[t,1]}$. Then we have since $c(t) \in B$ and $\exp_p : B_\varepsilon(0) \to B$ is a diffeomorphism,

$$c(t) = \exp_p(w(t)), \qquad w(t) \in T_pM$$

for a smooth $w: I \to B_\varepsilon(0)$. Since $c(t) \neq p$ we have $w(t) \neq 0$ for $t > 0$. Then define smooth maps

$$r(t) = |w(t)|\,, \qquad v(t) = \frac{w(t)}{r(t)}.$$

Now define

$$f: [0, \varepsilon) \times [0,1] \to M, \qquad f(r,t) = \exp_p(rv(t)).$$

Then

$$c(t) = f(r(t), t),$$

so

$$c'(t) = \frac{\partial f}{\partial r} \cdot r' + \frac{\partial f}{\partial t},$$

and thus

$$|c'(t)|^2 = \left|\frac{\partial f}{\partial r}\right|^2 |r'|^2 + \left|\frac{\partial f}{\partial t}\right|^2 + 2r'\left\langle \frac{\partial f}{\partial r}, \frac{\partial f}{\partial t}\right\rangle.$$

Notice that

$$\begin{aligned}\left\langle \frac{\partial f}{\partial r}, \frac{\partial f}{\partial t}\right\rangle &= \left\langle \frac{d}{dr}\exp(rv(t)), \frac{d}{dt}\exp(rv(t))\right\rangle \\ &= \left\langle d(\exp)_{rv(t)}(v(t)), d(\exp)_{rv(t)}(rv'(t))\right\rangle \\ &= r\langle v(t), v'(t)\rangle_p\end{aligned}$$

where the last equality is due to Gauss' lemma. This is equal to

$$= \frac{r}{2}\frac{d}{dt}\langle v(t), v(t)\rangle = 0,$$

since $|v(t)| = 0$ and thus its derivative is zero.

Thus we have

$$|c'(t)|^2 = \left|\frac{\partial f}{\partial r}\right|^2 |r'|^2 + \left|\frac{\partial f}{\partial t}\right|^2 \geq \left|\frac{\partial f}{\partial r}\right|^2 |r'|^2.$$

Furthermore, let $\beta(r) = \exp(rv(t))$ so it is a geodesic with initial velocity $v(t)$, and

$$\frac{\partial f}{\partial r} = \left.\frac{\partial}{\partial r}\right|_{(r,t)} \exp(rv(t)) = \beta'(r).$$

Since geodesics have constant speed,

$$\left|\frac{\partial f}{\partial r}\right| = |\beta'(r)| = |\beta'(0)| = |v(t)| = 1.$$

Thus

$$|c'(t)| \geq |r'|.$$

Furthermore we have

$$\gamma(1) = c(1) = \exp_p(r(1)v(1)),$$

and since γ is a geodesic we must then have

$$\gamma(t) = \exp_p(tr(1)v(1)) \Longrightarrow |\gamma'(t)| = |\gamma'(0)| = r(1)v(1).$$

Thus we have

$$L_g(\gamma) = \int_0^1 |r(1)v(1)| = |r(1)| = r(1).$$

And (recall that the following is an improper integral)

$$L_g(c) = \int_0^1 |c'(t)| \geq \int_0^1 |r'| \geq \int_0^1 r' = r(1) - \lim_{t\to 0} r(t) = r(1).$$

So we have the desired inequality.

If we have equality then we must have that $\left|\frac{\partial f}{\partial t}\right| = 0$ so $\frac{\partial f}{\partial t} = 0$. That is

$$0 = \frac{\partial f}{\partial t} = d\exp_p(rv'(t))$$

which implies, because $\exp_p$ is a diffeomorphism, that $v'(t) = 0$. Thus $v(t) = v(1)$ is constant. Furthermore, we must have (by considering our inequalities above) that $|r'| = r'$ so we must have $r' \geq 0$. So

$$c(t) = \exp_p(r(t)v(1)).$$

And since $\gamma(t) = \exp_p(tr(1)v(1))$ this implies $c(t) = \gamma(r(t)/r(1))$ and $r(t)/r(1)$ has nonnegative derivative and thus is nondecreasing, as desired.

Finally in the case that c leaves the ball B, let $t_1 \in [0,1]$ be the first value such that $c(t) \in \overline{B} - B$ (by compactness one exists). Thus $L_g(c|_{[0,t_1)})$ is greater than the length of the radial geodesic from p to $c(t_1)$ by our above proof. This has length equal to the radius of B (as it is radial), and since γ lies within B, its length is less than the radius. ∎

Notice that this gives another proof of the Riemannian distance being a metric. Because if $p \neq q$ then any curve connection p and q has curve length greater than a geodesic defined in a geodesic ball around p, which is nonzero.

Recall that the topology on M is equivalent to the metric topology induced by the metric d_g. Thus d_g is a continuous map $M \times M \to \mathbb{R}$.

18.3.7 Corollary

If B is a geodesic ball around p, then for every $q \in B$ the unique minimizing curve (up to reparameterization) from p to q is given by the radial geodesic from p to q.

Note:

Since $q \in B$ we have that $q = \exp_p(w)$ for some $w \in B_\varepsilon(0)$ (the ball mapped to B diffeomorphically by $\exp_p$). The **radial geodesic** from p to q is the geodesic $\gamma\colon [0,1] \to M$ given by $t \mapsto \exp_p(tw)$.

18.3.8 Definition

A **uniformly normal neighborhood** of $p \in M$ is an open neighborhood $p \in W$ such that there is a $\delta > 0$ such that for all $q \in W$, $W \subseteq \exp_q(B_\delta(0))$.

18.3.9 Lemma

Every point in M has a uniformly normal neighborhood.

Proof

For $p \in M$, an open neighborhood $p \in V$, and $\varepsilon > 0$ define

$$U_{V,\varepsilon} = \{(q,v) \mid q \in V, v \in T_qM, |v| < \varepsilon\} \subseteq TM|_V.$$

For small enough ε, exp is defined on all of $U_{V,\varepsilon}$. So the following map is smooth

$$F\colon U_{V,\varepsilon} \to M \times M, \qquad (q,v) \mapsto (q, \exp_q(v)).$$

Let $\varphi: U \to M$ be a coordinate chart which we identify with its coordinate chart $\varphi: U \times \mathbb{R}^n \to TM$ of TM:

$$\varphi(u, v) = (\varphi(u), v^i \partial_i|_u).$$

Then

$$F \circ \varphi(u, v) = (\varphi(u), \exp_{\varphi(u)}(v^i \partial_i|_u)).$$

Thus

$$d(F \circ \varphi)_0 = \begin{pmatrix} d\varphi_0 & 0 \\ d\varphi_0 & d\varphi_0 \end{pmatrix},$$

which is invertible, and thus by the inverse function theorem there is a neighborhood of $(p, 0)$ such that F is a diffeomorphism onto its image.

Since $U_{V,\varepsilon}$ form a local basis around $(p, 0) \in TM$, we can choose this neighborhood to be $U_{V,\delta}$. So $F(U_{V,\delta})$ is open in $M \times M$, there is an open neighborhood $p \in W$ such that $W \times W \subseteq F(U_{V,\delta})$. So for every $q \in W$ we have that

$$\{q\} \times W \subseteq F(U_{V,\delta}) \cap (\{q\} \times M) = \{q\} \times \exp_q(B_\delta(0)).$$

Thus $W \subseteq \exp_q(B_\delta(0))$ as desired. ∎

18.3.10 Definition

An admissible curve γ is **minimizing** if $L_g(\gamma) \leq L_g(\widetilde{\gamma})$ for every other admissible curve $\widetilde{\gamma}$ with the same endpoints.

And $\gamma: I \to M$ is **locally minimizing** if for every $t_0 \in I$ there is a neighborhood $t_0 \in I_0 \subseteq I$ such that whenever $a < b$ are in I_0 then $\gamma|_{[a,b]}$ is minimizing.

Note that $\gamma: [a, b] \to M$ is minimizing iff $d_g(\gamma(a), \gamma(b)) = L_g(\gamma)$ by definition of d_g.

18.3.11 Lemma

Let $p \in M$ and B be a geodesic ball around p. Then for every $q \in B$, the radial geodesic from p to q is the unique minimizing curve from p to q.

Proof

Suppose $q = \exp_p(cv)$ for $v \in T_pM$ a unit vector. Then $\gamma: [0, c] \to M$ given by $\gamma(t) = \exp_p(tv)$ is the radial geodesic from p to q and it has arc length c.

Now let $\sigma: [0, b] \to M$ be admissible from p to q which we can assume is parameterized by arc length. Let $a_0 \in [0, b]$ be the last time $\sigma(t) = p$, and $b_0 \in [0, b]$ the first time after a_0 that $\sigma(t)$ meets the geodesic sphere S_c of radius c around p. Then

18.3.12 Theorem

Every geodesic is locally minimizing. Conversely, every admissible minimizing curve is a geodesic (in particular it is smooth).

Proof

Let $\gamma: I \to M$ be a geodesic, which we take to be defined on an open interval. Let $t_0 \in I$ and let W be a uniformly normal neighborhood of $\gamma(t_0)$. And let $I_0 \subseteq I$ be the connected component of $\gamma^{-1}W$ containing t_0. If $a < b$ are in I_0 then the definition of a uniformly normal neighborhood implies $\gamma(b)$ is contained in a geodesic ball centered at $\gamma(a)$. Thus there is a unique minimizing curve from $\gamma(a)$ to $\gamma(b)$ given by a radial geodesic. Since $\gamma|_{[a,b]}$ is also a geodesic segment from $\gamma(a)$ to $\gamma(b)$ lying in the same geodesic ball, it must coincide with this radial geodesic. So $\gamma|_{[a,b]}$ is minimizing for all $a < b$ in I_0, and thus γ is locally minimizing.

Conversely suppose $\gamma\colon I = [a,b] \to M$ is a minimizing admissible curve. Then for every $t_0 \in I$ as before we can find a connected neighborhood I_0 of t_0 such that $\gamma(I_0)$ is contained in a uniformly normal neighborhood W. Then for every $a_0, b_0 \in I_0$, the unique minimizing curve $\gamma(a_0)$ to $\gamma(b_0)$ is a radial geodesic. Since $\gamma|_{[a_0,b_0]}$ is minimizing, it must coincide with this radial geodesic. So γ is a geodesic in a neighborhood of t_0 for all $t_0 \in I$, since being a geodesic is a local property, γ is a geodesic. ■

Note that there does not need to exist an admissible minimizing curve between two points, even if the manifold is connected.

18.4 Sectional curvature

18.4.1 Lemma

Let $p \in M$ and $\sigma \subseteq T_pM$ be a two-dimensional subspace. Then for every basis x, y of σ, define

$$K(x,y) = \frac{\mathrm{Rm}(x,y,x,y)}{\langle x,x\rangle\langle y,y\rangle - \langle x,y\rangle^2}.$$

This is a constant, independent of choice of basis.

Proof

Every basis of σ can be obtained from x, y by elementary operations:

$$x, y \mapsto y, x, \qquad x, y \mapsto x, \lambda y, \qquad x, y \mapsto x + \lambda y, y.$$

Since Rm is antisymmetric in both its first two and last two coordinates, K is invariant under the first elementary operation (swapping). And by multilinearity it is routine to check that it is invariant under the second two elementary operations. Thus K is constant. ■

Note that $\langle x,x\rangle\langle y,y\rangle - \langle x,y\rangle^2$ is the square of the area of parallelogram spanned by x, y and is nonzero when x, y are linearly independent.

18.4.2 Definition

For a Riemannian manifold (M, g), we define its **sectional curvature** at (p, σ) for $\sigma \subseteq T_pM$ a two dimensional subspace by

$$K_p(\sigma) = K(x,y) \qquad \text{for any basis } x, y \in \sigma.$$

Note:

While sign conventions for R (and thus for Rm) vary, sign conventions for K do not. If R is defined with the opposite sign convention, then we could define K by

$$K_p(\sigma) = \frac{\mathrm{Rm}(x,y,y,x)}{\langle x,x\rangle\langle y,y\rangle - \langle x,y\rangle^2},$$

see e.g. Lee. Sectional curvature is a generalization of Gaussian curvature, so its sign is kept constant with Gaussian curvature.

18.4.3 Lemma

Let V be a finite-dimensional vector space and $F, G\colon V^4 \to \mathbb{R}$ multilinear functionals. Further suppose that F, G satisfy the symmetries of Riemannian curvature. If $F(x, y, x, y) = G(x, y, x, y)$ for all $x, y \in V$ then $F = G$.

Proof

Substitute $y + z$ for y in $F(x, y, x, y) = G(x, y, x, y)$ and use the symmetries to obtain

$$F(x, y, x, z) = G(x, y, x, z).$$

Then substitute $x + w$ for x to obtain

$$F(x, y, w, z) = G(x, y, w, z). \qquad \blacksquare$$

In particular this means that sectional curvature determines Riemannian curvature. That is, if we are given a Riemannian manifold, it is enough to know either the Riemannian or sectional curvature to obtain the other.

18.4.4 Corollary

Let M be a Riemannian manifold, then define a 4-tensor on M by

$$\mathrm{Rm}'(X, Y, Z, W) = \langle X, W\rangle\langle Y, Z\rangle - \langle Y, W\rangle\langle X, Z\rangle.$$

Then M has constant sectional curvature K_0 at p iff $\mathrm{Rm} = K_0\mathrm{Rm}'$ at p.

Proof

Note that Rm' satisfies the symmetries of Riemannian curvature, then $\mathrm{Rm}(X, Y, X, Y) = K_0\mathrm{Rm}'(X, Y, X, Y)$ by definition. By the above lemma this means $\mathrm{Rm} = K_0\mathrm{Rm}'$ as desired, and the converse is immediate. $\blacksquare$

18.5 Jacobi fields

Recall that a **Jacobi field** is a vector field J along a geodesic γ satisfying the **Jacobi equation**:

$$D_t^2 J + R(\gamma', J)\gamma' = 0.$$

18.5.1 Proposition

Let $\gamma\colon I \to M$ be a geodesic on a Riemannian manifold M. Let $t_0 \in I$ and $p = \gamma(t_0)$, then for every $v, w \in T_pM$ there is a unique Jacobi field J along γ such that

$$J(a) = v, \qquad D_t J(a) = w.$$

Proof

Take a parallel orthonormal frame (E_i) along γ (obtained by parallel transport of an orthonormal basis), and write $v = v^i E_i(a)$ and $w = w^i E_i(a)$, and $\gamma' = y^i E_i$. Then $J \in \mathfrak{X}(\gamma)$ can be written as $J = J^j E_j$, so that

$D_t^2 J = \ddot{J}^j E_j$ since the E_j are paralell. The Jacobi equation becomes

$$0 = \ddot{J}^i E_i + R(J^j E_j, y^k E_k)(y^\ell E_\ell) = \ddot{J}^i E_i + J^j y^k y^\ell R^i_{jk\ell}(\gamma(t)) E_i,$$

where $R^i_{jk\ell}(\gamma)$ are the components of R in the frame E_i. This is zero iff for all i,

$$0 = \ddot{J}^i + J^j y^k y^\ell R^i_{jk\ell}(\gamma),$$

Notice that this is all smooth, and thus a smooth second-order linear ODE, which has a unique solution along all of I. ■

Notice that for a geodesic γ, $\dot{\gamma}$ itself is a Jacobi field along γ. Indeed, $\dot{\gamma}$ is the variation field of $\Gamma(s,t) = \gamma(s)$, and variation fields of variations through geodesics are Jacobi.

Further note that the collection of Jacobi fields along γ form a subspace of $\mathfrak{X}(\gamma)$, by the linearity of the Jacobi field equation.

18.5.2 Corollary

Let (M,g) be a Riemannian manifold of dimension n, and γ a geodesic on M. The set of Jacobi fields along γ form a $2n$-dimensional subspace of $\mathfrak{X}(\gamma)$.

Proof

For any $a \in I$, let $p = \gamma(a)$, and consider the map from Jacobi fields along γ to $T_pM \oplus T_pM$ given by $J \mapsto (J(a), D_t J(a))$. The above proposition shows that this is a bijection. ■

We consider the case of a Riemannian manifold with constant sectional curvature. For $K \in \mathbb{R}$ let $s_K\colon \mathbb{R} \to \mathbb{R}$ be defined by

$$s_K(t) = \begin{cases} t & \text{if } K = 0 \\ R \sin t/R & \text{if } K = 1/R^2 > 0 \\ R \sinh t/R & \text{if } K = -1/R^2 < 0. \end{cases}$$

18.5.3 Lemma

Suppose (M,g) is a Riemannian manifold with constant sectional curvature K. Let γ be a unit-speed geodesic in M. Then the Jacobi fields J orthogonal to $\dot{\gamma}$ which vanish at $t = 0$ are given by

$$J(t) = c \cdot s_K(t) E(t),$$

where c is an arbitrary constant, and $E \in \mathfrak{X}(\gamma)$ is any vector field parallel along γ and orthogonal to $\dot{\gamma}$.

Proof

Since M has constant curvature, we have that

$$\mathrm{Rm}(X,Y,Z,W) = K(\langle X,W\rangle\langle Y,Z\rangle - \langle Y,W\rangle\langle X,Z\rangle).$$

So we have

$$R(X,Y)Z = K(\langle Y,Z\rangle X - \langle X,Z\rangle Y).$$

Substituting into the Jacobi equation we get

$$0 = D_t^2 J + K(\langle \gamma', \gamma'\rangle J - \langle J, \gamma'\rangle \gamma') = D_t^2 J + KJ.$$

Choosing $J(t) = u(t)E(t)$, the ODE becomes

$$u''(t) + Ku(t) = 0,$$

whose solutions given $u(0) = 0$ are multiples of s_K. Thus all fields of the form in the lemma are Jacobi fields.

Now, the dimension of the space of Jacobi fields vanishing at 0 orthogonal to γ' is $n-1$ (the space of vanishing Jacobi fields has dimension n, by considering the isomorphism $J \mapsto (J(0), D_t J(0))$, and we reduce the dimension by one by considering the orthogonal subspace to γ'). The space of vector fields parallel along γ orthogonal to $\dot{\gamma}$ also has dimension $n-1$ (mapping E to $E(0) \in T_pM$), so this gives all such Jacobi fields.

■

18.6 The second fundamental form

18.6.1 Definition

The **curvature** of a vector bundle $E \to M$ with connection ∇ is the section

$$R \in \Gamma\big(\Lambda^2 T^*M \otimes \operatorname{end}(E)\big)$$

given by

$$R(X,Y)s = -\nabla_X \nabla_Y s + \nabla_Y \nabla_X s + \nabla_{[X,Y]} s, \qquad X, Y \in \mathfrak{X}(M),\ s \in \Gamma(E).$$

The curvature is seen to be a section because the formula is easily seen to be $C^\infty(M)$-multilinear and alternating.

Suppose we have a smooth map $f\colon M \to N$ and a connection $\nabla\colon \mathfrak{X}(N) \times \Gamma(E) \to \Gamma(E)$. Then f pulls back the connection to a connection $f^*\nabla = \nabla^*\colon \mathfrak{X}(M) \times \Gamma(f^*E) \to \Gamma(f^*E)$. Let R be the curvature of ∇ and R^* be ∇^*'s. Then we see for $s \in \Gamma(E)$,

$$\begin{aligned} R^*(X,Y)(s \circ f) &= -\nabla^*_X \nabla^*_Y (s \circ f) + \nabla^*_Y \nabla^*_X (s \circ f) + \nabla^*_{[X,Y]}(s \circ f) \\ &= -\nabla^*_X (\nabla_{df(Y)} s) \circ f + \nabla^*_Y (\nabla_{df(X)} s) \circ f + \nabla_{df[X,Y]} s \circ f \\ &= \big(-\nabla_{df(X)} \nabla_{df(Y)} s + \nabla_{df(Y)} \nabla_{df(X)} s + \nabla_{[df(X), df(Y)]} s\big) \circ f \\ &= (R(df \circ X, df \circ Y)s) \circ f. \end{aligned}$$

We summarize this in the following lemma.

18.6.2 Lemma

Let $f\colon M \to N$ be smooth and ∇ a connection on $E \to N$. Define ∇^* to be the pullback of ∇ along f, and let R and R^* be the curvatures of ∇ and ∇^* respectively. Then for every $X, Y \in \mathfrak{X}(M)$ and $s \in \Gamma(E)$ we have

$$R^*(X,Y)(s \circ f) = (R(df \circ X, df \circ Y)s) \circ f.$$

Explicitly for a point $p \in M$ we have

$$R^*(X,Y)(s \circ f)|_p = R(df_p(X_p), df_p(Y_p)) s_{fp}.$$

18.6.3 Definition

A **metric** on a vector bundle $E \to M$ is a section

$$h \in \Gamma(E^* \otimes E^*),$$

such that $h(s,t) = h(t,s)$ for all $s,t \in \Gamma(E)$, and $h(s,s)_p > 0$ if $s_p \neq 0$. Equivalently it is a section in $\Gamma(\mathrm{Sym}^2 E^*)$ which is positive definite on each fiber.

A **metric connection** relative to h is a connection ∇ such that

$$d(h(s,t)) = h(\nabla s, t) + h(s, \nabla t)$$

for all sections s,t, that is to say for all $X \in \mathfrak{X}(M)$,

$$X(h(s,t)) = h(\nabla_X s, t) + h(s, \nabla_X t).$$

If ∇ is a metric connection with respect to h, we define its **Riemannian curvature** to be

$$\mathrm{Rm}(X,Y,s,t) = h(R(X,Y)s,t).$$

Riemannian curvature is a section $\mathrm{Rm} \in \Gamma(T^*M \otimes T^*M \otimes E^* \otimes E^*)$.

If $S \subseteq E$ is a subvector bundle and h is a metric on E, then define $S^\perp \subseteq E$ to be the subvector bundle whose fibers are

$$(S^\perp)_p = (S_p)^\perp.$$

If $\overline{\nabla}$ is a metric connection on E, let $P\colon E \to S$ and $P^\perp\colon E \to S^\perp$ be orthogonal projections (on each fiber), which are vector bundle morphisms. Then we have $E = P \oplus P^\perp$, and we define connections ∇ on S and $\nabla^\perp$ on $S^\perp$ by

$$\nabla s = P(\overline{\nabla} s) \text{ for } s \in \Gamma(S), \qquad \nabla^\perp s = P^\perp(\overline{\nabla} s) \text{ for } s \in \Gamma(S^\perp).$$

Note that $\overline{\nabla} s$ is defined at a point p to be $\overline{\nabla}\widetilde{s}$ where $\widetilde{s}$ is an extension of s in a neighborhood of p to a section $\Gamma(E)$. So we get connections

$$\nabla\colon \Gamma(S) \to \Gamma(T^*M \otimes S), \qquad \nabla^\perp\colon \Gamma(S^\perp) \to \Gamma(T^*M \otimes S^\perp).$$

18.6.4 Definition

Let $E \to M$ be a vector bundle equipped with a metric h and metric connection $\overline{\nabla}$ For a subvector bundle $S \subseteq M$, consider the orthogonal projection of the connection to ∇ as above. Then the **second fundamental form** of S is the map

$$\mathrm{II}\colon \mathfrak{X}(M) \otimes \Gamma(S) \to \Gamma(S^\perp), \qquad \mathrm{II}(X \otimes s) = \overline{\nabla}_X s - \nabla_X s.$$

We also write $\mathrm{II}(X)s$ instead of $\mathrm{II}(X \otimes s)$.

18.6.5 Lemma

The second fundamental form is $C^\infty(M)$-linear, and thus defines a morphism of vector bundles

$$\mathrm{II}\colon TM \otimes S \to S^\perp.$$

Proof

It is sufficient to show that the map $(X,s) \mapsto \overline{\nabla}_X s - \nabla_X s$ is $C^\infty(M)$-bilinear. This is clearly additive and $C^\infty(M)$-linear over X. For $f \in C^\infty(M)$ we have

$$\mathrm{II}(X \otimes fs) = \overline{\nabla}_X(fs) - \nabla_X(fs) = f\overline{\nabla}_X s + (Xf)s - f\nabla_X s - (Xf)s = f(\overline{\nabla}_X s - \nabla_X s) = f\mathrm{II}(X \otimes s)$$

as desired. ■

Since $P + P^\perp = \mathrm{id}$ we have that $\mathrm{id} - P = P^\perp$ so $\mathrm{II}(X)s = P^\perp \circ \overline{\nabla}_X s$.

The second fundamental form of $S^\perp$ is denoted by $\mathrm{II}^\perp$, and is equal to

$$\mathrm{II}^\perp(X)s = \overline{\nabla}_X s - \nabla^\perp_X s = P \circ \overline{\nabla}_X s.$$

So we have $\mathrm{II} + \mathrm{II}^\perp = \overline{\nabla}$.

Let us denote by $\overline{R}$ the curvature relative to $\overline{\nabla}$ on E, R relative to ∇, and $R^\perp$ relative to $\nabla^\perp$. And $\overline{\mathrm{Rm}}, \mathrm{Rm}, \mathrm{Rm}^\perp$ similarly for Riemannian curvature.

18.6.6 Theorem (Gauss)

For all $X, Y \in \mathfrak{X}(M)$ and $s \in \Gamma(S)$ we have the following equality

$$P \circ \overline{R}(X,Y)s = R(X,Y)s - \mathrm{II}^\perp(X) \circ \mathrm{II}(Y)s + \mathrm{II}^\perp(Y) \circ \mathrm{II}(X)s.$$

Proof

We have by definition

$$\overline{\nabla}_X s = \nabla_X s + \mathrm{II}(X)s,$$

and thus

$$\overline{\nabla}_Y \overline{\nabla}_X s = \overline{\nabla}_Y \nabla_X s + \overline{\nabla}_Y \mathrm{II}(X)s.$$

Now $\nabla_X s$ is a section of S, while $\mathrm{II}(X)s$ is a section of $S^\perp$, so we get

$$= \nabla_Y \nabla_X s + \mathrm{II}(Y)\nabla_X s + \nabla^\perp_Y \mathrm{II}(X)s + \mathrm{II}^\perp(Y)\mathrm{II}(X)s.$$

The first and last terms are sections of S, while the middle terms are sections of $S^\perp$. The projecting them orthogonally via P yields

$$P \circ \overline{\nabla}_Y \overline{\nabla}_X s = \nabla_Y \nabla_S x + \mathrm{II}^\perp(Y)\mathrm{II}(X)s.$$

Similarly we get

$$P \circ \overline{\nabla}_X \overline{\nabla}_Y s = \nabla_X \nabla_Y s + \mathrm{II}^\perp(X)\mathrm{II}(Y)s,$$

and by definition we have

$$P \circ \overline{\nabla}_{[X,Y]} s = \nabla_{[X,Y]} s.$$

Putting this all together we get the desired result.

Note:

If curvature is defined with the opposite sign convention, then this result is

$$P \circ \overline{R}(X,Y)s = R(X,Y)s - \mathrm{II}^\perp(Y) \circ \mathrm{II}(X)s + \mathrm{II}^\perp(X) \circ \mathrm{II}(Y)s.$$

18.6.7 Lemma

For a vector field $X \in \mathfrak{X}(M)$ and sections $s \in \Gamma(S)$ and $t \in \Gamma(S^\perp)$ then

$$h(\mathrm{II}(X)s, t) = -h(s, \mathrm{II}^\perp(X)t).$$

Proof

We know $h(s,t) = 0$ and since $\overline{\nabla}$ is metric,

$$0 = X(h(s,t)) = h(\overline{\nabla}_X s, t) + h(s, \overline{\nabla}_X t).$$

Since $P, P^\perp$ are orthogonal projections they don't affect the metric when one of the inputs are in $S, S^\perp$ respectively. Thus this is equal to

$$= h(P^\perp \overline{\nabla}_X s, t) + h(s, P\overline{\nabla}_X t) = h(\mathrm{II}(X)s, t) + h(s, \mathrm{II}^\perp(X)t),$$

so we get the desired result. ■

18.6.8 Corollary

For $X, Y \in \mathfrak{X}(M)$, $s, t \in \Gamma(S)$,

$$\overline{\mathrm{Rm}}(X,Y,s,t) = \mathrm{Rm}(X,Y,s,t) - h(\mathrm{II}(X)s, \mathrm{II}(Y)t) + h(\mathrm{II}(Y)s, \mathrm{II}(X)t).$$

Proof

We have

$$\overline{\mathrm{Rm}}(X,Y,s,t) = h(\overline{R}(X,Y)s,t) = h(P \circ \overline{R}(X,Y)s,t),$$

where the last equality is due to t being a section of s. By Gauss's theorem, this is equal to

$$= h(R(X,Y)s,t) - h(\mathrm{II}^\perp(X) \circ \mathrm{II}(Y)s,t) + h(\mathrm{II}^\perp(Y) \circ \mathrm{II}(X)s,t),$$

and by the lemma above this is equal to

$$= \mathrm{Rm}(X,Y,s,t) - h(\mathrm{II}(Y)s, \mathrm{II}(X)t) + h(\mathrm{II}(X)s, \mathrm{II}(Y)t). \quad ■$$

Note:

For the opposite sign convention, we get

$$\overline{\mathrm{Rm}}(X,Y,s,t) = \mathrm{Rm}(X,Y,s,t) + h(\mathrm{II}(X)s, \mathrm{II}(Y)t) - h(\mathrm{II}(Y)s, \mathrm{II}(X)t).$$

If $f\colon (M,g) \to (\overline{M}, \overline{g})$ is a Riemannian embedding then let $\nabla, \overline{\nabla}$ be their respective Levi-Civita connections. Now we consider the vector bundle $T\overline{M}|_M$ on M, which splits as

$$T\overline{M}|_M = TM \oplus \nu_M,$$

where ν_M is the orthogonal complement of TM, and we identify TM with $df(TM) \subseteq T\overline{M}|_M$.
Now let $\nabla^* = f^*\overline{\nabla}$ be the pullback connection along f, so ∇^* is a connection on $\mathfrak{X}(f) = \Gamma(f^*T\overline{M}) \cong \Gamma(T\overline{M}|_M)$,

$$\nabla^*\colon \mathfrak{X}(M) \times \Gamma(T\overline{M}|_M) \to \Gamma(T\overline{M}|_M).$$

If we consider $P \circ \nabla^*\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$, we see that this is a symmetric metric connection, and thus equal to ∇. Recall though that we are identifying $df(TM)$ with TM.
Let $R, \overline{R}, R^*$ be the curvatures of $\nabla, \overline{\nabla}, \nabla^*$ respectively. Then we have for $X, Y \in \mathfrak{X}(M)$ and $Z \in \mathfrak{X}(\overline{M})$,

$$R^*(X,Y)(Z \circ f) = (\overline{R}(df(X), df(Y))Z) \circ f.$$

In particular if we consider $Z = \partial_i$, then for $Z \in \mathfrak{X}(M)$ we see that

$$R^*(X,Y)(df(Z)) = Z^i \frac{\partial f^j}{\partial_i} R^*(X,Y)(\partial_j \circ f) = \overline{R}(df(X), df(Y))(df(Z)).$$

If we let $\mathrm{Rm}, \overline{\mathrm{Rm}}, \mathrm{Rm}^*$ be the Riemannian curvatures, then we see

$$\begin{aligned}\mathrm{Rm}^*(X, Y, Z \circ f, W \circ f) &= \langle R^*(X,Y)(Z \circ f), W \circ f\rangle_{\overline{M}} \\ &= \left\langle \overline{R}(df(X), df(Y))Z, W\right\rangle_{\overline{M}} \circ f \\ &= \overline{\mathrm{Rm}}(df(X), df(Y), Z, W) \circ f,\end{aligned}$$

and again

$$\mathrm{Rm}^*(X, Y, df(Z), df(W)) = \overline{\mathrm{Rm}}(df(X), df(Y), df(Z), df(W))$$

when $X, Y, Z, W \in \mathfrak{X}(M)$.

Now let us assume that f is simply the inclusion, so we are viewing M as a submanifold of $\overline{M}$. So we have a literal equality $P \circ \nabla^* = \nabla$, and the formulas above give

$$\mathrm{Rm}^*(X, Y, Z, W) = \overline{\mathrm{Rm}}(X, Y, Z, W), \qquad X, Y, Z, W \in \mathfrak{X}(M).$$

Let II be TM's second fundamental form relative to $T\overline{M}|_M$ with connection ∇^*. So a bundle morphism $\mathrm{II}\colon TM \otimes TM \to \nu_M$. Then the above results give us for $X, Y, Z, W \in \mathfrak{X}(M)$,

$$\mathrm{Rm}^*(X, Y, Z, W) = \mathrm{Rm}(X, Y, Z, W) + \langle \mathrm{II}(X)Z, \mathrm{II}(Y)W\rangle - \langle \mathrm{II}(Y)W, \mathrm{II}(X)Z\rangle,$$

the metric being $\overline{M}$'s.

We summarize this,

18.6.9 Corollary

If M is a Riemannian submanifold of $\overline{M}$, then

$$\overline{\mathrm{Rm}}(X, Y, Z, W) = \mathrm{Rm}(X, Y, Z, W) - \langle \mathrm{II}(X)Z, \mathrm{II}(Y)W\rangle + \langle \mathrm{II}(Y)Z, \mathrm{II}(X)W\rangle.$$

We continue this special case where we have a Riemannian embedding of our manifold into a larger one. For the second fundamental form of $M \subseteq \overline{M}$ let us write $\mathrm{II}(X, Y)$ instead of $\mathrm{II}(X)Y$.

18.6.10 Lemma

The second fundamental form is symmetric.

Proof

Let $\widehat{X}, \widehat{Y}$ be extensions of X, Y in a common neighborhood. So by definition

$$\mathrm{II}(X, Y) = \nabla^*_X \widehat{Y} - \nabla_X Y,$$

and around a point this is equal to

$$= \overline{\nabla}_{\widehat{X}} \widehat{Y} - \nabla_X Y$$

by the definition of the pullback connection. By symmetry,

$$= \overline{\nabla}_{\widehat{Y}} \widehat{X} + [\widehat{X}, \widehat{Y}] - \nabla_Y X - [X, Y],$$

since $\widehat{X}, \widehat{Y}$ are extensions of X, Y in a neighborhood of our point in M, their commutators agree on this point so this is equal to

$$= \overline{\nabla}_{\widehat{Y}} \widehat{X} - \nabla_Y X = \mathrm{II}(Y, X)$$

as desired. ■

18.6.11 Corollary

If $M \subseteq \overline{M}$ is embedded, then for $p \in M$ and $\sigma \subseteq T_pM \subseteq T_p\overline{M}$ with orthonormal basis v, w we have

$$\overline{K}_p(\sigma) = K_p(\sigma) - \langle \mathrm{II}(v,v), \mathrm{II}(w,w)\rangle + \langle \mathrm{II}(v,w), \mathrm{II}(v,w)\rangle.$$

Proof

When v, w are orthonormal $\overline{K}_p(\sigma) = \overline{\mathrm{Rm}}(v,w,v,w)$ and the result follows from the previous corollary. ■

We now consider a special case of this analysis, when $M \subseteq \overline{M}$ is a hypersurface: $\dim \overline{M} = \dim M + 1$. For example, $\overline{M} = \mathbb{R}^{n+1}$ and $M = S^n$. Further suppose that there is a nonvanishing section $\eta \in \nu_M$, so we can normalize it and assume $\langle \eta, \eta\rangle = 1$. This section is called the **Gauss map**. Because $\dim \nu_M = 1$, η is unique up to sign.

Define the $(0,2)$-tensor

$$b\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to C^\infty(M), \qquad b(X,Y) = \langle \mathrm{II}(X,Y), \eta\rangle.$$

Then we see that as II lies in ν_M, it is a scalar multiple of η, meaning $\mathrm{II}(X,Y) = \langle \mathrm{II}(X,Y), \eta\rangle\eta$. Thus we see

$$\overline{K}_p(\sigma) = K_p(\sigma) - b(v,v)b(w,w) + b(v,w)^2.$$

Furthermore let us define $\beta = b^\flat$, so the map $\beta\colon TM \to TM$ characterized by

$$\langle \beta(X), Y\rangle = b(X,Y).$$

Then

$$b(X,Y) = \langle \mathrm{II}(X)Y, \eta\rangle = -\left\langle Y, \mathrm{II}^\perp(X)\eta\right\rangle,$$

thus $\langle \beta(X), Y\rangle = -\left\langle \mathrm{II}^\perp(X)\eta, Y\right\rangle$ so that $\beta(X) = -\mathrm{II}^\perp(X)\eta$.

Now let $P_\sigma\colon T_pM \to \sigma$ be the orthogonal projection onto σ. Then $P_\sigma(x) = \langle x, v\rangle v + \langle x, w\rangle w$ and so

$$P_\sigma \circ \beta(x) = \langle \beta(x), v\rangle v + \langle \beta(x), w\rangle w = b(x,v)v + b(x,w)w.$$

Thus

$$\det(P_\sigma \circ \beta|_\sigma) = b(v,v)b(w,w) - b(v,w)^2,$$

so that

$$\overline{K}_p(\sigma) = K_p(\sigma) - \det(P_\sigma \circ \beta|_\sigma).$$

We also have

$$\beta(X) = -\mathrm{II}^\perp(X)\eta = -\overline{\nabla}_X\eta + \nabla^\perp_X\eta.$$

Consider the second term: since the connection is metric so

$$0 = X\langle \eta, \eta\rangle = 2\left\langle \nabla^\perp_X\eta, \eta\right\rangle,$$

and η spans the normal bundle so $\nabla^\perp_X\eta = 0$. Thus

$$\beta(X) = -\overline{\nabla}_X\eta.$$

Note that since $-\eta$ is also a unit tangent to TM, we get the same result with $\beta(X) = \overline{\nabla}_X\eta$. Indeed, because σ is two-dimensional $\det(\beta|_\sigma) = \det(-\beta|_\sigma)$. Thus β is the the covariant derivative of η: $\beta = \nabla\eta$.

Further suppose that $\overline{M}$ has zero sectional curvature: $\overline{K} = 0$. Thus we have that

$$K_p(\sigma) = \det(P_\sigma \circ \beta|_\sigma),$$

sometime people write $\beta|_\sigma$ as a shorthand for $P_\sigma \circ \beta|_\sigma$, so this identity is written as $K_p(\sigma) = \det \beta|_\sigma$.

18.6.12 Example

Take for example $M = S^n_r \subseteq \mathbb{R}^{n+1}$ the sphere of radius r. Then $\eta \in \Gamma(\nu_M)$ is given by mapping $p \in S^n_r$ to $\frac{1}{r}p \in T_p\mathbb{R}^{n+1}$, and this is orthogonal to the tangent space of the sphere. Thus $\eta(x) = \frac{1}{r}x^i\partial_i$, so that its covariant derivative is

$$\beta(\partial_j) = \nabla_{\partial_j}\eta(x) = \frac{1}{r}(\partial_j x^i)\partial_i = \frac{1}{r}\partial_j$$

so that $\beta(x) = \frac{1}{r}x$. So $\beta = \frac{1}{r}\mathrm{id}$, and restricting this to any two-dimensional subspace has determinant $1/r^2$. Thus the n-sphere of radius r has constant sectional curvature of $1/r^2$.

18.7 Mean curvature

Recall that given an isometric embedding $f\colon M \to \overline{M}$, we have $T\overline{M}|_M = TM \oplus TM^\perp$. This gives us:

(1) the Levi-Civita connection $\overline{\nabla}$ on $\Gamma(T\overline{M}|_M) = \mathfrak{X}(f)$ and the Levi-Civita connection ∇ on $\Gamma(TM) = \mathfrak{X}(M)$.

(2) orthogonal projections

$$P\colon T\overline{M}|_M \to TM, \qquad P^\perp\colon T\overline{M}|_M \to TM^\perp.$$

(3) The **second fundamental form**

$$\mathrm{II}\colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \Gamma(TM^\perp), \qquad \mathrm{II}(X,Y) = \mathrm{II}(X)Y = \overline{\nabla}_X\overline{Y} - \overline{\nabla_X Y} = P^\perp(\overline{\nabla}_X\overline{Y}),$$

where $\overline{Y} \in \mathfrak{X}(f)$ is $df(Y)$ (not to be confused with the covector field df, this is the vector field along f given by $df(Y)_p = df_p(Y)$). Similarly we have

$$\mathrm{II}^\perp\colon \mathfrak{X}(M) \times \Gamma(TM^\perp) \to \mathfrak{X}(M), \qquad \mathrm{II}^\perp(X,\xi) = \mathrm{II}^\perp(X)\xi = \overline{\nabla}_X\xi - \nabla^\perp_X\xi = P(\overline{\nabla}_X\xi).$$

18.7.1 Lemma

For $X, Y \in \mathfrak{X}(M)$, we have

$$\overline{\nabla}_X\overline{Y} - \overline{\nabla}_Y\overline{X} = \overline{[X,Y]}.$$

Proof

We can assume that $M \subseteq \overline{M}$ is an embedded submanifold and $f = \mathrm{id}$. Then $\overline{X}$ is just the same as X. For any point $p \in M$ we can extend X, Y to vector fields of M locally around p, and then by symmetry

$$\overline{\nabla}_X\overline{Y} - \overline{\nabla}_Y\overline{X} = \overline{\nabla}_X\widetilde{Y} - \overline{\nabla}_Y\widetilde{X} = [\widetilde{X},\widetilde{Y}]$$

where $\widetilde{X}, \widetilde{Y}$ are the local extensions of X, Y. On the other hand, $[\widetilde{X},\widetilde{Y}]$ is a local extension of $[X,Y]$ so we have the desired result. ■

18.7.2 Corollary

The second fundamental form is symmetric.

Proof

For $X, Y \in \mathfrak{X}(M)$,

$$\mathrm{I\!I}(X,Y) - \mathrm{I\!I}(Y,X) = \overline{\nabla}_X \overline{Y} - \overline{\nabla_X Y} - \overline{\nabla}_Y \overline{X} + \overline{\nabla_Y X} = \left(\overline{\nabla}_X \overline{Y} - \overline{\nabla}_Y \overline{X}\right) - \overline{\left(\nabla_X Y - \nabla_Y X\right)},$$

by the above lemma and symmetry of ∇, both terms are equal to $\overline{[X,Y]}$ and so this is zero. ■

Let $(\overline{M}, \overline{g})$ be a Riemannian manifold, and $F: M \times [0,1] \to \overline{M}$ be a smooth map such that for every $t \in [0,1]$, $f_t = F(\bullet, t)$ is an embedding. For each t, let $g_t = f_t^* \overline{g}$ be the pullback of $\overline{g}$ to a Riemannian metric on M. Define

$$Z = \frac{\partial F}{\partial t} = dF\left(\frac{\partial}{\partial t}\right).$$

This is a vector field along F, so $Z \in \mathfrak{X}(F)$, which is means it is a section of the pullback $F^* T\overline{M}$.

Each $t \in [0,1]$ defines a metric g_t, and thus a connection ∇^t, a second fundamental form $\mathrm{I\!I}_t$, and orthogonal projections $P_t, P_t^\perp$.

18.7.3 Lemma

For $X, Y \in \mathfrak{X}(M)$,

$$\frac{\partial}{\partial t} g_t(X,Y) = -2\langle \mathrm{I\!I}_t(X,Y), Z\rangle_{\overline{g}} + \left\langle \nabla_X^t P_t(Z), Y\right\rangle_{g_t} + \left\langle X, \nabla_Y^t P_t(Z)\right\rangle_{g_t}.$$

Proof

Notice that

$$g_t(X,Y) = (f_t^* \overline{g})(X,Y) = \overline{g}(df_t(X), df_t(Y)),$$

where $df_t(X) = \overline{X}$ as in the previous lemma.

At each point p, $df_t(X), df_t(Y)$ are vector fields along the curve $f^{(p)}: [0,1] \to \overline{M}$ given by $f^{(p)}(t) = f_t(p)$. Since $\overline{\nabla}$ is metric,

$$\frac{\partial}{\partial t} g_t(X,Y) = \frac{\partial}{\partial t} \overline{g}(df_t(X), df_t(Y)) = \overline{g}(D_t df_t(X), df_t(Y)) + \overline{g}(df_t(X), D_t df_t(Y)).$$

By definition of Z this is equal to

$$= \overline{g}(\overline{\nabla}_Z df_t(X), df_t(Y)) + \overline{g}(df_t(X), \overline{\nabla}_Z df_t(Y)).$$

Since $Z = df_t(\partial_t)$, and $[X, \partial_t] = 0$ and similar for Y, by the above lemma, this is equal to

$$= \overline{g}(\overline{\nabla}_X Z, df_t(Y)) + \overline{g}(df_t(X), \overline{\nabla}_Y Z).$$

Decomposing Z as $P(Z) + P^\perp(Z)$, this is equal to

$$= \overline{g}(\overline{\nabla}_X P(Z), df_t(Y)) + \overline{g}(\overline{\nabla}_X P^\perp(Z), df_t(Y)) + \overline{g}(df_t(X), \overline{\nabla}_Y P(Z)) + \overline{g}(df_t(X), \overline{\nabla}_Y P^\perp(Z)).$$

By definition of $\mathrm{I\!I}^\perp$, this is equal to

$$= \overline{g}(\mathrm{I\!I}^\perp(X) P^\perp(Z), Y) + \overline{g}(\mathrm{I\!I}^\perp(Y) P^\perp(Z), X) + \overline{g}(\nabla_X P(Z), Y) + \overline{g}(\nabla_Y P(Z), X).$$

Since the adjoint of $\mathrm{I\!I}^\perp(X)$ is $-\mathrm{I\!I}(X)$ (showed previously), we get

$$= -\overline{g}(P^\perp(Z), \mathrm{I\!I}(X)Y) - \overline{g}(P^\perp(Z), \mathrm{I\!I}(Y)X) + \overline{g}(\nabla_X P(Z), Y) + \overline{g}(\nabla_Y P(Z), X).$$

Since $\mathrm{I\!I}(X)Y = \mathrm{I\!I}(X,Y) = \mathrm{I\!I}(Y)X$ by symmetry, and $\mathrm{I\!I}(X,Y)$ is orthogonal to TM, we get the desired result. ■

Recall that for a differentiable map $A_t : \mathbb{R} \to \mathrm{Mat}_n(\mathbb{R})$, we have

$$\frac{d}{dt}\det(A_t) = \mathrm{tr}\left(\mathrm{adj}(A_t)\frac{dA_t}{dt}\right),$$

if A_t is invertible then this is equal to

$$\frac{d}{dt}\det(A_t) = \det(A_t)\mathrm{tr}\left(A_t^{-1}\frac{dA_t}{dt}\right).$$

18.7.4 Definition

Let (M, g) be an embedded Riemannian submanifold of $(\overline{M}, \overline{g})$. The second fundamental form is a tensor $\mathrm{II} : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \Gamma(TM^\perp)$, via the musical isomorphisms this corresponds to a tensor $\mathrm{II}^\sharp : \mathfrak{X}(M) \times \mathfrak{X}^*(M) \to \Gamma(TM^\perp)$, which we can then take the trace of.

Let $H = \mathrm{tr}\mathrm{II}^\sharp = \mathrm{tr}_g\mathrm{II}$ be this trace, it is a section in $\Gamma(TM^\perp)$ called the **mean curvature** of M.

Let us unpack this definition. We can view the second fundamental form as a tensor $\mathrm{II} \in \mathfrak{X}(M) \otimes \mathfrak{X}(M) \otimes \Gamma(TM^\perp)^*$. So if we take a local frame (E_i) of $\mathfrak{X}(M)$ with dual coframe (ε^i), and a coframe (ν^j) of $\Gamma(TM^\perp)^*$, then

$$\mathrm{II} = \mathrm{II}_{i,j,k}\varepsilon^i \otimes \varepsilon^j \otimes \nu^k.$$

Because II is symmetric, $\mathrm{II}_{i,j,k} = \mathrm{II}_{j,i,k}$ so raising either the first or second index results in the same tensor. So then

$$\mathrm{II}^\sharp = \mathrm{II}^i_{j,k} E_i \otimes \varepsilon^j \otimes \nu^k = g^{i\ell}\mathrm{II}_{\ell,j,k}E_i \otimes \varepsilon^j \otimes \nu^k.$$

We can then take the trace of this tensor relative to the contravariant and first covariant index. That is,

$$\mathrm{tr}_g\mathrm{II} = \mathrm{tr}(\mathrm{II}^\sharp) = g^{i\ell}\mathrm{II}_{\ell,i,k}\nu^k.$$

We reindex this to give $\mathrm{tr}_g\mathrm{II} = g^{ij}\mathrm{II}_{i,j,k}\nu^k$. By definition $\mathrm{II}_{i,j,k}$ is the component of ν^k in $\mathrm{II}(E_i, E_j)$, so this corresponds to the section $H \in \Gamma(TM^\perp)$ given by

$$H = g^{ij}\mathrm{II}(E_i, E_j).$$

In particular for an orthonormal basis of tangent vectors $e_1, \ldots, e_n \in T_pM$,

$$H(p) = \sum_i \mathrm{II}(e_i, e_i).$$

Denote by ω_g the volume form of (M, g). Then as in the previous case, for a family of smooth embeddings f_t we have metrics g_t on M and thus also mean curvatures H_t.

18.7.5 Theorem

If $f_t : M \to \overline{M}$ is a family of smooth embeddings into a Riemannian manifold, with M compact (and without boundary), then

$$\frac{d}{dt}\int_M \omega_{g_t} = -\int_M \langle H_t, Z\rangle\omega_{g_t}.$$

Proof

We can swap integration and differentiation,

$$\frac{d}{dt}\int_M \omega_{g_t} = \int_M \frac{d}{dt}\omega_{g_t}.$$

We claim that
$$\frac{d}{dt}\omega_{g_t} = -2\langle H_t, Z\rangle\omega_{g_t} + 2\mathrm{Div}(P(Z))\omega_{g_t}.$$
This is sufficient, since by the divergence theorem as M has no boundary,
$$\int_M \mathrm{Div}(P(Z))\omega_{g_t} = 0.$$
Since our claim is local, we work in an oriented coordinate chart $(U,(x^i))$ so that $\omega_g = \sqrt{\det(g_{ij})}\,dx^1 \wedge \cdots \wedge dx^n$. Thus
$$\frac{d}{dt}\omega_{g_t} = \frac{1}{2}\frac{1}{\sqrt{\det(g_{ij})}}\frac{d}{dt}\det(g_{ij})\,dx^1 \wedge \cdots \wedge dx^n.$$
And the derivative of a determinant gives (we focus on the function),
$$= \frac{1}{2}\frac{1}{\sqrt{\det(g_{ij})}}\det(g_{ij})\mathrm{tr}\Big(g^{ik}\frac{d}{dt}g_{ij}\Big).$$
If we let $b_{ij} = \mathrm{II}(\partial_i, \partial_j)$ then by the above lemma this is equal to
$$= \frac{1}{2}\sqrt{\det g_{ij}}\mathrm{tr}\big(g^{ik}(2\langle b_{ij}, Z\rangle + 2\langle\nabla_{\partial_i}P(Z), \partial_j\rangle\big).$$
The trace is just the sum as $k = j$ so
$$= \sqrt{\det g_{ij}}g^{ij}(\langle b_{ij}, Z\rangle + \langle\nabla_{\partial_i}P(Z), \partial_j\rangle),$$
and this is just
$$= \sqrt{\det g_{ij}}(\langle H, Z\rangle + \mathrm{Div}(P(Z))),$$
which is precisely what we want. ■

18.7.6 Definition

A submanifold $M \subseteq \overline{M}$ is **totally geodesic** if the geodesics of M are geodesics of $\overline{M}$.

18.7.7 Proposition

A submanifold $M \subseteq \overline{M}$ is totally geodesic iff its second fundamental form is zero.

Proof

Note that since II is symmetric, it is zero iff it is also antisymmetric which is iff $\mathrm{II}(x,x) = 0$ for all $x \in T_pM$. Let γ be the geodesic in M with $\dot\gamma(0) = x$, and let X be an extension of x to a local vector field of M. Then by tensorality
$$\mathrm{II}(x,x) = P \circ \overline{\nabla}_X X = P \circ \overline{\nabla}_{\dot\gamma(0)}\dot\gamma(0) = \overline{D}_t\dot\gamma(0).$$
If M is totally geodesic then $\dot\gamma$ is a geodesic of $\overline{M}$, so this is zero. Thus $\mathrm{II} = 0$ as desired.
Conversely if $\mathrm{II} = 0$ then as above
$$\overline{D}_t\dot\gamma(t) = \mathrm{II}(\dot\gamma(t), \dot\gamma(t)) = 0.$$
Thus $\dot\gamma$ is a geodesic of $\overline{M}$ as desired. ■

18.7.8 Definition

A submanifold $M \subseteq \overline{M}$ is **minimal** if its mean curvature is zero.

Because the mean curvature is the trace of the second fundamental form, if $\mathrm{II} = 0$ then $H = 0$.

18.7.9 Corollary

A totally geodesic submanifold is minimal.

19 The Topology of Smooth Manifolds

19.1 Calculus of variations

In this section we discuss the calculus of variations, which doesn't quite align with the implied purpose of this section, the topology of smooth manifolds, but it does provide a powerful tool which will be of much help.

Recall the following definition.

19.1.1 Definition

A family of curves $\Gamma\colon J \times [a,b] \to M$ is **piecewise smooth** if J is an open interval, and there is a partition $a = t_0 < t_1 < \cdots < t_k = b$ such that Γ is smooth on $J \times [t_i, t_{i+1}]$ for all i, and $\Gamma(0,t) = \gamma(t)$.

Let $\gamma\colon [a,b] \to M$ be piecewise smooth. Then a **variation** of γ is a piecewise smooth family of curves Γ such that $\gamma(t) = \Gamma(a,t)$.

Note:

We will sometimes introduce a piecewise smooth family of curves, or curve, along with "its" partition to mean that the given partition partitions the interval into rectangles (resp. intervals) over which the family (resp. curve) is smooth.

Given a piecewise smooth family of curves $\Gamma\colon J \times [a,b] \to M$ we define curves

$$\Gamma_s\colon [a,b] \to M, \qquad \Gamma^{(t)}\colon J \to M, \qquad \Gamma_s(t) = \Gamma^{(t)}(s) = \Gamma(s,t).$$

These are called the **main** and **transverse** curves, respectively. By assumption, the transverse curves are smooth.

19.1.2 Definition

Let Γ be piecewise smooth. Then we define vector fields along curves

$$\partial_t\Gamma(s,t) = (\Gamma_s)'(t) \in T_{\Gamma(s,t)}M, \qquad \partial_s\Gamma(s,t) = (\Gamma^{(t)})'(s) \in T_{\Gamma(s,t)}M.$$

The $\partial_t\Gamma$ is defined on the entire interval only when Γ is smooth.

Henceforth, we will assume that our piecewise smooth curves γ start at 0 for simplicity.

19.1.3 Definition

Let Γ be a variation of γ. Its **variation field** is the smooth vector field $V \in \mathfrak{X}(\gamma)$ given by

$$V(t) = \partial_s \Gamma(0,t).$$

Recall we proved the following.

19.1.4 Lemma

If γ is piecewise smooth, and V is a piecewise smooth vector field along γ, then V is the variation field of some variation of γ.

Proof

We showed that $\Gamma(s,t) = \exp_{\gamma(t)}(sV(t))$ is defined on $(-\varepsilon, \varepsilon) \times [a,b]$ for some $\varepsilon > 0$ by compactness of $[a,b]$, and it is piecewise smooth for the same partition of γ. It is a variation of γ, and its variation is V. ∎

19.1.5 Definition

Let $\gamma \colon [0,a] \to M$ be piecewise smooth, with M Riemannian. Define its **length** and **energy**, respectively, as

$$L(\gamma) = \int_0^a \left|\frac{d\gamma}{dt}\right| dt, \qquad E(\gamma) = \int_0^a \left|\frac{d\gamma}{dt}\right|^2 dt.$$

Note:
$L(\gamma)$ has previously been defined, as $L_g(\gamma)$.

19.1.6 Lemma

For any piecewise smooth γ,

$$L(\gamma)^2 \leq aE(\gamma).$$

There is equality iff $\left|\frac{d}{d\gamma} t\right|$ is constant.

Proof

Cauchy-Schwarz tells us

$$\left(\int fg\right)^2 \leq \left(\int f^2\right)\left(\int g^2\right)$$

with equality iff f, g are linearly dependent. Taking $f = 1$ and $g = \left|\frac{d\gamma}{dt}\right|$, we get the desired result. ∎

19.1.7 Lemma

Let $\gamma \colon [0,a] \to M$ be a minimizing geodesic. Then $E(\gamma) \leq E(c)$ for all piecewise smooth $c \colon [0,a] \to M$ whose endpoints are the same as γ. There is equality iff c is also a minimizing geodesic.

Proof

By the above lemma,

$$aE(\gamma) = L(\gamma)^2 \leq L(c)^2 \leq aE(c),$$

where the first equality is because γ is a geodesic and thus its derivative has constant norm; the first inequality is because γ is minimizing.

If $E(\gamma) = E(c)$ then $L(\gamma) = L(c)$ so that c is minimizing and thus a geodesic. Conversely if c is a minimizing geodesic, then $L(\gamma) = L(c)$ and $L(c)^2 = aE(c)$. ■

Given a piecewise smooth family $\Gamma\colon J \times [0, a]$, define

$$E_\Gamma(s) = E(\Gamma_s)\colon J \to \mathbb{R}.$$

When Γ is understood, we may omit the subscript.

19.1.8 Lemma

Let $\Gamma\colon J \times [0, a]$ be piecewise smooth, with partition $0 = t_0 < \cdots < t_{k+1} = a$.

$$\frac{1}{2}E'(s) = \sum_{i=0}^{k+1} \langle \partial_s\Gamma(s,t), \partial_t\Gamma(s,t)\rangle\big|_{t_i}^{t_{i+1}} - \int_0^a \langle \partial_s\Gamma, D_t\partial_t\Gamma\rangle\, dt.$$

If V is a variation field of Γ, letting $\gamma = \Gamma_0$, then in particular

$$\frac{1}{2}E'(0) = -\sum_{i=1}^{k}\left\langle V(t_i), \frac{d\gamma}{dt}(t_i^+) - \frac{d\gamma}{dt}(t_i^-)\right\rangle + \left\langle V(a), \frac{d\gamma}{dt}(a)\right\rangle - \left\langle V(0), \frac{d\gamma}{dt}(0)\right\rangle - \int_0^a \left\langle V, D_t\frac{d\gamma}{dt}\right\rangle dt.$$

Where $\frac{d\gamma}{dt}(x^\pm)$ denote one-sided derivatives. The integrals are summed over the intervals $[t_i, t_{i+1}]$.

Proof

By definition

$$E(s) = \int_0^a \langle \partial_t\Gamma(s,t), \partial_t\Gamma(s,t)\rangle\, dt,$$

and then we can swap the integral and derivative to get for $s \in [t_i, t_{i+1}]$ (potentially taking a one-sided derivative),

$$E'(s) = \partial_s E(s) = \int_{t_i}^{t_{i+1}} 2\langle D_s\partial_t\Gamma, \partial_t\Gamma\rangle.$$

Recall by symmetry $D_s\partial_t\Gamma = D_t\partial_s\Gamma$ so this is equal to

$$\frac{1}{2}E'(s) = \int_{t_i}^{t_{i+1}} \langle D_t\partial_s\Gamma, \partial_t\Gamma\rangle.$$

Now,

$$\frac{\partial}{\partial t}\langle \partial_s\Gamma, \partial_t\Gamma\rangle = \langle D_t\partial_s\Gamma, \partial_t\Gamma\rangle + \langle \partial_s\Gamma, D_t\partial_t\Gamma\rangle.$$

Thus

$$\frac{1}{2}E'(s) = \int_{t_i}^{t_{i+1}} \partial_t\langle \partial_s\Gamma, \partial_t\Gamma\rangle - \int_{t_i}^{t_{i+1}} \langle \partial_s\Gamma, D_t\partial_t\Gamma\rangle.$$

Then the claim follows by the fundamental theorem of calculus. ■

19.1.9 Definition

A **proper variation** of a piecewise smooth γ is a variation Γ of γ such that $\Gamma(s,0)=\gamma(0)$ and $\Gamma(s,a)=\gamma(a)$ for all s.

A vector field V along γ is **proper** if $V(0)$ and $V(a)$ are zero.

If V is a proper vector field along γ, it is the variation field of a proper variation of γ. This is because in our construction above,

$$\Gamma(s,t)=\exp_{\gamma(t)}(sV(t))$$

Γ is proper:

$$\Gamma(s,0)=\exp_{\gamma(0)}(0)=\gamma(0), \qquad \Gamma(s,a)=\exp_{\gamma(a)}(0)=\gamma(a).$$

Conversely if Γ is proper, then its variation field is proper:

$$V(0)=\partial_s\Gamma(0,0)=\partial_s\gamma(0)=0,$$

similarly for a.

19.1.10 Lemma

A piecewise smooth $\gamma\colon[0,a]\to M$ is a geodesic iff $E'_\Gamma(0)=0$ for all proper variations of γ.

Proof

Suppopse γ is a geodesic. If V is a variation field of a proper Γ it is proper itself. Since γ is smooth, $\frac{d\gamma}{dt}(t_i^+)=\frac{d\gamma}{dt}(t_i^-)$, and V is proper so $V(a),V(0)$ are zero. Since γ is a geodesic $D_t\frac{d\gamma}{d=}0$. Thus all the terms in the formula in the above lemma are zero, so $E'(0)=0$.

Conversely, let $g\colon[0,a]\to M$ be such that $g(t)>0$ for $t\neq t_i$ and $g(t)=0$ when $t=t_i$. This can be constructed explicitly. Then define

$$V(t)=g(t)D_t\frac{d\gamma}{dt}.$$

This is proper, since $g(0)=g(a)=0$, so let Γ be a proper variation of γ whose variation field is V. Then by the previous lemma since $V(t_i)=0$ for all i,

$$0=\frac{1}{2}E'(0)=-\sum_{i=0}^{k}\int_{t_i}^{t_{i+1}}g(t)\left\langle D_t\frac{d\gamma}{dt},D_t\frac{d\gamma}{dt}\right\rangle.$$

These are integrals of nonnegative continuous maps, so it is zero iff $D_t\frac{d\gamma}{dt}=0$ over $[t_i,t_{i+1}]$ for all i. So $\gamma|_{[t_i,t_{i+1}]}$ are geodesic segments.

We just need to show that γ is smooth. Let $\overline{V}\in\mathfrak{X}(\gamma)$ be such that $\overline{V}(0),\overline{V}(a)$ are both zero and $\overline{V}(t_i)=\frac{\partial\gamma}{\partial t}(t_i^+)-\frac{\partial\gamma}{\partial t}(t_i^-)$ for all other i. Let $\overline{\Gamma}$ be a proper variation of γ with variation field $\overline{V}$. Then

$$0=\frac{1}{2}E'(0)=-\sum\left|\frac{d\gamma}{dt}(t_i^+)-\frac{d\gamma}{dt}(t_i^-)\right|^2.$$

Thus $\frac{d\gamma}{dt}(t_i^+)=\frac{d\gamma}{dt}(t_i^-)$ for all i, so it is differentiable at these points. So it is continuously differentiable over $[0,a]$.

We further know that $D_t\frac{d\gamma}{dt}=0$ in $[t_i,t_{i+1}]$ and by continuity of the second derivatives in (t_i,t_{i+1}) we see that these second derivatives are defined at the t_i. So γ is continuously twice differentiable in $[0,a]$ and satisfies the ODE $D_t\frac{d\gamma}{dt}=0$. By uniqueness of solutions, it is smooth. ∎

19.1.11 Lemma

Let Γ be a variation of a geodesic γ, with energy $E(s)$. Then

$$\begin{aligned}\frac{1}{2}E''(0) &= \langle D_s\partial_s\Gamma(a), \dot\gamma(a)\rangle - \langle D_s\partial_s\Gamma(0), \dot\gamma(0)\rangle \\ &\quad + \langle V(a), D_tV(a)\rangle - \langle V(0), D_tV(0)\rangle + \sum_{i=1}^{k}\big\langle V(a), D_tV(t_i^+) - D_tV(t_i^-)\big\rangle \\ &\quad - \sum_{i=0}^{k}\big\langle V(t_i), D_tV(t_i^+) - D_tV(t_i^-)\big\rangle - \int_0^a \langle V(t), V'' + R(\dot\gamma(t), V(t))\dot\gamma(t)\rangle\, dt.\end{aligned}$$

Proof

Previously we showed

$$\frac{1}{2}E'(s) = \sum_{i=0}^{k}\langle\partial_s\Gamma, \partial_t\Gamma\rangle|_{t_i}^{t_{i+1}} - \sum_{i=0}^{k}\int_{t_i}^{t_{i+1}}\langle\partial_s\Gamma, D_t\partial_t\Gamma\rangle\, dt.$$

Then we have

$$\begin{aligned}\frac{1}{2}E''(0) = \sum_{i=0}^{k}&\langle D_s\partial_s\Gamma, \partial_t\Gamma\rangle + \langle\partial_s\Gamma, D_s\partial_t\Gamma\rangle|_{t_i}^{t_{i+1}} - \\ &\sum_{i=0}^{k}\int_{t_i}^{t_{i+1}}\langle D_s\partial_s\Gamma, D_t\partial_t\Gamma\rangle + \langle\partial_s\Gamma, D_sD_t\partial_t\Gamma\rangle\, dt.\end{aligned}$$

The first term is

$$\begin{aligned}\sum_{i=0}^{k}\langle D_s\partial_s\Gamma, \partial_t\Gamma\rangle|_{t_i}^{t_{i+1}}(0) &= \left\langle D_s\partial_s\Gamma(a), \frac{\partial\gamma}{\partial t}(a)\right\rangle - \left\langle D_s\partial_s\Gamma(0), \frac{\partial\gamma}{\partial t}(0)\right\rangle \\ &\quad - \sum_{i=1}^{k}\left\langle D_s\partial_s\Gamma(t_i), \frac{\partial\gamma}{\partial t}(t_i^+) - \frac{\partial\gamma}{\partial t}(t_i^-)\right\rangle.\end{aligned}$$

The last sum is zero since γ is smooth and thus its partial derivatives are continuous.
The second term is

$$\begin{aligned}\sum_{i=0}^{k}\langle\partial_s\Gamma, D_s\partial_t\Gamma\rangle|_{t_i}^{t_{i+1}}(0) &= \sum_{i=0}^{k}\langle\partial_s\Gamma, \partial_sD_t\Gamma\rangle|_{t_i}^{t_{i+1}}(0) \\ &= \langle V(a), D_tV(a)\rangle - \langle V(0), D_tV(0)\rangle + \sum_{i=1}^{k}\big\langle V(a), D_tV(t_i^+) - D_tV(t_i^-)\big\rangle.\end{aligned}$$

Since $\partial_t\Gamma = \gamma$, $D_t\partial_t\Gamma = 0$ so the third term is zero.
Recall that $D_sD_t\partial_t\Gamma - D_tD_s\partial_t\Gamma = R(\partial_t\Gamma, \partial_s\Gamma)\partial_t\Gamma$. With this in mind, the fourth term is

$$\begin{aligned}\sum_{i=0}^{k}\int_{t_i}^{t_{i+1}}\langle\partial_s\Gamma, D_sD_t\partial_t\Gamma\rangle(0) &= \sum\int\langle V, D_tD_s\partial_t\Gamma + R(\dot\gamma, V)\dot\gamma\rangle \\ &= \sum\int\langle V, V'' + R(\dot\gamma, V)\dot\gamma\rangle.\end{aligned}$$

And we are finished. ■

19.2 Completeness

19.2.1 Definition

A Riemannian manifold M is **geodesically complete** iff every maximal geodesic on it is defined over all of $\mathbb{R}$.

Recall that a smooth manifold is metrizable, so we can talk about bounded sets, and also **metrically complete** manifolds which are of course smooth manifolds which are complete metric spaces.

19.2.2 Lemma

Let (M, g) be a connected Riemannian manifold such that there is a $p \in M$ for which $\exp_p$ is defined on all of T_pM. Then for every $q \in M$ there is a minimizing geodesic from p to q.

Proof

Consider the geoedesic sphere $S_\delta(p)$ about p of radius δ. This is the image of a compact subset of T_pM under $\exp_p$ and is thus compact. Thus $d(\bullet, q)$ must attain a minimum on $S_\delta(p)$, let it be at x_0.

Let $v \in T_pM$ of norm 1 be such that $\exp_p(\delta v) = x$. Then define the geodesic $\gamma(t) = \exp(tv)$, and we claim that γ_v is a minimizing geodesic from p to q.

We prove this as follows: let $r = d(p, q)$ then we claim that $\gamma(r) = q$. Let

$$A = \{s \in [0, r] \mid d(\gamma(s), q) = r - s\}.$$

This is nonempty as it contains 0. Furthermore, A is closed as it is the set of points where two continuous maps (to a Hausdorff space) agree. We want to show that $r \in A$ since then $d(\gamma(r), q) = 0$ and thus $r = q$.

We now claim that for every $s \in A$ less than r, there is an $\varepsilon > 0$ such that $s + \varepsilon \in A$. This is sufficient: let $s = \sup A$ and since A is closed necessarily $s \in A$. If s were not r, then there is some $s+\varepsilon \in A$, contradicting s's maximality.

Now we know that for $s \in A$,

$$d(\gamma(s), q) = r - s = d(p, q) - s \Longrightarrow s = d(\gamma(s), q) - d(p, q) \leq d(p, \gamma(s)),$$

where the final inequality is due to the triangle inequality. And since γ forms a curve of arclength s_0 between p and s_0, we see that $d(p, \gamma(s)) = s$ for all $s \in A$ so γ is minimizing in A.

So let $s_0 \in A$, and form a geodesic sphere $S_{\delta'}(\gamma(s_0))$. As before let $x_0' \in S_{\delta'}(\gamma(s_0))$ be such that $d(x_0', q)$ is minimal. Now,

$$d(\gamma(s_0), q) = r - s_0 = d(p, q) - d(p, \gamma(s_0)).$$

And

$$d(\gamma(s_0), q) = d(\gamma(s_0), x_0') + d(x_0', q),$$

as every curve from $\gamma(s_0)$ to q has to interset $S_{\delta'}(\gamma(s_0))$ and x_0' has minimal distance to q. Thus we see

$$d(x_0', q) = r - s_0 - \delta'.$$

Now we claim that $\gamma(s_0+\delta') = x_0$ and this will complete the proof (of $(*)$) since this implies $d(\gamma(s_0+\delta'), q) = r - (s_0 + \delta')$ and so $s_0 + \delta' \in A$ as desired.

So we want to show $\gamma(s_0 + \delta') = x_0'$. Notice that by the triangle inequality,

$$d(p, x_0') + d(x_0', q) \geq d(p, q) = r \Longrightarrow d(p, x_0') \geq r - d(x_0', q) = s_0 + \delta'.$$

Let β be the piecewise smooth curve obtained from $\gamma|_{[0,s_0]}$ and the radial geodesic from $\gamma(s_0)$ to x_0'. Then we know as $s_0 \in A$,

$$L_g(\beta) = L_g(\gamma|_{[0,s_0]}) + \delta' = s_0 + \delta' \leq d(p, x_0').$$

So there must be equality, and thus it is a minimizing curve, thus a geodesic and by uniqueness $\beta = \gamma|_{[0,s_0+\delta']}$. And thus $\gamma(s_0 + \delta') = x_0$. ∎

19.2.3 Theorem (Hopf-Rinow)

Let (M, g) be a connected Riemannian manifold. Then the following are equivalent:

(1) There is a point $p \in M$ for which $\exp_p$ is defined on all of T_pM,

(2) (Heine-Borel) if $K \subseteq M$ is closed and bounded (relative to the metric), then it is compact,

(3) M is metrically complete,

(4) M is geodesically complete.

Proof

Suppose (1) and $K \subseteq M$ be closed and bounded, so $K \subseteq B_r(p)$ for some $r > 0$ in the metric. For $x \in K$ we can connect it to p via a minimizing geodesic $\exp_p(tv)$. Thus $x = \exp_p(v)$ for $v \in T_pM$ and since this geodesic is minimizing $|v| = d(x, p) < r$. Thus K is contained in $\exp_p(\overline{B}_r(0))$ and $\overline{B}_r(0) \subseteq T_pM$ is compact so its image is compact, and since K is closed it is compact. So we have shown Heine-Borel.

(2) implies (3) is standard for metric spaces. Let $\{p_n\}$ be a Cauchy sequence in M, then it is bound so $\overline{\{p_n\}}$ is closed and bound and thus compact by (2). Since compact metric spaces are sequentially compact, p_n has a convergent subsequence. Cauchy sequences with convergent subsequences converge, so we see that M is metrically complete.

Now suppose M is metrically complete and let γ be a maximal geodesic. If it is not defined on all of $\mathbb{R}$, suppose its domain is bound from above by $s_0 \in \mathbb{R}$. Let s_i increase to s_0, and let $x_i = \gamma(s_i)$, then $d(x_i, x_j) \leq L_g(\gamma|_{[s_i,s_j]}) = |\gamma'|\,|s_i - s_j|$. Since s_i is convergent, it is Cauchy, and thus so is x_i. By metric completeness, x_i converges to some $p_0 \in M$.

Let W be a uniformly normal neighborhood of p_0 for parameter $\delta > 0$. Take j large enough so that $s_j > s_0 - \delta$ and $x_j \in W$. So $B_\delta(x_j)$ is a geodesic ball, so let $\sigma(t) = \exp_{x_j}(t\gamma'(s_j))$, i.e. it is the radial geodesic with initial velocity $\gamma'(s_j)$ starting at x_j. As it lies in $B_\delta(x_j)$ it is defined for $t \in (-\delta, \delta)$.

By uniqueness of geodesics, $\gamma(t + s_j) = \sigma(t)$ where defined. Thus

$$\widetilde{\gamma}(t) = \begin{cases} \gamma(t) & t < s_0 \\ \sigma(t - s_j) & t \in (s_j - \delta, s_j + \delta) \end{cases}$$

is smooth, and a geodesic. So it extends γ to $s_j + \delta > s_0$, contradicting γ's maximality. Thus M is geodesically complete.

Clearly (4) implies (1). ∎

Note:

We say that a manifold M is **complete** to mean it is geodesically complete. If M is connected, then this equivalent to being metrically complete. Otherwise this is equivalent to M being metrically complete on each of its connected components, as completeness is equivalent to completeness on each of its components.

19.2.4 Corollary

A complete manifold has minimizing geodesics between any two points.

19.2.5 Corollary

Compact manifolds are complete.

Proof

We can assume the manifold is connected, in which case this follows from Heine-Borel which is trivially satisfied as every closed subset of the manifold is compact. ■

19.3 Smooth covering maps and orientability

19.3.1 Definition

A surjective continuous map between topological spaces $\pi \colon \widetilde{X} \to X$ is a **covering map** if for every $p \in X$ there is a neighborhood $p \in U$, called an **evenly covered neighborhood**, such that $\pi^{-1}U = \bigsqcup_i \widetilde{U}_i$ for $\widetilde{U}_i \subseteq \widetilde{X}$ open, and $\pi|_{\widetilde{U}_i} \colon \widetilde{U}_i \to U$ is a homeomorphism for all i. The $\widetilde{U}_i$ are called **sheets**.

A **smooth covering map** is a covering map between smooth manifolds which is smooth, and the homeomorphisms from the sheets are diffeomorphisms.

19.3.2 Proposition

Let $\pi \colon \widetilde{X} \to X$ be a topological covering.

(1) For any evenly covered neighborhood U of $p \in X$, the number of sheets of U is uniquely determined and is equal to the cardinality of the fiber $\pi^{-1}p$.

(2) If X is connected, then the fibers of π all have the same cardinality. Thus the number of sheets of any evenly covered neighborhood is a constant, equal to the cardinality of the fibers of π. This cardinality is called the **degree** of π.

Proof

Let $\pi^{-1}U = \bigsqcup_i \widetilde{U}_i$ be the sheets of U. Then every sheet $\widetilde{U}_i$ contains a single element of $\pi^{-1}p$. Indeed, $\pi|_{\widetilde{U}_i}$ is a homeomorphism to U, so $p \in U$ has a unique preimage in $\widetilde{U}_i$. And since $\pi^{-1}p \subseteq \pi^{-1}U$, this gives a one-to-one correspondence between sheets of U and $\pi^{-1}p$, showing (1).

For (2), we note that the map which maps $p \in X$ to the cardinality of the fiber $\pi^{-1}p$ is locally constant. Indeed for U evenly covered, by (1) for every $p \in U$ the cardinality of $\pi^{-1}p$ is equal to the number of sheets of U (which is then uniquely determined). Thus the cardinality of fibers is constant on U, and since the evenly covered neighborhoods cover X, this shows the map is locally constant.

If X is connected, then locally constant maps out of X are constant. So $p \mapsto \left|\pi^{-1}p\right|$ is constant, as desired. ■

19.3.3 Proposition

(1) Every topological covering map is a local homeomorphism, and a quotient map. A smooth covering map is a local diffeomorphism.

(2) Every smooth covering map is a surjective submersion.

(3) An injective smooth covering map is a diffeomorphism.

Proof

(2) and (3) follow clearly from (1). We note that in general covering maps are open: let $U \subseteq \widetilde{X}$ be open and let $p \in \pi(U)$. Take a neighborhood $p \in V$ such that $\pi^{-1}V = \bigsqcup_i \widetilde{V}_i$. Consider then $\bigsqcup_i \widetilde{V}_i \cap U$ which is open, and since restricting π to $\widetilde{V}_i$ is a homeomorphism, $\pi(\widetilde{V}_i \cap U)$ is open. Now, as π is bijective on $\widetilde{V}_i$,

$$\pi\left(\bigsqcup_i \widetilde{V}_i \cap U\right) = \bigcup \pi(\widetilde{V}_i \cap U) = \bigcup \pi(\widetilde{V}_i) \cap \pi(U) = V \cap \pi(U)$$

is open, and it is a neighborhood of p in $\pi(U)$ so that $\pi(U)$ is open. As covering maps are surjective, they are quotient maps.

Let $q \in \widetilde{M}$ and $p = \pi(q)$. Take a neighborhood V of p where $\pi^{-1}V = \bigsqcup_i \widetilde{V}_i$. Then $q \in \widetilde{V}_i$ for some i, and π is a homeo/diffeomorphism $\widetilde{V}_i \to U$, showing it is a local homeo/diffeomorphism. ∎

Thus an injective covering map is a homeomorphism (or diffeomorphism in the smooth case), as bijective local homeomorphisms (diffeomorphisms) are globally so. A covering map is injective iff it has degree 1 when the target is connected.

A classic result regarding covering maps is the following.

19.3.4 Theorem

If $\pi\colon \widetilde{X} \to X$ is a covering map and X is simply connected (i.e. has trivial fundamental group), then π has degree 1. Thus π is a homeomorphism, and a diffeomorphism if π is smooth.

19.3.5 Proposition

Suppose M is a connected smooth n-manifold, and $\pi\colon E \to M$ is a topological covering map. Then E is a topological n-manifold, and there is a unique smooth structure on it making π a smooth covering map.

Proof

Because π is a local homeomorphism, E is locally Euclidean. We show it is Hausdorff: for $p_1 \neq p_2 \in E$, if $\pi(p_1) = \pi(p_2)$ then they are either in the same sheet which is Hausdorff (as it is homeomorphic to a subset of M) or in different sheets which are disjoint. If $\pi(p_1) \neq \pi(p_2)$ then since M is Hausdorff they have disjoint neighborhoods, and their preimages are disjoint neighborhoods of p_1, p_2.

Now we claim that each fiber of π is countable. Let $q \in M$ and $p_0 \in \pi^{-1}q$, we construct a surjective $\beta\colon \pi_1(M, q) \to \pi^{-1}q$. Since the fundamental group of a manifold is countable, this is sufficient. For every $[\gamma] \in \pi_1(M, q)$ we can lift γ to $\widetilde{\gamma}\colon [0,1] \to E$, and the endpoint of $\widetilde{\gamma}$ is independent on choice of homotopy representative. So map $\beta[\gamma] = \widetilde{\gamma}(1)$, and this is surjective: for $\pi(p) = q$ take a path $\widetilde{\gamma}$ from p_0 to p then $\pi \circ \widetilde{\gamma}$ lifts to $\widetilde{\gamma}$ and is then mapped to p.

The collection of all evenly covered open subsets (subsets whose preimage decomposes to sheets) cover M, and by second countability of M has a countable subcover $\{U_i\}$. For any i, each sheet of $\pi^{-1}U_i$ has a single element from each fiber over U_i, so there can be only be countably many sheets. Each sheet is second countable, and thus we cover E with countably many second countable subsets and thus is second countable itself.

To form a smooth structure, let $p \in E$ and U an evenly covered neighborhood of $\pi(p)$. After shrinking it potentially, we may assume that U is in the domain of a coordinate map $\varphi\colon U \to \mathbb{R}^n$. Then the sheets of U form coordinate charts via $\varphi \circ \pi$. The transition functions of these coordinate charts are just the transition functions of the coordinate maps of M which are diffeomorphisms, so this is smooth.

And π is a smooth covering map because its coordinate representation relative to these constructed coordinate maps is just the identity. ∎

Since manifolds are locally simply connected, they admit an **universal covering**. This is a simply connected covering space $\pi\colon \widetilde{M} \to M$. It is unique as for any other simply connected covering $\pi'\colon E \to M$, there is a homeomorphism $\Phi\colon \widetilde{M} \to E$ such that $\pi' \circ \Phi = \pi$.

In lieu of the above proposition, $\widetilde{M}, E$ can be made into smooth manifolds. Then Φ is a homeomorphism and $\pi' \circ \Phi = \pi$, since π', π are local diffeomorphisms this means Φ is smooth. As $\pi \circ \Phi^{-1} = \pi'$, so is Φ^{-1} so it is a diffeomorphism.

$\widetilde{M}$ is called the **universal covering manifold** of M.

19.3.6 Corollary

If M is a connected smooth manifold, then there exists a simply connected smooth manifold $\widetilde{M}$, the **universal covering manifold** of M, with smooth covering $\pi: \widetilde{M} \to M$. The universal covering map is unique as for any other smooth covering $\pi': \widetilde{M}' \to M$ there is a unique diffeomorphism $\Phi: \widetilde{M} \to \widetilde{M}'$ such that $\pi' \circ \Phi = \pi$.

19.3.7 Definition

Let $\pi: E \to X$ be a topological covering map, then an **automorphism** of π is a homeomorphism $\varphi: E \to E$ such that $\pi \circ \varphi = \pi$. The automorphisms of π form a group, the **automorphism group** of π denoted $\mathrm{Aut}_\pi(E)$.

A covering map π is **normal** if $\mathrm{Aut}_\pi(E)$ acts transitively on each of π's fibers. This can be shown to be equivalent to $\pi_*(\pi_1(E, q))$ being a normal subgroup of $\pi_1(M, \pi(q))$ for all $q \in E$.

If $\pi: E \to M$ is a smooth covering map then its automorphisms are smooth. Indeed, if $\varphi: E \to E$ is an automorphism of π, then $\pi \circ \varphi = \pi$ implies φ is smooth as π is a local diffeomorphism. Thus $\mathrm{Aut}_\pi(E)$ acts smoothly on E.

19.3.8 Lemma

If $\pi: E \to M$ is a smooth covering map, then $\mathrm{Aut}_\pi(E)$ is a zero-dimensional Lie group acting smoothly and freely on E.

Proof

Suppose $\varphi \in \mathrm{Aut}_\pi(E)$ fixes some $p \in E$. Now, $\pi \circ \varphi = \pi$ meaning that φ can be viewed as a lift of π. The identity is another lift of π which agrees with φ at p, and by uniqueness we have that $\varphi = \mathrm{id}$. So the action is free, and we already showed it is smooth.

To show that $\mathrm{Aut}_\pi(E)$ is a Lie group, we just need to show it is countable. Let $q \in E$ and $p = \pi(q) \in M$, and let $p \in U$ be an evenly covered neighborhood of p. Because E is second countable, $\pi^{-1}U$ has countably many sheets, and thus $\pi^{-1}p$ is countable. Let $\theta^{(p)}: \mathrm{Aut}_\pi(E) \to E$ be given by $\theta^{(q)}(\varphi) = \varphi(q)$. This maps $\mathrm{Aut}_\pi(E)$ into $\pi^{-1}p$ and since the action is free this is an injection. So $\mathrm{Aut}_\pi(E)$ is indeed countable.

■

Now we claim that the action of $\mathrm{Aut}_\pi(E)$ on E is in fact proper as well, which allows us to quotient E by $\mathrm{Aut}_\pi(E)$. This would then allow us to quotient E by the automorphism group, which we could naturally compare to M.

19.3.9 Lemma

Suppose a discrete Lie group Γ acts continuously and freely on a manifold E. Then the action is proper iff the following conditions hold:

(1) Every point $p \in E$ has a neighborhood U such that for all $g \in \Gamma$, $(gU) \cap U = \emptyset$ unless $g = e$.

(2) If $p, p' \in E$ are not in the same Γ-orbit, there exist neighborhoods $p \in V$ and $p' \in V'$ such that $(gV) \cap V' = \emptyset$ for all $g \in \Gamma$.

Proof

First assume the action is free and proper, then consider the quotient map $\pi\colon E \to E/\Gamma$. If $p, p' \in E$ are not in the same orbit their images under π are different. Since E/Γ is Hausdorff, there are neighborhoods $\pi(p) \in W, \pi(p') \in W'$ such that $W \cap W' = \emptyset$. Then let $V = \pi^{-1}W$ and $V' = \pi^{-1}W'$; we see that if $p \in V$ such that $gp \in V'$ we have $\pi(gp) = \pi(p) \in W \cap W'$ a contradiction. This shows (2).

For (1), let $p \in E$ and let V be a precompact neighborhood of p. Then $\Gamma_{\overline{V}} = \{g \in \Gamma \mid (g\overline{V}) \cap \overline{V} \neq \emptyset\}$ is compact (since an action on G is proper iff G_K is compact for all K compact). Since Γ is discrete, $\Gamma_{\overline{V}}$ is then finite, so w write $\Gamma_{\overline{V}} = \{e, g_1, \ldots, g_n\}$. By potentially shrinking V we may assume that $g_i^{-1}p \notin \overline{V}$ for all i. Let

$$U = V - (g_1\overline{V} \cup \cdots \cup g_n\overline{V}).$$

This is an open neighborhood of p and for $e \neq g \in \Gamma$ we have that either $g = g_i$, in which case $g_iU \subseteq g_i\overline{V}$ does not intersect U. Otherwise $g \notin \Gamma_{\overline{V}}$ so $(gU) \cap U \subseteq (g\overline{V}) \cap \overline{V} = \emptyset$. Thus U satisfies (1).

Conversely suppose (1) and (2) hold. Let (p_i) be a sequence in M and (g_i) be a sequence in Γ such that (p_i) and (g_ip_i) both converge. We want to show that a subsequence of (g_i) converges.

Suppose $p_i \to p$ and $g_ip_i \to p'$. If p, p' lie in different orbits, then there are V, V' as in (2). For large enough i we have $p_i \in V$ and $g_ip_i \in V'$, which contradicts $(g_iV) \cap V' = \emptyset$. So p, p' lie in the same orbit, so $gp = p'$ for some $g \in \Gamma$, and so $g^{-1}g_ip_i \to p$.

Choose a neighborhood U of p as in (1), and let i be large enough such that $g^{-1}g_ip_i \in U$ and $p_i \in U$. This means that $(g^{-1}g_iU) \cap U \neq \emptyset$, which implies $g = g_i$, so eventually g_i is constant and thus convergent. ∎

19.3.10 Proposition

Let $\pi\colon E \to M$ be a smooth covering map. Then $\mathrm{Aut}_\pi(E)$ acts smoothly, freely, and properly on E.

Proof

We just need to show properness, for which we use the previous lemma. For $p \in E$, choose $W \subseteq M$ an even covering of $\pi(p)$. Then take the sheet of p in $\pi^{-1}W$, call it U. Then for an automorphism φ, if $\varphi(q) \in U$ for $q \in U$ then we have $\pi \circ \varphi(q) = \pi(q)$. But π is a diffeomorphism from U, so $\varphi(q) = q$. As $\pi \circ \varphi = \pi$, φ is a lift of π which agrees with id on q; since lifts by a covering map are unique, $\varphi = \mathrm{id}$. Thus $\varphi(U) \cap U = \emptyset$ for $\varphi \neq \mathrm{id}$, so $\mathrm{Aut}_\pi(E)$ satisfies the first condition in the above lemma.

If $p, p' \in E$ are in different orbits, take disjoint neighborhoods $\pi(p) \in W, \pi(p') \in W'$. Then their preimages satisfy the second condition. ∎

19.3.11 Theorem

Let $\pi\colon E \to M$ be a smooth normal covering map, and let $\Gamma = \mathrm{Aut}_\pi(E)$. Then the quotient manifold E/Γ is diffeomorphic to M.

Proof

The quotient exists by the previous proposition and the quotient manifold theorem. Let $\Pi\colon E \to E/\Gamma$ be the quotient map, which is a smooth submersion. Now there is an obvious map $\Phi\colon E/\Gamma \to M$ given by $\Phi(\Gamma p) = \pi(p)$. This is well-defined as if $\Gamma p = \Gamma q$ for $p, q \in E$ then $q = \varphi(p)$ for some automorphism φ. Then

$$\Phi(\Gamma q) = \pi(q) = \pi \circ \varphi(p) = \pi(p) = \Phi(\Gamma p),$$

so this is well-defined. Thus $\pi = \Phi \circ \Pi$, and since Π is a smooth submersion and π is smooth, Φ is smooth. Since π is surjective, so too is Φ. It is also injective, since if $\pi(p) = \pi(q)$ then since π is normal, there is an automorphism φ mapping p to q, so $\Gamma p = \Gamma q$. So Φ is invertible and its inverse satisfies $\Phi^{-1} \circ \pi = \Pi$. As π is a local diffeomorphism, Φ^{-1} is also smooth, so Φ is a diffeomorphism as desired. ∎

We are also interested in how covering maps interplay with orientation.

19.3.12 Proposition

If $\pi: E \to M$ is a smooth covering map and M is orientable, then so is E.

Proof

As π is a local diffeomorphism, this is immediate. ■

19.3.13 Definition

If G is a Lie group acting smoothly on a smooth oriented manifold M, we say that action is **orientation-preserving** if each $p \mapsto gp$ is orientation-preserving.

19.3.14 Theorem

Suppose E is a connected, oriented, smooth manifold with or without boundary, and $\pi: E \to M$ is a smooth normal covering map, with M connected. Then M is orientable iff the action of $\mathrm{Aut}_\pi(E)$ on E is orientation-preserving.

Proof

Denote the orientation of E by $\mathcal{O}_E$. Note that as M, E are connected, they only have two potential orientations. This is because for any two orientation forms, they are multiples of one another by a non-zero smooth function. By connectivity, this function is either always positive or always negative.

Now let $q \in E$ be arbitrary, then one of the orientations makes $d\pi_q: T_qE \to T_{\pi q}M$ orientation-preserving. Then $\mathcal{O}_E$ and $\pi^*\mathcal{O}_M$ are both orientations of E agreeing at q, so they must be the same orientation. Now suppose $\varphi \in \mathrm{Aut}_\pi(E)$ then $\pi \circ \varphi = \pi$ so

$$\varphi^*\mathcal{O}_E = \varphi^*(\pi^*\mathcal{O}_M) = (\pi \circ \varphi)^*\mathcal{O}_M = \pi^*\mathcal{O}_M = \mathcal{O}_E$$

so that φ is orientation-preserving as desired.

Conversely suppose $\mathrm{Aut}_\pi(E)$ is orientation-preserving, and let $p \in M$. If $U \subseteq M$ is an evenly covered neighborhood of p, take a section $\sigma: U \to E$ (the inverse of one of the restrictions of π to a sheet). This then induces an orientation $\sigma^*\mathcal{O}_E$ on U.

Now let $\sigma_1: U \to E$ be another section. Since π is normal, $\mathrm{Aut}_\pi(E)$ acts transitively on the fiber $\pi^{-1}p$, so there is an automorphism φ such that $\sigma_1(p) = \varphi(\sigma(p))$. So $\varphi \circ \sigma$ is a local section of π which agrees with σ_1 at p, and since sections are inverses of restrictions of π, it follows that $\varphi \circ \sigma = \sigma_1$. Thus since φ is orientation-preserving

$$\sigma_1^*\mathcal{O}_E = (\varphi \circ \sigma)^*\mathcal{O}_E = \sigma^*(\varphi^*\mathcal{O}_E) = \sigma^*\mathcal{O}_E,$$

so all sections generate the same orientation.

So at every point $p \in M$ we can uniquely define an orientation in a neighborhood around p. By uniqueness, these orientations must agree on their overlaps and thus form an orientation of M. ■

This theorem allows us to show that certain manifolds are not orientable.

19.3.15 Example

For $n \geq 1$, consider the smooth covering $\pi: S^n \to \mathbb{RP}^n$ (which maps a point in $S^n \subseteq \mathbb{R}^{n+1} - 0$ to its equivalence class). This is a bijection and thus normal. The only non-trivial automorphism of π is the antipodal map $\alpha(x) = -x$. It can be shown that this is orientation-preserving iff n is odd (see homework).

Thus $\mathbb{RP}^n$ is orientable iff n is odd.

19.3.16 Definition

Let M be a connected, smooth, positive-dimensional manifold (with or without boundary). Define $\widehat{M}$ to be the set of all orientations of the tangent spaces of M:

$$\widehat{M} = \{(p, \mathcal{O}_p) \mid p \in M \text{ and } \mathcal{O}_p \text{ is an orientation of } T_pM\}.$$

There is an obvious projection $\widehat{\pi}\colon \widehat{M} \to M$ mapping $(p, \mathcal{O}_p) \mapsto p$. Since every tangent space has two orientations, the fibers of $\widehat{\pi}$ all have cardinality 2.
We call $\widehat{\pi}\colon \widehat{M} \to M$ is the **orientation covering** of M.

19.3.17 Lemma

If $\pi\colon X \to Y$ is a topological covering map with Y connected, then π restricts to a covering map on each of X's connected components.

Proof

This is clear. ∎

19.3.18 Proposition

Suppose M is a connected, smooth, positive-dimensional manifold. Let $\widehat{\pi}\colon \widehat{M} \to M$ be its orientation covering. Then $\widehat{M}$ can be given the structure of a smooth, oriented manifold without boundary, such that:

(1) $\widehat{\pi}$ is a normal covering map,

(2) A connected open subset $U \subseteq M$ is evenly covered by $\widehat{\pi}$ iff U is oriented,

(3) If $U \subseteq M$ is evenly covered, then every orientation of U is the pullback orientation induced by a local section of $\widehat{\pi}$ over U.

Proof

We first topologize $\widehat{M}$. For each pair $(U, \mathcal{O})$ where U is an open subset of M and $\mathcal{O}$ an orientation of U, define $\widehat{U}_{\mathcal{O}} \subseteq \widehat{M}$ as

$$\widehat{U}_{\mathcal{O}} = \left\{(p, \mathcal{O}_p) \in \widehat{M} \,\middle|\, p \in U \text{ and } \mathcal{O}_p \text{ is the orientation of } T_pM \text{ determined by } \mathcal{O}\right\}.$$

We claim that this forms a basis of a topology on $\widehat{M}$.
Let $(p, \mathcal{O}_p) \in \widehat{M}$, we can take an orientable neighborhood U of p (e.g. a Euclidean neighborhood). Then $\mathcal{O}_p$ must be determined by some orientation $\mathcal{O}$ of U (if $\mathcal{O}$ doesn't determine $\mathcal{O}_p$, then $-\mathcal{O}$ does). So $(p, \mathcal{O}_p) \in \widehat{U}_{\mathcal{O}}$, so these cover $\widehat{M}$.
Now suppose $(p, \mathcal{O}_p)$ is in both $\widehat{U}_{\mathcal{O}}$ and $\widehat{U}'_{\mathcal{O}'}$. So $p \in U \cap U'$ and $\mathcal{O}_p$ is determined by both $\mathcal{O}$ and $\mathcal{O}'$. Take a connected neighborhood $p \in V \subseteq U \cap U'$, then $\mathcal{O}, \mathcal{O}'$ both restrict to orientations of V, and since V is connected and they agree at p, they give the same orientation of V. So $\widehat{V}_{\mathcal{O}} = \widehat{V}_{\mathcal{O}'} \subseteq \widehat{U}_{\mathcal{O}} \cap \widehat{U}'_{\mathcal{O}'}$, meaning these $\widehat{U}_{\mathcal{O}}$ do indeed form a topological basis.

Note that $\widehat{\pi}$ maps $\widehat{U}_{\mathcal{O}}$ bijectively onto U, so it is an open map. This is continuous, as for $V \subseteq U$ open, its preimage is $\widehat{V}_{\mathcal{O}|_V}$ open. Thus its restriction to $\widehat{U}_{\mathcal{O}}$ is a homeomorphism into U, meaning $\widehat{\pi}$ is a local homeomorphism.

Now we show that $\widehat{\pi}$ is a covering map. Suppose that $U \subseteq M$ is open conencted and orientable with orientation $\mathcal{O}$. Then $\widehat{\pi}^{-1}U$ is the disjoint union of $\widehat{U}_{\mathcal{O}}$ and $\widehat{U}_{-\mathcal{O}}$, and $\widehat{\pi}$ restricts to a homeomorphism on both of them. So $\widehat{\pi}$ is indeed a covering map.

Thus it restricts to a covering map from $\widehat{M}$'s connected components to M, and thus each connected component has a unique smooth structure making $\widehat{\pi}$ a smooth covering map. This defines a smooth structure on $\widehat{M}$ making $\widehat{\pi}$ a smooth covering map.

We also show it is normal. Let us define $\alpha \colon \widehat{M} \to \widehat{M}$ which maps $(p, \mathcal{O}_p)$ to $(p, -\mathcal{O}_p)$. Clearly we have $\widehat{\pi} \circ \alpha = \widehat{\pi}$, and since α is its own inverse, to show it is an automorphism it is sufficient to show that α is open. For basic open $\widehat{U}_{\mathcal{O}}$, we see that $\alpha(\widehat{U}_{\mathcal{O}}) = \widehat{U}_{-\mathcal{O}}$, so it is indeed open. For $p \in M$, its fiber consists of just $(p, \mathcal{O}_p)$ and $(p, -\mathcal{O}_p)$, and α brings one to the other, showing that $\mathrm{Aut}_\pi(\widehat{M})$ acts transitively on all the fibers of $\widehat{\pi}$. Thus $\widehat{\pi}$ is a normal smooth cover, showing (1).

Now we orient $\widehat{M}$. Let $\widehat{p} = (p, \mathcal{O}_p)$, then $\mathcal{O}_p$ is an orientation of T_pM. So give $T_{\widehat{p}}\widehat{M}$ the unique orientation making $d\pi_{\widehat{p}}$ orientation-preserving. This defines a pointwise orientation $\widehat{\mathcal{O}}$ of $\widehat{M}$. On a basic open subset $\widehat{U}_{\mathcal{O}}$, $\widehat{\mathcal{O}}$ agrees with the pullback orientation given by $\widehat{\pi}$ on $(U, \mathcal{O})$ so it is continuous.

For point (2), we already showed that if $U \subseteq M$ is oriented then it is evenly covered. If $U \subseteq M$ is evenly covered, then take a smooth local section $\sigma \colon U \to \widehat{M}$ of $\widehat{\pi}$. This pulls back an orientation to U.

Now if U is evenly covered and $\mathcal{O}$ an orientation of U, define a section $\sigma \colon U \to \widehat{M}$ by $\sigma(p) = (p, \mathcal{O}_p)$. This is continuous, as if $\widehat{U}'_{\mathcal{O}'}$ is basic open of $\widehat{M}$, then for each component V of $U \cap U'$, the restrictions of $\mathcal{O}|_V$ and $\mathcal{O}|_{V'}$ must either agree or be opposites. Then $\sigma^{-1}\widehat{U}_{\mathcal{O}'}$ is the union of the components V where $\mathcal{O}$ and $\mathcal{O}'$ agree, and thus is open. And σ induces the orientation $\mathcal{O}$. ■

19.3.19 Theorem

Suppose M is connected, and let $\widehat{\pi} \colon \widehat{M} \to M$ be its orientation covering.

(1) If M is orientable, then $\widehat{M}$ has two components. The restriction of $\widehat{\pi}$ to each component is a diffeomorphism.

(2) If M is non-orientable, then $\widehat{M}$ is connected, and $\widehat{\pi}$ is a two-sheet normal covering map.

Proof

If M is orientable, then by the previous proposition, M is evenly covered. Then it has two sheets, $\widehat{M}_{\mathcal{O}}, \widehat{M}_{-\mathcal{O}}$, which by definition map diffeomorphically to M. They are connected as M is, so $\widehat{M}$ has two components.

Now if M is non-orientable, we show $\widehat{M}$ is connected. Let W be a component of $\widehat{M}$, so $\widehat{\pi}$ restricts to a covering $W \to M$. Its fibers then all have the same cardinality. We showed above that $\widehat{\pi}^{-1}p$ has cardinality 2, so the fibers in W have cardinality 1 or 2. If 2, then $W = \widehat{M}$ and we are finished. Otherwise $\widehat{\pi}$ is an injection, and an injective smooth covering map is a diffeomorphism, and its inverse would then induce an orientation on M, a contradiction.

So $\widehat{M}$ is connected, and thus the number of sheets is constant, and so must be two. ■

The orientation covering is unique in that if $\widetilde{\pi} \colon \widetilde{M} \to M$ is any other two-sheeted smooth covering map with $\widetilde{M}$ oriented, then there is a unique orientation-preserving diffeomorphism $\varphi \colon \widetilde{M} \to \widehat{M}$ such that $\widehat{\pi} \circ \varphi = \widetilde{\pi}$.

19.4 The Hadamard theorem

19.4.1 Definition

A smooth covering map is a **Riemannian covering map** if it is between Riemannian manifolds and also a local isometry.

19.4.2 Theorem

Suppose $(\widetilde{M}, \widetilde{g})$ and (M, g) are connected Riemannian manifolds with $\widetilde{M}$ complete, and $\pi \colon \widetilde{M} \to M$ is a local isometry. Then M is complete and π is a Riemannian covering map.

Proof

Recall the path-lifting property of covering spaces: for a continuous path $\gamma \colon I \to M$, there is a lift to $\widetilde{\gamma} \colon I \to \widetilde{M}$ such that $\pi \circ \widetilde{\gamma} = \gamma$. We first show that π has the path lifting property for geodesics: for $\pi(\widetilde{p}) = p$ and $\gamma \colon I \to M$ is a geodesic starting at p, we can lift γ uniquely to a geodesic $\widetilde{\gamma} \colon I \to \widetilde{M}$ starting at $\widetilde{p}$.

Let $v = \gamma'(0)$ and $\widetilde{v} = d\pi_{\widetilde{p}}^{-1} v \in T_{\widetilde{p}} \widetilde{M}$. This is well-defined since π is a diffeomorphism in a neighborhood of $\widetilde{p}$ and so $d\pi_{\widetilde{p}}$ is an isomorphism. Let $\widetilde{\gamma}$ be the geodesic in $\widetilde{M}$ starting at $\widetilde{p}$ with initial velocity $\widetilde{v}$. If a lift exists, this must be our lift since

$$d\pi_{\widetilde{p}} \circ \widetilde{\gamma}'(0) = \gamma'(0) = v.$$

Because $\widetilde{M}$ is complete, $\widetilde{\gamma}$ is defined on all of $\mathbb{R}$, and since π is a local isometry $\pi \circ \widetilde{\gamma}$ is a geodesic. It has the same starting point and velocity as γ, so they are equal as desired.

Now to show that M is complete, let p be a point in π's image. If $\gamma \colon I \to M$ is any geodesic starting at p, then lift γ to $\widetilde{\gamma}$, which has an extension to all of $\mathbb{R}$ as $\widetilde{M}$ is complete. Then $\pi \circ \widetilde{\gamma}$ is an extension of γ to all of $\mathbb{R}$. By Hopf-Rinow this means M is complete (since $\exp_p$ is defined on all of $T_p M$).

Now we show π is surjective. Let $\widetilde{p} \in \widetilde{M}$ and let $p = \pi(\widetilde{p})$ and let $q \in M$ be arbitrary. Since M is complete and connected, there is a unique minimizing unit-speed geodesic γ from p to q. Lift this to a geodesic $\widetilde{\gamma}$ starting at $\widetilde{p}$, and let $r = d_g(p, q)$. Then $\pi(\widetilde{\gamma}(r)) = \gamma(r) = q$ so π is surjective.

Let $p \in M$ and $U = B_\varepsilon(p)$ be the geodesic ball centered at p. Write $\pi^{-1} p = \{\widetilde{p}_i\}_{i \in A}$, and for each i let $\widetilde{U}_i$ denote the metric ball of radius ε around $\widetilde{p}_i$. We claim that $\widetilde{U}_i$ are disjoint. For $i \neq j$ let $\widetilde{\gamma}$ be a unit-speed minimizing geodesic from $\widetilde{p}_i$ to $\widetilde{p}_j$. Then $\gamma = \pi \circ \widetilde{\gamma}$ is a geodesic starting and ending at p whose length is the same as $\widetilde{\gamma}$'s. Since U is a geodesic ball, γ must leave and reenter it, as all geodesics lying in U are radial and thus injective. So its length must be at least 2ε, and so $\widetilde{\gamma}$'s length is at least 2ε so the distance between $\widetilde{p}_i$ and $\widetilde{p}_j$ is at least 2ε, so $\widetilde{U}_i, \widetilde{U}_j$ are disjoint.

Now we claim $\pi^{-1} U = \bigsqcup_i \widetilde{U}_i$. If $\widetilde{q} \in \widetilde{U}_i$ then there is a geodesic of length $< \varepsilon$ from $\widetilde{p}_i$ to $\widetilde{q}$, and $\gamma = \pi \circ \widetilde{\gamma}$ is a geodesic of the same length from p to $\pi(\widetilde{q})$, which shows that $\pi(\widetilde{q}) \in U$. Thus $\bigsqcup_i \widetilde{U}_i$ is contained in $\pi^{-1} U$. Conversely if $q = \pi(\widetilde{q}) \in U$, let γ be a minimizing geodesic from q to p which is of length $r < \varepsilon$. Lift γ to $\widetilde{\gamma}$ starting at $\widetilde{q}$, then $\pi(\widetilde{\gamma}(r)) = \gamma(r) = p$ so that $\widetilde{\gamma}(r) = \widetilde{p}_i$ for some i. Then $\widetilde{q} \in \widetilde{U}_i$ as desired.

Now we claim $\pi|_{\widetilde{U}_i} \colon \widetilde{U}_i \to U$ is a diffeomorphism. Since π is a local diffeomorphism, clearly this restriction is. It is also bijective: for $q \in U$ consider the radial geodesic p to q, lift it to $\widetilde{M}$ and map q to its endpoint. Since the length of this geodesic is $d_g(p, q) < \varepsilon$, this is well-defined and a clear inverse. As a bijective local diffeomorphism, the restriction is a diffeomorphism. ∎

19.4.3 Corollary

Suppose $\widetilde{M}, M$ are connected Riemannian manifolds with $\pi \colon \widetilde{M} \to M$ a Riemannian covering map. Then M is complete iff $\widetilde{M}$ is.

Proof

If $\widetilde{M}$ is complete then by the above theorem so is M. Conversely if M is complete, let $\widetilde{p} \in \widetilde{M}$ and $\widetilde{v} \in T_p\widetilde{M}$ be arbitrary. Let $p = \pi(\widetilde{p})$ and $v = d\pi_{\widetilde{p}}(\widetilde{v})$, then the geodesic γ starting at p with initial velocity v is defined over all $\mathbb{R}$. Its lift to $\widetilde{M}$ is also a geodesic defined over all $\mathbb{R}$, and it starts at $\widetilde{p}$ with initial velocity $\widetilde{v}$. ∎

19.4.4 Definition

Let γ be a geodesic segment from p to q in M, say $\gamma(a) = p$ and $\gamma(b) = q$. We say that p and q are **conjugate** along γ if there is a nonzero Jacobi field along γ which vanishes at $t = a$ and $t = b$.

19.4.5 Proposition

Let $p \in M$ and $v \in T_pM$ be in the domain of $\exp_p$, and let $q = \exp_p(v)$. Then v is a critical point of $\exp_p$ iff q and p are conjugate along $\gamma = \gamma_v$.

Proof

If v is a critical point of $\exp_p$, then there is a nonzero $w \in T_v(T_pM)$ such that $d(\exp_p)_v(w) = 0$. Identifying, as usual, $T_v(T_pM) \cong T_pM$, we take $w \in T_pM$. Let Γ be the variation of γ defined by

$$\Gamma(s,t) = \exp_p(t(v + sw)),$$

and $J(t) = \partial_s\Gamma(0,t)$ be its variation field. As Γ is a variation through geodesics, J is a Jacobi field. And,

$$J(1) = \partial_s\Gamma(0,1) = \left.\frac{\partial}{\partial s}\right|_{s=0} \exp_p(v + sw) = d(\exp_p)_v(w) = 0.$$

So J is a non-trivial Jacobi field along γ vanishing at 0 and 1, so p and q are conjugate.
Conversely if p and q are conjugate, then let J be a nonzero Jacobi field along γ with $J(0) = 0$ and $J(1) = 0$. We showed (see homework) that J is a variation field of a variation of γ through geodesics,

$$\Gamma(s,t) = \exp_{\sigma(s)}(tV(s)),$$

for $\sigma(s)$ a smooth curve and $V(s)$ a vector field along σ, where $\sigma(0) = p$ and $\sigma'(0) = J(0) = 0$, $V(0) = \gamma'(0) = v$ and $D_sV(0) = D_tJ(0) = w$. We can then take $\sigma(s) = p$ and $V(s) = v + sw \in T_pM$ so

$$\Gamma(s,t) = \exp_p(t(v + sw)).$$

Since $J(t) = \partial_s\Gamma(0,t)$, we have that $w \neq 0$ as otherwise J would be zero. Then by the same computation as before, $0 = J(1) = d(\exp_p)_v(w)$, so v is a critical point of $\exp_p$. ∎

19.4.6 Lemma

Suppose M is a Riemannian manifold with nonpositive sectional curvature. Then no two points of M are conjugate along any geodesic.

Proof

Let γ be a geodesic segment from p to q in M, say $\gamma(0) = p$ and $\gamma(a) = q$. Let J be a nonzero Jacobi field along γ with $J(0) = 0$. By the Jacobi equation,

$$D_t^2 J = -R(\dot{\gamma}, J)\dot{\gamma}.$$

Consider then the sectional curvature in the direction of J and $\dot{\gamma}$ for any t, this is nonpositive by assumption so

$$\left\langle D_t^2 J, J \right\rangle = -\langle R(\dot{\gamma}, J)\dot{\gamma}, J\rangle = -\mathrm{Rm}(\dot{\gamma}, J, \dot{\gamma}, J) \geq 0.$$

Now consider the smooth map $j\colon [0, a] \to \mathbb{R}$ given by $j = \langle J, J\rangle$. Then $j(0) = 0$ and

$$j' = 2\langle D_t J, J\rangle, j'' = 2\left\langle D_t^2 J, J\right\rangle + 2\langle D_t J, D_t J\rangle > 0.$$

So $j'(0) = 0$ and since $j'' > 0$ we see that j' is increasing and thus always positive. Thus j is strictly increasing, which means that since $j(t) > 0$ for all $t > 0$, in particular $J(t) \neq 0$ for all $t > 0$. ■

19.4.7 Theorem (Hadamard)

If (M, g) is a complete connected Riemannian manifold with nonpositive sectional curvature, then for every $p \in M$, the map $\exp_p\colon T_pM \to M$ is a smooth covering map. Thus if M is simply connected it is diffeomorphic to $\mathbb{R}^n$.

Proof

Combining the previous two results, $\exp_p$ has no critical points and is thus an immersion. Since the dimension of T_pM and M are the same, this means $\exp_p$ is a local diffeomorphism. If we pull back the metric of M to T_pM, this gives a metric to T_pM for which $\exp_p$ is a local isometry.

The straight lines $t \mapsto tv$ in T_pM are geodesics, which means that T_pM is complete. This is because for any $w \in T_pM$ and $v \in T_pM \cong T_wT_pM$, the geodesic tv is defined for all $t \in \mathbb{R}$, so the restricted exponential is defined on all of T_wT_pM.

So T_pM is complete, $\exp_p\colon T_pM \to M$ is a local isometry, and thus it is a Riemannian covering map. If M is simply connected, then covering maps are diffeomorphisms so $M \cong T_pM \cong \mathbb{R}^n$. ■

19.4.8 Corollary

The higher homotopy groups of a complete connected Riemannian manifold are trivial.

Proof

As $\exp_p\colon T_pM \to M$ is a covering map, it induces an isomorphism on higher homotopy groups. The homotopy groups of $T_pM \cong \mathbb{R}^n$ are all trivial as $\mathbb{R}^n$ is contractible. ■

Note:

A topological space X such that $\pi_n(X) = G$ and $\pi_k(X) = 0$ for all $k \neq n$ is called a $K(G, n)$-space, or an **Eilenberg-MacLane space**. Eilenberg-MacLane spaces always exist (if G is Abelian for $n > 1$), and are unique up to homotopy equivalence.

We have thus shown that complete connected Riemannian manifolds are Eilenberg-MacLane spaces for $n = 1$.

19.5 Bonnet-Myers and Synge-Weinstein

19.5.1 Definition

We define the **Ricci curvature** of a Riemannian n-manifold M to be the covariant 2-tensor

$$\mathrm{Rc}(X,Y) = \frac{1}{n-1}\mathrm{tr}(Z \mapsto R(X,Z)Y).$$

For a single vector X, we define

$$\mathrm{Rc}(X) = \mathrm{Rc}(X,X).$$

In terms of a local frame, the components of Rc are denoted R_{ij} and satisfy

$$R_{ij} = \frac{1}{n-1}R_{ikj}^{k} = \frac{1}{n-1}g^{km}R_{ikjm}.$$

In terms of an orthonormal frame (E_i), we see that

$$\mathrm{Rc}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}\langle R(X,E_i)Y,E_i\rangle.$$

For $\mathrm{Rc}(X) = \mathrm{Rc}(X,X)$, if $E_1,\dots,E_{n-1}$ are orthonormal and orthogonal to X then

$$\mathrm{Rc}(X) = \frac{1}{n-1}\sum_{i=1}^{n}\langle R(X,E_i)X,E_i\rangle = \frac{1}{n-1}\sum_{i=1}^{n-1}\langle R(X,E_i)X,E_i\rangle.$$

since $X = cE_n$ and $R(E_n,E_n)E_n = 0$ by anti-symmetry. We see that this is equal to

$$\mathrm{Rc}(X) = \frac{1}{n-1}\sum_{i=1}^{n}K(X,E_i)$$

where K is the sectional curvature.

19.5.2 Definition

We define the **scalar curvature** of a Riemannian n-manifold M as the smooth map

$$K = \frac{1}{n}\mathrm{tr}\mathrm{Rc}^{\sharp}.$$

More explicitly, Rc is a map $\mathfrak{X}(M) \to \mathfrak{X}^*(M)$, then we raise an index via the musical isomorphism to get a map $\mathfrak{X}(M) \to \mathfrak{X}(M)$, and at each point this gives an endomorphism of T_pM of which we take its trace.

In terms of a local frame, the scalar curvature is equal to

$$K = R_i^i = g^{ij}R_{ij}.$$

Note:

As a rule of thumb, Ricci curvature controls the growth of the volume of metric balls. More explicitly, consider the unit tangent $N\colon \partial B_r(p) \to TM|_{\partial B_r(p)}$ then

$$\frac{d}{dr}\mathrm{Vol}(B_r(p)) \sim \int \mathrm{Rc}(N).$$

We will not prove this.

19.5.3 Theorem (Bonnet-Myers)

Let (M, g) be a complete connected Riemannian manifold such that for every point $p \in M$ and tangent vector $v \in T_pM$

$$\frac{|v|^2}{r^2} \leq \mathrm{Rc}_p(v)$$

for some $r > 0$ constant. Then M is compact and satisfies $\mathrm{diam}(M) \leq \pi r$.

Proof

Note that M's diameter being bounded implies compactness by Hopf-Rinow: M is then closed and bounded and thus compact. So for $p, q \in M$ we want to show that $d(p, q) \leq \pi r$ and this is sufficient.

Take $\gamma \colon [0, 1] \to M$ a minimizing geodesic segment from p to q. We construct orthonormal vector fields $e_1, \ldots, e_{n-1}$ along γ which are orthogonal to $\dot{\gamma}$ and satisfy $D_t e_i = 0$. This is done by taking such vectors at T_pM and taking their parallel transport along γ. Let $\ell = |\dot{\gamma}|$, which is constant and equal to γ's length. Define $e_n = \gamma/\ell$ so we have an orthonormal frame along γ.

Define

$$V_i(t) = \sin(\pi t) e_i(t),$$

and since $D_t e_i(t) = 0$ this has second derivative

$$V_i''(t) = -\pi^2 \sin(\pi t) e_i(t).$$

Furthermore, $V_i(0)$ and $V_i(1)$ are both zero, so V_i is a proper vector field along γ so there is a proper variation Γ_i of γ with variation field V_i. Denote by E_i the energy of Γ_i:

$$E_i(s) = E_{\Gamma_i}(s) = E_i((\Gamma_i)_s).$$

Since γ is a minimizing geodesic and $(\Gamma_i)_0 = \gamma$, $E(\gamma)$ has minimal energy so $E_i''(0) > 0$ as $E_i(0)$ is a minimum (since Γ_i is proper, so each $(\Gamma_i)_s$ is a curve with the same endpoints as γ).

By the formula for the second derivative of energy, since V_i is smooth and proper,

$$\frac{1}{2} E_i''(0) = -\int_0^1 \langle V_i, V_i'' + R(\dot{\gamma}, V_i)\dot{\gamma} \rangle \, dt.$$

We have that $\dot{\gamma} = \ell e_n$ and V_i is a multiple of e_i so this is equal to

$$= -\int_0^1 \sin^2(\pi t)(\pi^2 - \ell^2 \langle R(e_n, e_i)e_n, e_i \rangle).$$

Summing over i shows that

$$0 < (n-1) \int_0^1 \sin^2(\pi t)(\pi^2 - \ell^2 \mathrm{Rc}(e_n)) \leq (n-1) \left(\pi^2 - \frac{\ell^2}{r}^2 \right) \int_0^1 \sin^2(\pi t),$$

and since the integral of $\sin^2$ is positive, we see that

$$0 \leq \pi^2 - \frac{\ell^2}{r^2} \implies \ell \leq \pi r.$$

Since $\ell = d(p, q)$ we are finished. ■

19.5.4 Corollary

If (M, g) is complete, connected, and $\mathrm{Rc} \geq 1/r^2$ for some $r > 0$, then M's fundamental group is finite.

Proof

Let $\pi\colon \widetilde{M} \to M$ be the universal covering of M. Then $\widetilde{M}$ is a smooth manifold and π a local diffeomorphism. Pulling back g to $\widetilde{M}$ via π, $\widetilde{g} = \pi^* g$, we see that $(\widetilde{M}, \widetilde{g})$ is complete and $\mathrm{Rc}_{\widetilde{M}} \geq 1/r^2$ since π is a local isometry. By Bonnet-Myers, $\widetilde{M}$ is compact.

For every $p \in M$, since $\widetilde{M}$ is a cover, $\pi^{-1}p$ is discrete, and by compactness this means $\pi^{-1}p$ is finite (it is compact as a closed subset of compact space, and discrete compact spaces are finite). But we know that $\pi^{-1}p$ and $\pi_1(M)$ have the same cardinality, so we are finished. ∎

19.5.5 Lemma

Suppose $A\colon \mathbb{R}^{n-1} \to \mathbb{R}^{n-1}$ is an orthogonal matrix with determinant $\det(A) = (-1)^n$. Then A has a nonzero fixed point.

Proof

Since A is real, its complex eigenvalues come in pairs with their conjugate, and all of its eigenvalues have modulus 1 (by orthogonality). If n is even, then $n-1$ is odd and so there must be at least one real eigenvalue. Let $\sigma(A)$ denote A's spectrum, repeated with multiplicity, so

$$1 = \det(A) = \prod_{\lambda \in \sigma(A)} \lambda.$$

Complex eigenvalues cancel out with their conjugates, so

$$1 = \prod_{\lambda \in \sigma(A) \cap \mathbb{R}} \lambda.$$

These real eigenvalues are all ± 1, and since there is an even number of complex eigenvalues there is an odd number of real ones and thus at least one must be 1 (otherwise the product is -1).

If n is odd, then there is an even number of real eigenvalues and

$$-1 = \prod_{\lambda \in \sigma(A) \cap \mathbb{R}} \lambda.$$

So at least one must be 1. ∎

19.5.6 Theorem (Synge-Weinstein)

Let (M, g) be a complete oriented Riemannian n-manifold such that its sectional curvature is bounded below by some positive $\varepsilon > 0$ (e.g. when M is compact with strictly positive sectional curvature).

Let $\varphi\colon M \to M$ be an isometry which reverses orientation if n is odd, and preserves it when n is even. Then φ has a fixed point.

Proof

If sectional curvature is bound, then so is Ricci curvature. So by Bonnet-Myers, M is compact. So there is a $p \in M$ such that $d(p, \varphi(p))$ is minimal; assume it is nonzero.

Then by Hopf-Rinow let us take $\gamma\colon [0,\ell] \to M$ a minimizing geodesic with unit velocity from p to $\varphi(p)$. Let $p' = \gamma(t')$ for $t' \in (0,\ell)$, then

$$d(p',\varphi(p')) \leq d(p',\varphi(p)) + d(\varphi(p),\varphi(p')) = d(p',\varphi(p)) + d(p,p').$$

Since γ mninimizes length, and all these points lie on it, this is equal to $d(p,\varphi(p))$. Because p was chosen minimally this means that these must all be equalities

$$d(p',\varphi(p')) = d(p',\varphi(p)) + d(\varphi(p),\varphi(p')) = d(p,\varphi(p)).$$

$\varphi \circ \gamma|_{[0,t']}$ is a minimizing geodesic from $\varphi(p)$ to $\varphi(p')$. Thus concatenating $\varphi \circ \gamma|_{[0,t']}$ and $\gamma|_{[t',\ell]}$ gives a curve from p' to $\varphi(p')$, of length $d(p',\varphi(p)) + d(\varphi(p),\varphi(p')) = d(p',\varphi(p'))$. So it is minimizing and thus it must be a smooth geodesic. By uniqueness of geodesics, $\varphi \circ \gamma$ must be a shift of γ by ℓ, so

$$d\varphi(\dot\gamma(0)) = (\varphi \circ \gamma)'(0) = \dot\gamma(\ell).$$

Let $P\colon T_{\varphi(p)} \to T_pM$ be the parallel transport along γ (or rather its inverse). Consider then the endomorphism of T_pM, $P \circ d\varphi_p$. We have, since $\dot\gamma$ is parallel,

$$P^{-1}\dot\gamma(0) = \dot\gamma(\ell) \Longrightarrow \dot\gamma(0) = P\dot\gamma(\ell).$$

P is an orthogonal map, $d\varphi_p$ is an orthogonal map since φ is an isometry, and so $P \circ d\varphi_p$ restricts to an orthogonal endomorphism of $\dot\gamma(0)^\perp \cong \mathbb{R}^{n-1}$, call it A.

The assumption on φ regarding preserving/reversing orientation dependent on n implies $\det(A) = (-1)^n$. So there is an $e_1 \in \dot\gamma(0)^\perp$ such that $Ae_1 = e_1$ by the above lemma. Via parallel transport extend e_1 to a parallel vector field $e_1(t)$ along γ.

As $Ae_1 = e_1$ we have

$$Pd\varphi_p(e_1) = e_1 \Longrightarrow d\varphi_p(e_1) = P^{-1}e_1 = e_1(\ell),$$

as P^{-1} is parallel transport to ℓ along γ.

Now let Γ be the variation of γ given by

$$\Gamma(s,t) = \exp_{\gamma(t)}(se_1(t)).$$

Then $\beta(s) = \Gamma(s,0) = \exp_p(se_1)$ is the geodesic starting at p with initial velocity e_1. And $\Gamma(s,\ell) = \exp_{\varphi(p)}(sd\varphi_p(e_1))$ is the geodesic starting at $\varphi(p)$ with initial velocity $d\varphi_p(e_1) = d\varphi_p(\beta(0))$. So

$$\Gamma(s,\ell) = \varphi \circ \beta(s).$$

Let $E(s) = E_\Gamma(s)$, then

$$\frac{1}{2}E''(0) = -\int_0^\ell \langle e_1, e_1'' + R(\dot\gamma, e_1)\dot\gamma\rangle,$$

and since e_1 is parallel, $e_1'' = 0$. Furthermore, $|e_1| = |\dot\gamma| = 1$ so

$$\frac{1}{2}E''(0) = -\int_0^\ell \mathrm{Rm}(\dot\gamma, e_1, \dot\gamma, e_1) = -\int_0^\ell K(e_1,\dot\gamma)$$

which is then negative, as the sectional curvature is bounded below.

So there is an s with $E(s) < E(0)$, then

$$\ell L(\Gamma_{s'}) \leq E(\Gamma_{s'}) < E(\Gamma_0)E(\gamma) = \ell L(\gamma),$$

showing $L(\Gamma_{s'}) < L(\gamma)$. But $\Gamma_{s'}$ is a map curve from $\beta(s')$ to $\varphi \circ \beta(s')$ so this implies

$$d(\beta(s'),\varphi \circ \beta(s')) < d(p,\varphi(p)),$$

contradicting p's minimality. ■

The following is the classic formulation of the Synge-Weinstein theorem.

19.5.7 Corollary

Suppose (M, g) is a complete Riemannian n-manifold with sectional curvature bounded below by some positive constant.

(1) If n is even and M is orientable, then M is simply connected.

(2) If n is odd, then M is orientable.

Proof

For n even we proceed as follows. Let $\pi\colon \widetilde{M} \to M$ be the universal covering, $\widetilde{g} = \pi^* g$ so that $\widetilde{M}$ becomes a complete Riemannian n-manifold with positively-bounded sectional curvature. We also pull back M's orientation, as π is a local diffeomorphism. Furthermore $\pi_1(M)$ acts freely on $\widetilde{M}$ by isometries which preserve orientation. Since it acts freely, the only isometry in $\pi_1(M)$ with fixed points is the identity. By Synge-Weinstein, this can be the only isometry, so $\pi_1(M)$ is trivial.

Now suppose n is odd, and consider the oriented cover $\pi\colon \widehat{M} \to M$. Suppose that M is not orientable, so that $\widehat{M}$ is connected. Let $\varphi \in \mathrm{Aut}_\pi(\widehat{M})$ be an automorphism, then

$$\varphi^*(\pi^* g) = (\pi \circ \varphi)^* g = \pi^* g,$$

so automorphisms are isometries of $\pi^* g$. Since $\mathrm{Aut}_\pi(\widehat{M})$ acts freely on $\widehat{M}$, no non-trivial automorphisms can have fixed points. Furthermore $\widehat{M}$ is oriented, so by Synge-Weinstein the non-trivial automorphisms cannot reverse orientation. Since $\widehat{M}$ is connected, this means they all preserve orientation.

Thus $\widehat{M} \to M$ is an orientation-preserving normal smooth covering, which implies M is orientable (a contradiction). ■

19.6 Chern-Gauss-Bonnet

There is a classical result in differential geometry is the **Gauss-Bonnet theorem**, as follows.

19.6.1 Theorem (Gauss-Bonnet)

Let M be a compact 2-dimensional Riemannian manifold without boundary, with scalar curvature K. Then

$$\int_M K \, dV = 4\pi\chi(M)$$

where $\chi(M)$ is the **Euler characteristic** of M.

TODO: This